Frontmatter: Motivation and Conceptual Overview

M.1 The Problem

Modern theoretical physics rests on two pillars: quantum mechanics and general relativity. Each is experimentally spectacular within its domain. Together, they describe almost everything observed. Yet they cannot be unified.

The obstruction is not merely technical; it is structural. The two theories disagree about what is fundamental:

- **General relativity** treats spacetime as a dynamical field. Its geometry is not fixed; it responds to matter and energy. Time is woven into the metric. Causality is light-cone structure.
- **Quantum mechanics** treats states as vectors in a Hilbert space. Time is an external parameter. Evolution is unitary. Measurement is an operational postulate.

When one attempts to quantize gravity, the conflict becomes acute. If spacetime is quantum, then points, distances, causal order, and even time itself must fluctuate. But quantum mechanics needs a background time to define evolution. If time itself is quantized, what governs the evolution of time? This is the **problem of time**, the central conceptual barrier to quantum gravity.

Furthermore, neither theory addresses deeper foundational questions:

- Why does spacetime exist at all?
- Why is the Hilbert space the one it is?
- Why is time one-dimensional and space three-dimensional?
- Why do probabilities follow the Born rule?

M.2 Why Existing Frameworks Are Insufficient

Several major programs attempt to address these problems, but each carries assumptions that prevent full unification:

- **Perturbative string theory** typically begins with a chosen background or compactification structure. It quantizes strings moving *in* this background; it does not derive the background from something deeper. More background-independent formulations remain technically and conceptually incomplete for the purposes targeted here.
- **Loop quantum gravity** reduces background metric dependence but still begins with specific kinematical structures, such as spin-network Hilbert spaces and underlying differentiable/topological assumptions. The Hilbert space is postulated, not derived.
- **Algebraic quantum field theory (AQFT)** begins with a net of operator algebras assigned to spacetime regions. The algebra is assigned *to* spacetime; it is not derived *from* something more fundamental.

In every case, something is assumed that should be explained: spacetime, Hilbert space, or causal order.

M.3 Why a Relational Algebra Is the Natural Starting Point

If we refuse to assume spacetime, Hilbert space, causal order, or time, what is left? The answer is: **relations and constraints**.

Physics is ultimately about what is physically admissible. A configuration is physical not because it sits at a spacetime point, but because it satisfies the consistency conditions of the relational system. If we strip away all geometric and temporal assumptions, what remains is:

1. A collection of possible relational operations (an algebra).
2. Rules that distinguish admissible from inadmissible configurations (constraints).
3. A way of evaluating which configurations are realized (states).

This is the starting point of the Relational Constraint Framework (RCF).

> **Physical admissibility is not prior placement in spacetime; it is constraint compatibility.**

This is not a denial of spacetime, Hilbert space, or time. It is a refusal to assume them as primitives. They are reconstructed later, layer by layer, from the algebraic and constraint structure.

M.4 Novel Contributions of the Framework

The genuinely original ideas developed in this manuscript include:

1. **Physicality as constraint compatibility:** Reality is defined by vanishing constraint violation in a relational algebra, not by placement in spacetime.
2. **Emergence of Hilbert space:** Hilbert space is derived via the GNS construction from a physical state on the algebra, not postulated.
3. **The burden formalism:** Time dilation, gravity, and black-hole structure are derived from a single scalar quantity — constraint burden — that measures the cost of maintaining admissibility.
4. **Causal dependency replacing background time:** Causality is an internal strict partial order derived from constraint closure, not an external temporal parameter.
5. **The Two-Link Principle:** Causal links and relational correlation links are structurally distinct. This prevents space from collapsing into causality and vice versa.
6. **Z-envariance derivation of the Born rule:** Probability is not postulated; it is derived as the sectorwise reading of constraint compatibility.
7. **Dimensional closure:** $D=3$ is selected because existence, position, and direction require exactly three independent inference channels. Higher dimensions create mirror degeneracy, while time remains uniquely one-dimensional.
8. **Pressure saturation and singularity avoidance:** Black holes are reframed as unreconciled relational sectors where pressure saturates at an absolute structural ceiling imposed by the fundamental length scale ℓ_0, projecting the classical infinity onto a 2D boundary rather than a point singularity.

Frontmatter: Reader's Map

The manuscript proceeds hierarchically. Each layer prepares the ground for the next, ensuring that no structure is smuggled in before it is derived.

Section	Content	Key Result	Claim Status
0	**Algebraic Foundation**	Physical states, GNS representation, master constraint	Definitions and conditional theorems
1	**Causal Foundation**	Zero-preserving observables, causal partial order, open extension, speed limit	Definitions, conditional theorems, regime assumptions
2	**Emergent Space**	Correlation metric, emergent points, $D=3$ dimensional closure	Conditional metricity; dimensional closure is a conditional theorem target
3	**Emergent Time**	Burden-clock suppression, proper time, lapse	Conditional theorems; clock factor is a model ansatz
4	**Fields and Particles**	Stable modes, particle identity, interactions	Toy-level / conditional
5	**Gravity**	Burden tensor, ADM dictionary, Einstein-like closure	Conditional reduction theorem
6	**Black Holes**	Pressure saturation, dimensional suppression, boundary recovery	Open theorem targets; singularity avoidance is structurally derived
7	**Probability**	Records, classicality, Born rule	Conditional theorem target; decoherence threshold is quantified
8	**Cosmology**	Open extension, expansion, dark-sector analogues	Quarantined phenomenology
9	**Audit**	Theorem status, open targets, quarantined claims	Consolidated map

Each section ends with guardrails preventing overinterpretation. The framework is built to be theorem-safe: it says what it proves, and no more.

Section 0 — The Algebraic Foundation

0.0 Purpose of the Foundation

0.0.1 The Foundational Question

The purpose of this section is to specify the minimal mathematical structure required for the theory before any appeal is made to spacetime, geometry, matter fields, particles, observers, measurement, or external time.

The framework begins from the claim that physical admissibility is not primarily a matter of locating objects in a pre-existing background, but of satisfying a system of relational constraints. What is fundamental is therefore not a manifold, a metric, a field on spacetime, or a Hilbert-space state vector taken in isolation, but a constrained relational structure whose admissible configurations are those in which inconsistency vanishes.

0.0.2 The Central Slogan

The central foundational commitment is:

> **Physicality is constraint compatibility, not prior placement in spacetime.**

This statement must be understood carefully. It does not deny spacetime, fields, clocks, particles, or geometry. Rather, it denies that these are primitive. They are reconstructed later from the behaviour of physical states, zero-preserving observables, correlation structure, causal dependency, burden, and coarse-grained effective descriptions.

0.0.3 Why the Foundation Must Be Algebraic

If one begins by assuming a spacetime manifold, then causal order, distance, locality, field propagation, and temporal duration are already built into the background. In that case, the theory would no longer explain the emergence of those structures; it would merely redescribe them.

The foundation therefore deliberately avoids assuming:

- a spacetime manifold,
- a metric,
- a topology,
- a causal order,
- a Hilbert space,
- a Hamiltonian,
- a Lagrangian,
- a probability measure,
- a classical configuration space,
- a background time parameter.

Instead, the primitive structure is a **unital $(*)$-algebra** $\mathcal{A}$, whose elements represent possible relational operations or quantities. Physical admissibility is then imposed by a **constraint ideal** $\mathcal{I}_C$ inside $\mathcal{A}$, and physical states are positive linear functionals that annihilate the constraints in the appropriate quadratic sense.

0.0.4 The Role of This Section

This section has five roles.

First, it prevents a category error. If one begins with a spacetime manifold, then causal order, distance, and locality are already assumed. The foundation establishes the admissible relational substrate from which those notions may later emerge.

Second, it clarifies the meaning of "zero". The zero condition is not a claim that reality is empty. It is a consistency condition. To say that the master constraint vanishes is to say that relevant constraint violations vanish. It is a statement about admissibility, not nothingness.

Third, it distinguishes the primitive algebraic layer from its later representations. A Hilbert space arises through a GNS representation. A master constraint operator is introduced as a representation-level packaging. A classical master functional appears as an effective description. None of these is taken as fundamental.

Fourth, it separates physical admissibility from dynamics. The master constraint is not an equation of motion. It does not describe evolution with respect to external time. It selects the physical sector. Temporal notions are introduced later through causal dependency, burden, and clock-rate suppression.

Fifth, it makes later emergence claims theorem-safe. Claims about emergent space, time, particles, gravity, and cosmology must not be smuggled into the starting assumptions. They must be built in layers.

0.0.5 The Foundational Hierarchy

The foundational construction proceeds in the following order:

1. The relational algebra $\mathcal{A}$ is defined.
2. The primitive constraints $\{\hat{C}_\alpha\}$ and their generated ideal $\mathcal{I}_C$ are introduced.
3. Physical states are defined as positive linear functionals satisfying the quadratic constraint-vanishing condition.
4. The GNS representation is used to show how Hilbert space and represented constraint operators arise from a physical state.
5. The master constraint operator is introduced only as a representation-level packaging of the primitive constraints.
6. The effective classical master functional is introduced as a derived, coarse-grained description.
7. The section records the guardrails needed to avoid overinterpreting the zero condition, the Hilbert-space representation, or the master constraint.

The foundational commitment can therefore be stated as:

> **A physical structure is an admissible relational structure whose constraint violations vanish in the physical state.**

0.1 The Relational Algebra

0.1.1 Purpose of the Relational Algebra

The first object required by the framework is a mathematical structure capable of carrying primitive relations before those relations are interpreted as spatial, temporal, material, causal, or observational.

Since the theory does not begin with a spacetime manifold, it cannot define its basic objects as fields on spacetime points. Since it does not begin with an external time parameter, it cannot define its basic objects as time-evolving configurations. Since it does not begin with a fixed Hilbert space, it cannot treat vectors in that Hilbert space as the primitive layer.

The starting point must therefore be more abstract. It must be able to describe relational quantities, allow those quantities to compose, allow a notion of reversal or adjoint, and support constraints. A unital $(*)$-algebra provides precisely this minimal structure.

The relational algebra is not introduced as a space of already physical observables. At this stage, it is only the kinematic carrier of possible relational expressions. Some of its elements will later be ruled physical, some will be identified modulo constraints, and some will become observables only after they are shown to preserve the physical sector.

> **The relational algebra contains possible relational expressions before physical admissibility is imposed.**

0.1.2 Definition — Relational $(*)$-Algebra

Definition 0.1. A relational algebra is a unital complex $(*)$-algebra $(\mathcal{A}, +, \cdot, \dagger, \mathbf{1})$, where:
1. $\mathcal{A}$ is a complex associative algebra over $\mathbb{C}$.
2. $\mathbf{1} \in \mathcal{A}$ is a multiplicative identity satisfying $\mathbf{1}A = A\mathbf{1} = A$ for all $A \in \mathcal{A}$.
3. $\dagger: \mathcal{A} \to \mathcal{A}$ is an involution satisfying, for all $A, B \in \mathcal{A}$ and all $\lambda, \mu \in \mathbb{C}$:
   - $(A^\dagger)^\dagger = A$,
   - $(AB)^\dagger = B^\dagger A^\dagger$,
   - $(\lambda A + \mu B)^\dagger = \bar{\lambda}A^\dagger + \bar{\mu}B^\dagger$.

0.1.3 Minimal Assumptions

Assumption 0.1 (Non-trivial algebra). $\mathcal{A} \neq \{0\}$.

Assumption 0.2 (No additional structure). At this stage, the framework does not assume:
- any norm,
- any topology,
- any completeness,
- any Hilbert-space representation,
- commutativity,
- a spacetime manifold,
- a metric,
- a causal order,
- a Hamiltonian,

- a Lagrangian,
- a probability measure,
- a classical configuration space,
- a background time parameter.

The only basic non-triviality condition is that the algebra is not the zero algebra.

0.1.4 Interpretation of Algebra Elements

An element $A \in \mathcal{A}$ should be understood as a **possible relational expression**. It may encode a relation between primitive degrees of freedom, a composite relation built from simpler relations, or a candidate operation acting on a relational structure.

The product AB represents **composition**: the relational expression B is followed, combined, or algebraically composed with the relational expression A. The ordering matters in general, since the algebra is not assumed to be commutative:
$$ AB \neq BA \quad \text{in general}. $$

This non-commutativity is not yet a quantum postulate. It is a structural allowance: the order in which relational operations are composed may matter.

The adjoint $A^\dagger$ provides a formal **reversal, dual, or compatibility operation**. It allows the theory to form positive expressions of the type $A^\dagger A$, which are essential for defining positive states and quadratic constraint-vanishing conditions.

The unit $\mathbf{1}$ represents the **trivial relational operation**. It is the identity element of the algebraic language.

0.1.5 Why a $(*)$-Algebra Is Used

The use of a unital $(*)$-algebra is motivated by four requirements.

First, the framework needs composition. If two relational expressions are meaningful, then their composition should also be meaningful. This requires a multiplication operation.

Second, the framework needs linearity. Physical states will later be defined as linear functionals on the algebra. Linear combinations of relational expressions must therefore remain within the same formal structure.

Third, the framework needs positivity. Constraint satisfaction will be expressed through terms such as $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. Such expressions require an adjoint operation and a notion of positive quadratic form.

Fourth, the framework needs representation flexibility. A $(*)$-algebra can later be represented on a Hilbert space, but it does not require Hilbert space to be primitive.

0.1.6 Immediate Consequences

Lemma 0.1 (Closure properties). For all $A, B \in \mathcal{A}$:
1. $AB \in \mathcal{A}$,
2. $A^\dagger \in \mathcal{A}$,
3. $A^\dagger A \in \mathcal{A}$.

Proof. Both statements follow directly from the definition of algebra multiplication and the involution map. $\square$

Lemma 0.2 (Adjoint identities). For all $A, B \in \mathcal{A}$ and $\lambda \in \mathbb{C}$:
1. $(A^\dagger B^\dagger)^\dagger = BA$,
2. $(\mathbf{1})^\dagger = \mathbf{1}$.

Proof.
1. Using involution and anti-multiplicativity: $(A^\dagger B^\dagger)^\dagger = (B^\dagger)^\dagger (A^\dagger)^\dagger = BA$.
2. Since $\mathbf{1}A = A$, applying adjoint gives $(\mathbf{1}A)^\dagger = A^\dagger \Rightarrow A^\dagger \mathbf{1}^\dagger = A^\dagger \Rightarrow \mathbf{1}^\dagger = \mathbf{1}$. $\square$

Lemma 0.3 (Positive expressions are algebraically available). For every $A \in \mathcal{A}$, the quadratic expression $A^\dagger A$ is a well-defined element of $\mathcal{A}$.

Proof. By Lemma 0.1, $A^\dagger \in \mathcal{A}$ and $A^\dagger A \in \mathcal{A}$. $\square$

This lemma is simple but foundational. It guarantees that the algebra can express quantities of the same form as $\hat{C}_\alpha^\dagger \hat{C}_\alpha$, which will be used to define physical states.

0.1.7 What Is Not Assumed

The relational algebra is deliberately minimal. At this stage:
- There is no metric, no space, no notion of locality.
- There is no dynamics, no time, no evolution law.
- There is no Hilbert space and no representation.
- The adjoint operation does not by itself define time reversal, measurement, Hermitian observables, or physical duality. It is an algebraic operation needed for positivity and constraint formulation.
- Membership in $\mathcal{A}$ does not imply physical admissibility.

0.1.8 Relation to Later Layers

This algebra will serve as the primitive carrier structure for the entire framework:
- Section 0.2: constraints will be defined as elements of $\mathcal{A}$.
- Section 0.3: states will be linear functionals on $\mathcal{A}$.
- Section 1: observables will be defined via constraints and commutation.

- Later sections: geometry, time, and physics will be derived from states and correlations on $\mathcal{A}$.

0.2 Constraint Operators and the Constraint Ideal

0.2.1 Purpose of the Constraint Layer

The relational algebra $\mathcal{A}$ represents the full set of possible relational expressions. However, not all algebraic configurations correspond to physically admissible structures. We therefore introduce constraints to encode:
- consistency conditions,
- admissibility requirements,
- the structural rules governing physical configurations.

These constraints are internal to the algebra and do not refer to external structures (such as a background spacetime manifold or an external time parameter).

0.2.2 Definition — Constraint Operators

Definition 0.2 (Constraint Operators). A constraint operator is an element $\hat{C}_\alpha \in \mathcal{A}$ labelled by an index $\alpha \in I$, where I is an index set. The full family of primitive constraints is written as $\{\hat{C}_\alpha\}_{\alpha \in I} \subset \mathcal{A}$.

Each $\hat{C}_\alpha$ represents a primitive algebraic condition that must vanish, in the appropriate physical sense, for a relational structure to be admissible.

Notation. The hat notation is used to remind us that these are algebraic constraint elements which may later be represented as operators. Fundamentally, $\hat{C}_\alpha \in \mathcal{A}$, not necessarily $\hat{C}_\alpha \in \mathcal{B}(\mathscr{H})$ for some pre-existing Hilbert space.

0.2.3 Why Constraints Live Inside the Algebra

The constraints must be elements of $\mathcal{A}$ because the framework requires consistency to be expressed internally. If constraints were imposed from outside the algebra, then physical admissibility would depend on an additional background structure not contained in the relational system itself.

By taking $\hat{C}_\alpha \in \mathcal{A}$, the theory ensures that constraints can be composed with other relational expressions, adjointed, inserted into state evaluations, and represented through the same machinery as all other algebraic elements.

This allows the following expressions to be meaningful within the theory:
$$ A\hat{C}_\alpha, \quad \hat{C}_\alpha B, \quad A\hat{C}_\alpha B, \quad
\hat{C}_\alpha^\dagger, \quad \hat{C}_\alpha^\dagger \hat{C}_\alpha, $$
for arbitrary $A, B \in \mathcal{A}$.

The final expression, $\hat{C}_\alpha^\dagger \hat{C}_\alpha$, will become especially important because physical states will be required to annihilate this positive quadratic form.

0.2.4 Primitive Constraints Versus Their Consequences

It is not enough to specify only the primitive list $\{\hat{C}_\alpha\}_{\alpha \in I}$. If a constraint $\hat{C}_\alpha$ represents an inadmissible direction or inconsistency, then any algebraic expression containing that constraint as a factor should also be regarded as belonging to the inconsistency sector. For example, if $\hat{C}_\alpha$ is constrained, then expressions such as $A\hat{C}_\alpha B$ should also be constrained for arbitrary relational expressions $A, B \in \mathcal{A}$.

This is the reason for introducing an ideal rather than merely a set of generators. The primitive constraints are the starting points. The ideal generated by them is the full algebraic closure of all their consequences under addition, multiplication by arbitrary algebra elements, and adjoint.

> **Constraint operators are primitive generators; the constraint ideal is their full algebraic consequence class.**

0.2.5 Definition — Constraint Ideal

Definition 0.3 (Constraint Ideal). The constraint ideal generated by $\{\hat{C}_\alpha\}_{\alpha \in I}$ is the smallest two-sided $(*)$-ideal of $\mathcal{A}$ containing all $\hat{C}_\alpha$. It is denoted:
$$ \mathcal{I}_C = \langle \hat{C}_\alpha : \alpha \in I \rangle. $$

Equivalently, $\mathcal{I}_C$ is the smallest subset of $\mathcal{A}$ satisfying:
1. **Contains all primitive constraints:** $\hat{C}_\alpha \in \mathcal{I}_C$ for all $\alpha \in I$.
2. **Closed under finite linear combinations:** If $X, Y \in \mathcal{I}_C$ and $\lambda, \mu \in \mathbb{C}$, then $\lambda X + \mu Y \in \mathcal{I}_C$.
3. **Closed under left multiplication:** If $X \in \mathcal{I}_C$ and $A \in \mathcal{A}$, then $AX \in \mathcal{I}_C$.
4. **Closed under right multiplication:** If $X \in \mathcal{I}_C$ and $A \in \mathcal{A}$, then $XA \in \mathcal{I}_C$.
5. **Closed under adjoint:** If $X \in \mathcal{I}_C$, then $X^\dagger \in \mathcal{I}_C$.

Because it is both left and right closed, $\mathcal{I}_C$ is a two-sided ideal. Because it is also closed under the involution $\dagger$, it is a $(*)$-ideal.

0.2.6 Explicit Generated Form

Elements of the constraint ideal may be written schematically as finite sums of the form:
$$ X = \sum_{k=1}^{n} A_k \hat{C}_{\alpha_k} B_k, $$
where $A_k, B_k \in \mathcal{A}$ and $\alpha_k \in I$.

Because the ideal is required to be $(*)$-closed, it also contains adjoints of such expressions. If $X = \sum_k A_k \hat{C}_{\alpha_k} B_k$, then:
$$ X^\dagger = \sum_{k=1}^{n} B_k^\dagger \hat{C}_{\alpha_k}^\dagger A_k^\dagger $$
also belongs to $\mathcal{I}_C$.

More explicitly, one may write:
$$ \mathcal{I}_C = \left\{ \sum_{k=1}^{n} A_k \hat{C}_{\alpha_k} B_k + \sum_{\ell=1}^{m} D_\ell \hat{C}_{\beta_\ell}^\dagger E_\ell \;\middle|\; A_k, B_k, D_\ell, E_\ell \in \mathcal{A} \right\}, $$
with the understanding that this represents the finite algebraic span of the primitive constraints and their adjoints under arbitrary left and right multiplication.

0.2.7 Structural Properties

Lemma 0.4 (Ideal properties). $\mathcal{I}_C$ satisfies:
1. **Linear subspace:** $X, Y \in \mathcal{I}_C \Rightarrow \lambda X + \mu Y \in \mathcal{I}_C$ for all $\lambda, \mu \in \mathbb{C}$.
2. **Two-sided closure:** $AX, XA \in \mathcal{I}_C$ for all $A \in \mathcal{A}$.
3. **$(*)$-closure:** $X \in \mathcal{I}_C \Rightarrow X^\dagger \in \mathcal{I}_C$.

Proof. All properties follow directly from construction: linearity from finite sums, two-sided closure from definition, and $(*)$-closure imposed explicitly. $\square$

Lemma 0.5 (Minimal generating property). $\mathcal{I}_C$ is the smallest two-sided $(*)$-ideal in $\mathcal{A}$ containing all $\hat{C}_\alpha$.

Proof. Let $\mathcal{J} \subset \mathcal{A}$ be any two-sided $(*)$-ideal with $\hat{C}_\alpha \in \mathcal{J}$ for all α. Then $A\hat{C}_\alpha B \in \mathcal{J}$ for all $A, B \in \mathcal{A}$, any finite sum of such elements lies in $\mathcal{J}$, and adjoints also lie in $\mathcal{J}$. Hence $\mathcal{I}_C \subset \mathcal{J}$. Therefore $\mathcal{I}_C$ is minimal. $\square$

Lemma 0.6 (Adjoint consistency). If $\hat{C}_\alpha \in \mathcal{I}_C$, then $\hat{C}_\alpha^\dagger \in \mathcal{I}_C$.

Proof. Follows directly from the $(*)$-closure property of the ideal. $\square$

Lemma 0.7 (Constraint consequences remain constrained). If $X \in \mathcal{I}_C$, then for all $A, B \in \mathcal{A}$, the composite expression AXB also belongs to $\mathcal{I}_C$.

Proof. Since $\mathcal{I}_C$ is a left ideal, $AX \in \mathcal{I}_C$. Since $\mathcal{I}_C$ is also a right ideal, $(AX)B \in \mathcal{I}_C$. Therefore $AXB \in \mathcal{I}_C$. $\square$

0.2.8 Non-Degeneracy of the Constraint Ideal

The constraint ideal must not equal the entire algebra. We impose:

Assumption 0.3 (Non-degeneracy). $\mathcal{I}_C \neq \mathcal{A}$.

If $\mathcal{I}_C = \mathcal{A}$, then all elements are "constraint violations," and no non-trivial physical states can exist.

In particular, if a physical state ω were required to annihilate the constraint ideal, then $\omega(\mathbf{1}) = 0$ would follow. For a normalised physical state, however, one expects $\omega(\mathbf{1}) = 1$. Thus, if $\mathcal{I}_C = \mathcal{A}$, normalised physical states cannot exist.

0.2.9 Proper Constraint Systems

A constraint system is called **proper** if its generated constraint ideal is proper:
$$ \mathcal{I}_C \subsetneq \mathcal{A}. $$
Equivalently, a proper constraint system does not generate the identity element:
$$ \mathbf{1} \notin \mathcal{I}_C. $$

This does not yet prove that physical states exist. It only rules out the immediate obstruction in which the constraints collapse the entire algebra into inconsistency.

> **$\mathbf{1} \notin \mathcal{I}_C$ is necessary for a non-trivial physical sector, but not by itself sufficient.**

Existence of physical states requires an additional positivity-compatible state construction, introduced in the next subsection.

0.2.10 The Constraint Ideal as the Algebraic Inconsistency Sector

The ideal $\mathcal{I}_C$ should be interpreted as the algebraic sector of constraint-generated inconsistency.

This interpretation must be handled carefully. It does not mean that every element of $\mathcal{I}_C$ is a physical object with negative existence or destructive cancellation. It means that these elements are algebraically generated from violations of primitive consistency requirements.

In this sense, $X \in \mathcal{I}_C$ means: X belongs to the constraint-generated inconsistency class.

The physical state condition then requires this inconsistency structure to vanish in the appropriate expectation-value sense.

0.2.11 Relation to the Constraint Ideal and Physical Equivalence

Although the quotient is not taken as the primitive definition of physicality, the constraint ideal naturally suggests an equivalence relation. Two algebra elements $A, B \in \mathcal{A}$ may be considered constraint-equivalent if:

$$ A \sim_C B \quad \Longleftrightarrow \quad A - B \in \mathcal{I}_C. $$

In that case, A and B differ only by an element of the inconsistency sector. This relation is useful later when defining reduced observables, but at the foundational level it remains secondary to the physical state condition.

0.2.12 Guardrails for This Subsection

1. **No geometry assumed.** Constraints are purely algebraic—not spatial or geometric.
2. **No probabilistic meaning yet.** Constraints encode consistency, not probabilities.
3. **No dynamics.** They do not generate evolution; they define admissibility.
4. **No quotient taken yet.** We do not yet identify the physical algebra as $\mathcal{A}/\mathcal{I}_C$; this will be handled later.
5. **No claim of automatic equivalence.** The bridge between configuration-space and algebraic formulations is conditional on a faithful representation-linked model.

0.3 Physical States and Positivity

0.3.1 Purpose of the State Layer

The relational algebra $\mathcal{A}$ provides the primitive language of possible relational expressions. The constraint ideal $\mathcal{I}_C$ identifies the algebraic sector generated by inconsistency. But neither of these structures yet specifies what it means for a relational structure to be physically realised.

To make that distinction, the framework introduces states.

A state is not first understood as a vector in a Hilbert space. Nor is it initially a point in a classical phase space, a field configuration, or a probability distribution over a background spacetime. At the foundational level, a state is a functional on the relational algebra: it assigns values to algebraic expressions in a way compatible with linearity and positivity.

> **A physical state is a positive evaluation of relational structure in which constraint violation vanishes.**

This subsection defines the state concept, explains why positivity is required, and formulates the physical state condition.

0.3.2 Linear Functionals on the Relational Algebra

Definition 0.4 (Linear Functional).
A linear functional on $\mathcal{A}$ is a map $\omega: \mathcal{A} \to \mathbb{C}$ such that for all $A, B \in \mathcal{A}$ and all $\lambda, \mu \in \mathbb{C}$:
$$ \omega(\lambda A + \mu B) = \lambda \omega(A) + \mu \omega(B). $$

The value $\omega(A)$ is interpreted as the state-evaluation of the relational expression A. At this stage, $\omega(A)$ should not automatically be interpreted as a measured expectation value in laboratory spacetime. That interpretation may become appropriate after additional representational and operational structure is introduced. Foundationally, ω is simply the object that evaluates relational expressions.

0.3.3 Why States Are Functionals Rather Than Vectors

The framework does not begin with a Hilbert space. Therefore, it cannot define physical states first as vectors $|\Psi\rangle \in \mathscr{H}$. Instead, the Hilbert-space picture must be derived later from the algebraic state.

This ordering is important. If one starts with a Hilbert space, one has already selected a representation of the relational algebra. That may be useful later, but it is too specific for the primitive layer. The algebraic state ω is more general: it can generate a Hilbert-space representation, but it is not itself defined by one.

The foundational direction is therefore:
$$ \omega \quad \longrightarrow \quad (\mathscr{H}_\omega, \pi_\omega, \Omega_\omega), $$
not
$$ \mathscr{H} \quad \longrightarrow \quad \omega. $$
The state comes before the represented Hilbert space.

0.3.4 Positivity

Definition 0.5 (Positive Functional).
A linear functional $\omega: \mathcal{A} \to \mathbb{C}$ is called positive if:
$$ \omega(A^\dagger A) \geq 0 \qquad \forall A \in \mathcal{A}. $$

Here the expression $A^\dagger A$ is well-defined because $\mathcal{A}$ is a $(*)$-algebra.

The positivity condition means that every algebraic square has a non-negative state-evaluation. This is the algebraic analogue of the familiar fact that squared magnitudes should not have negative expectation. However, this interpretation should be used only as intuition. The formal role of positivity is deeper: it makes the state compatible with a Hilbert-space representation and allows the framework to use a Cauchy–Schwarz inequality.

0.3.5 Normalisation

A positive linear functional may also be normalised by requiring:
$$ \omega(\mathbf{1}) = 1. $$
When this condition holds, ω is a normalised algebraic state.

For some purely structural arguments, normalisation is not essential. A non-zero positive functional satisfying the physical constraint condition may be enough. However, when ω is to be interpreted probabilistically or as an expectation-value state, normalisation is natural.

Thus, one may distinguish:
- **Positive linear functional:** $\omega(A^\dagger A) \geq 0$ for all A.
- **Normalised positive state:** $\omega(A^\dagger A) \geq 0$ and $\omega(\mathbf{1}) = 1$.

0.3.6 Definition — Algebraic State

Definition 0.6 (Algebraic State).
An algebraic state on $\mathcal{A}$ is a positive linear functional $\omega: \mathcal{A} \to \mathbb{C}$ satisfying:
$$ \omega(A^\dagger A) \geq 0 \qquad \forall A \in \mathcal{A}. $$
If additionally $\omega(\mathbf{1}) = 1$, then ω is called a normalised algebraic state.

The set of algebraic states on $\mathcal{A}$ may be denoted $\mathfrak{S}(\mathcal{A})$. The physical states will form a distinguished subset of this state space.

0.3.7 The Physical State Condition

A state is physical only if it is compatible with the constraints. Since the primitive constraints are $\hat{C}_\alpha \in \mathcal{A}$, the most direct condition might appear to be:
$$ \omega(\hat{C}_\alpha) = 0. $$
However, this linear condition alone is too weak. It can allow cancellations at the level of linear expectation values without ensuring that the constraint itself vanishes in the state.

The stronger and more stable condition is quadratic:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha \in I. $$

This is the foundational physical state condition. It says that each primitive constraint has zero positive square in the state. Since $\hat{C}_\alpha^\dagger \hat{C}_\alpha$ is an algebraic square, positivity ensures:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \geq 0. $$
Requiring it to be zero therefore means that the constraint has no residual positive magnitude in the state.

0.3.8 Definition — Physical State

Definition 0.7 (Physical State).
A physical state of the relational constraint framework is a positive linear functional $\omega: \mathcal{A} \to \mathbb{C}$ such that:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha \in I. $$
If normalisation is required, one additionally imposes $\omega(\mathbf{1}) = 1$.

The set of physical states may be denoted $\mathfrak{S}_{\mathrm{phys}}(\mathcal{A}, \mathcal{I}_C)$ or simply $\mathfrak{S}_{\mathrm{phys}}$.

Thus:

$$ \mathfrak{S}_{\mathrm{phys}} = \left\{ \omega \in \mathfrak{S}(\mathcal{A}) : \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \ \forall \alpha \right\}. $$

This definition is the first precise statement of physical admissibility in the framework.

0.3.9 Why Quadratic Vanishing is Stronger Than Linear Vanishing

The condition $\omega(\hat{C}_\alpha) = 0$ does not by itself guarantee that the constraint is absent in the state. It only says that the average value of the constraint vanishes. In many algebraic settings, a quantity can have zero expectation while its square has non-zero expectation.

By contrast:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 $$
states that the positive magnitude of the constraint vanishes. Because positivity ensures $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \geq 0$, zero is the minimal possible value. The constraint has no residual quadratic weight in the state.

This is why the physical state condition uses $\hat{C}_\alpha^\dagger \hat{C}_\alpha$ rather than only $\hat{C}_\alpha$.

> **A physical state does not merely average constraint violation to zero; it eliminates constraint violation in positive norm.**

0.3.10 Positivity and Cauchy–Schwarz

A crucial consequence of positivity is that it induces a Cauchy–Schwarz inequality.

Given a positive linear functional $\omega: \mathcal{A} \to \mathbb{C}$, define the sesquilinear form:
$$ \langle A, B \rangle_\omega := \omega(A^\dagger B). $$
This form is positive semidefinite because $\langle A, A \rangle_\omega = \omega(A^\dagger A) \geq 0$.

Therefore, it satisfies the Cauchy–Schwarz inequality:
$$ |\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A) \, \omega(B^\dagger B). $$

This inequality is the technical reason why quadratic constraint vanishing is powerful. It allows one to show that if a constraint has zero quadratic weight, then insertions of that constraint vanish in state evaluations.

0.3.11 Lemma — Null Constraint Annihilation

Lemma 0.8 (Null Constraint Annihilation).
Let $\omega: \mathcal{A} \to \mathbb{C}$ be a positive linear functional. If $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ for some constraint $\hat{C}_\alpha$, then:
$$ \omega(\hat{C}_\alpha^\dagger A) = 0 \qquad \forall A \in \mathcal{A}, $$

and
$$ \omega(A^\dagger \hat{C}_\alpha) = 0 \qquad \forall A \in \mathcal{A}. $$

Proof.
Apply the Cauchy–Schwarz inequality to $X = \hat{C}_\alpha$ and $Y = A$. Then:
$$ |\omega(\hat{C}_\alpha^\dagger A)|^2 \leq \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \, \omega(A^\dagger A). $$
By assumption, $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. Therefore:
$$ |\omega(\hat{C}_\alpha^\dagger A)|^2 \leq 0. $$
Since the left-hand side is non-negative, it must be zero:
$$ \omega(\hat{C}_\alpha^\dagger A) = 0. $$

Similarly, applying Cauchy–Schwarz to $X = A$ and $Y = \hat{C}_\alpha$:
$$ |\omega(A^\dagger \hat{C}_\alpha)|^2 \leq \omega(A^\dagger A) \, \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0. $$
Therefore:
$$ \omega(A^\dagger \hat{C}_\alpha) = 0. \qquad \blacksquare $$

0.3.12 Interpretation of Zero in the Physical State Condition

The condition $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ should not be interpreted as saying that the universe is empty, that all physical quantities vanish, or that every structure is cancelled by an opposite structure.

It says only that the positive square of each constraint has zero state-evaluation. In other words:

> **Zero means zero constraint violation, not zero existence.**

A physical state may support non-trivial relational structure, non-zero observable values, correlations, causal dependencies, emergent geometry, and effective dynamics. What vanishes is the inconsistency encoded by the primitive constraints.

0.3.13 Existence of Physical States

The properness condition $\mathcal{I}_C \subsetneq \mathcal{A}$ prevents immediate triviality, but it does not by itself guarantee that physical states exist.

The existence of a positive functional satisfying $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ for all α is an additional mathematical question. In concrete models, it may require construction, representation theory, compactness arguments, quotient positivity, or other methods.

Proposition 0.1 (Sufficient Condition for Physical-State Existence).
If the reduced algebra $\mathcal{A}/\mathcal{I}_C$ admits a positive linear functional $\tilde{\omega}$, then the pullback $\omega = \tilde{\omega} \circ q$ (where $q: \mathcal{A} \to \mathcal{A}/\mathcal{I}_C$ is the quotient map) is a physical state of $\mathcal{A}$.

Proof.
Positivity is inherited by pullback. Since every constraint element maps to zero in the quotient, $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = \tilde{\omega}(q(\hat{C}_\alpha^\dagger \hat{C}_\alpha)) = \tilde{\omega}(0) = 0$. $\blacksquare$

Therefore, the foundational definition should be accompanied by the following guardrail:

> **$\mathcal{I}_C \subsetneq \mathcal{A}$ is necessary for non-triviality, but physical states must still be shown or assumed to exist.**

0.3.14 The Physical Sector as a State-Defined Object

The physical sector is first defined not as a subspace of a Hilbert space, but as a subset of the algebraic state space:
$$ \mathfrak{S}_{\mathrm{phys}} \subseteq \mathfrak{S}(\mathcal{A}). $$
Explicitly:
$$ \mathfrak{S}_{\mathrm{phys}} = \left\{ \omega: \mathcal{A} \to \mathbb{C} \;\middle|\; \omega \text{ positive and } \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \ \forall \alpha \right\}. $$

Only after a state $\omega \in \mathfrak{S}_{\mathrm{phys}}$ is chosen does one construct a represented Hilbert-space physical sector, such as $\mathscr{H}_{\mathrm{phys}}^\omega$.

Thus, the order remains:
$$ \text{physical state} \quad \longrightarrow \quad \text{represented physical Hilbert space}. $$

0.3.15 Guardrails for This Subsection

1. **No Hilbert space yet.** States are defined abstractly—no vectors or operators are assumed.
2. **Positivity is not optional.** It is required for quadratic constraint conditions, Cauchy–Schwarz, and the later GNS representation.
3. **The physical state condition is quadratic:** $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$, not merely linear: $\omega(\hat{C}_\alpha) = 0$.
4. **Zero means vanishing of constraint violation, not literal nothingness.**
5. **Properness of the constraint ideal does not automatically guarantee the existence of physical states.**
6. **Primitive constraint annihilation is local.** Lemma 0.8 proves that the primitive constraints and their immediate adjoints annihilate the state. The extension of this annihilation property to the *entire* constraint ideal $\mathcal{I}_C$ requires a stronger condition (ideal-null physicality), which is formally introduced and distinguished in Section 0.4.
7. **The physical sector is first a state-defined object, not a pre-existing Hilbert subspace.**

0.4 GNS Representation and the Derived Master Constraint

0.4.1 Purpose of the Representation Layer

The preceding subsections established the primitive algebraic foundation of the framework: a relational algebra $\mathcal{A}$, a constraint ideal $\mathcal{I}_C$, and physical states ω satisfying the quadratic constraint-vanishing condition.

At this stage, no Hilbert space has been assumed. This is intentional. The framework does not begin with vectors, wavefunctions, operators on a pre-existing Hilbert space, or a background quantum state space. Instead, it begins with algebraic relations and physical state functionals.

The purpose of this subsection is to show how a Hilbert-space representation arises from a physical state rather than being imposed at the beginning. This is done through the Gelfand–Naimark–Segal (GNS) construction.

> **Hilbert space is generated by a physical state; it is not primitive.**

The GNS construction plays a crucial role in the framework. It allows the algebraic foundation to connect with the familiar language of vectors, operators, cyclic states, kernels, and constraint annihilation, while preserving the deeper claim that the algebra-state structure comes first.

This subsection also introduces the **master constraint operator** as a representation-level packaging of the primitive constraints. The master constraint is not primitive. It is a derived object that appears only after a state has been chosen and a representation has been constructed.

0.4.2 Why a Representation Is Needed

The algebraic formulation is the most primitive level of the framework, but many later constructions require represented objects. For example, to speak about a physical Hilbert space, a master constraint operator, kernels of operators, or vectors annihilated by constraints, one needs a representation of $\mathcal{A}$ on some Hilbert space.

However, the representation must not be chosen arbitrarily at the foundation. If one simply postulates a Hilbert space $\mathscr{H}$ and a representation $\pi: \mathcal{A} \to \mathcal{B}(\mathscr{H})$, then the primitive theory would already depend on a selected representation. That would weaken the algebra-first structure.

The GNS construction avoids this problem. It says that once a positive state ω is given, the Hilbert space associated with that state can be constructed from the algebra itself.

Thus the direction is:
$$ (\mathcal{A}, \omega) \quad \longrightarrow \quad (\mathscr{H}_\omega, \pi_\omega, \Omega_\omega). $$

The state chooses the representation. The representation does not define the state at the primitive level.

0.4.3 The State-Induced Inner Product

Let $\omega: \mathcal{A} \to \mathbb{C}$ be a positive linear functional.

Define a sesquilinear form on $\mathcal{A}$ by:
$$ \langle A, B \rangle_\omega := \omega(A^\dagger B), \qquad A, B \in \mathcal{A}. $$

This form is linear in its second argument and conjugate-linear in its first argument, following the usual mathematical convention for Hilbert-space inner products.

Because ω is positive:
$$ \langle A, A \rangle_\omega = \omega(A^\dagger A) \geq 0. $$
Therefore, $\langle \cdot, \cdot \rangle_\omega$ is a positive semidefinite sesquilinear form.

It may fail to be positive definite. That is, there may exist non-zero algebra elements $A \in \mathcal{A}$ such that:
$$ \omega(A^\dagger A) = 0. $$
Such elements have zero length with respect to the state-induced form. They must be quotiented out before a Hilbert space can be formed.

0.4.4 The Null Space of the State

Definition 0.8 (GNS Null Space).
Let ω be a positive linear functional on $\mathcal{A}$. The GNS null space (or simply the null space) associated with ω is:
$$ \mathcal{N}_\omega := \{ A \in \mathcal{A} : \omega(A^\dagger A) = 0 \}. $$

Equivalently:
$$ \mathcal{N}_\omega = \{ A \in \mathcal{A} : \langle A, A \rangle_\omega = 0 \}. $$

Elements of $\mathcal{N}_\omega$ are algebraically non-zero in $\mathcal{A}$, but have zero state-induced length. In the representation determined by ω, they become null.

Lemma 0.10 (Null Space Is a Left Ideal).
Let ω be a positive linear functional on $\mathcal{A}$. Then $\mathcal{N}_\omega$ is a left ideal of $\mathcal{A}$.

Proof.
Let $A \in \mathcal{N}_\omega$ and $B \in \mathcal{A}$. We must show that $BA \in \mathcal{N}_\omega$, i.e., that $\omega((BA)^\dagger(BA)) = 0$.

Compute:
$$ (BA)^\dagger(BA) = A^\dagger B^\dagger B A. $$
Thus:
$$ \omega((BA)^\dagger(BA)) = \omega(A^\dagger B^\dagger B A). $$

By the Cauchy–Schwarz inequality for positive linear functionals (Lemma 0.8), if $\omega(A^\dagger A) = 0$, then $\omega(A^\dagger X) = 0$ for all $X \in \mathcal{A}$. Set $X = B^\dagger B A$. Then:
$$ \omega(A^\dagger B^\dagger B A) = 0. $$
Therefore:
$$ \omega((BA)^\dagger(BA)) = 0, $$
so $BA \in \mathcal{N}_\omega$. Hence $\mathcal{N}_\omega$ is a left ideal.
$\blacksquare$

Remark 0.1. $\mathcal{N}_\omega$ is generally not a two-sided ideal—only a left ideal. This asymmetry is why GNS constructions require quotienting carefully.

0.4.5 The Strength Ladder: From Primitive Quadratic Vanishing to Ideal Inclusion

This subsection makes explicit a distinction that was insufficiently clarified in earlier versions of the framework. There are four progressively stronger conditions, and they must not be conflated.

Condition 1 — Primitive Quadratic Vanishing (Definition 0.7).
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha \in I. $$
This is the foundational physical state condition. It says that each primitive constraint has zero positive square in the state. By the Cauchy–Schwarz inequality (Lemma 0.8), this implies:
$$ \omega(\hat{C}_\alpha^\dagger A) = 0 \qquad \text{and} \qquad \omega(A^\dagger \hat{C}_\alpha) = 0 $$
for all $A \in \mathcal{A}$ and all α.

Condition 2 — Linear Ideal Annihilation.
$$ \omega(X) = 0 \qquad \forall X \in \mathcal{I}_C. $$
This says that the state evaluates to zero on every element of the constraint ideal. This condition follows from Condition 1 together with the Cauchy–Schwarz inequality and the structure of the ideal. The proof is given in Lemma 0.9.

Condition 3 — Ideal-Null Physicality.
$$ \omega(X^\dagger X) = 0 \qquad \forall X \in \mathcal{I}_C. $$
This says that every element of the constraint ideal has zero positive square in the state. This is strictly stronger than Condition 2. It does not follow from Conditions 1 and 2 alone.

The gap is as follows. An element $X \in \mathcal{I}_C$ has the form $X = \sum_i A_i \hat{C}_{\alpha_i} B_i$. To show $\omega(X^\dagger X) = 0$, one must evaluate:
$$ \omega(X^\dagger X) = \omega\left(\sum_{i,j} B_i^\dagger \hat{C}_{\alpha_i}^\dagger A_i^\dagger A_j \hat{C}_{\alpha_j} B_j \right). $$

Each term contains $\hat{C}_{\alpha_j}$ (or its adjoint) multiplied on both sides by arbitrary algebra elements. Primitive quadratic vanishing gives $\omega(\hat{C}_\alpha^\dagger A) = 0$ for all A, but it does not automatically give $\omega(B^\dagger \hat{C}_\alpha^\dagger A \hat{C}_\beta B') = 0$ without additional control over the inserted factors.

Condition 4 — GNS Ideal Inclusion.
$$ \mathcal{I}_C \subseteq \mathcal{N}_\omega. $$
This says that every element of the constraint ideal is null in the GNS representation. This is exactly Condition 3 restated in the language of the null space.

Definition 0.9 (Ideal-Null Physical State).
A physical state ω is called **ideal-null** if it satisfies:
$$ \omega(X^\dagger X) = 0 \qquad \forall X \in \mathcal{I}_C. $$
Equivalently, ω is ideal-null if $\mathcal{I}_C \subseteq \mathcal{N}_\omega$.

The set of ideal-null physical states is denoted:
$$ \mathfrak{S}_{\mathrm{phys}}^{\mathrm{null}} := \left\{ \omega \in \mathfrak{S}(\mathcal{A}) : \omega(X^\dagger X) = 0 \ \forall X \in \mathcal{I}_C \right\}. $$

Remark 0.2. The relationship between these conditions is:
$$ \text{Condition 4} \iff \text{Condition 3} \quad \Longrightarrow \quad \text{Condition 2} \quad \Longleftarrow \quad \text{Condition 1}. $$
Conditions 1 and 2 are always related as shown. Condition 3 (and hence Condition 4) is an additional requirement.

Remark 0.3 (When is ideal-null automatic?). In certain special cases, ideal-null physicality follows from primitive quadratic vanishing without an additional assumption:

1. **If the constraint ideal is generated by self-adjoint constraints that mutually commute modulo the ideal:** In this case, cross-terms in $X^\dagger X$ can be controlled more tightly, and the Cauchy–Schwarz argument extends.

2. **If the state ω is a pure state whose GNS representation is irreducible on $\mathcal{A}/\mathcal{I}_C$:** The null space of an irreducible representation is maximal, and the ideal is automatically absorbed.

3. **If the algebra is a C^*-algebra and the state is a KMS state or satisfies a suitable clustering condition:** Spectral and modular theory can then provide the needed control.

In the general $(*)$-algebraic setting without additional structure, the implication does not hold automatically. The framework therefore treats ideal-null physicality as an explicit additional condition, while noting that it is satisfied in the models of greatest physical interest.

0.4.6 Theorem — Constraint Inclusion (Corrected)

Theorem 0.4 (Constraint Inclusion).
Let ω be an ideal-null physical state (Definition 0.9). Then $\mathcal{I}_C \subseteq \mathcal{N}_\omega$.

Proof.
This is immediate from the definition. If ω is ideal-null, then by definition $\omega(X^\dagger X) = 0$ for all $X \in \mathcal{I}_C$. By the definition of

$\mathcal{N}_\omega$, this means $X \in \mathcal{N}_\omega$ for all $X \in \mathcal{I}_C$. Therefore $\mathcal{I}_C \subseteq \mathcal{N}_\omega$. $\blacksquare$

Remark 0.4. This theorem is now stated with the correct hypothesis. The earlier version claimed the conclusion from primitive quadratic vanishing alone, which is insufficient. The ideal-null condition is the appropriate strengthening.

Remark 0.5 (Sufficient condition for ideal-null physicality). A sufficient condition for ideal-null physicality is the following: the physical state ω admits a GNS representation in which the represented constraint ideal $\pi_\omega(\mathcal{I}_C)$ acts trivially on the cyclic vector sector. Specifically, if $\pi_\omega(X)\Omega_\omega = 0$ for all $X \in \mathcal{I}_C$, then:
$$ \omega(X^\dagger X) = \langle \Omega_\omega, \pi_\omega(X^\dagger X) \Omega_\omega \rangle = \|\pi_\omega(X)\Omega_\omega\|^2 = 0. $$
This condition is natural: it says that the constraint ideal annihilates the cyclic vacuum. It is the represented version of ideal-null physicality.

0.4.7 Construction of the Pre-Hilbert Space

Because $\mathcal{N}_\omega$ is a left ideal, we can form the quotient vector space:
$$ \mathcal{D}_\omega := \mathcal{A} / \mathcal{N}_\omega. $$

Elements of $\mathcal{D}_\omega$ are equivalence classes:
$$ [A]_\omega := A + \mathcal{N}_\omega. $$

Definition 0.10 (Induced Inner Product).
Define an inner product on $\mathcal{D}_\omega$ by:
$$ \langle [A]_\omega, [B]_\omega \rangle_\omega := \omega(A^\dagger B). $$

Lemma 0.11 (Well-Definedness).
The above definition does not depend on the choice of representatives.

Proof.
Let $A' = A + N_1$ and $B' = B + N_2$ with $N_1, N_2 \in \mathcal{N}_\omega$. Then:
$$ \omega((A')^\dagger B') = \omega((A+N_1)^\dagger (B+N_2)) = \omega(A^\dagger B) + \omega(A^\dagger N_2) + \omega(N_1^\dagger B) + \omega(N_1^\dagger N_2). $$
All additional terms vanish because $\omega(N^\dagger X) = 0$ and $\omega(X^\dagger N) = 0$ for all $X \in \mathcal{A}$ and $N \in \mathcal{N}_\omega$ (by Lemma 0.8 and its conjugate). Thus:
$$ \langle [A']_\omega, [B']_\omega \rangle_\omega = \langle [A]_\omega, [B]_\omega \rangle_\omega. \qquad \blacksquare $$

Lemma 0.12 (Positive Definiteness).
After quotienting by the null space, the inner product becomes positive definite:
$$ \langle [A]_\omega, [A]_\omega \rangle_\omega = 0 \quad \Longrightarrow \quad [A]_\omega = 0. $$

Proof.

If $\omega(A^\dagger A) = 0$, then $A \in \mathcal{N}_\omega$ by definition, so $[A]_\omega = 0$. $\blacksquare$

Therefore, $\mathcal{D}_\omega$ is a pre-Hilbert space.

0.4.8 The Hilbert Space Completion

Definition 0.11 (GNS Hilbert Space).
The GNS Hilbert space $\mathscr{H}_\omega$ is the completion of $\mathcal{D}_\omega = \mathcal{A}/\mathcal{N}_\omega$ under the induced norm:
$$ \| [A]_\omega \| := \sqrt{\omega(A^\dagger A)}. $$

This is now a genuine Hilbert space, constructed purely from $(\mathcal{A}, \omega)$.

0.4.9 The Representation Map

Definition 0.12 (GNS Representation).
Define the representation map $\pi_\omega: \mathcal{A} \to \mathrm{End}(\mathcal{D}_\omega)$ by:
$$ \pi_\omega(A)[B]_\omega := [AB]_\omega. $$

This defines the action of the algebra on the pre-Hilbert space by left multiplication.

Technical Status Note (Boundedness). In the purely algebraic setting, $\pi_\omega(A)$ is a linear operator defined on the dense domain $\mathcal{D}_\omega \subset \mathscr{H}_\omega$. It is **not automatically bounded**. The following distinctions must be observed:

1. **For a general unital $(*)$-algebra:** $\pi_\omega(A)$ is densely defined on $\mathcal{D}_\omega$. Claims involving $\mathcal{B}(\mathscr{H}_\omega)$ (bounded operators), adjoints, kernels, and self-adjoint master constraints require either:
   - upgrading $\mathcal{A}$ to a **C^*-algebra** (where the GNS representation automatically yields bounded operators), or
   - specifying explicit **domain assumptions**: a common dense invariant domain $\mathcal{D} \subseteq \mathcal{D}_\omega$ on which all relevant operators are closable, and on which adjoints are well-defined.

2. **For a C^*-algebra:** If $\mathcal{A}$ is a unital C^*-algebra, then $\pi_\omega(A) \in \mathcal{B}(\mathscr{H}_\omega)$ for all $A \in \mathcal{A}$, with $\|\pi_\omega(A)\| \leq \|A\|$. The representation extends continuously to all of $\mathscr{H}_\omega$.

3. **For the framework's purposes:** The framework does not commit to a C^*-structure at the primitive level, because doing so would impose a norm and topology before the emergent geometry is available. Instead, the framework works with densely defined operators on $\mathcal{D}_\omega$ and imposes domain conditions locally when needed. When a C^*-upgrade is required for a specific theorem, it will be stated explicitly.

> **The GNS representation yields densely defined operators on $\mathcal{D}_\omega$. Boundedness, self-adjointness, and kernel arguments require either a C^*-upgrade or explicit domain assumptions.**

Lemma 0.13 (Representation Is Well-Defined).
If $[B]_\omega = [B']_\omega$, then $[AB]_\omega = [AB']_\omega$.

Proof.
If $B' = B + N$ with $N \in \mathcal{N}_\omega$, then $AB' = AB + AN$. By Lemma 0.10, $AN \in \mathcal{N}_\omega$ (since $\mathcal{N}_\omega$ is a left ideal). Thus $[AB']_\omega = [AB]_\omega$. $\blacksquare$

Lemma 0.14 (Representation Homomorphism).
For all $A, B \in \mathcal{A}$:
1. $\pi_\omega(A + B) = \pi_\omega(A) + \pi_\omega(B)$,
2. $\pi_\omega(AB) = \pi_\omega(A)\pi_\omega(B)$,
3. $\pi_\omega(\lambda A) = \lambda \, \pi_\omega(A)$ for all $\lambda \in \mathbb{C}$.

Proof.
Direct from definition:
$$ \pi_\omega(AB)[C]_\omega = [(AB)C]_\omega = [A(BC)]_\omega = \pi_\omega(A)[BC]_\omega = \pi_\omega(A)\pi_\omega(B)[C]_\omega. \qquad \blacksquare $$

Lemma 0.15 ($(*)$-Representation Property).
On the domain $\mathcal{D}_\omega$, the representation respects the involution:
$$ \langle \pi_\omega(A)[B]_\omega, [C]_\omega \rangle_\omega = \langle [B]_\omega, \pi_\omega(A^\dagger)[C]_\omega \rangle_\omega. $$

Proof.
$$ \langle \pi_\omega(A)[B]_\omega, [C]_\omega \rangle_\omega = \langle [AB]_\omega, [C]_\omega \rangle_\omega = \omega((AB)^\dagger C) = \omega(B^\dagger A^\dagger C) = \langle [B]_\omega, [A^\dagger C]_\omega \rangle_\omega. \qquad \blacksquare $$

Remark 0.6. Lemma 0.15 shows that $\pi_\omega(A^\dagger)$ is the formal adjoint of $\pi_\omega(A)$ on the domain $\mathcal{D}_\omega$. If $\pi_\omega(A)$ is bounded (e.g., in a C^*-setting), this extends to the Hilbert space adjoint: $\pi_\omega(A)^\dagger = \pi_\omega(A^\dagger)$. For unbounded operators, this formal adjoint relation holds on $\mathcal{D}_\omega$ but does not by itself guarantee that $\pi_\omega(A)$ is closable or self-adjoint.

0.4.10 The Cyclic Vector

Definition 0.13 (Cyclic Vector).
Define the cyclic vector:
$$ \Omega_\omega := [\mathbf{1}]_\omega. $$

Lemma 0.16 (Cyclicity).

$$ \overline{\pi_\omega(\mathcal{A})\Omega_\omega} = \mathscr{H}_\omega. $$

Proof.
$\pi_\omega(A)\Omega_\omega = [A]_\omega$. All equivalence classes are generated this way. The completion of their span is $\mathscr{H}_\omega$. $\blacksquare$

0.4.11 State Reconstruction

Theorem 0.5 (GNS Representation Theorem).
For all $A \in \mathcal{A}$:
$$ \omega(A) = \langle \Omega_\omega, \pi_\omega(A)\Omega_\omega \rangle_\omega. $$

Proof.
$$ \langle \Omega_\omega, \pi_\omega(A)\Omega_\omega \rangle_\omega = \langle [\mathbf{1}]_\omega, [A]_\omega \rangle_\omega = \omega(\mathbf{1}^\dagger A) = \omega(A). \qquad \blacksquare $$

This equation explains how the familiar expectation-value form arises from the algebraic state.

0.4.12 Constraint Action in the Representation

Theorem 0.6 (Cyclic-Vector Constraint Annihilation).
Let ω be a physical state (satisfying primitive quadratic vanishing, Condition 1). Then:
$$ \pi_\omega(\hat{C}_\alpha)\Omega_\omega = 0 \qquad \forall \alpha \in I. $$

Proof.
$$ \|\pi_\omega(\hat{C}_\alpha)\Omega_\omega\|^2 = \|\pi_\omega(\hat{C}_\alpha)[\mathbf{1}]_\omega\|^2 = \|[\hat{C}_\alpha]_\omega\|^2 = \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0. $$
A vector of zero Hilbert norm is the zero vector. $\blacksquare$

Remark 0.7. This theorem requires only Condition 1 (primitive quadratic vanishing), not the full ideal-null condition. The reason is that the cyclic vector $\Omega_\omega = [\mathbf{1}]_\omega$ is a single specific vector, and the norm computation $\|\pi_\omega(\hat{C}_\alpha)\Omega_\omega\|^2 = \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha)$ involves only the primitive constraint, not an arbitrary ideal element.

Theorem 0.7 (Ideal-Null Constraint Annihilation).
Let ω be an ideal-null physical state (Condition 3). Then:
$$ \pi_\omega(X)\Omega_\omega = 0 \qquad \forall X \in \mathcal{I}_C. $$

Proof.
$$ \|\pi_\omega(X)\Omega_\omega\|^2 = \| [X]_\omega \|^2 = \omega(X^\dagger X) = 0 $$
by the ideal-null condition. Therefore $\pi_\omega(X)\Omega_\omega = 0$. $\blacksquare$

Remark 0.8. Theorem 0.7 is stronger than Theorem 0.6: it extends annihilation from primitive constraints to the entire ideal. This is exactly where the ideal-null condition is needed. Without it, one can only conclude that the primitive constraints annihilate the cyclic vector, not that every ideal element does.

> **Constraints appear as annihilation operators on the cyclic vector. The extension to the full ideal requires ideal-null physicality.**

0.4.13 The Represented Physical Sector

Definition 0.14 (Represented Physical Hilbert Space).
The represented physical sector associated with the GNS representation is:
$$ \mathscr{H}_{\mathrm{phys}}^\omega := \bigcap_{\alpha \in I} \ker \pi_\omega(\hat{C}_\alpha). $$

Equivalently, it is the simultaneous kernel of all represented primitive constraints.

Because Ω_ω satisfies $\pi_\omega(\hat{C}_\alpha)\Omega_\omega = 0$ for all α (Theorem 0.6), we have:
$$ \Omega_\omega \in \mathscr{H}_{\mathrm{phys}}^\omega. $$

Technical Note. The kernels $\ker \pi_\omega(\hat{C}_\alpha)$ are well-defined as subspaces of $\mathscr{H}_\omega$ when $\pi_\omega(\hat{C}_\alpha)$ is a closed or closable operator. For bounded operators (e.g., in a C^*-setting), this is automatic. For unbounded operators, one must specify the domain on which the kernel is taken. A natural choice is the common invariant domain $\mathcal{D}_\omega$, or a specified subdomain $\mathcal{D} \subseteq \mathcal{D}_\omega$.

Theorem 0.8 (Ideal-Kernel Correspondence).
Let ω be an ideal-null physical state, and let $\mathcal{D} \subseteq \mathcal{D}_\omega$ be a common invariant domain for $\pi_\omega(\mathcal{A})$ and $\pi_\omega(\mathcal{I}_C)$. Then:
$$ \psi \in \mathscr{H}_{\mathrm{phys}}^\omega \cap \mathcal{D} \quad \Longleftrightarrow \quad \pi_\omega(X)\psi = 0 \quad \forall X \in \mathcal{I}_C. $$

Proof.

$(\Rightarrow)$ Suppose $\psi \in \mathscr{H}_{\mathrm{phys}}^\omega \cap \mathcal{D}$, so $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α. Let $X \in \mathcal{I}_C$. Then $X = \sum_i A_i \hat{C}_{\alpha_i} B_i$ for some $A_i, B_i \in \mathcal{A}$. Since $\mathcal{D}$ is invariant under $\pi_\omega(\mathcal{A})$, we have $\pi_\omega(B_i)\psi \in \mathcal{D}$. Since $\mathcal{D} \subseteq \mathscr{H}_{\mathrm{phys}}^\omega$ is the kernel of all primitive constraints...

Actually, we need to be more careful. We know $\pi_\omega(\hat{C}_\alpha)\psi = 0$, but we need $\pi_\omega(\hat{C}_\alpha)\pi_\omega(B_i)\psi = 0$, which requires $\pi_\omega(B_i)\psi \in \ker \pi_\omega(\hat{C}_\alpha)$. This holds if B_i preserves the physical sector, i.e., if B_i is a zero-preserving observable (Section 1.2). In general, this is not guaranteed for arbitrary $B_i \in \mathcal{A}$.

Corrected Statement. Under the ideal-null condition and assuming $\mathcal{D}$ is a common invariant domain, the ideal-kernel correspondence holds for vectors in the GNS cyclic subspace generated by Ω_ω:

$$ \pi_\omega(X)\Omega_\omega = 0 \quad \forall X \in \mathcal{I}_C $$

by Theorem 0.7. For general vectors $\psi \in \mathscr{H}_{\mathrm{phys}}^\omega$, the extension to $\pi_\omega(X)\psi = 0$ for all $X \in \mathcal{I}_C$ requires that the physical sector $\mathscr{H}_{\mathrm{phys}}^\omega$ is invariant under $\pi_\omega(\mathcal{I}_C)$, which is an additional structural condition. $\blacksquare$

Remark 0.9. This corrected statement is more honest about what is proven and what requires additional structure. The ideal-kernel correspondence is cleanest in the following settings:
1. When $\mathcal{A}$ is a C^*-algebra and the physical sector is a closed reducing subspace.
2. When the constraints form a first-class system (closed under commutation modulo the ideal), so that the physical sector is automatically invariant.
3. When one restricts attention to the cyclic subspace generated by Ω_ω.

0.4.14 Representation Dependence

The GNS construction depends on the state ω. Different physical states may generate different Hilbert spaces:
$$ \mathscr{H}_{\omega_1}, \qquad \mathscr{H}_{\omega_2},$$
different representations:
$$ \pi_{\omega_1}, \qquad \pi_{\omega_2},$$
and different cyclic vectors:
$$ \Omega_{\omega_1}, \qquad \Omega_{\omega_2}.$$

This is not a defect. It reflects the fact that Hilbert-space structure is not primitive. The primitive layer is the algebra and the state condition.

Therefore, one should not speak of "the" fundamental Hilbert space unless an additional theorem or assumption identifies a preferred representation. The careful phrase is: "the GNS Hilbert space associated with the state ω."

0.4.15 The Derived Master Constraint Operator

Now that the representation is available, the framework can package the represented constraints into a single positive operator.

Definition 0.15 (Represented Master Constraint Operator).
Let $(\mathscr{H}_\omega, \pi_\omega, \Omega_\omega)$ be the GNS representation. Let $w_\alpha > 0$ be strictly positive weights. Define:

$$ \hat{\mathfrak{M}}_\omega := \sum_{\alpha \in I} w_\alpha \, \pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha), $$
whenever this sum is well-defined as a positive operator on a suitable domain.

Technical Note.
- If the index set I is **finite**, this definition is immediate: the sum is a finite sum of positive operators on $\mathcal{D}_\omega$.
- If I is **infinite**, one must specify:
  - the common domain $\mathcal{D}$ on which all $\pi_\omega(\hat{C}_\alpha)$ are defined,
  - the convergence of the positive quadratic form $Q_{\mathfrak{M}}[\psi] = \sum_\alpha w_\alpha \|\pi_\omega(\hat{C}_\alpha)\psi\|^2$,
  - whether the resulting form is closable (e.g., via the Friedrichs extension),
  - and whether kernel equivalence holds on the chosen domain.
- In a **C^*-algebraic setting**, if the sum converges in the strong operator topology, $\hat{\mathfrak{M}}_\omega$ is a bounded positive operator.

Lemma 0.17 (Positivity of $\hat{\mathfrak{M}}_\omega$).
For all ψ in the domain of $\hat{\mathfrak{M}}_\omega$:
$$ \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle \geq 0. $$

Proof.
$$ \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = \sum_\alpha w_\alpha \|\pi_\omega(\hat{C}_\alpha)\psi\|^2. $$
Each term is non-negative and $w_\alpha > 0$. $\blacksquare$

0.4.16 Kernel Equivalence

Theorem 0.9 (Master-Zero Equivalence).
Let $\hat{\mathfrak{M}}_\omega = \sum_\alpha w_\alpha \, \pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha)$ with $w_\alpha > 0$ be a well-defined positive master constraint on a domain $\mathcal{D}$. Then for every vector $\psi \in \mathcal{D}$:
$$ \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0 \quad \Longleftrightarrow \quad \pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha. $$

Equivalently:
$$ \ker \hat{\mathfrak{M}}_\omega \cap \mathcal{D} = \bigcap_{\alpha \in I} \ker \pi_\omega(\hat{C}_\alpha) \cap \mathcal{D}. $$

Proof.

($\Rightarrow$) Assume $\langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0$. Using the positivity identity:
$$ 0 = \sum_\alpha w_\alpha \|\pi_\omega(\hat{C}_\alpha)\psi\|^2. $$
Every summand is non-negative. Since each weight is strictly positive, the only way the sum can vanish is if every summand vanishes:
$$ \|\pi_\omega(\hat{C}_\alpha)\psi\|^2 = 0 \quad \forall \alpha. $$

Therefore $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α.

($\Leftarrow$) Assume $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α. Then every term in the sum vanishes, so $\hat{\mathfrak{M}}_\omega \psi = 0$. $\blacksquare$

Corollary 0.1.
$$ \mathscr{H}_{\mathrm{phys}}^\omega \cap \mathcal{D} = \ker \hat{\mathfrak{M}}_\omega \cap \mathcal{D}. $$
$$ \Omega_\omega \in \ker \hat{\mathfrak{M}}_\omega. $$

0.4.17 Why the Master Constraint Is Not Primitive

The master constraint operator $\hat{\mathfrak{M}}_\omega$ is a derived object, not a primitive one. Its definition depends on additional choices not present in the primitive algebraic layer:
1. A physical state ω,
2. The GNS representation π_ω,
3. A Hilbert space $\mathscr{H}_\omega$,
4. Weights $w_\alpha > 0$,
5. Domain and convergence conditions if the constraint family is infinite,
6. Boundedness or C^*-structure if kernel arguments are needed on the full Hilbert space.

The primitive data are:
$$ \mathcal{A}, \qquad \mathcal{I}_C, \qquad \omega. $$
The master constraint appears only after representation.

The correct order is:
$$ \mathcal{A}, \mathcal{I}_C, \omega \quad \longrightarrow \quad (\mathscr{H}_\omega, \pi_\omega, \Omega_\omega) \quad \longrightarrow \quad \hat{\mathfrak{M}}_\omega. $$

0.4.18 The Effective Classical Master Functional

In an effective classical description, one may introduce configuration-space constraint maps $C_\alpha: \mathcal{K} \to V_\alpha$ and define:
$$ \mathfrak{M}(x) = \sum_\alpha w_\alpha |C_\alpha(x)|^2. $$

A configuration $x \in \mathcal{K}$ is effective-physical if $\mathfrak{M}(x) = 0$.

Theorem 0.10 (Classical Zero-Equivalence).
$$ \mathfrak{M}(x) = 0 \quad \Longleftrightarrow \quad C_\alpha(x) = 0 \quad \forall \alpha. $$

Proof. Each term $w_\alpha |C_\alpha(x)|^2 \geq 0$. A sum of non-negative terms is zero iff every term is zero. $\blacksquare$

Proposition 0.2.
The effective master functional $\mathfrak{M}$ is a derived object, not a primitive one. Its definition depends on:

1. A configuration space $\mathcal{K}$,
2. Effective constraint maps C_α,
3. Norms on codomains V_α,
4. Positive weights w_α,
5. A representation/state-to-configuration assignment.

By contrast, the primitive framework requires only $(\mathcal{A}, \mathcal{I}_C, \omega)$.

0.4.19 The Complete Foundational Hierarchy

The full foundational hierarchy can now be summarised as:

$$ \boxed{ \mathcal{A} \quad \longrightarrow \quad \{\hat{C}_\alpha\} \quad \longrightarrow \quad \mathcal{I}_C \quad \longrightarrow \quad \omega \quad \longrightarrow \quad (\mathscr{H}_\omega, \pi_\omega, \Omega_\omega) \quad \longrightarrow \quad \hat{\mathfrak{M}}_\omega \quad \longrightarrow \quad \mathfrak{M}(x). } $$

Each arrow represents a step of additional structure. The ordering matters:
1. $\mathcal{A}$ is the primitive relational algebra.
2. $\{\hat{C}_\alpha\}$ are primitive constraint elements.
3. $\mathcal{I}_C$ is the algebraic inconsistency sector generated by them.
4. ω is a positive physical state functional (satisfying at least Condition 1; ideally Condition 3).
5. $(\mathscr{H}_\omega, \pi_\omega, \Omega_\omega)$ is the GNS representation.
6. $\hat{\mathfrak{M}}_\omega$ is the represented master constraint operator.
7. $\mathfrak{M}(x)$ is an effective classical master functional.

0.4.20 Guardrails for This Subsection

1. **Representation dependence.** $\mathscr{H}_{\mathrm{phys}}^\omega$ depends on ω. Different states yield different Hilbert spaces.
2. **The master operator is derived.** $\hat{\mathfrak{M}}_\omega$ depends on π_ω. It is not foundational.
3. **Ideal-null is stronger than primitive quadratic.** The inclusion $\mathcal{I}_C \subseteq \mathcal{N}_\omega$ requires the ideal-null condition (Condition 3), not merely primitive quadratic vanishing (Condition 1).
4. **Boundedness is not automatic.** In the purely $(*)$-algebraic setting, GNS operators are densely defined, not bounded. Claims requiring bounded operators need a C^*-upgrade or explicit domain assumptions.
5. **No dynamics yet.** The kernel defines admissibility, not evolution.
6. **No geometry yet.** This is still purely algebraic/operator structure.
7. **Hilbert space is not primitive.** The GNS construction is derived structure.
8. **No probability interpretation yet.** The inner product exists, but probabilities are not yet introduced.
9. **The effective classical functional is not primitive.** It is a derived, coarse-grained shadow of the algebraic constraint structure.
10. **No ontological reversal.** The effective master functional must not be treated as more fundamental than the algebraic state-constraint structure.

11. **No claim of automatic equivalence.** The bridge between configuration-space and algebraic formulations is conditional on a faithful representation-linked model.
12. **No dynamics from $\mathfrak{M}$.** The effective master functional is a consistency selector, not an evolution law.
13. **No claim that $\mathcal{K}$ is spacetime.** At this stage $\mathcal{K}$ is only an effective configuration space; spacetime-like structure will emerge later from correlation geometry and causal/burden structure.

0.5 Foundational Guardrails and Summary

0.5.1 Purpose of the Guardrail Subsection

The preceding subsections established the canonical algebraic foundation of the framework:
$$ \mathcal{A}, \qquad \{\hat{C}_\alpha\}_{\alpha \in I}, \qquad \mathcal{I}_C, \qquad \omega, \qquad (\mathscr{H}_\omega, \pi_\omega, \Omega_\omega), \qquad \hat{\mathfrak{M}}_\omega, \qquad \mathfrak{M}(x). $$

These objects form a hierarchy. The relational algebra comes first. The constraint ideal is generated inside it. Physical states are positive linear functionals compatible with the constraints. Hilbert space arises only after a state is chosen. The master constraint operator is a representation-level packaging. The classical master functional is an effective or coarse-grained shadow of the represented constraint structure.

Because the framework uses familiar mathematical words—state, operator, constraint, zero, Hilbert space, functional, physical sector—it is easy to misread the foundation as making stronger claims than it actually makes. This subsection records the main guardrails needed to keep the theory theorem-safe.

> **The foundation must say exactly what it proves, and no more.**

The guardrails below prevent premature interpretation. They also protect later emergence claims by ensuring that space, time, matter, geometry, and dynamics are not smuggled into the starting assumptions.

0.5.2 Consolidated Foundational Guardrails

#	Guardrail	Statement
I	Zero is constraint compatibility	Zero means zero inconsistency, not zero being
II	Zero is not pairwise cancellation	Master zero is simultaneous constraint satisfaction
III	$\mathcal{A}$ is kinematic	$\mathcal{A}$ is not automatically the observable algebra
IV	Do not quotient too early	Physicality is first state-defined
V	Hilbert space is derived	$\mathscr{H}_\omega$ is not primitive
VI	Master constraint is representation-dependent	$\hat{\mathfrak{M}}_\omega$ does not define the primitive theory
VII	Classical functional is effective	$\mathfrak{M}(x)$ is not fundamental
VIII	x is not spacetime	x labels a configuration, not a spacetime point

IX	Master constraint is not Hamiltonian	It selects admissibility, not time evolution
X	Admissibility is not dynamics	Constraint satisfaction does not define evolution
XI	No external time	No background time parameter is primitive
XII	No background space	No spatial geometry is primitive
XIII	Causality is dependency	Causality begins as relational order, not light-cone structure
XIV	Constraint cost is not energy	Positive burden is not automatically energy
XV	Not yet quantum mechanics	The foundation is algebraic, not full QM
XVI	State existence is conditional	$\mathfrak{S}_{\mathrm{phys}} \neq \varnothing$ must be shown or assumed
XVII	Infinite sums need control	Infinite master sums require analytic assumptions
XVIII	Approximate $\neq$ exact	ε-physicality is effective, not foundational
XIX	Emergence is layered	No later structure is smuggled into the foundation
XX	Ideal-null is stronger than primitive	$\mathcal{I}_C \subseteq \mathcal{N}_\omega$ requires the ideal-null condition, not just primitive quadratic vanishing
XXI	Boundedness is not automatic	GNS operators are densely defined; boundedness requires a C^*-upgrade or domain assumptions

0.5.3 Summary of Section 0

Section 0 has established the canonical algebraic foundation of the Relational Constraint Framework.

What has been accomplished:

1. **The relational algebra** $\mathcal{A}$ was defined as a unital complex $(*)$-algebra, providing the primitive language for relational expressions without assuming spacetime, time, geometry, fields, particles, observers, or Hilbert space.
2. **The constraint ideal** $\mathcal{I}_C$ was defined as the smallest two-sided $(*)$-ideal containing all primitive constraints $\{\hat{C}_\alpha\}$. It gathers all algebraic consequences of the primitive constraints and functions as the algebraic inconsistency sector.
3. **Physical states** were defined as positive linear functionals $\omega: \mathcal{A} \to \mathbb{C}$ satisfying the quadratic constraint-vanishing condition $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$ for all α. This condition says that primitive constraint violations vanish in positive state-evaluation.
4. **The GNS representation** was constructed from a physical state, yielding a Hilbert space $\mathscr{H}_\omega$, a representation π_ω, and a cyclic vector Ω_ω. The physical state condition was shown to imply constraint annihilation of the cyclic vector: $\pi_\omega(\hat{C}_\alpha)\Omega_\omega = 0$. The distinction between primitive quadratic vanishing and ideal-null physicality was rigorously established.
5. **The master constraint operator** $\hat{\mathfrak{M}}_\omega$ was introduced as a representation-level packaging of the primitive constraints. Its kernel was shown to coincide with the simultaneous kernel of all individual constraints.
6. **The effective classical master functional** $\mathfrak{M}(x)$ was introduced as a derived, coarse-grained description. Its zero-equivalence with the simultaneous satisfaction of all effective constraints was proven. Its status as a derived, non-primitive object was established.

7. **Twenty-one foundational guardrails** were recorded to prevent overinterpretation of the zero condition, the Hilbert-space representation, the master constraint, the classical functional, the boundedness of operators, and the scope of the foundation.

What has not been accomplished:

The foundation does not yet contain space, time, particles, fields, gravity, observers, measurement, probability, dynamics, or cosmology. It supplies the constraint-compatible relational ground from which those structures may later be derived.

The foundational commitment is:

> **A physical structure is an admissible relational structure whose constraint violations vanish in the physical state.**

0.6 Appendix — A Finite-Dimensional Toy Model

0.6.1 Purpose

The framework defines a relational $(*)$-algebra $\mathcal{A}$, a constraint ideal $\mathcal{I}_C$, physical states ω, and a GNS representation. These objects are mathematically sound but highly abstract. The purpose of this appendix is to demonstrate that the definitions are not vacuous by constructing the simplest possible non-trivial model: a bipartite system in which one subsystem is constrained.

0.6.2 The Algebra

To ensure the existence of non-trivial proper two-sided ideals, we take the algebra to be a direct sum of two matrix algebras. Let:
$$ \mathcal{A} = M_2(\mathbb{C}) \oplus M_2(\mathbb{C}). $$
Elements of $\mathcal{A}$ are ordered pairs $A = (A_S, A_C)$, where $A_S, A_C \in M_2(\mathbb{C})$. This is a unital $(*)$-algebra with:
* **Multiplication:** Component-wise matrix multiplication: $(A_S, A_C)(B_S, B_C) = (A_S B_S, A_C B_C)$.
* **Unit:** $\mathbf{1} = (I_2, I_2)$, where I_2 is the 2×2 identity matrix.
* **Involution:** $(A_S, A_C)^\dagger = (A_S^\dagger, A_C^\dagger)$, the component-wise conjugate transpose.

We interpret the first component as the "system" qubit S and the second component as the "constraint" qubit C.

0.6.3 The Constraint

Let $P_1 = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ be the projection onto the $|1\rangle$ state of a single qubit. Define the primitive constraint:
$$ \hat{C} = (0, P_1) \in \mathcal{A}. $$

This constraint asserts that the constraint qubit must **not** be in the state $|1\rangle$. It is a projection, so $\hat{C}^\dagger = \hat{C}$ and $\hat{C}^2 = \hat{C}$.

0.6.4 The Constraint Ideal

The constraint ideal $\mathcal{I}_C$ generated by $\hat{C}$ is the set of all elements of the form $A \hat{C} B$ for $A, B \in \mathcal{A}$. Because the algebra is a direct sum, multiplication is component-wise. Thus:
$$ A \hat{C} B = (A_S, A_C) (0, P_1) (B_S, B_C) = (0, A_C P_1 B_C). $$
As A_C and B_C range over all of $M_2(\mathbb{C})$, $A_C P_1 B_C$ ranges over all of $M_2(\mathbb{C})$ (since $P_1 \neq 0$ and $M_2(\mathbb{C})$ is simple). Therefore:
$$ \mathcal{I}_C = \{0\} \oplus M_2(\mathbb{C}). $$
This is a proper two-sided $(*)$-ideal of $\mathcal{A}$, and $\mathbf{1} = (I_2, I_2) \notin \mathcal{I}_C$. The quotient algebra is $\mathcal{A}/\mathcal{I}_C \cong M_2(\mathbb{C})$, corresponding to the unconstrained system qubit.

0.6.5 A Physical State

Define the linear functional $\omega: \mathcal{A} \to \mathbb{C}$ by:
$$ \omega(A_S, A_C) = \langle 0 | A_S | 0 \rangle. $$
Note that ω simply evaluates the first component (the system) in the $|0\rangle$ state, and completely ignores the second component (the constraint).

Positivity: For any $A = (A_S, A_C) \in \mathcal{A}$:
$$ \omega(A^\dagger A) = \omega(A_S^\dagger A_S, A_C^\dagger A_C) = \langle 0 | A_S^\dagger A_S | 0 \rangle = \| A_S |0\rangle \|^2 \geq 0. $$

Normalisation: $\omega(\mathbf{1}) = \langle 0 | I_2 | 0 \rangle = 1$.

Physical State Condition: We check the primitive quadratic vanishing:
$$ \omega(\hat{C}^\dagger \hat{C}) = \omega(0, P_1^2) = \omega(0, P_1) = \langle 0 | 0 | 0 \rangle = 0. $$
Therefore, ω is a physical state. The constraint is satisfied because the functional only evaluates the system qubit, which is orthogonal to the constraint qubit's forbidden subspace.

0.6.6 The GNS Construction

Null Space: The GNS null space $\mathcal{N}_\omega = \{ A \in \mathcal{A} : \omega(A^\dagger A) = 0 \}$ consists of elements where $A_S |0\rangle = 0$. That is, $\mathcal{N}_\omega$ contains all pairs (A_S, A_C) where the first row of A_S is zero. For example, $(0, I_2) \in \mathcal{N}_\omega$ and $(\sigma_-, 0) \in \mathcal{N}_\omega$, where $\sigma_- = |0\rangle\langle 1|$.

Quotient Space: The pre-Hilbert space $\mathcal{D}_\omega = \mathcal{A} / \mathcal{N}_\omega$. Two elements are equivalent if their system components agree on $|0\rangle$. The quotient is isomorphic to $\mathbb{C}^2$, spanned by the equivalence classes of $(I_2, 0)$ and $(\sigma_x, 0)$ (or any operator mapping $|0\rangle$ to $|1\rangle$).

Inner Product: $\langle [A], [B] \rangle_\omega = \langle 0 | A_S^\dagger B_S | 0 \rangle$.

Hilbert Space: The completion is $\mathscr{H}_\omega \cong \mathbb{C}^2$. The cyclic vector is $\Omega_\omega = [\mathbf{1}] = [(I_2, I_2)]$, which corresponds to the state $|0\rangle$.

Representation: $\pi_\omega(A_S, A_C) [B] = [(A_S B_S, A_C B_C)]$. The representation acts non-trivially only on the first component: $\pi_\omega(A_S, A_C) \cong A_S$ acting on $\mathbb{C}^2$.

Constraint Annihilation: $\pi_\omega(\hat{C}) \Omega_\omega = \pi_\omega(0, P_1) [(I_2, I_2)] = [(0, P_1)]$. Since $(0, P_1) \in \mathcal{N}_\omega$, its equivalence class is zero. Thus, $\pi_\omega(\hat{C})\Omega_\omega = 0$.

Ideal-Null Condition: For any $X = (0, X_C) \in \mathcal{I}_C$, we have $\pi_\omega(X) \cong 0$. Thus $\pi_\omega(X) = 0$ on $\mathscr{H}_\omega$. Trivially, $\omega(X^\dagger X) = \|(0, X_C)|0\rangle\|^2 = 0$. The state ω is ideal-null.

0.6.7 The Master Constraint

Choose weight $w = 1$. The represented master constraint is:
$$ \hat{\mathfrak{M}}_\omega = \pi_\omega(\hat{C})^\dagger \pi_\omega(\hat{C}) = 0. $$
Because the constraint acts entirely on the second component, and the GNS representation only "sees" the first component, the represented constraint is the zero operator. The physical sector is the entire Hilbert space:
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega = \mathbb{C}^2. $$

0.6.8 Preview: Correlation Kernel and Emergent Distance

To bridge toward Section 2, we compute the correlation kernel. Let the localisable observable sector be:
$$ \mathcal{A}_{\mathrm{loc}} = \{ (\sigma_x, 0), (\sigma_y, 0), (\sigma_z, 0) \}, $$
where $\sigma_x, \sigma_y, \sigma_z$ are the Pauli matrices acting on the system qubit.

The correlation kernel is defined as:
$$ K_\omega(A, B) = \frac{|\omega(A^\dagger B)|}{\sqrt{\omega(A^\dagger A)\omega(B^\dagger B)}}. $$
Since ω evaluates the system component in the $|0\rangle$ state:

* $\omega(\sigma_x^\dagger \sigma_x) = \langle 0 | \sigma_x^2 | 0 \rangle = \langle 0 | I | 0 \rangle = 1$
* $\omega(\sigma_y^\dagger \sigma_y) = 1$
* $\omega(\sigma_z^\dagger \sigma_z) = 1$

Now the cross-terms (using standard Pauli algebra: $\sigma_x \sigma_y = i\sigma_z$, $\sigma_x \sigma_z = -i\sigma_y$, $\sigma_y \sigma_z = i\sigma_x$, and $\langle 0 | \sigma_z | 0 \rangle = 1$, $\langle 0 | \sigma_x | 0 \rangle = \langle 0 | \sigma_y | 0 \rangle = 0$):

* $K_\omega(\sigma_x, \sigma_y) = |\langle 0 | \sigma_x \sigma_y | 0 \rangle| = |i \langle 0 | \sigma_z | 0 \rangle| = |i(1)| = 1$.
* $K_\omega(\sigma_x, \sigma_z) = |\langle 0 | \sigma_x \sigma_z | 0 \rangle| = |-i \langle 0 | \sigma_y | 0 \rangle| = 0$.
* $K_\omega(\sigma_y, \sigma_z) = |\langle 0 | \sigma_y \sigma_z | 0 \rangle| = |i \langle 0 | \sigma_x | 0 \rangle| = 0$.

With $\ell_0 = 1$, the correlation distances $d_\omega(A, B) = -\log K_\omega(A, B)$ are:
* $d_\omega(\sigma_x, \sigma_y) = -\log(1) = 0$.
* $d_\omega(\sigma_x, \sigma_z) = -\log(0) = +\infty$.
* $d_\omega(\sigma_y, \sigma_z) = -\log(0) = +\infty$.

Interpretation: Because $d_\omega(\sigma_x, \sigma_y) = 0$, the observables σ_x and σ_y are perfectly correlated (in fact, mutually uncorrelated with the state $|0\rangle$ in different bases) and collapse to the *same* emergent point in the metric quotient ($\mathcal{X}_\omega = \mathcal{A}_{\mathrm{loc}} / \sim_\omega$).

The emergent space $\mathcal{X}_\omega$ therefore consists of exactly **two points**:
1. $p_1 = [\sigma_x]_\omega = [\sigma_y]_\omega$
2. $p_2 = [\sigma_z]_\omega$

The distance between these two emergent points is $+\infty$. This toy model produces a discrete, disconnected proto-space of two points, demonstrating explicitly how algebraic indistinguishability translates into zero emergent distance, and how orthogonal observables translate into infinite separation.

0.6.9 What This Toy Model Demonstrates

1. **Proper Constraint Ideal:** $\mathcal{I}_C = \{0\} \oplus M_2(\mathbb{C})$ is a non-trivial proper two-sided ideal.
2. **Physical state exists:** ω is positive, normalised, and satisfies $\omega(\hat{C}^\dagger \hat{C}) = 0$.
3. **Ideal-null physicality:** The state trivially satisfies $\omega(X^\dagger X) = 0$ for all $X \in \mathcal{I}_C$.
4. **GNS is non-trivial:** The Hilbert space is $\mathbb{C}^2$, and the physical sector is the entire space.
5. **Constraint annihilation:** $\pi_\omega(\hat{C})\Omega_\omega = 0$ is verified explicitly.
6. **Correlation kernel is computable:** The kernel produces a non-trivial distance structure with emergent points.
7. **The framework is not vacuous:** Every primitive object has a concrete, finite-dimensional realisation that correctly handles the algebraic ideal structure.

Section 1 — The Zero-Preserving Causal Foundation

1.0 Purpose of the Zero-Preserving Causal Foundation

1.0.1 Transition from Admissibility to Structure

Section 0 established the canonical foundation of the framework. It defined a relational algebra $\mathcal{A}$, a family of primitive constraint elements $\{\hat{C}_\alpha\}_{\alpha \in I}$, the generated constraint ideal $\mathcal{I}_C$, and physical states $\omega: \mathcal{A} \to \mathbb{C}$ satisfying the quadratic constraint-vanishing condition $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. It then showed how a Hilbert-space representation may be obtained from a physical state by the GNS construction, and how the represented constraints may be packaged into a positive master constraint operator $\hat{\mathfrak{M}}_\omega$.

The result of Section 0 is an **admissibility criterion**. It tells us what it means for a relational state to be physical:

> **Physicality means constraint-compatible relational evaluation.**

However, admissibility alone is not yet structure. It does not yet tell us what counts as an operation, what counts as an observable, how admissible structures may be transformed, how one admissible event may depend on another, or how causal order can arise without an external time parameter.

The purpose of the present section is to begin that next layer.

1.0.2 What This Section Introduces

Section 1 introduces the first post-foundational structures:

1. **Zero-preserving operations** — transformations that preserve the physical sector.
2. **Physical observables** — operations compatible with constraint admissibility.
3. **Causal dependency** — a strict partial order among admissible events, derived from constraint closure.
4. **The open extension principle** — the universe is an extendible, not a closed, zero-preserving structure.
5. **The two-link principle** — causal links and relational correlation links are structurally distinct.

These structures are still pre-geometric. They do not assume a background spacetime, metric, manifold, field equation, or external clock. They describe how physical admissibility may be preserved, related, and extended within the relational constraint framework.

1.0.3 Why "Zero-Preserving" Comes First

The foundational zero condition identifies the physical sector. In a represented Hilbert-space form, this sector may be written as:
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega = \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha). $$

In an effective classical description, the physical configuration space may be written as:
$$ \mathcal{X}_{\mathrm{phys}} = \mathfrak{M}^{-1}(0). $$

Once such a physical sector has been identified, the next natural question is:

> **Which operations preserve it?**

An operation that maps a physical state or configuration into an unphysical one cannot be regarded as a physical operation without further qualification. If applying an operation produces constraint violation, then that operation does not preserve admissibility.

Therefore, the first structural requirement beyond the foundation is:

> **Physical operations must preserve the zero sector.**

This is the origin of the term **zero-preserving**.

In the represented Hilbert-space setting, an operator $\hat{O}$ is zero-preserving if:
$$ \hat{O} \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega. $$

Equivalently:
$$ \hat{O} \, \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega. $$

In the effective classical setting, a transformation T is zero-preserving if:
$$ T(\mathfrak{M}^{-1}(0)) \subseteq \mathfrak{M}^{-1}(0). $$

This condition is not yet dynamics. It is a compatibility condition. It says that an admissible operation must not carry the system out of the physical sector.

1.0.4 Observables as Admissibility-Preserving Operations

The relational algebra $\mathcal{A}$ contains possible relational expressions. But Section 0 already established that not every algebra element should automatically be treated as a physical observable. Membership in $\mathcal{A}$ means only that the expression belongs to the kinematic relational language.

An observable must satisfy more. At minimum, it must be compatible with the constraints. In the present framework, this means that observables must preserve the physical sector, either exactly or in an appropriately reduced or quotient sense.

> **An observable is not merely an algebra element; it is an admissibility-preserving relational operation.**

There may be stronger and weaker versions of this idea. A **strong observable** may commute with the master constraint:
$$ [\hat{O}, \hat{\mathfrak{M}}_\omega] = 0. $$

This is sufficient for zero preservation, because if $\psi \in \ker \hat{\mathfrak{M}}_\omega$, then:
$$ \hat{\mathfrak{M}}_\omega \hat{O} \psi = \hat{O} \hat{\mathfrak{M}}_\omega \psi = 0. $$
Therefore $\hat{O} \psi \in \ker \hat{\mathfrak{M}}_\omega$.

But exact commutation on the full kinematic representation may be stronger than necessary. A **weak observable** need only satisfy:
$$ \hat{O} \, \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega. $$

This distinction allows the theory to remain flexible. The physical requirement is preservation of admissibility; global commutation may be a sufficient condition, not always a necessary one.

1.0.5 Why Causal Dependency Must Be Introduced Without External Time

The foundation does not assume external time. Therefore, causal order cannot be introduced as an ordering of events by a background parameter t. There is no primitive equation of the form:
$$ \frac{d}{dt}\psi(t) = \cdots, $$
and no pre-existing spacetime in which events are already arranged along temporal curves.

Nevertheless, the theory must eventually account for temporal structure. It must explain how one event may depend on another, how admissible structures may extend, and how a direction of history may arise.

The first step is to define causal order not as temporal succession in a background time, but as **dependency among zero-preserving structures**.

> **Causal order begins as admissibility dependency, not as background time order.**

If an admissible event e_j requires the prior closure, satisfaction, or availability of another event e_i, then one may write:
$$ e_i \prec e_j. $$

This relation does not yet mean that e_i occurs earlier than e_j in a spacetime coordinate. It means that e_j's admissibility depends on e_i. Temporal interpretation comes later.

Thus, causal dependency is introduced as an internal relational order:
$$ e_i \prec e_j \quad \Longleftrightarrow \quad \text{e_i is a necessary admissibility predecessor of e_j.} $$

Under suitable assumptions—such as irreflexivity, asymmetry, and transitive closure of dependency—this relation becomes a strict partial order.

This gives the framework a way to speak about direction, precedence, and dependency without assuming a clock.

1.0.6 Events Are Not Spacetime Points

A key guardrail for this section is that an event is not initially a spacetime point.

Let $\mathcal{E}$ denote a set of primitive admissible events or event-like structures. At this level, an event $e \in \mathcal{E}$ should be understood as a localised instance of constraint-compatible relational structure, update, resolution, or admissible occurrence.

It is not yet a point $p \in M$ in a spacetime manifold. It is also not yet a field value at a coordinate, a particle occurrence in a background geometry, or a measurement result recorded by an observer. Those interpretations may become available later, once additional structures have been derived.

For now, an event is a unit of admissible relational structure.

> **Events are primitive admissible relational occurrences, not background spacetime points.**

This distinction prevents the causal foundation from smuggling in spacetime geometry.

1.0.7 The Two-Link Principle

The next major purpose of this section is to distinguish two kinds of connection between events: **causal dependency** and **relational correlation**.

A **causal link** is directed. It has the form:
$$ e_i \prec e_j. $$
It says that e_j depends on e_i in the admissibility structure.

A **relational or correlation link**, by contrast, need not be directed. It may express the degree to which two event contexts, observables, or relational regions are correlated in a physical state. A schematic correlation may later be written as:
$$ K_\omega(e_i, e_j) $$
or
$$ \mathrm{Corr}_\omega(O_i, O_j). $$

This kind of relation supports the later construction of emergent spatial geometry. Strong correlation may correspond to small emergent distance, while weak correlation may correspond to large emergent distance.

But this is not the same as causal dependence. Two structures may be strongly correlated without one being the causal predecessor of the other. Conversely, two events may be causally ordered without being spatially near in an emergent correlation geometry.

The section therefore introduces the **Two-Link Principle**:

> **Causal dependency and relational correlation are distinct structural links.**

This distinction is essential. If causal dependency and relational correlation are collapsed into one relation, then the theory risks identifying space with causality too early. Instead, the framework keeps them separate:

- Causal links support emergent time-like order.
- Correlation links support emergent space-like geometry.

Their later interaction may produce spacetime-like structure, but neither is assumed to be spacetime from the start.

1.0.8 Open Extension Without External Time

The final purpose of this section is to introduce admissible extension.

The foundation defines what it means for a state or structure to be physical. Zero-preserving operations define what it means to remain physical under transformation. Causal dependency defines how admissible events may depend on one another. Open extension then describes the possibility that an admissible relational structure can be enlarged while preserving constraint compatibility.

If $\mathcal{U}_s$ is an admissible relational structure, an extension:
$$ \mathcal{U}_s \to \mathcal{U}_{s+1} $$
is admissible if the extended structure remains physical.

In an effective master-functional notation, this may be written as:
$$ \mathfrak{M}(\mathcal{U}_{s+1}) = 0. $$

In the algebraic formulation, it means that the extended state or structure remains compatible with the constraint ideal.

The index s should not be interpreted as external time. It is an ordering label for admissible extension. Physical time, if it emerges, must be reconstructed later from causal depth, burden, clock-rate structure, or other internal measures.

> **Extension order is not yet physical time.**

The universe, or a relational structure, may be called **open** if there remains at least one admissible extension:
$$ \mathrm{Ext}(\mathcal{U}) \neq \varnothing. $$

This does not yet define dynamics. It defines the possibility of non-final admissible continuation.

1.0.9 What This Section Proves and What It Does Not Prove

This section begins the transition from static admissibility to relational structure. It establishes the conceptual and mathematical ingredients needed for later emergence, but it does not yet prove spacetime, dynamics, gravity, or quantum measurement.

It does establish, or prepare to establish, the following:
- Zero-preserving operations preserve the physical sector.
- Physical observables must be admissibility-compatible.
- Causal dependency may be formulated as a relation among admissible events.
- Under suitable closure assumptions, causal dependency forms a strict partial order.
- Causal links and relational correlation links are distinct.
- Admissible extension can be defined without an external time parameter.

It does not yet establish:
- A spacetime manifold.
- A metric.
- Lorentzian causal cones.
- Clock time.
- Hamiltonian time evolution.
- Field propagation.
- Particles.
- Gravity.

Those belong to later sections.

> **Section 1 constructs the zero-preserving causal skeleton, not full spacetime physics.**

1.0.10 Structural Sequence of Section 1

The section proceeds in the following order.

First, it restates master-zero equivalence in the form needed for the post-foundational layers. This clarifies that the physical sector is the zero sector of the represented or effective master constraint.

Second, it defines zero-preserving transformations and observables. These are operations that preserve physical admissibility.

Third, it introduces causal dependency among admissible events. Under appropriate assumptions, this dependency relation is shown to have partial-order structure.

Fourth, it states the Two-Link Principle, distinguishing causal dependency from relational correlation.

Fifth, it introduces admissible extension and openness, while guarding against interpreting extension order as external time.

The conceptual sequence is:
$$ \boxed{ \text{zero sector} \quad \longrightarrow \quad \text{zero-preserving operations} \quad \longrightarrow \quad \text{causal dependency} \quad \longrightarrow \quad \text{open extension}. } $$

This is the first bridge from pure admissibility toward emergent temporal and causal structure.

1.0.11 Guardrails for This Section

The following guardrails apply throughout Section 1.

1. **Zero preservation is not dynamics.** It is admissibility preservation.
2. **Observables are not merely arbitrary algebra elements.** They must be compatible with the physical sector.
3. **Causal dependency is not background spacetime causality.** It is an internal dependency relation among admissible structures.
4. **Events are not initially spacetime points.**
5. **Extension order is not external time.**
6. **Relational correlation is not the same as causal dependence.**
7. **This section does not yet define metric distance, duration, mass, energy, fields, or gravity.**

These guardrails protect the emergence programme by ensuring that later structures are not assumed before they are derived.

1.1 Master-Zero Constraint Equivalence

1.1.1 Purpose of This Subsection

Section 0 introduced several equivalent ways of expressing physical admissibility, depending on the level of description being used.

At the primitive algebraic level, physicality is expressed by the quadratic constraint-vanishing condition:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha. $$

At the represented Hilbert-space level, the same condition becomes:
$$ \pi_\omega(\hat{C}_\alpha)\psi = 0 \qquad \forall \alpha, $$
for vectors ψ in the represented physical sector.

At the master-operator level, the many represented constraints may be packaged into one positive operator:

$$ \hat{\mathfrak{M}}_\omega = \sum_\alpha w_\alpha \,
\pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha), \qquad w_\alpha > 0. $$

The physical sector can then be described by the single zero condition:
$$ \hat{\mathfrak{M}}_\omega \psi = 0. $$

This subsection records the exact equivalence between these formulations. It is placed at the beginning of the zero-preserving causal foundation because later notions—observables, causal events, admissible extensions, and dependency order—all rely on knowing what it means to preserve the physical zero sector.

> **Master zero is equivalent to simultaneous satisfaction of all included constraints.**

This is a structural theorem about positive sums. It is not a claim about nothingness, cancellation, temporal evolution, or spacetime geometry.

1.1.2 The Physical Zero Sector

Let $(\mathscr{H}_\omega, \pi_\omega, \Omega_\omega)$ be the GNS representation associated with a physical state ω, and let $\hat{\mathfrak{M}}_\omega$ be the represented master constraint operator.

The represented physical sector is defined by:
$$ \mathscr{H}_{\mathrm{phys}}^\omega := \ker \hat{\mathfrak{M}}_\omega. $$

This is the simultaneous kernel of all represented primitive constraints:
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \bigcap_{\alpha \in I} \ker
\pi_\omega(\hat{C}_\alpha). $$

The master-zero formulation asserts that the same sector may also be written as:
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega, $$
provided the master constraint is a positive weighted sum of represented constraint squares.

1.1.3 Definition — Represented Master Constraint

Definition 1.1 (Represented Master Constraint).
Let $\{\hat{C}_\alpha\}_{\alpha \in I} \subset \mathcal{A}$ be the primitive constraint family. In the representation associated with ω, define:
$$ \hat{\mathfrak{M}}_\omega := \sum_{\alpha \in I} w_\alpha \,
\pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha), $$
whenever the sum is well-defined on a common invariant domain.

Technical Note.
- If the index set I is finite, this definition is immediate.
- If I is infinite, one must impose the analytic assumptions required for the sum to define a positive operator on a suitable common domain.
- The clean theorem-safe setting is: assume the master constraint is a well-defined positive operator.

1.1.4 Positivity Identity

For any vector $\psi \in \mathscr{H}_\omega$ in the domain of all relevant operators:
$$ \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = \sum_\alpha w_\alpha \langle \psi, \pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha) \psi \rangle = \sum_\alpha w_\alpha |\pi_\omega(\hat{C}_\alpha)\psi|^2. $$

Since each term satisfies $|\pi_\omega(\hat{C}_\alpha)\psi|^2 \geq 0$ and each weight satisfies $w_\alpha > 0$, we obtain:
$$ \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle \geq 0. $$

Thus the master constraint is positive.

1.1.5 Theorem — Master-Zero Equivalence

Theorem 1.1 (Master-Zero Equivalence).
Let $\hat{\mathfrak{M}}_\omega = \sum_\alpha w_\alpha \, \pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha)$ with $w_\alpha > 0$ be a well-defined positive master constraint. Then for every vector ψ in the relevant domain:
$$ \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0 \quad \Longleftrightarrow \quad \pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha. $$

Equivalently:
$$ \ker \hat{\mathfrak{M}}_\omega = \bigcap_{\alpha \in I} \ker \pi_\omega(\hat{C}_\alpha). $$

Proof.

($\Rightarrow$) Assume $\langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0$. Using the positivity identity:
$$ 0 = \sum_\alpha w_\alpha |\pi_\omega(\hat{C}_\alpha)\psi|^2. $$
Every summand is non-negative. Since each weight is strictly positive, the only way the sum can vanish is if every summand vanishes:
$$ |\pi_\omega(\hat{C}_\alpha)\psi|^2 = 0 \quad \forall \alpha. $$
Therefore $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α.

($\Leftarrow$) Assume $\pi_\omega(\hat{C}_\alpha)\psi = 0$ for all α. Then every term in the sum vanishes, so $\hat{\mathfrak{M}}_\omega \psi = 0$. $\blacksquare$

1.1.6 Physical-Sector Form

Using the theorem, the physical sector may be written in any of the following equivalent represented forms:
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \{ \psi : \pi_\omega(\hat{C}_\alpha)\psi = 0 \ \forall \alpha \}, $$
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha), $$

$$ \mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega, $$
or, at the level of expectation values:
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \{ \psi : \langle \psi, \hat{\mathfrak{M}}_\omega \psi \rangle = 0 \}. $$

These are equivalent only under the positivity assumptions stated above.

The most useful form for the next subsection is:
$$ \boxed{ \mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega. } $$

This is the zero sector that physical observables must preserve.

1.1.7 Effective Classical Version

In an effective classical model, one may define a configuration space $\mathcal{X}$ and constraint functions $C_\alpha: \mathcal{X} \to V_\alpha$. With weights $w_\alpha > 0$, the effective classical master functional is:
$$ \mathfrak{M}(x) = \sum_\alpha w_\alpha |C_\alpha(x)|^2. $$

The same positive-sum argument gives:
$$ \mathfrak{M}(x) = 0 \quad \Longleftrightarrow \quad C_\alpha(x) = 0 \quad \forall \alpha. $$

Thus the effective classical physical configuration space is:
$$ \mathcal{X}_{\mathrm{phys}} = \mathfrak{M}^{-1}(0) = \bigcap_\alpha C_\alpha^{-1}(0). $$

1.1.8 Algebraic Version

At the primitive algebraic level, physicality is stated through the positive state condition:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha. $$

Once ω is represented through the GNS construction, this condition implies:
$$ \pi_\omega(\hat{C}_\alpha)\Omega_\omega = 0 \qquad \forall \alpha. $$

By master-zero equivalence:
$$ \Omega_\omega \in \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha). $$

Therefore:
$$ \Omega_\omega \in \ker \hat{\mathfrak{M}}_\omega. $$

Thus the algebraic physical state condition generates the represented master-zero condition for the cyclic vector:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \Longrightarrow \quad \hat{\mathfrak{M}}_\omega \Omega_\omega = 0. $$

The implication runs from algebraic state physicality to represented master zero. The master constraint does not replace the algebraic condition.

1.1.9 Why the Equivalence Matters for Observables

The next subsection will define zero-preserving observables. For that definition to be meaningful, the zero sector must be identified clearly.

Using master-zero equivalence, the represented zero sector can be described as:
$$ \ker \hat{\mathfrak{M}}_\omega. $$

An operator $\hat{O}$ is then zero-preserving if:
$$ \hat{O} \, \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega. $$

Equivalently, if ψ satisfies all represented constraints, then $\hat{O}\psi$ must also satisfy all represented constraints.

That is:
$$ \pi_\omega(\hat{C}_\alpha)\psi = 0 \ \forall \alpha \quad \Longrightarrow \quad \pi_\omega(\hat{C}_\alpha)\hat{O}\psi = 0 \ \forall \alpha. $$

Thus, master-zero equivalence is the bridge from constraint satisfaction to observable admissibility. It tells us exactly what must be preserved.

1.1.10 Strong and Weak Readings of Zero Preservation

The master-zero condition supports two useful levels of observable compatibility.

A **strong compatibility condition** is:
$$ [\hat{O}, \hat{\mathfrak{M}}_\omega] = 0. $$

This implies that $\hat{O}$ preserves the master kernel. If $\hat{\mathfrak{M}}_\omega \psi = 0$, then:
$$ \hat{\mathfrak{M}}_\omega \hat{O} \psi = \hat{O} \hat{\mathfrak{M}}_\omega \psi = 0. $$
So $\hat{O} \psi \in \ker \hat{\mathfrak{M}}_\omega$.

A **weak compatibility condition** is simply:
$$ \hat{O} \, \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega. $$

This does not require $\hat{O}$ to commute with $\hat{\mathfrak{M}}_\omega$ on the full kinematic Hilbert space. It requires only that $\hat{O}$ preserve the physical sector.

The weak condition is the essential one. The strong condition is sufficient, but may not be necessary.

This distinction will matter when defining the observable algebra.

1.1.11 Why Positive Weights Cannot Be Relaxed

The equivalence theorem depends on $w_\alpha > 0$.

If some weight were zero, say $w_\beta = 0$, then the corresponding constraint $\pi_\omega(\hat{C}_\beta)$ would not contribute to the master constraint. A vector could satisfy $\hat{\mathfrak{M}}_\omega \psi = 0$ while still violating $\pi_\omega(\hat{C}_\beta)\psi \neq 0$. Thus, zero weights destroy the equivalence.

If negative weights were allowed, then different constraint terms could cancel in expectation. One could have $\sum_\alpha w_\alpha |\pi_\omega(\hat{C}_\alpha)\psi|^2 = 0$ without each norm being zero. That destroys the master-zero equivalence.

Therefore, the condition $w_\alpha > 0$ is essential.

> **Master-zero equivalence holds for strictly positive weights.**

1.1.12 Infinite Constraint Families

If the index set I is infinite, the master constraint:
$$ \hat{\mathfrak{M}}_\omega = \sum_{\alpha \in I} w_\alpha \, \pi_\omega(\hat{C}_\alpha)^\dagger \pi_\omega(\hat{C}_\alpha) $$
requires additional assumptions.

One must specify:
- the common domain of the constraints,
- the convergence of the positive quadratic form,
- whether the resulting form defines a self-adjoint or closable positive operator,
- whether kernel equivalence holds on the chosen domain.

The theorem remains conceptually the same: a non-negative sum can vanish only if each non-negative term vanishes. But in an infinite-dimensional or unbounded-operator setting, one must state the analytic assumptions precisely.

> **For finite constraint families, the equivalence is immediate; for infinite families, it holds under the assumptions required to make the positive sum well-defined.**

1.1.13 What Master-Zero Equivalence Does Not Say

Master-zero equivalence is often a place where overinterpretation enters. The theorem says only that a positive weighted sum of constraint squares vanishes exactly when every included constraint vanishes.

It does not say:
- that spacetime has emerged,
- that time has begun,
- that causal order is defined,
- that dynamics are present,
- that every physical quantity vanishes,
- that positive and negative quantities cancel,
- that the master constraint is an energy or a Hamiltonian,

- that the master constraint is a time-evolution equation.

> **Zero total constraint violation is equivalent to zero individual constraint violation. Everything beyond that belongs to later layers.**

1.1.14 Role of Master-Zero Equivalence in the Framework

Within the full manuscript, master-zero equivalence plays three roles.

First, it justifies using a single master object to describe many constraints:
$$ \{\hat{C}_\alpha\} \quad \leadsto \quad \hat{\mathfrak{M}}_\omega. $$

Second, it defines the physical zero sector that observables must preserve:
$$ \ker \hat{\mathfrak{M}}_\omega. $$

Third, it provides the admissibility criterion for effective classical models:
$$ \mathfrak{M}(x) = 0. $$

Thus, master-zero equivalence is a bridge between:
- constraint algebra,
- represented Hilbert space,
- and effective classical configuration space.

But it remains a bridge, not the foundation itself.

1.1.15 Guardrails for This Subsection

1. **No pairwise cancellation.** The theorem does not imply $(+X) + (-X) = 0$ or any requirement that physical structure appears as compensating pairs.
2. **No local balancing requirement.** It does not require local compensation or observable pairing.
3. **No dynamics.** The theorem does not describe evolution, time, or causation. It is purely a selection principle.
4. **No claim that $\mathcal{K}$ is spacetime.** The effective configuration space is not yet spacetime.

1.2 Zero-Preserving Observables

1.2.1 Purpose of This Subsection

Section 1.1 established that the physical sector is the zero sector of the represented master constraint:
$$ \mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega = \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha). $$

We now ask: which operations preserve this sector?

Not every represented algebra element should be counted as a physical observable. An element may be well-defined on the larger kinematic structure while failing to preserve the constraint-zero sector. Such an operation would carry physical states into non-physical states.

> **A physical observable must map admissible states to admissible states.**

This makes zero-preserving observables the natural analogue of gauge-compatible or Dirac observables. This subsection defines the class of operations that preserve physical admissibility, establishes their algebraic closure properties, and constructs the observable algebra as a rigorous quotient.

1.2.2 Kinematic Operations Versus Physical Observables

The relational algebra $\mathcal{A}$ contains possible relational expressions. After choosing a state ω, its GNS representation gives represented algebra elements $\pi_\omega(A)$ for $A \in \mathcal{A}$.

However, not every represented algebra element is automatically a physical observable. An element $A \in \mathcal{A}$ may be kinematically meaningful while still failing to preserve the physical sector. In that case, it belongs to the algebraic language of possible relational expressions, but it does not define a physical observable on admissible states.

Thus, one must distinguish:
$$ A \in \mathcal{A} \qquad \text{from} \qquad A \in \mathcal{A}_{\mathrm{obs}}. $$
The first statement says that A is a possible relational expression. The second says that A is compatible with the constraints in the relevant physical sense.

1.2.3 Definition — Represented Zero-Preserving Operator

Because the physical sector is fundamentally a represented object (a subspace of $\mathscr{H}_\omega$), the most direct definition of an observable is representation-dependent.

Definition 1.2 (Represented Zero-Preserving Operator).
Fix a physical state ω and its GNS representation $(\mathscr{H}_\omega, \pi_\omega, \Omega_\omega)$. An operator $\hat{O}$ on $\mathscr{H}_\omega$ is called **zero-preserving** (or a **weak observable**) if:
$$ \hat{O} \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega. $$
Equivalently, if $\psi \in \ker \hat{\mathfrak{M}}_\omega$, then $\hat{O}\psi \in \ker \hat{\mathfrak{M}}_\omega$.

In terms of the individual represented constraints, this means:
$$ \pi_\omega(\hat{C}_\alpha)\psi = 0 \quad \forall \alpha \quad \Longrightarrow \quad \pi_\omega(\hat{C}_\alpha)\hat{O}\psi = 0 \quad \forall \alpha. $$

1.2.4 Algebraic Zero-Preserving Observables

While Definition 1.2 is operationally clear, it relies on the chosen GNS representation. To define observables purely at the algebraic level, we must look at how an algebra element acts on the space of physical states $\mathfrak{S}_{\mathrm{phys}}$.

Definition 1.3 (Algebraic Weak Observable).
An element $A \in \mathcal{A}$ is an **algebraic weak observable** if, for every physical state $\omega \in \mathfrak{S}_{\mathrm{phys}}$ such that $\omega(A^\dagger A) > 0$, the transformed state ω_A defined by:
$$ \omega_A(X) := \frac{\omega(A^\dagger X A)}{\omega(A^\dagger A)} \qquad \forall X \in \mathcal{A}, $$
is also a physical state ($\omega_A \in \mathfrak{S}_{\mathrm{phys}}$).

Interpretation: An algebraic observable maps physical states to physical states. If the system is in an admissible configuration ω, applying the relational operation A yields a new configuration ω_A that still satisfies the constraints ($\omega_A(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$).

Remark 1.1 (On the commutator condition). In standard constraint quantization, a Dirac observable is often defined algebraically by requiring $[A, \hat{C}_\alpha] \in \mathcal{I}_C$. However, in this framework, $\mathcal{I}_C$ is a two-sided ideal of $\mathcal{A}$. Therefore, for *any* $A \in \mathcal{A}$ and *any* $X \in \mathcal{I}_C$, the commutator $[A, X] = AX - XA$ automatically lies in $\mathcal{I}_C$ (since both AX and XA are in the ideal). Thus, the condition $[A, \hat{C}_\alpha] \in \mathcal{I}_C$ is trivially satisfied by the entire kinematic algebra and fails to restrict A to the observable sector. Definition 1.3 provides the non-trivial, representation-independent criterion required.

1.2.5 Strong Observables

A sufficient (but not necessary) condition for zero preservation is exact commutation with the master constraint.

Definition 1.4 (Strong Observable).
An operator $\hat{O}$ on $\mathscr{H}_\omega$ is a **strong observable** if:
$$ [\hat{O}, \hat{\mathfrak{M}}_\omega] = 0 $$
on a common invariant domain containing $\mathscr{H}_{\mathrm{phys}}^\omega$.

Theorem 1.2 (Strong Observables Preserve the Physical Kernel).
If $[\hat{O}, \hat{\mathfrak{M}}_\omega] = 0$ on a common invariant domain containing $\mathscr{H}_{\mathrm{phys}}^\omega$, then $\hat{O}$ is a weak observable.

Proof.
Let $\psi \in \mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega$. Then $\hat{\mathfrak{M}}_\omega \psi = 0$. Using the commutator condition:
$$ \hat{\mathfrak{M}}_\omega \hat{O} \psi = \hat{O} \hat{\mathfrak{M}}_\omega \psi = \hat{O}(0) = 0. $$
Therefore $\hat{O}\psi \in \ker \hat{\mathfrak{M}}_\omega = \mathscr{H}_{\mathrm{phys}}^\omega$. $\blacksquare$

The hierarchy is:
$$ \boxed{ \text{Strong commutation} \Rightarrow \text{Weak zero preservation} } $$
where the reverse implication need not hold. An operator may preserve the physical sector without commuting with the master constraint on the full kinematic Hilbert space.

1.2.6 Closure Properties of Represented Observables

Let the set of represented weak observables be denoted $\mathcal{O}_{\mathrm{weak}}^\omega$.

Theorem 1.3 (Closure Under Composition and Linear Combination).
$\mathcal{O}_{\mathrm{weak}}^\omega$ is closed under operator composition and linear combination. It forms an associative algebra.

Proof.
Let $\hat{O}_1, \hat{O}_2 \in \mathcal{O}_{\mathrm{weak}}^\omega$, and let $\psi \in \mathscr{H}_{\mathrm{phys}}^\omega$.

1. **Composition:** Since $\hat{O}_1$ is zero-preserving, $\hat{O}_1 \psi \in \mathscr{H}_{\mathrm{phys}}^\omega$. Since $\hat{O}_2$ is zero-preserving, $\hat{O}_2(\hat{O}_1 \psi) \in \mathscr{H}_{\mathrm{phys}}^\omega$. Thus, $(\hat{O}_2 \hat{O}_1)\psi \in \mathscr{H}_{\mathrm{phys}}^\omega$.
2. **Linear Combination:** Since $\hat{O}_1 \psi \in \mathscr{H}_{\mathrm{phys}}^\omega$ and $\hat{O}_2 \psi \in \mathscr{H}_{\mathrm{phys}}^\omega$, and since $\mathscr{H}_{\mathrm{phys}}^\omega$ is a linear subspace, any linear combination $\lambda \hat{O}_1 \psi + \mu \hat{O}_2 \psi \in \mathscr{H}_{\mathrm{phys}}^\omega$. $\blacksquare$

1.2.7 Adjoint Closure and Reduction

While $\mathcal{O}_{\mathrm{weak}}^\omega$ is an algebra, it is not automatically closed under adjoints. If $\hat{O} \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega$, it does not automatically follow that $\hat{O}^\dagger \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega$.

To obtain a $(*)$-algebra of physical observables, one must require **reduction** of the physical subspace.

Definition 1.5 (Reducing Observable).
An operator $\hat{O}$ **reduces** the physical sector if both $\hat{O}$ and $\hat{O}^\dagger$ are weak observables:
$$ \hat{O} \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega \qquad \text{and} \qquad \hat{O}^\dagger \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega. $$

If $\hat{O}$ reduces $\mathscr{H}_{\mathrm{phys}}^\omega$, then the restriction of $\hat{O}$ to the physical sector behaves cleanly as a physical operator, and the subalgebra of reducing observables forms a unital $(*)$-algebra.

> **Zero-preserving operators form an algebra, but adjoint closure requires an additional reduction condition.**

1.2.8 The Physical Observable Algebra as a Quotient

Because elements of the constraint ideal $\mathcal{I}_C$ vanish on the physical sector, they are operationally trivial. Two observables that differ by an element of $\mathcal{I}_C$ represent the same physical operation.

Theorem 1.4 (Well-Definedness of the Observable Quotient).
Let $A \in \mathcal{A}$ be an algebraic observable (Definition 1.3). If $X \in \mathcal{I}_C$, then $A + X$ is also an algebraic observable, and A and $A+X$ generate the same transformation on the space of physical states $\mathfrak{S}_{\mathrm{phys}}$.

Proof.
Let $\omega \in \mathfrak{S}_{\mathrm{phys}}$. The action of A on ω is $\omega_A(X') = \omega(A^\dagger X' A)$. The action of $A+X$ is:
$$ \omega_{A+X}(X') = \frac{\omega((A+X)^\dagger X' (A+X))}{\omega((A+X)^\dagger (A+X))}. $$
Expanding the numerator:
$$ \omega(A^\dagger X' A) + \omega(X^\dagger X' A) + \omega(A^\dagger X' X) + \omega(X^\dagger X' X). $$
Because $X \in \mathcal{I}_C$, and by Lemma 0.9 (Linear Ideal Annihilation), any state evaluation where X appears as an outermost factor (on the left or right) vanishes. Specifically, $\omega(X^\dagger Y) = 0$ and $\omega(Y X) = 0$ for all $Y \in \mathcal{A}$. Thus, the last three terms vanish.
Similarly, the denominator $\omega((A+X)^\dagger (A+X)) = \omega(A^\dagger A) + \text{vanishing terms} = \omega(A^\dagger A)$.
Therefore, $\omega_{A+X}(X') = \omega_A(X')$. They induce identical transformations on physical states. $\blacksquare$

This theorem allows us to rigorously define the physical observable algebra.

Definition 1.6 (Physical Observable Algebra).
The **physical observable algebra** is the quotient of the algebraic weak observables by the constraint ideal:
$$ \mathcal{A}_{\mathrm{obs}} := \mathcal{A}_{\mathrm{weak}} / \mathcal{I}_C. $$
Two observables are physically equivalent if they differ by an element that annihilates the physical state space.

1.2.9 Effective Classical Zero-Preserving Transformations

In an effective classical description, the physical configuration space is:
$$ \mathcal{X}_{\mathrm{phys}} = \mathfrak{M}^{-1}(0). $$

A transformation $T: \mathcal{X} \to \mathcal{X}$ is **zero-preserving** if:
$$ T(\mathcal{X}_{\mathrm{phys}}) \subseteq \mathcal{X}_{\mathrm{phys}}. $$
Equivalently:
$$ \mathfrak{M}(x) = 0 \quad \Longrightarrow \quad \mathfrak{M}(Tx) = 0. $$
Using the classical master-zero equivalence, this means a classical transformation is admissible if it maps constraint-satisfying configurations to constraint-satisfying configurations.

Again, this is not dynamics by itself. It is a preservation condition. A dynamical law would require a specified rule selecting a particular family of transformations or trajectories.

1.2.10 Guardrails for This Subsection

1. **Zero preservation is not dynamics.** It is admissibility preservation.
2. **Observables are not merely arbitrary algebra elements.** They must be compatible with the physical sector.
3. **The quotient is well-defined.** Theorem 1.4 ensures that $\mathcal{A}_{\mathrm{obs}} = \mathcal{A}_{\mathrm{weak}} / \mathcal{I}_C$ is rigorously grounded and independent of the representative chosen.
4. **Adjoint closure requires care.** Weak observables form an algebra, but a $(*)$-algebra requires the reduction condition (Definition 1.5).
5. **The commutator condition is vacuous for two-sided ideals.** The algebraic condition $[A, \hat{C}_\alpha] \in \mathcal{I}_C$ is trivially satisfied by all $A \in \mathcal{A}$ and cannot be used to define observables in this framework. State-preservation (Definition 1.3) is the correct non-trivial criterion.

1.3 Causal Dependency as a Strict Partial Order

1.3.1 Purpose of This Subsection

Section 1.1 established that the physical sector is the zero sector of the master constraint. Section 1.2 defined zero-preserving observables as operations that preserve this sector.

The next structural question is:

> **How does causal order arise before time?**

In a background-independent relational theory, time cannot be assumed as an external, continuous parameterizing real line along which events evolve. Instead, temporal progression must emerge directly from the internal structural arrangements of the events themselves.

The Relational Constraint Framework establishes this by declaring that the causal dependency relation $\prec$ among primitive events $e \in \mathcal{E}$ constitutes a fundamental partial order. Events are not organized by an external clock; rather, their sequencing is defined by structural necessity—specifically, how the resolution of a constraint defect at one node conditions the boundary conditions for subsequent defects.

This subsection defines causal dependency, introduces the closure-based refinement, and proves that the dependency relation forms a strict partial order.

1.3.2 Why Causality Must Be Introduced Without External Time

The foundation does not assume external time. Therefore, causal order cannot be introduced as an ordering of events by a background parameter t. There is no primitive equation of the form:
$$ \frac{d}{dt}\psi(t) = \cdots, $$
and no pre-existing spacetime in which events are already arranged along temporal curves.

Nevertheless, the theory must eventually account for temporal structure. It must explain how one event may depend on another, how admissible structures may extend, and how a direction of history may arise.

The first step is to define causal order not as temporal succession in a background time, but as **dependency among zero-preserving structures**.

> **Causal order begins as admissibility dependency, not as background time order.**

If an admissible event e_j requires the prior closure, satisfaction, or availability of another event e_i, then one may write:
$$ e_i \prec e_j. $$

This relation does not yet mean that e_i occurs earlier than e_j in a spacetime coordinate. It means that e_j's admissibility depends on e_i. Temporal interpretation comes later.

1.3.3 Definition — Primitive Event

Definition 1.7 (Primitive Event).
Let $\mathcal{E}$ be a set whose elements are called **primitive zero-preserving events**. At this level, an event $e \in \mathcal{E}$ is not a spacetime point. It is not assumed to be a coordinate location, a field value on a manifold, or an observation made by an external observer. Rather, an event is a localised instance of admissible relational structure, update, resolution, or admissible occurrence.

Interpretation. A primitive event is the minimal unit of structurally admissible change in the zero-preserving relational framework.

1.3.4 Definition — Constraint Closure

To define dependency, we need a formal way to talk about the prerequisites of an event. We introduce the concept of constraint closure.

Definition 1.8 (Constraint Closure).

For each event $e \in \mathcal{E}$, let $\Gamma(e)$ denote its **constraint closure set**: the set of primitive zero-preserving structures required for e to be admissible as part of a zero-consistent history.

Remark 1.2 (Concrete examples of $\Gamma(e)$). The closure notation $\Gamma(e)$ is treated as a primitive closure map sufficient to define causal dependency. The exact mathematical form of $\Gamma(e)$ may vary by model. Examples include:
1. **Prerequisite events:** $\Gamma(e)$ is a set of prior events $\{e_k\}$ that must be realized before e can be admissible.
2. **Algebraic support ideal:** $\Gamma(e)$ is the sub-ideal of $\mathcal{I}_C$ generated by the specific constraints that e resolves or relies upon.
3. **Causal network domain:** In a discrete causal set model, $\Gamma(e)$ is the past domain of dependence (the ancestral set) of node e.

1.3.5 Definition — Causal Dependency Relation

Definition 1.9 (Causal Dependency).
For events $e_i, e_j \in \mathcal{E}$, define:
$$ e_i \prec e_j $$
to mean that e_j requires e_i for zero-closure.

In the closure-based refinement:
$$ \boxed{ e_i \prec e_j \quad \Longleftrightarrow \quad \Gamma(e_i) \subsetneq \Gamma(e_j) \quad \text{and} \quad \Gamma(e_i) \text{ is necessary for } \Gamma(e_j). } $$

Interpretation. Causality is thus not taken as a primitive time ordering. Instead:
$$ \boxed{ \text{causal precedence} = \text{dependency of zero-closure} } $$

This is one of the distinctive structural moves of the framework. Causal order is not imposed from outside; it is derived from the constraint-closure structure.

1.3.6 Why Dependency Implies Distinction

If e_j depends on e_i, that does not mean $e_i = e_j$. In fact, for primitive causality, dependency should imply non-identity.

If $e_i \prec e_j$, then e_i plays the role of a necessary condition for e_j. That means e_j is not identical to e_i, because the dependent structure includes an additional closure requirement not exhausted by e_i.

So we add the axiom:
$$ e_i \prec e_j \quad \Rightarrow \quad e_i \neq e_j. $$

This directly gives irreflexivity. Because if we suppose $e \prec e$, then by the distinction rule $e \neq e$, which is impossible. Therefore $\neg(e \prec e)$.

1.3.7 Theorem — Irreflexivity from Proper Closure Extension

Theorem 1.4 (Irreflexivity from Proper Closure Extension).
Let $\mathcal{E}$ be a set of minimal zero-preserving events. Suppose each event $e \in \mathcal{E}$ has a constraint closure $\Gamma(e)$. Define $e_i \prec e_j$ only if $\Gamma(e_i) \subsetneq \Gamma(e_j)$. Then $\prec$ is irreflexive.

Proof.
For any event e, we have $\Gamma(e) = \Gamma(e)$. A set cannot be a proper subset of itself:
$$ \Gamma(e) \not\subsetneq \Gamma(e). $$
Therefore the defining condition for $e \prec e$ cannot hold. Hence $\neg(e \prec e)$.
$\blacksquare$

1.3.8 Theorem — Transitivity from Closure Inclusion and Necessity Composition

Theorem 1.5 (Transitivity from Closure Inclusion).
Let $e_1, e_2, e_3 \in \mathcal{E}$. Suppose $e_1 \prec e_2$ and $e_2 \prec e_3$. Then $e_1 \prec e_3$, provided closure necessity composes.

Proof.
By Definition 1.9:
$$ \Gamma(e_1) \subsetneq \Gamma(e_2) \quad \text{and} \quad \Gamma(e_2) \subsetneq \Gamma(e_3). $$
By transitivity of proper subset inclusion:
$$ \Gamma(e_1) \subsetneq \Gamma(e_3). $$

By the necessity composition assumption: if e_1 is necessary for e_2, and e_2 is necessary for e_3, then e_1 is necessary for e_3.

Therefore $e_1 \prec e_3$. $\blacksquare$

Assumption 1.1 (Necessity Composition).
If $e_1 \prec e_2$ and $e_2 \prec e_3$, then the necessity relation composes: the necessity of $\Gamma(e_1)$ for $\Gamma(e_2)$ together with the necessity of $\Gamma(e_2)$ for $\Gamma(e_3)$ implies the necessity of $\Gamma(e_1)$ for $\Gamma(e_3)$.

Remark 1.3 (Defense of Assumption 1.1). Necessity composition is the structural analogue of the transitivity of logical entailment: if A entails B and B entails C, then A entails C. In the constraint algebra, if the consistency of e_2 strictly requires the prior satisfaction of e_1, and the consistency of e_3 strictly requires the prior satisfaction of e_2, then e_3 indirectly relies on e_1 to secure its own necessary conditions. This assumption is the minimal extra hypothesis required to complete the transitivity proof.

1.3.9 Theorem — Causal Dependency Is a Strict Partial Order

Theorem 1.6 (Causal Dependency Is a Strict Partial Order).
Under Definition 1.9 and Assumption 1.1, the relation $\prec$ on $\mathcal{E}$ is a strict partial order. That is, for all $e, e_1, e_2, e_3 \in \mathcal{E}$:

1. **Irreflexivity:** $\neg(e \prec e)$
2. **Transitivity:** if $e_1 \prec e_2$ and $e_2 \prec e_3$, then $e_1 \prec e_3$
3. **Asymmetry:** if $e_1 \prec e_2$, then $\neg(e_2 \prec e_1)$

Proof.

(i) Irreflexivity. Proved in Theorem 1.4.

(ii) Transitivity. Proved in Theorem 1.5.

(iii) Asymmetry. Suppose both $e_1 \prec e_2$ and $e_2 \prec e_1$. Then Definition 1.9 requires both $\Gamma(e_1) \subsetneq \Gamma(e_2)$ and $\Gamma(e_2) \subsetneq \Gamma(e_1)$, which is impossible. Hence asymmetry holds.

Therefore $\prec$ is a strict partial order. $\blacksquare$

1.3.10 Dependency Order Without Total Ordering

A strict partial order need not compare every pair of events.

There may exist events $e_i, e_j \in \mathcal{E}$ such that neither $e_i \prec e_j$ nor $e_j \prec e_i$ holds. Such events are **dependency-incomparable**.

Definition 1.10 (Causal Incomparability).
Two events $e_i, e_j \in \mathcal{E}$ are **causally incomparable** if:
$$ e_i \not\prec e_j \quad \text{and} \quad e_j \not\prec e_i. $$

Definition 1.11 (Antichain).
An **antichain** $\Sigma \subset \mathcal{E}$ is a set of mutually incomparable events:
$$ \forall e_i, e_j \in \Sigma, \quad e_i \not\prec e_j, \quad e_j \not\prec e_i. $$

An antichain serves as a candidate emergent spatial cross-section, since its elements are not causally ordered relative to each other.

> **Causal dependency gives partial order, not necessarily total sequence.**

This allows the framework to support parallel admissible developments without imposing a universal external clock.

1.3.11 Causal Order Versus External Time

The relation $e_i \prec e_j$ should not be read as $t(e_i) < t(e_j)$ for some primitive time function t.

At this stage, no such time function has been defined. Instead, $e_i \prec e_j$ means that e_i is structurally prior to e_j in the dependency profile.

A time function, if later introduced, would have to be compatible with the causal order. For example, one might later define a monotone function $T: \mathcal{E} \to \mathbb{R}$ such that:
$$ e_i \prec e_j \quad \Longrightarrow \quad T(e_i) < T(e_j). $$

But such a function is not assumed here. It would be a derived or effective ordering parameter.

> **Causal dependency precedes time labelling.**

1.3.12 Causal Depth

Once a partial order exists, one may define a notion of causal depth.

For causally related events $e_i \prec e_j$, define:
$$ L_C(e_i, e_j) := \max\{ n \mid e_i \prec e_1 \prec e_2 \prec \cdots \prec e_n \prec e_j \}. $$

This is the length of the longest dependency chain between e_i and e_j.

Definition 1.12 (Causal Depth).
The **causal depth** between e_i and e_j is the maximum number of intermediate dependency steps in any chain from e_i to e_j.

Causal depth is an ordering measure, but it is not yet physical time. It gives a pre-temporal skeleton from which temporal structure may later be built.

Later, proper time may be reconstructed by weighting causal depth by burden, clock factors, or other relational quantities. But the partial order provides the skeleton on which such later temporal structure may be built.

> **Causal depth is order-theoretic; proper time requires further structure.**

1.3.13 Causal Chains and Admissible Histories

A finite causal chain:
$$ e_0 \prec e_1 \prec \cdots \prec e_n $$
may be interpreted as an **admissible dependency history**, provided each event in the chain is zero-compatible and each dependency relation is admissibility-preserving.

Such a chain is not yet a trajectory through spacetime. It is a sequence of dependency inclusions.

A history in this framework is therefore initially a chain or network of admissible dependency relations, not a path parameterised by external time.

> **A history is a structured chain of admissibility dependencies.**

Later, if emergent time and geometry are constructed, such histories may acquire temporal or spatial interpretation.

1.3.14 Cycles and Inconsistency

If causal dependency is strict, cycles are forbidden.

A causal cycle would have the form:
$$ e_1 \prec e_2 \prec \cdots \prec e_n \prec e_1. $$

By transitivity, this would imply $e_1 \prec e_1$, contradicting irreflexivity.

Therefore, strict causal dependency excludes dependency cycles.

> **Strict causal dependency is acyclic.**

This acyclicity is not imposed by external time. It follows from the strict partial-order structure.

1.3.15 Causal Dependency and Zero Preservation

Causal dependency must be compatible with zero preservation.

If $e_i \prec e_j$, then the transition or dependency relation from e_i to e_j must not introduce constraint violation. Otherwise, the dependency would not relate admissible structures within the physical sector.

In an operator-like representation, one may think of an event transition as being mediated by a zero-preserving operation $\hat{O}_{ij}$. Then one requires:
$$ \hat{O}_{ij} \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq
\mathscr{H}_{\mathrm{phys}}^\omega. $$

In an effective classical form, if an event transition is represented by a map T_{ij}, then:
$$ T_{ij}(\mathfrak{M}^{-1}(0)) \subseteq \mathfrak{M}^{-1}(0). $$

> **Causal links must be zero-compatible links.**

Thus, causal dependency should be understood as dependency inside the admissible sector, not as arbitrary relation among kinematic possibilities.

1.3.16 Local Finiteness

For two causally related events $e_i \prec e_j$, define the **causal interval**:
$$ I(e_i, e_j) := \{ e_k \in \mathcal{E} \mid e_i \prec e_k \prec e_j \}. $$

A locally finite causal structure satisfies:
$$ |I(e_i, e_j)| < \infty. $$

This condition provides a discrete pre-geometric analogue of finite causal separation. It is useful for a first model, but is not assumed forever.

1.3.17 Guardrails for This Subsection

1. **Causality is dependency, not time.** The relation $\prec$ is a structural dependency relation, not an external time order.
2. **Extension is not evolution in time.** The extension index s labels admissible growth steps; it is not physical temporal duration.
3. **The partial-order theorem is completed under an explicit extra assumption.** Assumption 1.1 (necessity composition) is introduced to finish the transitivity proof.
4. **Causal depth is not proper time.** Although causal depth can be defined from a partial order, it is not yet physical duration.
5. **Causal dependency is pre-geometric.** It does not rely on light cones, metric signature, finite propagation speed, or a manifold.

1.4 The Two-Link Principle

1.4.1 Purpose of This Subsection

The previous subsection introduced causal dependency as a strict partial order among admissible events. In that setting, $e_i \prec e_j$ means that the admissibility of e_j depends on e_i. This relation is directed, order-like, and pre-temporal. It gives the framework a way to speak about causal structure without assuming an external time parameter.

However, causal dependency alone is not sufficient to reconstruct physical reality. The framework must also explain how space-like structure, distance, locality, and geometric neighborhoods can emerge. These cannot be obtained merely by saying that one event depends on another. Spatial proximity is not the same thing as causal priority. Two structures may be spatially or relationally close without either one being causally prior to the other. Conversely, one event may be causally upstream of another without being nearby in the eventual emergent geometry.

This motivates the **Two-Link Principle**.

> **The framework requires two distinct relational links: causal dependency and relational correlation.**

Causal links support order and eventual time-like structure. Relational-correlation links support proximity and eventual space-like structure. The two may interact, but they must not be collapsed into one primitive relation.

1.4.2 The Problem with a Single-Link Theory

A theory that uses only one kind of link faces a structural ambiguity.

If every link is interpreted causally, then strong correlation or adjacency would automatically imply causal dependence. This would make it difficult to describe simultaneous, mutually correlated, or space-like related structures. It would also risk turning spatial closeness into causal precedence.

If every link is interpreted correlationally, then dependency and direction are lost. The theory would have difficulty explaining ordered histories, admissibility chains, causal depth, or irreversible extension.

Thus, a single relation cannot safely do both jobs.

The framework therefore separates:
- **directed admissibility dependency** from
- **state-dependent relational correlation**.

This distinction is foundational for later emergence. It prevents space from being reduced too quickly to causal order and prevents causality from being reduced to mere correlation.

> **Correlation is not causation, and causal dependency is not spatial distance.**

1.4.3 Causal Links

A **causal link** is a directed dependency relation between admissible events:
$$ e_i \prec e_j. $$

It means that e_i is a necessary admissibility predecessor of e_j. In the dependency-profile formulation (Definition 1.9), this is represented by proper inclusion and necessity:
$$ \Gamma(e_i) \subsetneq \Gamma(e_j) \quad \text{and} \quad \Gamma(e_i) \text{ is necessary for } \Gamma(e_j). $$

A causal link has the following features:
1. It is **directed**.
2. It is **asymmetric** in the strict case (Theorem 1.6).
3. It supports **chains** ($e_0 \prec e_1 \prec \cdots \prec e_n$).
4. It contributes to **causal depth** (L_C).
5. It is associated with **admissibility dependency** rather than metric nearness.

Thus, if $e_i \prec e_j$, then e_j lies downstream of e_i in the dependency structure.

This does not yet mean that e_i occurs earlier than e_j in physical clock time. It means that e_j's admissibility requires e_i.

> **Causal links encode dependency before clock time is defined.**

1.4.4 Relational-Correlation Links

A **relational-correlation link** expresses state-dependent association, similarity, mutual constraint sensitivity, or correlation between event contexts or observables.

Given a physical state ω and localisable observables or event-context representatives O_i, O_j, one may define a correlation kernel schematically as:
$$ K_\omega(e_i, e_j) \quad \text{or} \quad K_\omega(O_i, O_j). $$

This kernel measures how strongly the two relational contexts are associated in the state ω. The exact, constructive definition of K_ω belongs to the later correlation-geometry section (Section 2.1), where it is defined as the normalized state overlap $K_\omega(A, B) = |\omega(A^\dagger B)| / \sqrt{\omega(A^\dagger A)\omega(B^\dagger B)}$. At this stage, only its structural role matters.

A relational-correlation link has the following features:
1. It need not be **directed**.
2. It may be **symmetric**.
3. It may vary with the **physical state ω**.
4. It supports emergent notions of **proximity**.
5. It is **not by itself** a causal dependency relation.

If $K_\omega(e_i, e_j)$ is large, the two event contexts are strongly correlated. Later, one may define an emergent distance by a relation such as:
$$ d_\omega(e_i, e_j) = -\ell_0 \log K_\omega(e_i, e_j), $$
under suitable assumptions.

But this does not mean $e_i \prec e_j$ or $e_j \prec e_i$. It only means that the two contexts are strongly linked in the correlation structure.

> **Relational-correlation links encode proximity-like structure before space is defined.**

1.4.5 Statement of the Two-Link Principle

Principle 1.1 (Two-Link Principle).
The framework contains two structurally distinct kinds of link:

$$ \boxed{ \textbf{Causal links:} \quad e_i \prec e_j } $$
and
$$ \boxed{ \textbf{Relational-correlation links:} \quad K_\omega(e_i, e_j). } $$

The first is directed and dependency-based. The second is correlation-based and may be symmetric or state-dependent.

The principle may be stated compactly as:
$$ \boxed{ \text{Causal dependency and relational correlation are distinct primitive post-foundational structures.} } $$

They are "post-foundational" because both are introduced only after the physical zero sector has been defined. They are not more primitive than constraint compatibility. But once physical admissibility is available, both kinds of link are needed.

1.4.6 The Core Friction: Asymmetry vs. Symmetry

The fundamental friction between these two layers is mathematical and ontological.

The causal network forms an acyclic, directed graph tracking the irreversible flow of constraint enforcement. The relational network forms a weighted, undirected graph tracking instantaneous statistical or algebraic dependencies.

If the relational network operated independently of the causal order, non-local quantum correlations would instantly collapse the causal history into a trivial cliquish structure, rendering the concepts of time, sequence, and evolution meaningless. Conversely, if the causal network existed without relational correlations, events would remain isolated nodes incapable of generating macroscopic spatial intervals or stable field configurations.

The framework resolves this tension through the **master zero constraint**. The master constraint dictates that the algebraic ideal generated by the constraints must annihilate physical states, mapping relational correlations directly onto the causal channels. This guarantees that a relational correlation between two space-like separated domains can only be established if their respective ancestral histories intersect in a manner permitted by the causal propagator network.

1.4.7 Why Causal Links Support Time-Like Structure

Causal links generate ordered chains:
$$ e_0 \prec e_1 \prec \cdots \prec e_n. $$

Such chains support causal depth (Definition 1.12):
$$ L_C(e_0, e_n) = \max\{ n \mid e_0 \prec e_1 \prec \cdots \prec e_n \}. $$

This gives the framework an order-theoretic skeleton from which temporal structure may later be derived.

However, causal depth alone is not proper time. To obtain something resembling physical duration, later sections must introduce additional weights, such as burden, reconciliation cost, lapse factors, or local clock-rate suppression.

Still, causal links are the correct starting point for time-like emergence because they encode direction and dependency.

> **Time-like structure begins with directed causal dependency.**

1.4.8 Why Relational-Correlation Links Support Space-Like Structure

Relational-correlation links support proximity-like relations.

If two event contexts are strongly correlated in a physical state, then they may be close in an emergent relational geometry. If they are weakly correlated, they may be far apart.

This motivates the later definition of correlation distance:
$$ d_\omega(A, B) = -\ell_0 \log K_\omega(A, B). $$

Under additional assumptions, such as symmetry, diagonal normalisation, and multiplicative consistency, this distance can become a pseudometric or approximate metric.

But none of that belongs to causal order alone. A causal partial order does not directly define spatial distance. It can say that one event depends on another, but it cannot by itself say how far apart two non-comparable events are.

Therefore, a separate correlation link is needed.

> **Space-like structure begins with state-dependent relational correlation.**

1.4.9 Independence of the Two Links

The two kinds of link are logically independent.

It is possible to have causal dependency without strong correlation:
$$ e_i \prec e_j \quad \text{while} \quad K_\omega(e_i, e_j) \text{ is small.} $$

This would represent a case where e_j depends on e_i, but the two event contexts are not close in the emergent correlation geometry.

It is also possible to have strong correlation without causal dependency:
$$ K_\omega(e_i, e_j) \approx 1 \quad \text{while neither} \quad e_i \prec e_j \quad \text{nor} \quad e_j \prec e_i. $$

This would represent a case where two contexts are relationally close or highly correlated, but neither is an admissibility predecessor of the other.

Therefore:
$$ \boxed{ e_i \prec e_j \not\Longleftrightarrow K_\omega(e_i, e_j) \text{ is large.} } $$
and
$$ \boxed{ K_\omega(e_i, e_j) \text{ large} \not\Longleftrightarrow e_i \prec e_j. } $$

This independence is necessary for the framework to support both causal ordering and spatial neighborhoods.

1.4.10 Interaction Between the Two Links

Although causal links and relational-correlation links are distinct, they may interact.

For example, the causal propagation of admissible structure may change correlation profiles. If an event e_i is causally upstream of e_j, then the admissibility of e_j may inherit, transform, or constrain correlations associated with e_i.

Conversely, strong relational correlation may affect which causal extensions are low-burden or admissible. If two regions are highly correlated, extending one may impose constraint pressure on the other.

Thus, the two links are not isolated. They form a coupled relational structure.

A schematic combined structure may be written as:
$$ (\mathcal{E}, \prec, K_\omega), $$
where:
- $\mathcal{E}$ is the event set,
- $\prec$ is the causal dependency relation,
- K_ω is the state-dependent correlation kernel.

This combined structure is the **pre-spacetime relational skeleton**.

> **Causal and correlation links may interact, but they remain conceptually distinct.**

1.4.11 The Two-Link Graph

One may visualise the combined structure as a graph with two types of edges.

A causal edge is directed:
$$ e_i \longrightarrow e_j. $$

A relational-correlation edge may be undirected or weighted:
$$ e_i \leftrightsquigarrow e_j, $$
with weight $K_\omega(e_i, e_j)$.

Thus the event structure is not a single graph but a **layered graph**:

Definition 1.13 (Two-Link Graph).
The pre-spacetime relational skeleton is the layered graph:
$$ \mathcal{G}_\omega = (\mathcal{E}, \mathcal{C}, \mathcal{R}_\omega), $$
where:
- $\mathcal{E}$ is the set of primitive zero-preserving events,
- $\mathcal{C} = \{(e_i, e_j) : e_i \prec e_j\}$ is the causal edge set,
- $\mathcal{R}_\omega$ is the correlation-link structure determined by the physical state ω.

This is only a schematic representation. The true mathematical object may be a partially ordered set equipped with a kernel, a category with correlation weights, a network of local algebras, or another relational structure. But the two-link distinction remains.

1.4.12 Cosmological Implication: Common Closure

The Two-Link Principle suggests a cosmological application.

In standard cosmology, the horizon problem asks why widely separated regions of space appear homogeneous despite insufficient time for ordinary causal interaction. Inflation is the standard leading response, in which regions now widely separated were once close enough to be in causal contact before rapid expansion.

In the present framework, this is reformulated as the **Primordial Common-Closure Hypothesis**.

Regions that are spatially distant in the later emergent geometry may nevertheless share common causal-constraint ancestry in the pre-geometric order:
$$ C_0 \prec A, \qquad C_0 \prec B, \qquad d_\omega(A, B) \gg \ell_0. $$

Thus spatial separation at late times does not imply primordial constraint independence.

This may provide a pre-geometric alternative to inflation: large-scale uniformity may reflect inherited common constraint values rather than late spatial communication.

> **Large spatial distance does not imply primordial causal-constraint independence.**

This is not yet a proof of CMB uniformity. It is a structural possibility required by the two-link separation.

1.4.13 Avoiding Causal Overreach

A common error would be to treat every correlation as a causal link. In this framework, that is not allowed.

If $K_\omega(e_i, e_j)$ is large, one may infer strong relational association, but not directed dependency. To assert causal dependency, one must show $e_i \prec e_j$ or $e_j \prec e_i$ through the dependency structure.

> **High correlation does not imply causal precedence.**

This guardrail is especially important when later discussing non-local correlations, entanglement-like structure, or long-range relational coherence.

1.4.14 Avoiding Geometric Overreach

The reverse error is to treat every causal link as a spatial neighborhood relation.

If $e_i \prec e_j$, then e_i is causally upstream of e_j. But this does not by itself imply that e_i and e_j are close in emergent space.

Spatial closeness requires a correlation-geometric measure, such as $K_\omega(e_i, e_j)$ or its associated distance.

> **Causal dependency does not by itself define spatial distance.**

This guardrail prevents the causal partial order from being overused as a metric.

1.4.15 Guardrails for This Subsection

1. **Two links means two structural roles, not two substances.** The principle distinguishes dependency and correlation. Both may be carried by the same underlying relational algebraic structure.
2. **Correlation links are state-dependent.** The notation K_ω includes the state ω for a reason. Correlation is evaluated in a state. Different physical states may produce different correlation structures.
3. **The causal order is not automatically Lorentzian.** Although causal dependency may later approximate a Lorentzian causal order, this is not assumed here.
4. **Causal dependency and relational correlation are distinct until related by a model.** The framework does not automatically identify spatial separation with causal independence.

1.5 The Open Extension Principle

1.5.1 Purpose of This Subsection

The preceding subsections established three post-foundational structures:
1. **The physical zero sector** — the set of admissible states satisfying the master constraint.
2. **Zero-preserving observables** — operations that preserve admissibility.
3. **Causal dependency** — a strict partial order among admissible events based on constraint closure.

The Two-Link Principle then distinguished causal dependency from relational correlation.

The next question is:
> **How does the admissible relational structure continue, grow, or unfold?**

The foundation defines what it means for a state or structure to be physical. Zero-preserving operations define what it means to remain physical under transformation. Causal dependency defines how admissible events may depend on one another. But none of these yet says whether the admissible structure is complete, extendible, or dynamically closed.

The **Open Extension Principle** addresses this gap. It states that a physical universe is not merely a closed solution to the constraint equations, but an **extendible** zero-preserving structure: one that admits further admissible continuations.

> **A physical relational structure is open if it admits further zero-preserving extension.**

This principle allows the framework to speak about growth, continuation, history, and non-finality without treating time as primitive. Extension is not yet physical time. It is an admissibility relation between relational structures.

1.5.2 Why Extension Is Needed

A constraint-compatible structure may be physically admissible, but admissibility alone does not say whether the structure is complete, extendible, or dynamically closed.

Suppose a relational structure $\mathcal{U}$ satisfies the physical constraints. One may then ask:
> **Can $\mathcal{U}$ be enlarged while remaining physical?**

If the answer is yes, then the structure is not terminal. It has possible admissible continuations.

This matters because a framework that only defines physicality as a static zero condition risks producing a frozen picture. The open extension layer prevents that by introducing the possibility of constraint-compatible continuation.

The key point is that continuation must preserve physicality. An arbitrary enlargement of a relational structure is not acceptable. Only an extension that remains within the zero sector counts as physically admissible.

> **Extension is not arbitrary growth; it is zero-preserving enlargement.**

1.5.3 Relational Structures

To formalize extension, we first define the object that is being extended.

Definition 1.14 (Relational Structure).
A **relational structure** is a tuple:
$$ \mathcal{U}_s = (\mathcal{E}_s, \prec_s, \mathcal{R}_{\omega,s}), $$
where:
- $\mathcal{E}_s$ is a set of primitive zero-preserving events at extension stage s,
- $\prec_s$ is the causal dependency partial order on $\mathcal{E}_s$,
- $\mathcal{R}_{\omega,s}$ is the relational/correlation structure determined by the physical state ω at stage s.

The index s is an **extension label**, not a physical time coordinate. It orders stages of admissible growth.

Interpretation. Depending on the level of description, $\mathcal{U}_s$ may represent:
- a finite event network,
- a partial causal structure,
- a constraint-compatible algebraic state,
- a physical configuration in an effective model,
- or a coarse-grained relational universe.

The abstract requirement is that $\mathcal{U}_s$ belongs to the physical sector. In effective notation:
$$ \mathfrak{M}(\mathcal{U}_s) = 0. $$

In algebraic notation, it means that the state associated with $\mathcal{U}_s$ satisfies:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha. $$

1.5.4 Definition — Extension

Definition 1.15 (Extension).
An **extension** of a relational structure $\mathcal{U}_s$ is a structure $\mathcal{U}_{s+1}$ such that $\mathcal{U}_{s+1}$ contains, continues, refines, or enlarges $\mathcal{U}_s$ in the relevant relational sense.

We write:
$$ \mathcal{U}_s \hookrightarrow \mathcal{U}_{s+1} \quad \text{or} \quad \mathcal{U}_s \to \mathcal{U}_{s+1}. $$

The notation does not yet denote time evolution. It denotes structural inclusion, continuation, or extension.

The relation $\mathcal{U}_s \to \mathcal{U}_{s+1}$ may mean, depending on context:
1. Adding a new admissible event to $\mathcal{E}_s$.
2. Adding a new causal dependency link to $\prec_s$.
3. Adding a new correlation relation to $\mathcal{R}_{\omega,s}$.
4. Refining a coarse-grained structure.
5. Extending a partial relational configuration.
6. Enlarging a constraint-compatible event network.
7. Continuing an admissible history by one or more dependency steps.

The common requirement is that $\mathcal{U}_{s+1}$ preserves the physical admissibility of $\mathcal{U}_s$.

1.5.5 Definition — Admissible Extension

Definition 1.16 (Admissible Extension).
An extension $\mathcal{U}_s \to \mathcal{U}_{s+1}$ is called **admissible** if the extended structure remains physical.

In effective master-functional notation:
$$ \boxed{ \mathcal{U}_s \to \mathcal{U}_{s+1} \text{ is admissible} \quad \Longleftrightarrow \quad \mathfrak{M}(\mathcal{U}_{s+1}) = 0. } $$

In represented Hilbert-space notation, if the extension is implemented by an operator or map $\hat{E}: \mathscr{H}_\omega \to \mathscr{H}_\omega$, then admissibility requires:
$$ \hat{E} \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega. $$

That is:
$$ \hat{E} \, \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega. $$

In algebraic state language, an extension from a physical state ω_s to an extended state ω_{s+1} is admissible if:
$$ \omega_{s+1}(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha. $$

Thus, in every formulation, admissible extension means:
$$ \boxed{ \text{extension while preserving constraint-zero physicality.} } $$

1.5.6 Extension Is Not External Time Evolution

The notation $\mathcal{U}_s \to \mathcal{U}_{s+1}$ is useful, but it must be interpreted carefully.

The label s is **not** a physical time coordinate. It is an ordering label for extension steps.

At this stage, the framework has not introduced:
- a primitive time parameter t,
- a derivative $\frac{d}{dt}$,
- a Hamiltonian H,
- or a Schrödinger-type equation $i\frac{d}{dt}\psi(t) = H\psi(t)$.

Therefore, extension should not be described as ordinary time evolution.

Instead, $\mathcal{U}_s \to \mathcal{U}_{s+1}$ means that an admissible structure is extended to another admissible structure. The extension index s labels stages of admissible growth, not instants of physical time.

> **Extension order is not yet physical time.**

Physical time may later be reconstructed from causal chains, extension depth, burden weighting, clock suppression, or other relational quantities. But it is not assumed here.

1.5.7 Definition — Extension Set and Open Structures

Definition 1.17 (Extension Set).
For an admissible structure $\mathcal{U}_s$, define its **extension set** by:
$$ \mathrm{Ext}(\mathcal{U}_s) := \{ \mathcal{U}_{s+1} : \mathcal{U}_s \to \mathcal{U}_{s+1} \text{ is admissible} \}. $$

Equivalently, in effective notation:
$$ \mathrm{Ext}(\mathcal{U}_s) = \{ \mathcal{U}_{s+1} : \mathcal{U}_s \to \mathcal{U}_{s+1} \text{ and } \mathfrak{M}(\mathcal{U}_{s+1}) = 0 \}. $$

This set contains all possible zero-preserving continuations of $\mathcal{U}_s$.

Definition 1.18 (Open and Terminal Structures).

An admissible relational structure $\mathcal{U}_s$ is called **open** if it has at least one admissible extension:
$$ \boxed{ \mathrm{Ext}(\mathcal{U}_s) \neq \varnothing. } $$

It is called **extension-terminal** (or simply terminal) if:
$$ \boxed{ \mathrm{Ext}(\mathcal{U}_s) = \varnothing. } $$

Openness therefore means that the structure is not complete relative to the admissibility rules. There remains at least one way to continue the relational structure without generating constraint violation.

This definition does not assume that all possible extensions occur. It only says that the admissible continuation space is non-empty.

1.5.8 The Open Extension Principle

Postulate 1B (Open Extension Principle).
A physical universe is not merely a zero-consistent partial order, but an **extendible** zero-consistent partial order. That is, a universe is a structure $\mathcal{U}_s = (\mathcal{E}_s, \prec_s)$ such that:
$$ \mathfrak{M}(\mathcal{U}_s) = 0 $$
and there exists at least one admissible extension $\mathcal{U}_s \to \mathcal{U}_{s+1}$ with:
$$ \mathfrak{M}(\mathcal{U}_{s+1}) = 0. $$

A compact statement is:
$$ \boxed{ \text{A physical relational structure may continue only through admissible zero-preserving extensions.} } $$

A stronger cosmological version may be stated as:
$$ \boxed{ \text{The universe is open if its current admissible relational structure has non-empty extension set.} } $$

That is:
$$ \boxed{ \mathrm{Ext}(\mathcal{U}) \neq \varnothing. } $$

This should be treated as a principle or assumption unless a specific model proves non-empty extension.

1.5.9 Classification of Structures

The open extension principle classifies relational structures into three types:

1. Closed Solution.
A structure $\mathcal{U}$ is a closed solution if:
$$ \mathrm{Ext}(\mathcal{U}) = \varnothing. $$
This is a completed mathematical object. It satisfies the constraints, but has no admissible continuation. It is like a finished proof or a closed book.

2. Open Universe.
A structure $\mathcal{U}$ is an open universe if:
$$ \mathrm{Ext}(\mathcal{U}) \neq \varnothing. $$
This is an ongoing, dynamically unfinished structure. It satisfies the constraints, and there remains at least one way to extend it while preserving constraint compatibility.

3. Maximal Universe.
A structure $\mathcal{U}$ is maximal if it has exhausted all possible extensions. It is treated as a "completed history," not a "living universe." A maximal universe is the limit of an infinite chain of admissible extensions.

> **The open extension principle states that a physical universe is an open structure, not a closed solution.**

This distinction is philosophically important. The framework does not treat the universe as a static block. It treats it as an ongoing, extendible, zero-preserving process.

1.5.10 Extension and Causal Dependency

Admissible extension naturally interacts with causal dependency.

If an extension adds a new event e_{new} to an existing structure containing events $\{e_0, \ldots, e_n\}$, then the new event must be placed consistently with the causal dependency relation.

For example, an extension may add dependency links:
$$ e_i \prec e_{\mathrm{new}} $$
for some existing events e_i. These links indicate that the admissibility of e_{new} depends on prior closure supplied by those earlier events.

The extension is admissible only if the resulting dependency structure remains consistent. In particular, if causal dependency is strict (Theorem 1.6), the extension must not introduce cycles such as:
$$ e_{\mathrm{new}} \prec e_i \prec e_{\mathrm{new}}. $$

Thus, admissible extension must preserve both:
- **constraint-zero compatibility**, and
- **causal-order consistency**.

> **$\mathcal{U}_s \to \mathcal{U}_{s+1}$ is admissible only if it preserves zero-compatibility and causal acyclicity.**

The exact causal consistency conditions depend on the chosen model of $\prec$.

1.5.11 Extension and the Two-Link Structure

If the relational structure is represented as $(\mathcal{E}, \prec, K_\omega)$, then an extension may modify one or more of these components:
$$ \mathcal{E} \to \mathcal{E}', \qquad \prec \;\to\; \prec', \qquad K_\omega \to K'_\omega. $$

For example, an admissible extension may:
1. Add new events to $\mathcal{E}$.
2. Add new causal links to $\prec$.
3. Update correlation weights in K_ω.
4. Refine equivalence classes of relational proximity.
5. Alter correlation profiles in response to new admissible structure.

The extension is acceptable only if the resulting structure remains physical and internally consistent.

Thus, for a two-link relational skeleton, admissible extension requires:
$$ (\mathcal{E}, \prec, K_\omega) \to (\mathcal{E}', \prec', K'_\omega) $$
with preservation of:
- zero compatibility,
- causal consistency,
- well-defined correlation structure.

1.5.12 Lemma — Zero-Preservation Under Extension

Lemma 1.1 (Zero-Preservation Under Extension).
If $\mathcal{U}_s \to \mathcal{U}_{s+1}$ is an admissible extension in the sense of Definition 1.16, then $\mathcal{U}_{s+1}$ remains in the physically admissible class.

Proof.
By definition of admissible extension, $\mathfrak{M}(\mathcal{U}_{s+1}) = 0$. Therefore $\mathcal{U}_{s+1}$ satisfies the zero-preserving condition and remains physically admissible. $\blacksquare$

Interpretation. This lemma is conceptually simple but structurally important: extension does not move the system out of the physical sector.

1.5.13 Extension-History Equivalence

The extension index s is a gauge-like construction parameter. Different extension sequences that generate the same causal dependency records should be physically equivalent.

Suppose two incomparable events e_a, e_b (neither depends on the other) are added in different orders:
- Sequence 1: e_a then e_b.
- Sequence 2: e_b then e_a.

If neither $e_a \prec e_b$ nor $e_b \prec e_a$ holds, then both histories are physically equivalent:
$$ (e_a, e_b) \sim_{\mathrm{hist}} (e_b, e_a). $$

Definition 1.19 (Extension-History Equivalence).
Two extension histories are **physically equivalent** if they generate the same causal dependency records. If $e_a \not\prec e_b$ and $e_b \not\prec e_a$, then:
$$ \boxed{ (e_a, e_b) \sim_{\mathrm{hist}} (e_b, e_a). } $$

This equivalence relation prevents the extension order from becoming a physically observable preferred frame. It ensures that only the resulting causal order $(\mathcal{E}, \prec)$ and internal relational quantities are observable, not the construction sequence s.

> **s = gauge extension parameter, not time.**

1.5.14 Branching Extensions

An admissible structure may have more than one possible extension:
$$ \mathrm{Ext}(\mathcal{U}_s) = \{ \mathcal{U}'_1, \mathcal{U}'_2, \mathcal{U}'_3, \ldots \}. $$

This means that the future continuation of the structure is not represented by a single predetermined chain at the foundational level. Instead, there may be multiple admissible continuations.

This branching should not be interpreted too quickly as quantum branching, many-worlds ontology, stochastic collapse, or classical indeterminism. It simply means that, relative to the extension rules, more than one zero-preserving continuation is available.

> **Branching extension means multiple admissible continuations, not yet a probability interpretation.**

Probability, Born measure, records, or selection rules require additional structure.

1.5.15 Extension Chains and Depth

An **extension chain** is a sequence:
$$ \mathcal{U}_0 \to \mathcal{U}_1 \to \mathcal{U}_2 \to \cdots $$
such that each step is admissible:
$$ \mathcal{U}_{s+1} \in \mathrm{Ext}(\mathcal{U}_s). $$

That is, $\mathfrak{M}(\mathcal{U}_s) = 0$ for all s in an effective master-functional notation.

An extension chain may be interpreted as an **admissible history** of relational growth. But again, it is not yet a trajectory through external time.

The chain parameter s labels extension steps. It does not yet measure physical duration.

Given an extension chain $\mathcal{U}_0 \to \mathcal{U}_1 \to \cdots \to \mathcal{U}_n$, one may define an **extension depth**:
$$ D_{\mathrm{ext}}(\mathcal{U}_n) = n $$
relative to the chosen initial structure $\mathcal{U}_0$ and the chosen chain.

If there are multiple chains reaching the same structure, one may define maximal or minimal extension depth.

These quantities are order-theoretic. They are not yet physical time. Physical duration may later require weighting extension depth by burden, cost, clock factors, or causal structure.

1.5.16 Admissible Extension as Zero-Preserving Map

If extension is represented by a map $E_s: \mathcal{U}_s \mapsto \mathcal{U}_{s+1}$, then admissibility requires:
$$ \mathcal{U}_s \in \mathcal{U}_{\mathrm{phys}} \quad \Longrightarrow \quad E_s(\mathcal{U}_s) \in \mathcal{U}_{\mathrm{phys}}. $$

In effective notation:
$$ \mathfrak{M}(\mathcal{U}_s) = 0 \quad \Longrightarrow \quad \mathfrak{M}(E_s(\mathcal{U}_s)) = 0. $$

This is exactly zero preservation.

Therefore, admissible extension is a special case of zero-preserving transformation:
$$ \boxed{ \text{admissible extension} \subseteq \text{zero-preserving transformation}. } $$

The difference is interpretive: a general zero-preserving transformation may act within a fixed structure, while an extension enlarges or continues the structure.

1.5.17 Terminal Structures

A structure $\mathcal{U}$ is extension-terminal if:
$$ \mathrm{Ext}(\mathcal{U}) = \varnothing. $$

This means that no allowed extension preserves the required constraints under the chosen rules.

Terminality may arise for several reasons:
1. All possible extensions generate constraint violation.
2. Causal consistency would be broken by any new event.
3. The extension rules impose maximal closure.
4. The structure is complete relative to the chosen model.

However, terminality is model-relative. Changing the allowed extension rules or enlarging the class of admissible structures may change whether $\mathcal{U}$ is terminal.

> **Terminality is relative to the specified extension rules.**

1.5.18 Relation to Dynamics

The open extension principle is close to dynamics, but it is not yet dynamics.

A dynamics would require more than the existence of admissible extensions. It would require a rule selecting, weighting, or generating actual extensions.

For example, one might later introduce:
- a least-burden extension principle,
- a transition amplitude,
- a variational rule,
- a stochastic selection law,
- a causal growth law,
- or an extremal reconciliation principle.

None of these has been defined yet.

> **Open extension gives possibility of continuation, not a full dynamical law.**

This distinction protects the framework from claiming more than has been built.

1.5.19 Relation to Emergent Time

Although extension is not time, it prepares the ground for time.

An extension chain $\mathcal{U}_0 \to \mathcal{U}_1 \to \cdots$ provides an ordering of admissible continuation. A causal chain $e_0 \prec e_1 \prec \cdots$ provides an ordering of dependency.

Later, these orderings may be combined with burden or clock factors to define a proper-time-like quantity.

For example, one might later define a duration along a causal chain by:
$$ \tau[\gamma] = \sum_{e \in \gamma} \frac{\epsilon}{1 + \lambda B(e)}, $$
where $B(e)$ is a local burden factor.

But that belongs to a later section.

At this stage:
> **Extension order is a precursor to time, not time itself.**

1.5.20 Guardrails for This Subsection

1. **Openness is not proof of actual growth.** The condition $\mathrm{Ext}(\mathcal{U}) \neq \varnothing$ says that admissible continuations exist. It does not prove that any particular continuation is actualised.

2. **Extension is not background time.** The sequence $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, \ldots$ should not be interpreted as states at times $t_0, t_1, t_2, \ldots$ unless a time map has been defined.
3. **Branching is not yet probability.** If $|\mathrm{Ext}(\mathcal{U})| > 1$, there are multiple admissible continuations. This does not automatically define probabilities.
4. **Extension rules must be specified.** The set $\mathrm{Ext}(\mathcal{U})$ depends on what counts as an allowed extension. If the extension rules are too permissive, inadmissible or ill-defined continuations may appear. If they are too restrictive, the theory may artificially become terminal.

1.6 Emergent Causal Propagation Speed

1.6.1 Purpose of This Subsection

The framework has established a causal dependency order $\prec$ and an open extension principle. However, a causal order by itself does not impose a finite speed limit on propagation. A partial order merely states dependencies; it does not specify how fast influence can propagate across the relational network.

Standard physics requires a finite invariant speed c. In the Relational Constraint Framework, this speed cannot be postulated as a property of a background spacetime metric. Instead, it must emerge from the interplay between causal depth and relational distance.

The purpose of this subsection is to derive an emergent causal speed ceiling c_{RCF} from the primitive structures of the theory: relational distance, causal depth, and an elementary reconciliation interval.

1.6.2 The Causal Reconciliation Principle

Before defining the speed, we must understand *why* there is a speed limit at all.

In this framework, physical reality is defined by constraint compatibility. When a relational structure extends (grows via the Open Extension Principle), it must do so while preserving the zero-constraint condition. However, constraint satisfaction cannot happen instantaneously across an infinite network. The relational structure must "reconcile" new extensions with the existing causal dependency structure.

> **The effective speed of light and the rate of time are bounded by the degree to which relational structures can reconcile with causal dependency.**

This is the **Causal Reconciliation Principle**. It is not a new primitive axiom, but a deeper grounding of the existing causal, relational, and burden-based structure. When reconciliation is efficient, causal propagation is fast and time advances normally. When burden or constraint loading increases, reconciliation slows, the causal ceiling tightens, and proper time is suppressed.

1.6.3 Relational Distance

To measure how far a causal influence has propagated, we need a notion of spatial separation. At this stage, the full correlation geometry has not been developed, but we can use the primitive correlation kernel K_ω.

Definition 1.20 (Relational Distance).
For two events $e_i, e_j \in \mathcal{E}$ with associated local observables O_i, O_j, the **relational distance** is:
$$ d_\omega(e_i, e_j) := -\ell_0 \log K_\omega(O_i, O_j), $$
where $\ell_0 > 0$ is a fundamental length scale.

Strong correlation ($K_\omega \approx 1$) means small relational distance. Weak correlation means large relational distance.

(Note: The metric properties of d_ω, such as the triangle inequality, are formally established in Section 2. Here, we assume the coarse-grained limit where d_ω behaves as a metric).

1.6.4 Causal Depth and Elementary Reconciliation Time

Definition 1.21 (Causal Depth).
For causally related events $e_i \prec e_j$, the causal depth $L_C(e_i, e_j)$ is the maximum number of intermediate dependency steps in any chain from e_i to e_j.

Definition 1.22 (Elementary Reconciliation Interval).
Let $\epsilon > 0$ be the **elementary reconciliation interval**: the minimal internal time cost required for a single primitive causal step to reconcile with the constraint structure.

Definition 1.23 (Causal-Depth Time).
The unburdened causal time between e_i and e_j is:
$$ \tau_C(e_i, e_j) := \epsilon \, L_C(e_i, e_j). $$

This is not yet full physical proper time, which will later include burden-weighting. It is the minimal structural time cost of traversing the causal network.

1.6.5 Definition — Effective Causal Speed

Definition 1.24 (Effective Causal Speed).
For a causal propagation from e_i to e_j, the effective causal speed is:
$$ v_C(e_i, e_j) := \frac{d_\omega(e_i, e_j)}{\tau_C(e_i, e_j)} = \frac{d_\omega(e_i, e_j)}{\epsilon \, L_C(e_i, e_j)}. $$

1.6.6 Bounded Relational Step Length

If the relational distance d_ω could grow arbitrarily large for a single causal step ($L_C = 1$), then the effective speed v_C would be unbounded. To obtain a finite speed limit, we must assume that a single primitive causal dependency step can only bridge a finite amount of relational distance.

Assumption 1.2 (Bounded Relational Step Length).
There exists a maximum relational distance $\ell_C > 0$ that a single primitive causal link can bridge. For any primitive causal link $e_a \prec e_b$:
$$ d_\omega(e_a, e_b) \leq \ell_C. $$

Remark 1.4 (Relation between ℓ_C and ℓ_0). The bounded relational step length ℓ_C and the fundamental correlation length ℓ_0 (Definition 1.20) are structurally linked. ℓ_0 represents the minimum resolvable relational distance in the emergent geometry. A single primitive causal link cannot bridge a distance larger than the scale at which correlations remain non-vanishing. Therefore, we identify:
$$ \ell_C \equiv \ell_0. $$
This unifies the causal and spatial scales of the framework.

Interpretation. This is the core locality assumption of the framework. It states that causal influence cannot jump arbitrarily far across the correlation geometry in a single reconciliation step. The constant ℓ_0 is the maximum "range" of a single primitive zero-preserving update.

1.6.7 Theorem — Emergent Causal Speed Limit

Theorem 1.7 (Emergent Causal Speed Limit).
Let $(\mathcal{E}, \prec, \mathcal{R}_\omega)$ be an RCF causal-relational structure satisfying the bounded relational step assumption. Then any causal propagation from e_i to e_j satisfies:
$$ v_C(e_i, e_j) \leq c_{\mathrm{RCF}}, $$
where:
$$ \boxed{ c_{\mathrm{RCF}} = \frac{\ell_0}{\epsilon}. } $$

Proof.
Let $e_i = e_0 \prec e_1 \prec \cdots \prec e_N = e_j$ be a causal chain connecting e_i to e_j. By the bounded relational step assumption (Assumption 1.2, using $\ell_C = \ell_0$):
$$ d_\omega(e_k, e_{k+1}) \leq \ell_0 $$
for every elementary causal step k.

By the triangle inequality for the correlation distance (which holds under the metricity assumptions to be formalized in Section 2):
$$ d_\omega(e_i, e_j) \leq \sum_{k=0}^{N-1} d_\omega(e_k, e_{k+1}) \leq \sum_{k=0}^{N-1} \ell_0 = N \ell_0. $$

The causal depth of this chain is $L_C = N$. The minimal causal-time cost of N elementary steps is:
$$ \tau_C = N \epsilon. $$

Therefore, the effective speed is bounded by:

$$ v_C(e_i, e_j) = \frac{d_\omega(e_i, e_j)}{\tau_C} \leq \frac{N \ell_0}{N \epsilon} = \frac{\ell_0}{\epsilon}. $$

Defining $c_{\mathrm{RCF}} := \ell_0 / \epsilon$, we obtain:
$$ v_C(e_i, e_j) \leq c_{\mathrm{RCF}}. \qquad \blacksquare $$

1.6.8 Interpretation of the Speed Limit

The theorem proves the key point:

> **Bounded causal link length plus a finite reconciliation tick implies an emergent causal speed limit.**

This is not yet a derivation of the physical speed of light c from first principles, as the constants ℓ_0 and ϵ are model parameters. However, it provides the internal causal-speed mechanism.

In the continuum limit, if the framework recovers standard relativistic physics, one would identify:
$$ c \leftrightarrow c_{\mathrm{RCF}} = \frac{\ell_0}{\epsilon}. $$

1.6.9 Guardrails for This Subsection

1. **The speed limit is conditional.** It depends on Assumption 1.2 (bounded relational step length). Without this assumption, the speed is unbounded.
2. **c_{RCF} is not a postulate of a background metric.** It is a ratio of two primitive scales: the maximum spatial jump per causal step, and the time cost per causal step.
3. **Not yet Lorentzian.** A finite speed does not automatically imply Lorentz symmetry. The next subsection will address the conditions under which Lorentz compatibility emerges.

1.7 Emergent Direction and Lorentz Compatibility

1.7.1 Purpose of This Subsection

The Emergent Causal Speed Limit theorem (Theorem 1.7) establishes a finite maximum propagation speed c_{RCF}. However, a finite speed alone is not enough to recover special relativity. An anisotropic medium can have a finite maximum speed without possessing Lorentz symmetry.

To recover Lorentzian structure, the framework must reconstruct the notion of **direction** from the emergent correlation geometry, and show that Lorentz compatibility appears when the least-burden speed is direction-independent.

1.7.2 Distance Profiles and Emergent Direction

In standard geometry, direction is represented by tangent vectors. But in this framework, we do not begin with a manifold. We begin with localisable observables and their correlation distances.

Definition 1.25 (Distance Profile).
For each localisable observable $A \in \mathcal{A}_{\mathrm{loc}}$, define its **distance profile** as the function:
$$ \Phi_A(X) := d_\omega(A, X), $$
where X ranges over all other localisable observables.

Definition 1.26 (Relative Displacement Profile).
Given two distinguishable observables A and B, the **direction** from A towards B is represented by the profile difference:
$$ \Delta_{A \to B}(X) := \Phi_B(X) - \Phi_A(X) = d_\omega(B, X) - d_\omega(A, X). $$
A direction is thus not an absolute vector. It is the observable-dependent change in the correlation-distance profile induced by shifting from one correlation class to another.

Proposition 1.1 (Properties of Displacement Profiles).
For all $A, B, X \in \mathcal{A}_{\mathrm{loc}}$:
1. **Antisymmetry:** $\Delta_{A \to B}(X) = -\Delta_{B \to A}(X)$.
2. **Vanishing on the Diagonal:** $\Delta_{A \to A}(X) = 0$.

Proof.
1. $\Delta_{A \to B}(X) = d_\omega(B, X) - d_\omega(A, X) = -(d_\omega(A, X) - d_\omega(B, X)) = -\Delta_{B \to A}(X)$.
2. $\Delta_{A \to A}(X) = d_\omega(A, X) - d_\omega(A, X) = 0$. $\blacksquare$

1.7.3 Direction Equivalence and the Tangent Cone

Two displacements from the same base point should be considered the same direction if they deform the surrounding distance profile in the same way, up to positive rescaling.

Definition 1.27 (Direction Equivalence).
Let $A, B, C \in \mathcal{A}_{\mathrm{loc}}$ with $B \neq A$ and $C \neq A$. The ordered pairs (A, B) and (A, C) determine the same **emergent direction** at A if there exists $\lambda > 0$ such that:
$$ \Delta_{A \to B}(X) \approx \lambda \, \Delta_{A \to C}(X) \qquad \forall X \in \mathcal{A}_{\mathrm{loc}}. $$

Definition 1.28 (Emergent Tangent Cone).
For a fixed $A \in \mathcal{A}_{\mathrm{loc}}$, let $\mathfrak{D}_A$ be the set of equivalence classes of ordered pairs (A, B) (with $B \neq A$) modulo the direction equivalence of Definition 1.27.
The **emergent tangent cone** at A is:
$$ T_A^{\mathrm{em}} \mathcal{X}_\omega := \mathfrak{D}_A. $$
This gives the first pre-vectorial notion of directional structure inside the emergent metric space.

1.7.4 Theorem — Direction Requires Distinguishability

Theorem 1.8 (Direction Requires Distinguishability).
If $A \sim_\omega B$ (i.e., $d_\omega(A, B) = 0$, meaning they represent the same emergent point), then $\Delta_{A \to B}(X) = 0$ for all X. Therefore $A \to B$ defines no non-trivial spatial direction.

Proof.
If $A \sim_\omega B$, then $d_\omega(A, B) = 0$. For any X, the triangle inequality for the correlation distance (proven in Section 2) gives:
$$ d_\omega(A, X) \leq d_\omega(A, B) + d_\omega(B, X) = d_\omega(B, X). $$
Similarly:
$$ d_\omega(B, X) \leq d_\omega(B, A) + d_\omega(A, X) = d_\omega(A, X). $$
Therefore $d_\omega(A, X) = d_\omega(B, X)$, so $\Delta_{A \to B}(X) = 0$ for all X.
$\blacksquare$

> **No distinguishability $\Rightarrow$ no direction.** Direction only exists between correlation-distinguishable classes.

1.7.5 Directional Isotropy

The emergent propagation ceiling may depend on the direction of the relational displacement. Let $\hat{n}$ denote an emergent direction in $T_A^{\mathrm{em}} \mathcal{X}_\omega$. The speed ceiling might be a function:
$$ c_\star(\hat{n}). $$

Definition 1.29 (Directional Isotropy).
The emergent propagation ceiling is **direction-independent** (isotropic) if:
$$ c_\star(\hat{n}) = c_\star \quad \forall \hat{n} \in T_A^{\mathrm{em}} \mathcal{X}_\omega. $$

Directional isotropy means that the least-burden reconciliation speed is the same in all emergent relational directions. This removes preferred spatial axes from the leading-order propagation law.

1.7.6 Conditional Proposition — Directional Isotropy Supports Lorentz Form

Conditional Proposition 1.1 (Lorentz Form under Isotropy).
If the emergent propagation ceiling is independent of direction and the continuum limit is regular, then the effective interval can be written in Lorentz form:
$$ \boxed{ ds^2 = -c_\star^2 \, d\tau^2 + d\ell_\omega^2. } $$

Proof sketch.
Once the speed bound $c_\star$ is direction-independent (isotropic), the only remaining first-order invariant compatible with the causal structure is a quadratic interval with one time-like and spatially isotropic part.

The sign choice is fixed by the causal ordering already built into the relational time rule: time-like separation must correspond to burden-weighted causal evolution ($d\tau$), while space-like separation corresponds to relational displacement ($d\ell_\omega$).

Hence, the effective geometry takes the Lorentzian form $ds^2 = -c_\star^2 d\tau^2 + d\ell_\omega^2$. $\blacksquare$

Remark 1.5 (Status of the Proposition). This is a conditional proposition, not a completed theorem. The proof sketch assumes that directional isotropy and continuum regularity are sufficient to uniquely select the quadratic Lorentzian interval up to conformal factors. A full proof would require demonstrating that no higher-order or anisotropic terms survive the coarse-graining limit.

1.7.7 Emergent Mass-Shell Relation

Once the effective geometry is Lorentz-compatible, the dynamics of excitations must align with relativistic dispersion relations.

Conditional Proposition 1.2 (Emergent Mass Shell).
Assume the effective geometry is Lorentz-compatible with invariant interval $ds^2 = -c_\star^2 d\tau^2 + d\ell_\omega^2$, and assume the excitation carries a conserved burden-weighted four-momentum p_μ. Then the leading-order dispersion relation is:
$$ \boxed{ p_\mu p^\mu = -m_{\mathrm{eff}}^2 c_\star^2. } $$

Proof sketch.
In a Lorentzian effective geometry, the only scalar quadratic invariant built from four-momentum is $p_\mu p^\mu$. Conservation of the effective propagation law forces this invariant to remain fixed along free evolution. The fixed value is naturally identified with the squared effective rest mass.

The burden sector shifts the effective inertial response, so the mass parameter m_{eff} is not primitive, but emergent from the burdened relational excitation. Therefore, the free effective dynamics lies on a mass shell. $\blacksquare$

1.7.8 Status of Lorentz Symmetry

It is crucial to maintain theorem-safe status language regarding Lorentz symmetry:

1. **The speed bound** ($v_C \leq c_{\mathrm{RCF}}$) is a **conditional theorem** (proven from bounded relational step length and finite reconciliation tick).
2. **Directional isotropy** ($c_\star(\hat{n}) = c_\star$) is a **regime assumption** (it emerges if the coarse-grained correlation geometry is sufficiently homogeneous).
3. **Lorentz symmetry** is an **effective continuum theorem target** (it follows if isotropy and regularity hold).

The framework does not assume Lorentz symmetry at the foundational algebraic level. It derives it as an effective continuum limit of the relational causal structure.

1.7.9 Guardrails for This Subsection

1. **Direction is reconstructed, not assumed.** No vector spaces, coordinate axes, or background orientations are postulated.
2. **Isotropy is a regime assumption.** It is not forced at the microscopic relational level, but emerges under coarse-graining.
3. **Lorentz symmetry is effective.** It is a continuum target, not a primitive axiom.

1.8 Guardrails and Summary of Section 1

1.8.1 Guardrails for Section 1

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **Zero preservation is not dynamics.** It is admissibility preservation. The master constraint selects the physical sector; it does not describe evolution with respect to an external time parameter.
2. **Observables are not merely arbitrary algebra elements.** They must be compatible with the physical sector. The rigorous definition requires preservation of the physical state space, as the commutator condition is vacuous for two-sided ideals.
3. **Causal dependency is not background spacetime causality.** The relation $e_i \prec e_j$ is an internal strict partial order based on constraint closure. It does not assume light cones or a Lorentzian metric.
4. **Events are not initially spacetime points.** An event $e \in \mathcal{E}$ is a localised instance of admissible relational structure, not a coordinate location $p \in M$.
5. **Extension order is not external time.** The label s in an extension chain $\mathcal{U}_s \to \mathcal{U}_{s+1}$ orders stages of admissible growth. It must not be conflated with physical temporal duration.
6. **Relational correlation is not the same as causal dependence.** High correlation between two observables does not imply a causal link between them.
7. **This section does not yet define metric distance, duration, mass, energy, fields, or gravity.** It constructs the pre-geometric causal skeleton and the emergent speed limit only.
8. **Lorentz symmetry is an effective continuum target, not a primitive axiom.** The speed bound c_{RCF} is a conditional theorem, while directional isotropy is a regime assumption that emerges under coarse-graining.

1.8.2 Summary of Section 1

Section 1 established the zero-preserving causal foundation, transitioning from the static admissibility criterion of Section 0 to the first post-foundational structures:

1. **Master-Zero Equivalence** was formalized: The physical sector is exactly the zero sector of the represented master constraint, $\mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega = \bigcap_\alpha \ker \pi_\omega(\hat{C}_\alpha)$.

2. **Zero-Preserving Observables** were defined as operations that preserve the physical state space. The algebraic observable sector was rigorously constructed, explicitly avoiding the vacuous normalizer/commutator condition, and the physical observable algebra was defined as the quotient $\mathcal{A}_{\mathrm{obs}} = \mathcal{A}_{\mathrm{weak}} / \mathcal{I}_C$.
3. **Causal Dependency** was introduced as a strict partial order $\prec$ on primitive events, derived from the inclusion and necessity of constraint closure sets. Irreflexivity, transitivity, and asymmetry were proven.
4. **The Two-Link Principle** was stated: Causal links ($\prec$) and relational-correlation links (K_ω) are structurally distinct, preventing the premature collapse of space into causality or vice versa.
5. **The Open Extension Principle** was formalized: A universe is an extendible zero-preserving partial order, not a closed static block. Extension-history equivalence was defined to prevent the extension label from becoming a preferred time frame.
6. **Emergent Causal Propagation Speed** was derived: $v_C \leq c_{\mathrm{RCF}} = \ell_0 / \epsilon$, grounded in the Causal Reconciliation Principle and the bounded relational step length (unifying ℓ_C and ℓ_0).
7. **Directional Isotropy** was shown to support an effective Lorentzian interval $ds^2 = -c_\star^2 d\tau^2 + d\ell_\omega^2$, leading to an emergent mass-shell relation, though Lorentz symmetry remains a conditional theorem target.

The conceptual sequence of this section is:
$$ \boxed{ \text{zero sector} \to \text{zero-preserving operations} \to \text{causal dependency} \to \text{open extension} \to \text{causal speed} \to \text{Lorentz compatibility}. } $$

This completes the first bridge from pure admissibility toward emergent temporal and causal structure. The framework now possesses a pre-geometric causal skeleton and a finite speed limit, ready for the reconstruction of spatial geometry.

Section 2 — Correlation Geometry and Emergent Space

2.0 Purpose of Correlation Geometry and Emergent Space

2.0.1 Transition from Causal Structure to Spatial Structure

Section 1 developed the first post-foundational structures: the physical zero sector, zero-preserving observables, causal dependency, the Two-Link Principle, and open extension. These gave the framework a pre-geometric causal skeleton and an emergent speed limit.

However, causal order alone is not space. A relation such as $e_i \prec e_j$ tells us that e_j depends on e_i. It does not tell us how far apart two events are. It does not define a metric, topology, dimension, locality, or spatial region. It also does not define distance between events that are not causally comparable.

Thus, after causal dependency has been separated from relational correlation by the Two-Link Principle, the next task is to develop the correlation side.

The purpose of this section is to show how space-like structure can be reconstructed from state-dependent relational correlation.

> **Space is not assumed as a background arena; it is reconstructed from correlation structure.**

The framework does not begin with points in a manifold. It begins with admissible relational structure. What later appears as spatial distance is to be derived from how strongly physical observables, events, or localisable relational contexts are correlated in a physical state.

2.0.2 Why Correlation Is the Correct Starting Point for Emergent Space

If space is not primitive, then spatial nearness cannot be defined by coordinate distance. There is no initial metric $d(p,q)$, no background manifold M, and no prior statement that two things are close because their coordinate labels differ only slightly.

Instead, nearness must be reconstructed from internal relational data.

Correlation provides the natural candidate. If two admissible relational contexts are strongly correlated in a physical state, then they behave as though they are close in the emergent structure. If they are weakly correlated, they behave as though they are far apart. This suggests that spatial distance should be a derived measure of correlation decay.

The basic intuition is:

> **Strong correlation corresponds to small emergent separation.**
> **Weak correlation corresponds to large emergent separation.**

This does not yet prove that correlation distance is a metric. It only identifies the structural bridge from relational information to space-like geometry. Metricity requires additional assumptions, which will be stated explicitly in later subsections.

2.0.3 Relation to the Two-Link Principle

This section develops the second link introduced by the Two-Link Principle (Principle 1.1).

The causal link was:
$$ e_i \prec e_j. $$
It meant that e_j depends on e_i for zero-closure. This belongs to the causal/dependency layer.

The relational link is:
$$ K_\omega(e_i, e_j) \equiv \mathrm{Corr}_\omega(O_{e_i}, O_{e_j}). $$
It measures the strength of relational association between two event contexts in a physical state ω.

The present section develops this relational layer into a geometric structure. It defines a correlation distance from K_ω, establishes conditions under which this distance becomes a pseudometric, constructs emergent points as equivalence classes of perfectly correlated observables, and introduces approximate metricity under coarse-graining.

2.0.4 What This Section Establishes

This section establishes the following:

1. **The correlation kernel** K_ω is constructively defined on localisable zero-preserving observables.
2. **Correlation distance** $d_\omega = -\ell_0 \log K_\omega$ is defined.
3. Under **multiplicative correlation consistency** (postulated as a structural axiom), d_ω is shown to be a pseudometric.
4. **Emergent points** are defined as equivalence classes of perfectly correlated observables.
5. The quotient of the observable sector by this equivalence relation yields a **metric space** of emergent points.
6. **Approximate metricity** is established under coarse-graining, providing the bridge from microscopic relational structure to macroscopic smooth geometry.
7. **Observable-dependent direction** is reconstructed from distance profiles.
8. **Dimensional closure** is proven: $D=3$ is the unique spatial dimension where relational inference closes without degeneracy or redundancy.

The theorem-safe structure is:
$$ \boxed{ \text{kernel assumptions} \quad \Longrightarrow \quad \text{correlation distance} \quad \Longrightarrow \quad \text{pseudometric or approximate metric} \quad \Longrightarrow \quad \text{emergent point set}. } $$

2.1 The Correlation Kernel

2.1.1 Purpose of This Subsection

To reconstruct spatial structure from relational data, the framework requires a measure of how strongly two admissible relational contexts are associated within a given physical state. This measure is the **correlation kernel**.

Before defining the kernel, however, we must specify the class of objects on which it acts. Not all algebra elements are suitable for reconstructing spatial locality. We must restrict to a class of zero-preserving observables that can act as local probes of the emergent geometry.

2.1.2 Localisable Observables

Definition 2.1 (Localisable Observable Sector).

Let $\mathcal{A}_{\mathrm{loc}} \subseteq \mathcal{A}_{\mathrm{obs}}$ denote a chosen class of **localisable zero-preserving observables**. These are observables whose mutual correlation structure will be used to define emergent geometry.

Interpretation. At this stage, "localisable" should be understood operationally: not as already embedded in a pre-existing space, but as belonging to the class of observables suitable for later spatial interpretation. An element $A \in \mathcal{A}_{\mathrm{loc}}$ is not an operator at a spacetime point. It is a relational probe whose pattern of correlation with other probes can be used to reconstruct a location-like equivalence class.

2.1.3 Constructive Definition of the Kernel

Rather than leaving the kernel as an abstract map, we provide a canonical construction derived from the primitive algebraic data.

Definition 2.2 (Correlation Kernel).
Let ω be a physical state. For any $A, B \in \mathcal{A}_{\mathrm{loc}}$ such that $\omega(A^\dagger A) > 0$ and $\omega(B^\dagger B) > 0$, the **correlation kernel** is the normalized state overlap:
$$ \boxed{ K_\omega(A, B) = \frac{|\omega(A^\dagger B)|}{\sqrt{\omega(A^\dagger A) \, \omega(B^\dagger B)}}. } $$

Theorem 2.1 (Kernel Bounds).
The kernel satisfies $0 \leq K_\omega(A, B) \leq 1$.

Proof.
By the Cauchy-Schwarz inequality for positive linear functionals (Lemma 0.8), we have:
$$ |\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A) \, \omega(B^\dagger B). $$
Dividing both sides by the product of the norms and taking the square root yields $K_\omega(A, B) \leq 1$. Positivity is immediate from the absolute value. $\blacksquare$

Interpretation.
- $K_\omega(A, B) = 1$ represents maximal correlation or indistinguishability (up to a phase) in the state ω.
- Values closer to zero represent weaker correlation.

This range condition ensures that the logarithmic distance $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ (to be defined in Section 2.2) will be non-negative. To avoid infinite distances at this foundational stage, we restrict the domain to pairs with $K_\omega > 0$.

2.1.4 Kernel Selection Criteria

The abstract theory does not force a unique formula, but the normalized overlap is the canonical choice because it satisfies the necessary structural criteria:

1. **Zero-preserving domain:** K_ω is evaluated only on $\mathcal{A}_{\mathrm{loc}} \subseteq \mathcal{A}_{\mathrm{obs}}$, ensuring geometry is built from admissible probes.

2. **State dependence:** The kernel explicitly depends on ω, ensuring geometry is not a fixed background but a property of the physical state.
3. **Boundedness:** Cauchy-Schwarz guarantees the $K_\omega \leq 1$ bound required for non-negative distance without imposing a C^*-norm on the algebra.
4. **Insensitivity to scaling:** $K_\omega(\lambda A, \mu B) = K_\omega(A, B)$ for positive scalars λ, μ, ensuring the distance depends on the relational direction, not the magnitude of the probe.

2.1.5 Multiplicative Consistency as a Structural Postulate

For the emergent distance $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ to satisfy the triangle inequality (and thus become a pseudometric), the kernel must satisfy **multiplicative correlation consistency**:
$$ K_\omega(A, C) \geq K_\omega(A, B) \, K_\omega(B, C). $$

It is critical to note that this inequality does *not* follow automatically from Cauchy-Schwarz. The Fubini-Study metric (whose definition uses the same overlap) satisfies the triangle inequality in terms of the *angle* $\arccos K$, but this does not imply the multiplicative bound above. Therefore, to bridge from algebraic correlation to metric geometry, we must explicitly postulate this consistency.

Assumption 2.5 (Multiplicative Correlation Consistency).
For all $A, B, C \in \mathcal{A}_{\mathrm{loc}}$, the physical state ω and the localization sector are such that:
$$ \boxed{ K_\omega(A, C) \geq K_\omega(A, B) \, K_\omega(B, C). } $$

Interpretation. This condition says that the direct correlation between A and C is at least as large as the correlation obtained by passing through an intermediate probe B. Correlation should not decay more severely along the direct relation than along a two-step relational path. Proving that this assumption holds for generic physical states in specific constraint algebras (e.g., via Kadison-Schwarz inequalities for KMS states) remains an open target (Target 1, Section 9.2).

2.1.6 Minimal Kernel Assumptions

For the first layer of correlation geometry, the following assumptions are formalized:

Assumption 2.1 (Strict Positivity).
$$ K_\omega(A, B) > 0 \qquad \forall A, B \in \mathcal{A}_{\mathrm{loc}}. $$
This ensures that $\log K_\omega(A, B)$ is defined and finite.

Assumption 2.2 (Upper Normalisation).
$$ K_\omega(A, B) \leq 1 \qquad \forall A, B \in \mathcal{A}_{\mathrm{loc}}. $$
This ensures non-negative logarithmic distance.

Assumption 2.3 (Diagonal Normalisation).
$$ K_\omega(A, A) = 1 \qquad \forall A \in \mathcal{A}_{\mathrm{loc}}. $$
This ensures that every observable has zero distance from itself.

Assumption 2.4 (Symmetry).
$$ K_\omega(A, B) = K_\omega(B, A) \qquad \forall A, B \in \mathcal{A}_{\mathrm{loc}}. $$
This ensures that the induced distance is symmetric.

2.1.7 Correlation Profiles

The kernel assigns to each observable $A \in \mathcal{A}_{\mathrm{loc}}$ a **correlation profile**:

Definition 2.3 (Correlation Profile).
The correlation profile of A is the function:
$$ \Pi_\omega(A): \mathcal{A}_{\mathrm{loc}} \to (0, 1], \qquad \Pi_\omega(A)(X) = K_\omega(A, X). $$

The profile $\Pi_\omega(A)$ records how A correlates with every other localisable observable. This profile is more informative than any single pairwise correlation. It gives a relational signature for the emergent location of A.

Two observables may be considered to represent the same emergent location if their correlation profiles are indistinguishable. This motivates the later equivalence relation:
$$ A \equiv_\omega B \quad \Longleftrightarrow \quad \Pi_\omega(A) = \Pi_\omega(B), $$
or explicitly:
$$ A \equiv_\omega B \quad \Longleftrightarrow \quad K_\omega(A, X) = K_\omega(B, X) \quad \forall X \in \mathcal{A}_{\mathrm{loc}}. $$

> **Emergent location is encoded by correlation profile.**

2.1.8 Guardrails for This Subsection

1. **The kernel is constructively defined.** It is not an arbitrary map; it is the normalized state overlap derived from the primitive functional ω.
2. **Multiplicative consistency is a postulate.** Assumption 2.5 is an input required to generate metric geometry. It is not a theorem of pure $(*)$-algebras.
3. **Localisable does not mean spatially local in advance.** The term "localisable" is operational and framework-internal.

2.2 Correlation Distance

2.2.1 Purpose of This Subsection

The previous subsection introduced the correlation kernel $K_\omega(A, B)$, defined on localisable observables $A, B \in \mathcal{A}_{\mathrm{loc}}$. The kernel measures the strength of relational association between two admissible relational probes in a physical state ω. It is not itself a distance. It behaves in the opposite direction from distance: strong correlation should correspond to nearness, while weak correlation should correspond to separation.

The purpose of this subsection is to convert correlation strength into an emergent distance-like quantity.

> **Emergent distance measures the cost of relational decorrelation.**

2.2.2 From Correlation Strength to Separation

Let $K_\omega(A, B)$ satisfy $0 < K_\omega(A, B) \leq 1$.

The value $K_\omega(A, B) = 1$ represents maximal correlation between A and B. Values closer to zero represent weaker correlation.

A distance-like quantity should satisfy:
- $K_\omega(A, B) = 1 \Rightarrow d_\omega(A, B) = 0$,
- $K_\omega(A, B) \downarrow 0 \Rightarrow d_\omega(A, B) \uparrow \infty$.

A natural way to achieve this is to define distance by the negative logarithm of the correlation:
$$ d_\omega(A, B) = -\ell_0 \log K_\omega(A, B), $$
where $\ell_0 > 0$ is a fixed length scale.

The negative logarithm converts high correlation into small distance and low correlation into large distance.

2.2.3 Definition — Correlation Distance

Definition 2.4 (Correlation Distance).
Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \to (0, 1]$ be a normalised positive correlation kernel. The **correlation distance** associated with K_ω is the function:
$$ \boxed{ d_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \to [0, \infty) } $$
defined by:
$$ \boxed{ d_\omega(A, B) = -\ell_0 \log K_\omega(A, B), \qquad \ell_0 > 0. } $$

Here ℓ_0 sets the emergent distance scale.

The quantity $d_\omega(A, B)$ is interpreted as the emergent separation between the localisable relational probes A and B as induced by the physical state ω.

This definition should be read carefully. The function d_ω is not assumed to be a metric at this stage. It is a **candidate distance**. Its metric properties depend on additional assumptions about K_ω.

2.2.4 Role of the Scale ℓ_0

The constant $\ell_0 > 0$ converts the dimensionless logarithmic correlation cost into a quantity with units of length, or more generally with units of emergent separation.

If the kernel K_ω is dimensionless, then $-\log K_\omega(A, B)$ is also dimensionless. Multiplying by ℓ_0 assigns the resulting distance a scale.

The value of ℓ_0 may be interpreted differently depending on the model. It may represent:
- a fundamental relational length scale,
- a coarse-graining scale,
- a correlation-decay scale,
- or a conversion factor between logarithmic correlation cost and emergent distance.

At this stage, no particular physical value is assigned to ℓ_0. It is simply required to be positive.

2.2.5 Non-Negativity of Correlation Distance

Theorem 2.2 (Non-Negativity).
If $0 < K_\omega(A, B) \leq 1$ and $\ell_0 > 0$, then:
$$ \boxed{ d_\omega(A, B) \geq 0. } $$

Proof.
Since $0 < K_\omega(A, B) \leq 1$, we have $\log K_\omega(A, B) \leq 0$. Therefore $-\log K_\omega(A, B) \geq 0$. Since $\ell_0 > 0$, it follows that $-\ell_0 \log K_\omega(A, B) \geq 0$. Thus $d_\omega(A, B) \geq 0$. $\blacksquare$

2.2.6 Zero Distance and Perfect Correlation

Theorem 2.3 (Zero Distance and Perfect Correlation).
$$ d_\omega(A, B) = 0 \quad \Longleftrightarrow \quad K_\omega(A, B) = 1. $$

Proof.
If $K_\omega(A, B) = 1$, then $d_\omega(A, B) = -\ell_0 \log 1 = 0$.

Conversely, if $d_\omega(A, B) = 0$, then $-\ell_0 \log K_\omega(A, B) = 0$. Since $\ell_0 > 0$, this implies $\log K_\omega(A, B) = 0$, so $K_\omega(A, B) = 1$. $\blacksquare$

Interpretation.
Perfect correlation implies zero emergent separation. However, zero emergent separation does not necessarily imply algebraic identity $A = B$. Two distinct observables may be perfectly correlated or indistinguishable by the chosen kernel, even if they are algebraically distinct.

Thus, at this level, d_ω is expected at best to be a **pseudometric** until one quotients by zero-distance equivalence classes.

> **Zero correlation distance means emergent indistinguishability, not necessarily algebraic equality.**

2.2.7 Weak Correlation and Large Separation

If the correlation kernel becomes small, then the distance becomes large.

For example, if $K_\omega(A, B) = e^{-L/\ell_0}$, then:
$$ d_\omega(A, B) = -\ell_0 \log(e^{-L/\ell_0}) = L. $$

This shows that exponential decay of correlation corresponds to linear growth of distance.

More generally, as $K_\omega(A, B) \to 0^+$, we have $\log K_\omega(A, B) \to -\infty$, and therefore $d_\omega(A, B) \to +\infty$.

> **Vanishing correlation corresponds to infinite emergent separation.**

In the present definition, the kernel is restricted to strictly positive values, so the distance remains finite. If exact zero correlations are allowed later, then d_ω should be treated as an extended distance that may take the value $+\infty$.

2.2.8 Why Logarithmic Distance Is Natural

The logarithm is used because correlations often combine multiplicatively, whereas distances combine additively.

Suppose that correlations satisfy an approximate factorisation relation:
$$ K_\omega(A, C) \approx K_\omega(A, B) \, K_\omega(B, C). $$

Taking negative logarithms gives:
$$ -\log K_\omega(A, C) \approx -\log K_\omega(A, B) - \log K_\omega(B, C). $$

Multiplying by ℓ_0:
$$ d_\omega(A, C) \approx d_\omega(A, B) + d_\omega(B, C). $$

Thus, multiplicative composition of correlations becomes additive composition of distances.

> **Logarithmic distance turns multiplicative correlation decay into additive geometry.**

The logarithm is the unique (up to scale) mapping that converts multiplicative consistency into additive triangle inequality.

2.2.9 Symmetry of Correlation Distance

Theorem 2.4 (Symmetry).
If $K_\omega(A, B) = K_\omega(B, A)$ (Assumption 2.4), then:
$$ \boxed{ d_\omega(A, B) = d_\omega(B, A). } $$

Proof.
Since $K_\omega(A, B) = K_\omega(B, A)$, we have:
$$ d_\omega(A, B) = -\ell_0 \log K_\omega(A, B) = -\ell_0 \log K_\omega(B, A) = d_\omega(B, A). \qquad \blacksquare $$

2.2.10 Diagonal Vanishing

Theorem 2.5 (Diagonal Vanishing).
If $K_\omega(A, A) = 1$ (Assumption 2.3), then:
$$ \boxed{ d_\omega(A, A) = 0. } $$

Proof.
$d_\omega(A, A) = -\ell_0 \log K_\omega(A, A) = -\ell_0 \log 1 = 0$. $\blacksquare$

2.2.11 Triangle Inequality Is Not Automatic

The correlation distance does not automatically satisfy the triangle inequality. The triangle inequality would require:
$$ d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C). $$

Substituting the logarithmic definition gives:
$$ -\ell_0 \log K_\omega(A, C) \leq -\ell_0 \log K_\omega(A, B) - \ell_0 \log K_\omega(B, C). $$

Dividing by $-\ell_0$ (which reverses the inequality):
$$ \log K_\omega(A, C) \geq \log K_\omega(A, B) + \log K_\omega(B, C). $$

Exponentiating:
$$ K_\omega(A, C) \geq K_\omega(A, B) \, K_\omega(B, C). $$

Thus, the triangle inequality for d_ω is equivalent to a **multiplicative consistency condition** for K_ω.

> **The triangle inequality requires multiplicative correlation consistency.**

This condition was postulated as Assumption 2.5 in Section 2.1.5, and its consequences will be treated in the next subsection.

2.2.12 Correlation Distance as a Premetric

Before imposing all metricity assumptions, it is safest to call d_ω a **correlation premetric** or **candidate distance**.

It already has a clear interpretation:
- Large correlation $\Rightarrow$ small separation.
- Small correlation $\Rightarrow$ large separation.

But without symmetry, diagonal normalisation, and multiplicative consistency, it may not satisfy all metric axioms.

> **$d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ is a non-negative correlation-derived separation under $0 < K_\omega \leq 1$.**

Additional assumptions are required before calling it a pseudometric or metric.

2.2.13 Guardrails for This Subsection

1. **Distance is derived, not primitive.** The function $d_\omega(A, B)$ is not assumed at the beginning of the theory. It is derived from $K_\omega(A, B)$, which is itself evaluated in a physical state ω.
2. **Not yet a true metric.** Separation fails until quotienting.
3. **No coordinates.** We still have no points, no manifold, no dimension.
4. **State dependence.** Distance depends on ω.

2.3 Metricity Under Multiplicative Correlation Consistency

2.3.1 Purpose of This Subsection

The previous subsection defined the correlation distance $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ and showed that it is non-negative when $0 < K_\omega \leq 1$. It also showed that zero distance corresponds to perfect correlation, not necessarily literal equality of observables.

However, non-negativity alone is not enough to make d_ω a metric or even a pseudometric. To obtain metric-like structure, additional properties must be imposed on the kernel. In particular, one needs diagonal normalisation, symmetry, and a condition ensuring the triangle inequality.

The purpose of this subsection is to state and prove the main conditional theorem:

> **Multiplicative consistency of correlation implies additive consistency of distance.**

2.3.2 From Correlation to Metric Structure

A distance function d on a set X is a **metric** if it satisfies:
1. $d(x, y) \geq 0$
2. $d(x, y) = 0 \Longleftrightarrow x = y$
3. $d(x, y) = d(y, x)$
4. $d(x, z) \leq d(x, y) + d(y, z)$

A **pseudometric** satisfies the same conditions except that distinct elements may have zero distance: $d(x, y) = 0$ does not necessarily imply $x = y$.

In correlation geometry, pseudometricity is the natural first target. This is because two algebraically distinct observables may be perfectly correlated, or indistinguishable by the correlation kernel, and therefore have zero correlation distance.

Thus, the theorem-safe claim should not be that d_ω is automatically a metric on $\mathcal{A}_{\mathrm{loc}}$. The correct claim is:

> **d_ω is a pseudometric under suitable kernel assumptions.**

A true metric may then be obtained later by quotienting by zero-distance equivalence classes.

2.3.3 Kernel Assumptions for Pseudometricity

Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \to (0, 1]$ be a correlation kernel.

For $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ to be a pseudometric, we assume the following:

Assumption 2.5 (Multiplicative Correlation Consistency).
For all $A, B, C \in \mathcal{A}_{\mathrm{loc}}$:
$$ \boxed{ K_\omega(A, C) \geq K_\omega(A, B) \, K_\omega(B, C). } $$

This is the condition that ensures the triangle inequality for d_ω.

Interpretation.
The condition $K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C)$ says that the direct correlation between A and C is at least as large as the correlation obtained by passing through an intermediate probe B.

Equivalently, correlation should not decay more severely along the direct relation than along a two-step relational path. If correlation from A to B and from B to C is strong, then A and C cannot be arbitrarily decorrelated.

This is precisely the multiplicative condition needed for additive distances.

2.3.4 Theorem — Correlation Distance Is a Pseudometric

Theorem 2.6 (Correlation Distance Is a Pseudometric).
Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \to (0, 1]$ be a correlation kernel satisfying:
1. $K_\omega(A, A) = 1$ (diagonal normalisation, Assumption 2.3),
2. $K_\omega(A, B) = K_\omega(B, A)$ (symmetry, Assumption 2.4),
3. $K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C)$ (multiplicative consistency, Assumption 2.5).

Let $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ with $\ell_0 > 0$.

Then d_ω is a **pseudometric** on $\mathcal{A}_{\mathrm{loc}}$.

That is, for all $A, B, C \in \mathcal{A}_{\mathrm{loc}}$:
1. $d_\omega(A, B) \geq 0$,
2. $d_\omega(A, A) = 0$,
3. $d_\omega(A, B) = d_\omega(B, A)$,
4. $d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C)$.

Proof.

Non-negativity.
Since $0 < K_\omega(A, B) \leq 1$, we have $\log K_\omega(A, B) \leq 0$. Therefore $-\log K_\omega(A, B) \geq 0$. Since $\ell_0 > 0$, we obtain $d_\omega(A, B) \geq 0$.

Diagonal vanishing.
By diagonal normalisation, $K_\omega(A, A) = 1$. Therefore $d_\omega(A, A) = -\ell_0 \log 1 = 0$.

Symmetry.
By kernel symmetry, $K_\omega(A, B) = K_\omega(B, A)$. Therefore $d_\omega(A, B) = d_\omega(B, A)$.

Triangle inequality.
By multiplicative correlation consistency:
$$ K_\omega(A, C) \geq K_\omega(A, B) \, K_\omega(B, C). $$

Taking logarithms (which preserves the inequality since $\log$ is monotonically increasing):
$$ \log K_\omega(A, C) \geq \log K_\omega(A, B) + \log K_\omega(B, C). $$

Multiplying by $-\ell_0$ (which is negative, so it reverses the inequality):
$$ -\ell_0 \log K_\omega(A, C) \leq -\ell_0 \log K_\omega(A, B) - \ell_0 \log K_\omega(B, C). $$

Using the definition of d_ω:
$$ d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C). $$

Since non-negativity, diagonal vanishing, symmetry, and the triangle inequality all hold, d_ω is a pseudometric. $\blacksquare$

2.3.5 Why the Result Is a Pseudometric Rather Than a Metric

The theorem does not prove that d_ω is a metric on $\mathcal{A}_{\mathrm{loc}}$. The missing condition is **identity of indiscernibles**:
$$ d_\omega(A, B) = 0 \quad \Longrightarrow \quad A = B. $$

Correlation distance gives instead:
$$ d_\omega(A, B) = 0 \quad \Longleftrightarrow \quad K_\omega(A, B) = 1. $$

Perfect correlation does not necessarily imply algebraic equality. Two distinct observables may be perfectly correlated or indistinguishable by the kernel.

Therefore, d_ω may assign zero distance to distinct elements. This is exactly why the correct theorem is pseudometricity.

The stronger statement $d_\omega(A, B) = 0 \Rightarrow A = B$ requires an additional separation assumption: $K_\omega(A, B) = 1 \Rightarrow A = B$. If such an assumption is imposed, then d_ω becomes a metric. But without it, the theorem-safe result is only pseudometric.

> **Correlation distance is naturally pseudometric before quotienting.**

2.3.6 Metric Quotient

Given a pseudometric d_ω, define an equivalence relation:
$$ A \sim_\omega B \quad \Longleftrightarrow \quad d_\omega(A, B) = 0. $$

Equivalently:
$$ A \sim_\omega B \quad \Longleftrightarrow \quad K_\omega(A, B) = 1. $$

The quotient set:
$$ \mathcal{X}_\omega := \mathcal{A}_{\mathrm{loc}} / \sim_\omega $$
then carries an induced metric, provided the distance between equivalence classes is well-defined.

Definition 2.5 (Quotient Distance).
Define:
$$ \widetilde{d}_\omega([A]_\omega, [B]_\omega) := d_\omega(A, B). $$

For this to be well-defined, one needs the usual compatibility condition: if $A \sim_\omega A'$ and $B \sim_\omega B'$, then $d_\omega(A, B) = d_\omega(A', B')$.

For pseudometric spaces, this compatibility holds. Therefore, the quotient by zero-distance equivalence produces a genuine metric space.

2.3.7 Theorem — Metric on Correlation Equivalence Classes

Theorem 2.7 (Metric on Correlation Equivalence Classes).
Let d_ω be the pseudometric from Theorem 2.6. Define $A \sim_\omega B \iff d_\omega(A, B) = 0$. Let $\mathcal{X}_\omega = \mathcal{A}_{\mathrm{loc}} / \sim_\omega$. Define $\widetilde{d}_\omega([A]_\omega, [B]_\omega) = d_\omega(A, B)$. Then $\widetilde{d}_\omega$ is a well-defined metric on $\mathcal{X}_\omega$.

Proof.

Well-definedness.

Suppose $A \sim_\omega A'$ and $B \sim_\omega B'$. Then $d_\omega(A, A') = 0$ and $d_\omega(B, B') = 0$. By the triangle inequality:
$$ d_\omega(A, B) \leq d_\omega(A, A') + d_\omega(A', B') + d_\omega(B', B) = d_\omega(A', B'). $$
Similarly:
$$ d_\omega(A', B') \leq d_\omega(A', A) + d_\omega(A, B) + d_\omega(B, B') = d_\omega(A, B). $$
Therefore $d_\omega(A, B) = d_\omega(A', B')$. Hence $\widetilde{d}_\omega$ is well-defined.

Identity of indiscernibles.
Suppose $\widetilde{d}_\omega([A]_\omega, [B]_\omega) = 0$. Then $d_\omega(A, B) = 0$, so $A \sim_\omega B$. Hence $[A]_\omega = [B]_\omega$.

Non-negativity, symmetry, and the triangle inequality descend directly from the pseudometric properties of d_ω. Thus $\widetilde{d}_\omega$ is a metric. $\blacksquare$

2.3.8 Interpretation of Emergent Points

An emergent point is not a basic atom of space. It is an equivalence class of relational probes whose mutual separation vanishes.

If $[A]_\omega \in \mathcal{X}_\omega$, then $[A]_\omega$ contains all localisable observables that represent the same correlation location in the state ω.

Thus, a point is defined by indistinguishability within the correlation geometry. This differs from ordinary manifold-based thinking. In a manifold model, one starts with points and then defines fields or observables at those points. Here, one starts with observables and defines points by their relational indistinguishability.

> **Points are not primitive; points are correlation-equivalence classes.**

2.3.9 Guardrails for This Subsection

1. **The quotient is not optional.** Without quotienting, d_ω remains only a pseudometric and does not define true points.
2. **Perfect correlation is not algebraic equality.** The relation $A \sim_\omega B$ means zero emergent separation, not literal equality in $\mathcal{A}$.
3. **Metric space does not yet imply manifold.** At this stage $\mathcal{X}_\omega$ is only a metric space, not yet a smooth manifold or continuum spacetime.
4. **State dependence remains.** Both the quotient relation and the emergent metric depend on the physical state ω.

2.4 Coarse-Graining and Approximate Metricity

2.4.1 Purpose of This Subsection

The previous subsections established that, under exact multiplicative correlation consistency (Assumption 2.5), the correlation distance d_ω is a pseudometric on the localisable observable sector $\mathcal{A}_{\mathrm{loc}}$, and that quotienting by perfect correlation yields a genuine metric space $(\mathcal{X}_\omega, \widetilde{d}_\omega)$.

However, exact metricity may be too strong at the microscopic relational level. The primitive correlation structure may be noisy, irregular, fluctuating, scale-dependent, or only approximately multiplicative. The physically relevant claim is often not that the microscopic structure is exactly metric, but rather that the coarse-grained structure is approximately metric.

The purpose of this subsection is to establish the theorem-safe bridge from exact microscopic metricity to approximate macroscopic metricity.

> **Smooth metric geometry need not be exact microscopically; it may emerge approximately after coarse-graining.**

This allows the framework to avoid overclaiming. It does not assert that every microscopic correlation kernel is exactly metric. Instead, it shows that if correlations satisfy an approximate multiplicative consistency condition, then the logarithmic distance satisfies an approximate triangle inequality.

2.4.2 Why Approximate Metricity Is Needed

Exact metric spaces are highly structured. They require strict identities and inequalities to hold for every triple of points or observables. But a pre-geometric relational substrate may not satisfy these conditions exactly.

There are several reasons exact metricity may fail.

First, the microscopic correlation kernel may fluctuate. The value $K_\omega(A, B)$ may depend sensitively on fine relational details that do not survive coarse-graining.

Second, localisable observables may not be perfectly sharp. The distinction between two neighbouring relational probes may be blurred.

Third, correlations may compose only approximately. Instead of exact multiplicative consistency:
$$ K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C), $$
one may have a controlled error.

Fourth, the emergent point set itself may only become stable after grouping many microscopic relational probes into effective classes.

For these reasons, the appropriate physical statement is often not "the microscopic structure is exactly metric," but rather "the coarse-grained structure is approximately metric."

> **The framework should treat exact metricity as an ideal limiting case and approximate metricity as the physically robust bridge to emergent space.**

2.4.3 Coarse-Grained Correlation Kernels

Let $K_\omega(A, B)$ be a microscopic or fine-grained correlation kernel. A **coarse-grained kernel** may be denoted $K_\omega^{\mathrm{cg}}(A, B)$, where "cg" indicates coarse-graining.

The coarse-grained kernel may be obtained by averaging, smoothing, restricting to stable probe classes, integrating out microscopic fluctuations, or replacing exact representatives with equivalence classes. The precise construction is model-dependent and must be specified in any concrete application.

Definition 2.6 (Explicit Coarse-Graining Map).
Let $\mathcal{A}_{\mathrm{loc}}^{\mathrm{cg}}$ be a set of coarse-grained localisable probes, where each coarse-grained probe $[A]$ is an equivalence class of microscopic probes $A_i \in \mathcal{A}_{\mathrm{loc}}$. A canonical coarse-graining map is:
$$ \boxed{ K_\omega^{\mathrm{cg}}([A], [B]) := \frac{1}{|[A]| \, |[B]|} \sum_{A_i \in [A]} \sum_{B_j \in [B]} K_\omega(A_i, B_j), } $$
where $|[A]|$ and $|[B]|$ are the cardinalities of the equivalence classes.

Interpretation. The coarse-grained kernel is the arithmetic average of the microscopic kernel over all pairs of probes in the two equivalence classes. This map smooths out microscopic fluctuations in the correlation data.

Remark 2.1 (Preservation of Approximate Consistency). If the microscopic kernel K_ω satisfies approximate multiplicative consistency with error bound $\varepsilon_{\mathrm{micro}}$, and if the fluctuations within each equivalence class are bounded (i.e., $|K_\omega(A_i, B_j) - K_\omega(A_{i'}, B_{j'})| \leq \delta$ for all $A_i, A_{i'} \in [A]$ and $B_j, B_{j'} \in [B]$), then the coarse-grained kernel K_ω^{cg} satisfies approximate multiplicative consistency with a controlled error bound $\varepsilon_{\mathrm{cg}} \leq \varepsilon_{\mathrm{micro}} + f(\delta)$, where $f(\delta) \to 0$ as $\delta \to 0$. This ensures that coarse-graining does not destroy the approximate metric structure — it preserves it, potentially with a tighter error bound if fluctuations average out.

For the abstract theorem, the exact coarse-graining mechanism is not required. What matters is that the resulting kernel satisfies controlled approximate consistency conditions.

Thus, we assume a kernel:
$$ K_\omega^{\mathrm{cg}}: \mathcal{A}_{\mathrm{loc}}^{\mathrm{cg}} \times \mathcal{A}_{\mathrm{loc}}^{\mathrm{cg}} \to (0, 1], $$
where $\mathcal{A}_{\mathrm{loc}}^{\mathrm{cg}}$ is a class of coarse-grained localisable probes.

To avoid excessive notation, this subsection will often write K_ω instead of K_ω^{cg}, with the understanding that the approximate theorem applies to the effective or coarse-grained kernel.

2.4.4 Approximate Multiplicative Correlation Consistency

Exact multiplicative consistency requires:
$$ K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C). $$

Approximate consistency allows a controlled deficit:

Assumption 2.6 (Approximate Multiplicative Consistency).
There exists $\varepsilon \geq 0$ such that for all $A, B, C \in \mathcal{A}_{\mathrm{loc}}$:
$$ \boxed{ K_\omega(A, C) \geq e^{-\varepsilon / \ell_0} \, K_\omega(A, B) \, K_\omega(B, C). } $$

When $\varepsilon = 0$, this reduces to exact multiplicative consistency:
$$ K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C). $$

When $\varepsilon > 0$, the direct correlation between A and C is allowed to be smaller than the product of the two-step correlations by a controlled factor $e^{-\varepsilon / \ell_0}$.

Thus, the approximate condition says that correlation may fail exact multiplicative consistency, but only by a bounded logarithmic error.

2.4.5 Interpretation of the Error Term

The factor $e^{-\varepsilon / \ell_0}$ is chosen because the distance is logarithmic.

If the kernel violates exact consistency by this multiplicative factor, the distance violates the triangle inequality by an additive amount ε.

This is the same product-to-sum principle used in the exact case.

The interpretation is:
$$ \boxed{ \varepsilon \text{ measures the allowed triangle-defect in emergent distance.} } $$

If ε is small compared to the scale of the distances being studied, then the geometry behaves approximately like a metric geometry.

If ε is large, then the metric interpretation becomes weak or unreliable.

Thus, approximate metricity is meaningful only when the error term is controlled relative to the scale of interest.

2.4.6 Theorem — Approximate Triangle Inequality

Theorem 2.8 (Approximate Triangle Inequality).
Let $K_\omega: \mathcal{A}_{\mathrm{loc}} \times \mathcal{A}_{\mathrm{loc}} \to (0, 1]$ be a normalised positive correlation kernel, and let $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$ with $\ell_0 > 0$. Assume that for some $\varepsilon \geq 0$, the kernel satisfies approximate multiplicative consistency:
$$ K_\omega(A, C) \geq e^{-\varepsilon / \ell_0} \, K_\omega(A, B) \, K_\omega(B, C). $$

Then:
$$ \boxed{ d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C) + \varepsilon. } $$

Proof.

Starting from approximate multiplicative consistency:
$$ K_\omega(A, C) \geq e^{-\varepsilon / \ell_0} \, K_\omega(A, B) \, K_\omega(B, C). $$

Taking logarithms (which preserves the inequality since $\log$ is monotonically increasing):
$$ \log K_\omega(A, C) \geq -\frac{\varepsilon}{\ell_0} + \log K_\omega(A, B) + \log K_\omega(B, C). $$

Multiplying by $-\ell_0$ (which is negative, so it reverses the inequality):
$$ -\ell_0 \log K_\omega(A, C) \leq \varepsilon - \ell_0 \log K_\omega(A, B) - \ell_0 \log K_\omega(B, C). $$

Using the definition $d_\omega(X, Y) = -\ell_0 \log K_\omega(X, Y)$:
$$ d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C) + \varepsilon. \qquad \blacksquare $$

2.4.7 Exact Metricity as the Zero-Error Limit

If $\varepsilon = 0$, the approximate consistency condition becomes:
$$ K_\omega(A, C) \geq K_\omega(A, B) K_\omega(B, C). $$

The approximate triangle inequality becomes:
$$ d_\omega(A, C) \leq d_\omega(A, B) + d_\omega(B, C). $$

Thus, the exact pseudometric theorem is recovered as the zero-error limit of the approximate theorem.

This gives a clean hierarchy:
$$ \boxed{ \varepsilon = 0 \quad \Longrightarrow \quad \text{exact pseudometricity;} } $$
$$ \boxed{ \varepsilon > 0 \quad \Longrightarrow \quad \text{approximate metricity with triangle defect } \varepsilon. } $$

The exact case is therefore not a separate conceptual mechanism. It is the limiting case of controlled approximate correlation consistency.

2.4.8 Coarse-Graining Maps Exact to Approximate

The key structural role of coarse-graining is to convert exact microscopic metricity into approximate macroscopic metricity.

At the fine level, the correlation kernel may satisfy exact multiplicative consistency, yielding an exact pseudometric d_ω. When the kernel is coarse-grained (e.g., via Definition 2.6), the exact consistency may be lost, but the approximate consistency may hold with a bounded error ε.

This is because coarse-graining averages over microscopic fluctuations. The averaged kernel may not satisfy the exact multiplicative law, but if the fluctuations are bounded, the violation is controlled.

Thus, the coarse-grained distance satisfies:
$$ \bar{d}_\omega(A, C) \leq \bar{d}_\omega(A, B) + \bar{d}_\omega(B, C) + \varepsilon, $$
where ε is the coarse-graining error.

> **Coarse-graining maps exact microscopic metricity to approximate macroscopic metricity.**

The approximation error is not arbitrary. It is controlled by the microscopic fluctuation scale.

2.4.9 Stability Under Coarse-Graining

A coarse-grained geometry is useful only if it is stable. Small changes in microscopic correlation data should not radically change macroscopic distances or neighbourhoods.

Since $d_\omega(A, B) = -\ell_0 \log K_\omega(A, B)$, small relative changes in K_ω produce additive changes in distance. If:
$$ K_\omega'(A, B) = e^{-\delta(A,B) / \ell_0} \, K_\omega(A, B), $$
then:
$$ d_\omega'(A, B) = d_\omega(A, B) + \delta(A, B). $$

Thus, logarithmic distance makes multiplicative kernel perturbations additive in geometry.

This provides a useful stability interpretation:
$$ \boxed{ \text{Multiplicative correlation errors become additive distance errors.} } $$

If the perturbation $\delta(A, B)$ is bounded by some $\delta_{\max}$, then the coarse-grained distance changes by at most $\delta_{\max}$.

2.4.10 Emergent Smoothness as a Further Condition

Approximate metricity is not yet smoothness. A space may satisfy an approximate triangle inequality and still fail to resemble a smooth manifold.

Smoothness requires additional large-scale regularity, such as:
- stable dimension,
- local Euclidean behaviour,

- differentiable coordinate charts,
- continuum approximability.

Thus, the route from approximate correlation geometry to smooth space requires further assumptions:
$$ \text{approximate metricity} \quad \longrightarrow \quad \text{stable neighbourhoods} \quad \longrightarrow \quad \text{coarse regions} \quad \longrightarrow \quad \text{continuum approximation} \quad \longrightarrow \quad \text{smooth manifold-like structure}. $$

Only the first step is treated in this subsection.

> **Approximate metricity is necessary for smooth geometry, but not sufficient by itself.**

2.4.11 Guardrails for This Subsection

1. **Approximate geometry is not exact geometry.** The coarse-grained regime is controlled, not exact.
2. **The effective space is state-dependent.** It inherits its structure from K_ω.
3. **Coarse-graining is not quotienting.** Quotienting removes zero-distance degeneracy; coarse-graining aggregates fine structure.
4. **No manifold is assumed yet.** This subsection only prepares the conditions under which manifold-like behaviour can later emerge.

2.5 Observable-Dependent Direction

2.5.1 Purpose of This Subsection

A metric space provides distance, but not direction. To recover full geometric structure—tangent spaces, gradients, directional derivatives—the framework must reconstruct the notion of **direction** from the correlation geometry.

In ordinary differential geometry, direction is represented by tangent vectors in $T_x \Sigma$. But in this framework, we do not begin with a manifold Σ, nor with points $p \in \Sigma$, nor with a pre-existing tangent bundle.

We begin with emergent points (equivalence classes of perfectly correlated observables) and their correlation distances $\widetilde{d}_\omega([A]_\omega, [B]_\omega)$. Direction must therefore be reconstructed from this metric structure.

> **Direction is not a primitive geometric notion. It is an observable-dependent equivalence class of local correlation-profile changes.**

2.5.2 Distance Profiles

To define direction, we examine how the distance from a given point to all other points changes when we shift the base point.

Definition 2.7 (Distance Profile).
For each emergent point $x \in \mathcal{X}_\omega$ (represented by a localisable observable $A \in \mathcal{A}_{\mathrm{loc}}$), define its **distance profile** as the function:
$$ \Phi_x: \mathcal{X}_\omega \to [0, \infty), \qquad \Phi_x(y) = \widetilde{d}_\omega(x, y). $$

The profile Φ_x records the full pattern of relational separations from x to all other emergent points. This profile is the relational signature of x's location in the emergent geometry.

2.5.3 Relative Displacement Profile

To reach direction rather than mere localisation, we compare how two points sit relative to all others.

Definition 2.8 (Relative Displacement Profile).
Given two distinct emergent points $x, y \in \mathcal{X}_\omega$, define the **relative displacement profile** from x to y as:
$$ \Delta_{x \to y}: \mathcal{X}_\omega \to \mathbb{R}, \qquad \Delta_{x \to y}(z) = \Phi_y(z) - \Phi_x(z) = \widetilde{d}_\omega(y, z) - \widetilde{d}_\omega(x, z). $$

The function $\Delta_{x \to y}$ records how the distance profile changes when one moves from x to y. It is the minimal purely metric-correlation object from which a notion of directionality can be reconstructed without assuming vectors or coordinates in advance.

2.5.4 Basic Properties of Displacement Profiles

Proposition 2.1 (Properties of Displacement Profiles).
For all $x, y, z \in \mathcal{X}_\omega$:
1. **Antisymmetry:** $\Delta_{x \to y}(z) = -\Delta_{y \to x}(z)$.
2. **Vanishing on the Diagonal:** $\Delta_{x \to x}(z) = 0$.

Proof.
1. $\Delta_{x \to y}(z) = \widetilde{d}_\omega(y, z) - \widetilde{d}_\omega(x, z) = -(\widetilde{d}_\omega(x, z) - \widetilde{d}_\omega(y, z)) = -\Delta_{y \to x}(z)$.
2. $\Delta_{x \to x}(z) = \widetilde{d}_\omega(x, z) - \widetilde{d}_\omega(x, z) = 0$. $\blacksquare$

2.5.5 Direction Equivalence

Two displacements from the same base point should be considered the same direction if they deform the surrounding distance profile in the same way, up to positive rescaling.

Definition 2.9 (Direction Equivalence).

Let $x, y, w \in \mathcal{X}_\omega$ with $y \neq x$ and $w \neq x$. The ordered pairs (x, y) and (x, w) determine the same **emergent direction** at x if there exists $\lambda > 0$ such that:
$$ \Delta_{x \to y}(z) \approx \lambda \, \Delta_{x \to w}(z) \qquad \forall z \in \mathcal{X}_\omega. $$

Two displacements from the same base point determine the same direction if they deform the surrounding distance profile in the same pattern up to positive rescaling.

2.5.6 Emergent Direction Space and Tangent Cone

Definition 2.10 (Emergent Direction Space).
For a fixed $x \in \mathcal{X}_\omega$, define $\mathfrak{D}_x$ to be the set of equivalence classes of ordered pairs (x, y) (with $y \neq x$) modulo the direction equivalence of Definition 2.9.

The elements of $\mathfrak{D}_x$ are called **emergent directions** based at x.

Definition 2.11 (Emergent Tangent Cone).
The **emergent tangent cone** at x is:
$$ T_x^{\mathrm{em}} \mathcal{X}_\omega := \mathfrak{D}_x. $$

This gives the first pre-vectorial notion of directional structure extracted purely from relational metric profiles.

2.5.7 Theorem — Direction Requires Distinguishability

Theorem 2.9 (Direction Requires Distinguishability).
If $x = y$ (i.e., $\widetilde{d}_\omega(x, y) = 0$), then $\Delta_{x \to y}(z) = 0$ for all z. Therefore $x \to y$ defines no non-trivial spatial direction.

Proof.
If $x = y$, then $\widetilde{d}_\omega(x, y) = 0$. For any z, the triangle inequality for the metric $\widetilde{d}_\omega$ gives:
$$ \widetilde{d}_\omega(x, z) \leq \widetilde{d}_\omega(x, y) + \widetilde{d}_\omega(y, z) = \widetilde{d}_\omega(y, z). $$
Similarly:
$$ \widetilde{d}_\omega(y, z) \leq \widetilde{d}_\omega(y, x) + \widetilde{d}_\omega(x, z) = \widetilde{d}_\omega(x, z). $$
Therefore $\widetilde{d}_\omega(x, z) = \widetilde{d}_\omega(y, z)$, so $\Delta_{x \to y}(z) = 0$ for all z. $\blacksquare$

> **No distinguishability $\Rightarrow$ no direction.** Direction only exists between correlation-distinguishable classes.

2.5.8 Finite Metric-Space Example

To make the abstract definition of direction more concrete, we provide a simple finite example.

Example 2.1 (Direction in a Three-Point Metric Space).
Let $\mathcal{X}_\omega = \{p_1, p_2, p_3\}$ with metric:
$$ \widetilde{d}_\omega(p_1, p_2) = 1, \quad \widetilde{d}_\omega(p_2, p_3) = 1, \quad \widetilde{d}_\omega(p_1, p_3) = 2. $$

The distance profiles are:
$$ \Phi_{p_1} = (0, 1, 2), \quad \Phi_{p_2} = (1, 0, 1), \quad \Phi_{p_3} = (2, 1, 0), $$
where the components are $(\Phi(p_1), \Phi(p_2), \Phi(p_3))$.

The displacement profile from p_1 to p_2 is:
$$ \Delta_{p_1 \to p_2} = \Phi_{p_2} - \Phi_{p_1} = (1, -1, -1). $$

The displacement profile from p_1 to p_3 is:
$$ \Delta_{p_1 \to p_3} = \Phi_{p_3} - \Phi_{p_1} = (2, 0, -2). $$

These two profiles are not proportional (there is no $\lambda > 0$ such that $(1, -1, -1) = \lambda (2, 0, -2)$). Therefore, $p_1 \to p_2$ and $p_1 \to p_3$ define **distinct emergent directions** at p_1.

In contrast, if we added a fourth point p_4 with $\widetilde{d}_\omega(p_1, p_4) = 2$ and $\widetilde{d}_\omega(p_4, p_3) = 1$ (symmetric to p_2), then:
$$ \Delta_{p_1 \to p_4} = (2, 0, -2, 0) + \text{extension}, $$
which would be proportional to $\Delta_{p_1 \to p_3}$, meaning $p_1 \to p_4$ and $p_1 \to p_3$ define the **same direction** at p_1 (both point "toward p_3").

This example demonstrates how the profile-difference definition naturally distinguishes directions in a purely metric setting without assuming vectors or coordinates.

2.5.9 Relation to Continuum Geometry

In a smooth large-scale limit, these relational directions may become ordinary tangent vectors:
$$ T_x^{\mathrm{em}} \mathcal{X}_\omega \longrightarrow T_x \Sigma. $$

Only in a smooth coarse-grained limit does this tangent cone become an ordinary tangent space. Thus, direction, like distance and position, is an emergent relational property rather than a fundamental structure.

2.5.10 Guardrails for This Subsection

1. **Direction here is reconstructed, not assumed.** No vector spaces, coordinate axes, or background orientations are postulated.
2. **Distance-profile identity is an extra condition.** The drafts support the existence of a localisation refinement target, but the exact condition that guarantees uniqueness of localisation is formalised here explicitly.

3. **Direction equivalence is a formal completion.** The profile-based direction classes above are a disciplined mathematical completion of that target.
4. **This is still metric-profile geometry, not smooth differential geometry.** Further coarse-graining and continuum-limit work would be needed before claiming tangent spaces or manifold structure.

2.6 Dimensional Closure: Relational Inference Selects Three Spatial Dimensions

2.6.1 Purpose of This Subsection

The framework does not assume a pre-existing 3D space or 4D spacetime manifold. Space must emerge from the zero-preserving relational structure. Before we can ask *why* the universe is three-dimensional, we must first establish *how* an emergent correlation graph can possess any stable dimension at all.

The purpose of this subsection is to establish the conditions under which the emergent spatial dimension is uniquely selected as $D=3$ by the closure requirements of relational inference.

2.6.2 The Relational Inference Argument

In this framework, local existence, position, direction, distance, and duration are not primitives. They are all inferred only through relations among configurations in the emergent correlation geometry.

Definition 2.12 (Relational Inference Map).
Let $\mathcal{X}_\omega$ be the emergent point set. Define the **relational inference map** at a point $x \in \mathcal{X}_\omega$:
$$ \mathcal{I}_x: \mathcal{N}(x) \to \mathbb{R}^4, $$
where $\mathcal{N}(x)$ is the local correlation neighbourhood. The four output components encode:
1. **Existence**: distinguishability of x from the null configuration (binary test).
2. **Position**: separation of x from neighbouring configurations (scalar measurement d_ω).
3. **Direction**: orientation of x relative to surrounding configurations (vectorial profile difference $\Delta_{A \to B}$).
4. **Duration**: causal-depth ordering of x relative to its predecessor and successor chain.

The **inference rank** at x is:
$$ r(x) = \dim(\operatorname{Im}(\mathcal{I}_x)). $$

2.6.3 Spatial vs. Temporal Channels

Duration is supplied entirely by the causal layer (Section 1), not the spatial correlation layer. The temporal inference channel is a single scalar parameter derived from the strict partial order $\prec$ and burden-weighted proper time τ. Thus, the temporal dimension is mathematically pinned to $D_t = 1$.

Consequently, the spatial correlation geometry must independently supply three channels: existence, position, and direction. In a D-dimensional correlation geometry, the maximum number of independent spatial relational channels at x is bounded by D:
$$ r_{\mathrm{spatial}}(x) \leq D. $$

2.6.4 Assumption — Spatial Inference Independence

To rigorously bound the dimensionality, we must assume that the spatial inference channels are structurally independent.

Assumption 2.7 (Spatial Inference Independence).
Existence, Position, and Direction define three independent spatial inference channels that cannot be losslessly encoded into fewer than three correlation-geometric degrees of freedom.

These three tests are of different algebraic types:
- **Existence** is a binary relational test.
- **Position** is a scalar relational comparison (d_ω).
- **Direction** is a vectorial relational comparison (profile difference $\Delta_{A \to B}$).

Because of this algebraic incommensurability, they cannot collapse into fewer than three independent channels without losing structural distinguishability.

2.6.5 Theorem — Relational Closure Selects Three Spatial Dimensions

Theorem 2.10 (Conditional Closure Theorem).
Under Assumption 2.7, let local existence, position, direction, and duration be inferred only through relations among configurations in the emergent correlation geometry. Then the smallest nondegenerate relational network that allows these inferences to close without collapse or redundancy is $D=3$.

Proof.

We prove by contradiction in three cases, defining the total closure defect as $\Xi_C(D) = \Xi_{\mathrm{under}}(D) + \Xi_{\mathrm{over}}(D)$.

Case 1: $D < 3$ (Under-closure).
Suppose the emergent correlation geometry has effective dimension $D < 3$.
By Assumption 2.7, the three spatial inference channels require three independent degrees of freedom. However, in a D-dimensional geometry:
$$ r_{\mathrm{spatial}}(x) \leq D < 3. $$
Therefore, at least one inference channel must collapse. For instance, if $D=2$, the geometry can support position (scalar) and existence (binary), but the vectorial direction channel cannot support 3 independent components.
This generates a positive under-closure defect:
$$ \Xi_{\mathrm{under}}(D) = 3 - D > 0. $$
Hence:

$$ \boxed{D < 3 \quad \Rightarrow \quad \Xi_C(D) > 0 \quad \Rightarrow \quad D \notin \mathfrak{D}_{\mathrm{adm}}.} $$

Case 2: $D > 3$ (Over-closure).
Suppose the emergent correlation geometry has effective dimension $D > 3$.
The minimal inference closure rank is $r_{\min} = 3$. When $D > 3$, the geometry supplies excess independent relational directions:
$$ r_{\mathrm{spatial}}(x) = D > 3 = r_{\min}. $$
The excess direction count is $x_D(x) = D - 3 > 0$.
This overcomplete structure introduces degeneracy in the inverse inference problem: given the observed relational data, the underlying configuration cannot be uniquely recovered because multiple relational paths produce the same inference signature. Position, direction, and distance cease to remain cleanly distinguishable because multiple relational paths encode the same observables, creating redundancy.
Unless all excess directions are pure gauge (which would collapse the effective dimension back to $D=3$), this generates a positive over-closure defect:
$$ \Xi_{\mathrm{over}}(D) = D - 3 > 0. $$
Hence:
$$ \boxed{D > 3 \quad \Rightarrow \quad \Xi_C(D) > 0 \quad \Rightarrow \quad D \notin \mathfrak{D}_{\mathrm{adm}}.} $$

Case 3: $D = 3$ (Balanced Closure).
Suppose the emergent correlation geometry has effective dimension $D = 3$.
Then $r_{\mathrm{spatial}}(x) = 3 = r_{\min}$.
The under-closure defect vanishes: $\Xi_{\mathrm{under}}(3) = 0$.
The over-closure defect vanishes: $\Xi_{\mathrm{over}}(3) = 0$.
Thus $\Xi_C(3) = 0$.
Hence:
$$ \boxed{D = 3 \quad \Rightarrow \quad \Xi_C(3) = 0 \quad \Rightarrow \quad 3 \in \mathfrak{D}_{\mathrm{adm}}.} $$

Combining all three cases:
$$ \Xi_C(D) = 0 \iff D = 3. $$
Since the master constraint contains $\Xi_C(D)$ non-negatively, physical admissibility selects $D=3$. Combined with the emergent proper-time parameter from the causal layer ($D_t=1$), the selected spacetime phase is $3+1$. $\blacksquare$

2.6.6 Volumetric Projection of Excess Spatial Dimensions

To clarify the nature of the over-closure defect ($D > 3$), we analyze the specific case of $D=4$. A 3D observer possesses exactly three inference channels. A single excess scalar degree of freedom cannot attach to Existence (which is binary) and does not constitute a full independent vectorial channel. Therefore, it attaches to the scalar distance metric d_ω underlying the Position channel.

Because Direction is computed via profile differences of d_ω (Definition 2.8), this single positional ambiguity propagates identically to direction readings. The projection of the

4D relational data onto the 3D quotient yields a two-valued ($\pm$) ambiguity regarding the depth of a localized entity along the missing axis.

Conjecture 2.2 (Generic Two-Sheet Apparent Projection).
Under a generic, convex one-extra-channel projection ($P_{4 \to 3}$), a higher-dimensional relational entity may yield two lower-dimensional *apparent* correlation profiles. These profiles are spatially distinguishable in the projected geometry (representing the near and far preimages along the missing axis) but remain one causal identity if they share the same causal ancestry and open-extension history.

Interpretation (Corrected).
The "mirror image" generated by an excess dimension is an *apparent* projection driven by a single shared $\pm$ displacement, not a literal duplication of an extended volumetric solid. One scalar excess coordinate provides one bit of ambiguity about an entity's depth. For the whole entity to duplicate as two solid copies, every constituent point would need its own independent excess coordinate, which contradicts the premise of a single excess dimension.

Instead, intact existence plus one ambiguous scalar results in one ambiguous depth reading on an otherwise single-valued object. The two apparent sheets are unified by the temporal layer. Because $D_t = 1$, the causal dependency relation $\prec$ and the burden-weighted proper time τ act as a single scalar filtration on the higher-dimensional structure. If the higher-dimensional entity moves, it traverses both inference paths simultaneously. The two apparent projections share the exact same causal ancestry and extension history. They are structurally two spatial images, but causally one entity.

2.6.7 Guardrails for This Subsection

1. **The closure-defect route is conditionally proven.** Theorem 2.10 establishes that $D \neq 3$ produces a positive closure defect, strictly relying on Assumption 2.7 (Inference Independence).
2. **The uniqueness lemma is conditionally closed.** The proof depends on the structural independence of existence, position, and direction as inference channels, and on the over-closure argument for $D > 3$.
3. **The two-sheet projection is a conjecture.** Conjecture 2.2 requires genericity and convexity assumptions to yield exactly two preimages. Non-convex topologies may yield more.
4. **This is a relational argument, not a geometric counting trick.** The dimension is not chosen because of ball-volume scaling or aesthetic preference. It is enforced by the closure requirements of relational inference.

2.7 Guardrails and Summary of Section 2

2.7.1 Guardrails for Section 2

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **Correlation is pre-geometric.** The kernel K_ω is an algebraic/state evaluation. It is not a distance, metric, or coordinate chart until the specific logarithmic and consistency axioms are applied.
2. **Localisable does not mean spatially local in advance.** The term "localisable" is operational and framework-internal. It identifies observables capable of supporting emergent localisation, not objects placed in a pre-existing space.
3. **Approximate metricity is a formal completion.** Exact theorem-backed metricity requires the multiplicative consistency condition. In physical regimes, this is relaxed to an approximate condition with a controlled defect ε.
4. **Perfect correlation is not algebraic identity.** The relation $A \sim_\omega B$ expresses zero emergent separation (indistinguishability by the kernel), not literal equality in the algebra $\mathcal{A}$.
5. **Direction is reconstructed, not assumed.** No vector spaces, coordinate axes, or background orientations are postulated. Direction is strictly an equivalence class of distance-profile deformations.
6. **Dimensional closure is a relational argument.** $D=3$ emerges from the structural independence of existence, position, and direction inference channels, not from geometric counting tricks or topological necessities.

2.7.2 Summary of Section 2

Section 2 reconstructed spatial structure purely from state-dependent relational correlation:

1. **The correlation kernel** K_ω was constructively defined on localisable zero-preserving observables $\mathcal{A}_{\mathrm{loc}}$, measuring the strength of relational association in a physical state ω.
2. **Correlation distance** $d_\omega = -\ell_0 \log K_\omega$ was defined, proving that strong correlation corresponds to small emergent separation.
3. Under **multiplicative correlation consistency** ($K_\omega(A,C) \geq K_\omega(A,B)K_\omega(B,C)$), d_ω was rigorously proven to be a pseudometric.
4. **Emergent points** were defined as equivalence classes of perfectly correlated observables: $\mathcal{X}_\omega = \mathcal{A}_{\mathrm{loc}} / \sim_\omega$.
5. The quotient was shown to carry a genuine **metric** $\widetilde{d}_\omega$, establishing a true metric space $(\mathcal{X}_\omega, \widetilde{d}_\omega)$.
6. **Approximate metricity** was established under coarse-graining, proving that a bounded logarithmic error in the kernel translates to an additive triangle defect ε in the distance, bridging microscopic noise to macroscopic smooth geometry.
7. **Observable-dependent direction** was reconstructed from distance-profile differences, defining the emergent tangent cone $T_x^{\mathrm{em}} \mathcal{X}_\omega$ without assuming a manifold.
8. **Dimensional closure** was proven: $D=3$ is the unique spatial dimension where relational inference of existence (binary), position (scalar), and direction (vectorial) closes without degeneracy or redundancy. Excess dimensions ($D>3$) generate volumetric mirror degeneracy, while deficient dimensions ($D<3$) collapse necessary inference channels.

The conceptual chain of this section is:

$$ \boxed{ \text{correlation kernel} \quad \longrightarrow \quad \text{correlation distance} \quad \longrightarrow \quad \text{pseudometric} \quad \longrightarrow \quad \text{emergent points} \quad \longrightarrow \quad \text{approximate metric space} \quad \longrightarrow \quad \text{direction} \quad \longrightarrow \quad D=3. } $$

This completes the reconstruction of proto-space from relational correlation structure. The framework now possesses an emergent spatial metric and a causal skeleton, setting the stage for the emergence of temporal duration and gravitational geometry.

Section 3 — Emergent Time and Burden-Clock Geometry

3.0 Purpose of Emergent Time and Burden-Clock Geometry

3.0.1 Transition from Space to Time

Section 1 established the causal dependency order $\prec$ and the open extension principle. Section 2 reconstructed emergent spatial structure from the correlation kernel K_ω, proving that correlation distance becomes a pseudometric under multiplicative consistency, and that emergent points arise as equivalence classes of perfectly correlated observables. Section 2 also proved that $D=3$ is the unique spatial dimension where relational inference closes without degeneracy.

However, space is not time. A correlation distance tells us which relational contexts are near or far in an emergent spatial sense. It does not by itself tell us how duration is measured, why clocks may run at different rates, how temporal ordering becomes quantitative, or how time-dilation-like behaviour could arise.

The causal dependency relation $e_i \prec e_j$ provides order. It says that one admissible event is dependency-prior to another. But a partial order alone does not define proper time. It gives sequence, precedence, or causal depth, but not duration.

Thus, after constructing emergent space from correlation, the next task is to reconstruct emergent time from causal structure together with a local measure of relational difficulty.

That measure is called **burden**.

3.0.2 Why Causal Order Alone Is Not Time

A causal chain has the form:
$$ e_0 \prec e_1 \prec \cdots \prec e_n. $$

This chain tells us that each event depends on the previous structure. It gives an order-theoretic notion of depth:
$$ L_C(e_0, e_n) = \max\{ n \mid e_0 \prec e_1 \prec \cdots \prec e_n \}. $$

But this depth is only a count of dependency steps. It does not yet tell us how much physical time passes along the chain.

For example, two chains may have the same number of dependency steps but differ in relational difficulty. One may pass through a low-burden region, where extension is easy and clock rate is high. Another may pass through a high-burden region, where maintaining admissibility is costly and clock rate is suppressed.

Therefore, causal depth must be weighted.

> **Causal order gives before/after; burden-weighted causal order gives duration.**

This section develops that second structure.

3.0.3 The Role of Burden

Burden measures the local relational load associated with preserving admissibility.

At the foundational level, physicality means constraint compatibility:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha. $$

At the level of extension, admissibility means that a structure may continue only through zero-preserving enlargement:
$$ \mathfrak{M}(\mathcal{U}_{s+1}) = 0. $$

However, not all admissible extensions are equally easy. Some regions of relational structure may require more constraint reconciliation, more dependency closure, more correlation adjustment, or more structural coordination in order to remain physical.

This local difficulty is what burden represents.

> **Burden is not constraint failure; it is the cost of maintaining constraint compatibility.**

A high-burden region is not necessarily a region of constraint violation. Rather, it is a region where preserving constraint-zero admissibility is costly. This distinction is crucial. If a structure is physical, its constraints still vanish in the appropriate sense. Burden measures the internal load required to keep them vanishing under extension, interaction, localisation, or deformation.

3.0.4 Burden and Clock Rate

The central claim of the burden-time relation is that higher burden suppresses local clock rate.

A simple functional form is:
$$ \alpha(B) = \frac{1}{1 + \lambda B}, \qquad \lambda > 0, $$
where B is a non-negative burden measure and $\alpha(B)$ is a local clock factor.

If $B = 0$, then $\alpha(0) = 1$. The clock is unsuppressed relative to the chosen reference scale.

If $B > 0$, then $0 < \alpha(B) < 1$. The local clock contribution is reduced.

If burden increases, the clock factor decreases:
$$ B_1 > B_2 \quad \Longrightarrow \quad \alpha(B_1) < \alpha(B_2). $$

> **Higher burden means slower internal clock rate.**

This provides the first theorem-safe bridge toward time-dilation-like behaviour.

3.0.5 Proper Time as Weighted Causal Depth

Let γ be a causal chain:
$$ \gamma: e_0 \prec e_1 \prec \cdots \prec e_n. $$

An unweighted depth count would assign a duration proportional to n. But if each event or step carries burden, then the contribution of each step should be weighted by the local clock factor.

A burden-weighted proper-time functional may be written as:
$$ \tau[\gamma] = \sum_{e \in \gamma} \epsilon \, \alpha(e) = \sum_{e \in \gamma} \frac{\epsilon}{1 + \lambda B(e)}. $$

This formula expresses proper time as accumulated causal depth, suppressed by local burden.

When burden is low, each causal step contributes nearly ϵ.

When burden is high, each causal step contributes less than ϵ.

> **Proper time is accumulated admissible causal succession, locally slowed by burden.**

3.0.6 Why This Resembles Gravitational Time Dilation

In familiar gravitational physics, clocks deeper in a gravitational potential run more slowly relative to clocks far away. The present framework does not assume spacetime gravity at the foundation. However, the burden-clock relation produces an analogous structure.

If a region carries high burden, then $B(e)$ is large. Therefore $\alpha(e) = 1/(1 + \lambda B(e))$ is small. The accumulated proper time along chains passing through that region is reduced.

This gives a relational explanation of clock suppression:

> **Clocks slow where maintaining relational admissibility is more burdensome.**

This is not yet a full theory of gravity. It is a clock-rate bridge. Later sections may connect burden density, burden flux, and burden stress to effective metric structure. But the first step is simply the monotonic relation between burden and clock rate.

3.0.7 The Causal Reconciliation Principle

The burden-clock relation is grounded in a deeper principle.

> **The effective speed of light and the rate of time are bounded by the degree to which relational structures can reconcile with causal dependency.**

This is the **Causal Reconciliation Principle**. It is not a new primitive axiom, but a deeper grounding of the existing causal, relational, and burden-based structure.

When reconciliation is efficient, causal propagation is fast and time advances normally. When burden or constraint loading increases, reconciliation slows, the causal ceiling tightens, and proper time becomes increasingly suppressed.

In compact form:
$$ c_{\mathrm{eff}} \sim \frac{\ell_0}{\tau_{\mathrm{rec}}}, \qquad d\tau \sim \frac{d\sigma}{1 + \lambda B}, \qquad \tau_{\mathrm{rec}} \uparrow \Rightarrow c_{\mathrm{eff}} \downarrow, \; d\tau \downarrow. $$

This principle unifies the causal speed limit (Section 1) with the burden-clock suppression (this section). The same mechanism that bounds propagation speed also bounds the rate of temporal flow.

3.0.8 What This Section Establishes

This section establishes the following:

1. **Constraint burden** $B_R(e)$ is defined rigorously from the primitive algebraic data as a scalar functional measuring the commutator cost of localisation.
2. A **local clock factor** $\alpha(B) = 1/(1 + \lambda B)$ is introduced.
3. **Burden-weighted proper time** $\tau[\gamma] = \sum_{e \in \gamma} \epsilon \, \alpha(e)$ is defined.
4. The theorem **"burden suppresses clock rate"** is proven: $B_1 > B_2 \Rightarrow \alpha(B_1) < \alpha(B_2)$.
5. **Non-uniform burden** is shown to induce non-uniform clock accumulation, producing temporal geometry.
6. An **effective potential** $\Phi_C = c_{\mathrm{eff}}^2 \log \alpha$ is derived.
7. The **weak-burden limit** $\Phi_C \approx -c_{\mathrm{eff}}^2 \lambda B$ is established.
8. The **scalar burden-to-lapse bridge** $d\tau = \alpha(x) \, d\sigma$ is formalised.
9. The relation between burden-clock geometry and **emergent gravity** is prepared.

The theorem-safe structure is:

$$ \boxed{ \text{causal order} + \text{constraint burden} \quad \Longrightarrow \quad \text{burden-weighted proper time} \quad \Longrightarrow \quad \text{temporal geometry}. } $$

3.0.9 Guardrails for This Section

1. **Time is still emergent.** Even after defining burden-weighted proper time, time is not a primitive background parameter. It is reconstructed along causal chains.
2. **Burden is not violation.** $B > 0 \not\Rightarrow \mathfrak{M} > 0$. A region may be exactly physical and still carry high burden.
3. **Clock suppression is not yet a full metric equation.** The formula $\alpha(B) = 1/(1+\lambda B)$ defines a clock factor. It does not by itself define the full metric tensor.
4. **The burden-clock relation is a model choice unless derived.** The rational form $\alpha(B) = 1/(1+\lambda B)$ is mathematically simple and monotone. Other monotone decreasing functions could serve similar roles. The theorem-safe statement is: any positive decreasing clock factor yields burden-induced clock suppression.

3.1 Constraint Burden

3.1.1 Purpose of This Subsection

Before defining emergent proper time, the framework needs a measure of how much relational cost is associated with maintaining admissibility at a given event or region. This measure is **constraint burden**.

The purpose of this subsection is to define burden precisely from the primitive algebraic data, distinguish it from constraint violation, and establish its role as the local reconciliation cost that will later weight the flow of proper time.

3.1.2 The Concept of Reconciliation Cost

At the foundational level, physicality means constraint compatibility:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \qquad \forall \alpha. $$

At the level of open extension, admissibility means that a structure may continue only through zero-preserving enlargement:
$$ \mathfrak{M}(\mathcal{U}_{s+1}) = 0. $$

However, not all admissible extensions are equally easy. Some regions of relational structure may require more constraint reconciliation, more dependency closure, more correlation adjustment, or more structural coordination in order to remain physical. This local difficulty is what burden represents.

Definition 3.1 (Constraint Burden).
Let $R \subseteq \mathcal{E}$ be a subsystem or coarse-grained region. Let $\Pi_R \in \mathcal{A}$ be a localisation or projection operator associated with region R. The

constraint burden of R in the physical state ω is a non-negative scalar functional:
$$ \boxed{ B(R) = \sum_{\alpha \in I} w_\alpha \, \omega\left([\hat{C}_\alpha, \Pi_R]^\dagger [\hat{C}_\alpha, \Pi_R] \right), } $$
where $w_\alpha > 0$ are strictly positive weights, and $[\hat{C}_\alpha, \Pi_R] = \hat{C}_\alpha \Pi_R - \Pi_R \hat{C}_\alpha$ is the commutator measuring the failure of the region to decouple from the primitive constraint.

Interpretation.
Burden measures the local load of preserving admissibility. It is not the amount by which physicality fails (which is measured by $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha)$). A physical event may have non-zero burden while still satisfying all constraints perfectly.

> **$B(R) > 0$ means the region is costly to maintain admissibly, not that it is unphysical.**

3.1.3 Classification of the Burden Object

It is crucial to classify the mathematical nature of burden:
- **It is a scalar functional:** $B(R)$ maps the algebra to a non-negative real number $\mathbb{R}_{\geq 0}$.
- **It is state-dependent:** It is evaluated in the physical state ω.
- **It is gauge invariant:** Because it is constructed from a positive operator $[\hat{C}_\alpha, \Pi_R]^\dagger [\hat{C}_\alpha, \Pi_R]$ and evaluated in a physical state, it is invariant under physical equivalence transformations.
- **It is additive/subadditive depending on correlations:** For disjoint, uncorrelated regions, burdens add. For interacting regions, non-additive terms appear (which form the basis of interactions, see Section 4.3).

3.1.4 Theorem — Positivity and Monotonicity

For burden to function as a physical cost, it must be non-negative, and it must increase (or remain constant) as the region grows.

Theorem 3.1 (Positivity and Monotonicity of Burden).
Let ω be a physical state. The burden functional $B(R)$ satisfies:
1. **Positivity:** $B(R) \geq 0$ for all R.
2. **Monotonicity under extension:** If $R_1 \subseteq R_2$, then $B(R_1) \leq B(R_2)$.

Proof.

Positivity:
The operator $X_\alpha = [\hat{C}_\alpha, \Pi_R]$ is an element of $\mathcal{A}$. Therefore $X_\alpha^\dagger X_\alpha$ is a positive operator. Since ω is a positive linear functional, $\omega(X_\alpha^\dagger X_\alpha) \geq 0$. As the sum of positive terms with positive weights, $B(R) \geq 0$.

Monotonicity:

Let $R_1 \subseteq R_2$. In a standard projection algebra, we may take $\Pi_{R_2} = \Pi_{R_1} + \Pi_{R_2 \setminus R_1}$, where Π_{R_1} and $\Pi_{R_2 \setminus R_1}$ are orthogonal ($\Pi_{R_1} \Pi_{R_2 \setminus R_1} = 0$).
The commutator expands as:
$$ [\hat{C}_\alpha, \Pi_{R_2}] = [\hat{C}_\alpha, \Pi_{R_1}] + [\hat{C}_\alpha, \Pi_{R_2 \setminus R_1}]. $$
The burden of R_2 is:
$$ B(R_2) = \sum_\alpha w_\alpha \omega\left(\left([\hat{C}_\alpha, \Pi_{R_1}] + [\hat{C}_\alpha, \Pi_{R_2 \setminus R_1}] \right)^\dagger \left([\hat{C}_\alpha, \Pi_{R_1}] + [\hat{C}_\alpha, \Pi_{R_2 \setminus R_1}] \right) \right). $$
Expanding the product yields:
$$ B(R_2) = B(R_1) + B(R_2 \setminus R_1) + \sum_\alpha w_\alpha \omega\left([\hat{C}_\alpha, \Pi_{R_2 \setminus R_1}]^\dagger [\hat{C}_\alpha, \Pi_{R_1}] \right) + \text{c.c.} $$
Because the cross-terms are state evaluations of operators, and because the physical state ω is positive, the cross-terms do not reduce the total sum below the isolated sums. In the worst case (complete decoherence between the regions), the cross-terms vanish, yielding $B(R_2) = B(R_1) + B(R_2 \setminus R_1) \geq B(R_1)$. $\blacksquare$

3.1.5 The Three-Component Decomposition

The framework refines burden from a single scalar into a source-origin decomposition with three distinct channels. This decomposition is not arbitrary; it falls out directly from expanding the commutator $[\hat{C}_\alpha, \Pi_R]$ when R is composed of interacting subsystems.

Definition 3.2 (Component Burden Decomposition).
For a composite region $R = R_{\text{mode}} \cup R_{\text{int}} \cup R_{\text{rel}}$, the total burden density $\rho_B(x)$ decomposes as:
$$ \boxed{ \rho_B(x) = \rho_{\mathrm{mode}}(x) + \rho_{\mathrm{int}}(x) + \rho_{\mathrm{rel}}(x). } $$

Theorem 3.2 (Derivation of the Decomposition).
The three components arise structurally from the algebra of projections.

Proof Sketch.
Consider two interacting subsystems A and B. The localization operator for the union is $\Pi_{A \cup B} = \Pi_A + \Pi_B + \Pi_{AB}^{\text{bind}}$, where Π_{AB}^{bind} represents the non-additive correlation structure (the binding network).
Expanding the commutator:
$$ [\hat{C}_\alpha, \Pi_{A \cup B}] = \underbrace{[\hat{C}_\alpha, \Pi_A] + [\hat{C}_\alpha, \Pi_B]}_{\text{Mode Sector}} + \underbrace{[\hat{C}_\alpha, \Pi_{AB}^{\text{bind}}]}_{\text{Interaction Sector}} + \underbrace{[\hat{C}_\alpha, \Pi_{\text{network}}]}_{\text{Relational Sector}}. $$
When inserted into the burden functional $B(R) = \sum w_\alpha \omega(\cdot^\dagger \cdot)$:

1. **Mode burden (ρ_{mode}):** Generated by the terms $[\hat{C}_\alpha, \Pi_A]$ and $[\hat{C}_\alpha, \Pi_B]$. This is the burden of the independent, stable configurations existing in the region.
2. **Interaction burden (ρ_{int}):** Generated by the cross-terms and the binding operator $[\hat{C}_\alpha, \Pi_{AB}^{\text{bind}}]$. This measures the non-additive composition cost. A neutron star carries more interaction burden than diffuse gas because its binding network is denser.
3. **Relational burden (ρ_{rel}):** Generated by the commutator of the constraint with the global correlation network operator Π_{network}. This measures the burden required to maintain relational consistency across the background correlation web, independent of the localized objects. $\blacksquare$

Interpretation.
These three components are structurally distinct source channels:
- **Mode burden** is generated by the existence and occupation of stable field or particle-like modes.
- **Interaction burden** arises from non-additivity in composition ($B_{\mathrm{int}} = B(A \cup B) - B(A) - B(B)$).
- **Relational burden** is generated by the correlation network itself. Unlike the first two, ρ_{rel} does not belong to a single localized object; it belongs to the network topology.

3.1.6 Burden Is Not Constraint Violation

This distinction must remain absolutely clear throughout the manuscript.

A physical state satisfies:
$$ \omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0. $$
A represented physical vector satisfies:
$$ \hat{\mathfrak{M}}_\omega \psi = 0. $$
An effective physical configuration satisfies:
$$ \mathfrak{M}(x) = 0. $$

These are zero-violation conditions. Burden, by contrast, may be non-zero even when the structure is physical. It measures the load of preserving physicality, not the failure of physicality.

In a physical but burdened region, one may have:
$$ \mathfrak{M} = 0 \quad \text{while} \quad B > 0. $$

> **$\mathfrak{M}$ measures constraint violation; B measures constraint-maintenance load.**

3.1.7 Guardrails for This Subsection

1. **Burden is not energy.** At this stage, burden is a relational cost measure. Energy requires additional structure (time translation, Hamiltonian, conservation). Burden may become energy-like under additional assumptions, but that identification is not automatic.

2. **Burden is not violation.** $B > 0 \not\Rightarrow \mathfrak{M} > 0$. A region may be exactly physical and still carry high burden.
3. **The three-component decomposition is structural.** Mode, interaction, and relational burden are not interchangeable. They represent distinct source channels that may activate differently in different physical regimes.
4. **The definition is representation-dependent.** The functional $B(R)$ requires a localization operator Π_R and a state ω. In a purely abstract algebraic setting without a chosen GNS representation, Π_R is a formal element; its physical meaning as a "region" requires the emergent correlation geometry of Section 2.

3.2 Local Clock Factor and Burden Suppression

3.2.1 Purpose of This Subsection

The previous subsection defined constraint burden $B(R)$ rigorously from the primitive algebraic data as a scalar functional measuring the commutator cost of localization. The next step is to connect burden to the rate at which proper time accumulates.

The central claim is that higher burden suppresses local clock rate. This subsection formalises that claim through the definition of a **local clock factor** and proves the monotonic suppression theorem.

3.2.2 Definition — Local Clock Factor

Definition 3.3 (Local Clock Factor).
Let $B \geq 0$ be a burden measure and $\lambda > 0$ a coupling constant. The **local clock factor** is:
$$ \boxed{ \alpha(B) = \frac{1}{1 + \lambda B}. } $$

Since $B \geq 0$ and $\lambda > 0$, we have:
$$ 1 + \lambda B \geq 1, $$
and therefore:
$$ 0 < \alpha(B) \leq 1. $$

Interpretation.
- If $B = 0$, then $\alpha(0) = 1$. The clock is unsuppressed.
- If $B > 0$, then $0 < \alpha(B) < 1$. The local clock contribution is reduced.
- As $B \to \infty$, $\alpha(B) \to 0$. The clock factor approaches zero.

The clock factor $\alpha(B)$ represents the fraction of the elementary causal-step time that accumulates at a given burden level.

> **$\alpha(B)$ is the local reconciliation rate: the fraction of the elementary tick that survives under burden loading.**

3.2.3 Why a Rational Form?

The rational form $\alpha(B) = 1/(1 + \lambda B)$ is chosen for four reasons:

1. **Positivity:** $\alpha(B) > 0$ for all finite $B \geq 0$. The clock never literally stops for finite burden.
2. **Boundedness:** $\alpha(B) \leq 1$. The clock never runs faster than the unsuppressed rate.
3. **Monotonicity:** $\alpha'(B) < 0$. Higher burden always suppresses the clock.
4. **Simplicity:** It is the simplest non-trivial function satisfying all three conditions above.

However, the essential structure only requires a positive decreasing function. The theorem-safe general statement is:

> **Any positive decreasing clock factor $\alpha(B)$ with $\alpha(0) = 1$ and $\alpha'(B) < 0$ yields burden-induced clock suppression.**

The rational form is one canonical ansatz. Other forms (e.g., $\alpha(B) = e^{-\lambda B}$) could serve similar roles. The manuscript uses the rational form because it gives simple algebraic expressions and clear monotonic behaviour.

3.2.4 Theorem — Burden Suppresses Clock Rate

Theorem 3.3 (Burden Suppresses Clock Rate).
Let $\alpha(B) = 1/(1 + \lambda B)$ with $\lambda > 0$. If $B_1 > B_2 \geq 0$, then:
$$ \boxed{ \alpha(B_1) < \alpha(B_2). } $$

Proof.

Since $B_1 > B_2$ and $\lambda > 0$:
$$ \lambda B_1 > \lambda B_2. $$

Adding 1 to both sides:
$$ 1 + \lambda B_1 > 1 + \lambda B_2. $$

Both sides are positive. Taking reciprocals reverses the inequality:
$$ \frac{1}{1 + \lambda B_1} < \frac{1}{1 + \lambda B_2}. $$

Therefore:
$$ \alpha(B_1) < \alpha(B_2). \qquad \blacksquare $$

3.2.5 Corollary — Limiting Behaviour

Corollary 3.1.
$$ \lim_{B \to \infty} \alpha(B) = 0. $$

Proof.
As $B \to \infty$, $1 + \lambda B \to \infty$, so $1/(1 + \lambda B) \to 0$. $\blacksquare$

Interpretation.

In the limit of infinite burden, the clock factor approaches zero. This means proper time accumulation flattens toward zero. However, for any finite burden $B < \infty$, the clock factor remains strictly positive: $\alpha(B) > 0$.

> **For finite burden, $\alpha > 0$; the clock slows but does not literally stop.**

This is the guardrail against interpreting black-hole-like regimes as places where "time ends." Time does not end; it is suppressed toward zero in the exterior projection. The interior relational structure continues, but its reconciliation rate becomes extremely slow.

3.2.6 Lemma — Clock Factor Bounds

Lemma 3.1.
For all $B \geq 0$ and $\lambda > 0$:
$$ 0 < \alpha(B) \leq 1. $$
Moreover:
$$ \alpha(B) = 1 \quad \Longleftrightarrow \quad B = 0. $$

Proof.
Since $B \geq 0$ and $\lambda > 0$, we have $1 + \lambda B \geq 1 > 0$. Therefore $\alpha(B) = 1/(1+\lambda B) > 0$ and $\alpha(B) \leq 1$.

Equality $\alpha(B) = 1$ holds if and only if $1 + \lambda B = 1$, which requires $\lambda B = 0$. Since $\lambda > 0$, this occurs exactly when $B = 0$. $\blacksquare$

3.2.7 Guardrails for This Subsection

1. **The clock factor is not a full metric.** $\alpha(B)$ is a scalar suppression factor. It does not by itself define the full metric tensor $g_{\mu\nu}$.
2. **The rational form is a model choice.** The specific function $\alpha(B) = 1/(1+\lambda B)$ is used for simplicity. The theorem-safe content is: any positive, bounded, strictly decreasing $\alpha(B)$ with $\alpha(0) = 1$ produces clock suppression.
3. **$\alpha \to 0$ does not mean $\alpha = 0$.** For finite burden, the clock factor is strictly positive. The approach to zero is asymptotic, not literal.

3.3 Burden-Weighted Proper Time

3.3.1 Purpose of This Subsection

The previous subsection defined the local clock factor $\alpha(B)$ and proved that burden suppresses it monotonically. The next step is to use this clock factor to weight causal depth, producing a proper-time functional along causal chains.

The purpose of this subsection is to define **burden-weighted proper time** rigorously and establish its basic properties.

3.3.2 Definition — Burden-Weighted Proper Time

Definition 3.4 (Burden-Weighted Proper Time).
Let $\gamma = (e_0, e_1, \ldots, e_n)$ be a finite causal chain with $e_0 \prec e_1 \prec \cdots \prec e_n$. Let $\epsilon > 0$ be the elementary reconciliation interval. Let $B_R(e_k)$ be the constraint burden of subsystem R at event e_k. Let $\lambda > 0$.

The **burden-weighted proper time** along γ is:
$$ \boxed{ \tau_R[\gamma] = \sum_{k=0}^{n} \frac{\epsilon}{1 + \lambda B_R(e_k)}. } $$

Equivalently, using $\alpha(B) = 1/(1 + \lambda B)$:
$$ \boxed{ \tau_R[\gamma] = \sum_{k=0}^{n} \epsilon \, \alpha(B_R(e_k)). } $$

Interpretation.
Each causal step e_k contributes an amount $\epsilon \cdot \alpha(B_R(e_k))$ to the accumulated proper time. When burden is low, the contribution is close to ϵ. When burden is high, the contribution is reduced.

> **Proper time is accumulated causal depth, locally suppressed by burden.**

3.3.3 Basic Bounds

Theorem 3.4 (Basic Bounds).
For any causal chain $\gamma = (e_0, \ldots, e_n)$ with $n+1$ events:
$$ 0 < \tau_R[\gamma] \leq (n+1)\epsilon. $$

Proof.
Since $0 < \alpha(B_R(e_k)) \leq 1$ for all k (Lemma 3.1), each term satisfies:
$$ 0 < \epsilon \, \alpha(B_R(e_k)) \leq \epsilon. $$

Summing over $k = 0, \ldots, n$:
$$ 0 < \sum_{k=0}^{n} \epsilon \, \alpha(B_R(e_k)) \leq \sum_{k=0}^{n} \epsilon = (n+1)\epsilon. $$

Therefore:
$$ 0 < \tau_R[\gamma] \leq (n+1)\epsilon. \qquad \blacksquare $$

Interpretation.
Burden-weighted proper time is always strictly positive (for finite burden) and never exceeds the unburdened causal-step time. Burden can only reduce clock accumulation relative to the unburdened count.

> **Burden can only reduce clock accumulation relative to the unburdened causal count.**

3.3.4 Constant-Burden Case

If the burden is constant along the chain, $B_R(e_k) = B_0$ for all k, then:

$$ \tau_R[\gamma] = \sum_{k=0}^{n} \frac{\epsilon}{1 + \lambda B_0} = \frac{(n+1)\epsilon}{1 + \lambda B_0}. $$

Thus, constant burden rescales causal depth by a uniform suppression factor:
$$ \boxed{ \tau = \alpha(B_0) \times \text{unweighted causal-step time}. } $$

This is the simplest burden-time relation. The clock runs at a uniformly reduced rate.

3.3.5 Variable-Burden Case

If burden varies along the chain, then:
$$ \tau_R[\gamma] = \sum_{k=0}^{n} \frac{\epsilon}{1 + \lambda B_R(e_k)}. $$

There is no single suppression factor. An effective average clock factor may be defined as:
$$ \bar{\alpha}_\gamma = \frac{1}{n+1} \sum_{k=0}^{n} \alpha(B_R(e_k)), $$
so that:
$$ \tau_R[\gamma] = (n+1)\epsilon \, \bar{\alpha}_\gamma. $$

However, because $\alpha(B) = 1/(1+\lambda B)$ is nonlinear, the effective average burden is not generally the arithmetic average of $B_R(e_k)$.

> **Variable burden produces genuinely path-dependent proper time.**

This is the origin of path-dependent temporal geometry in the framework.

3.3.6 Theorem — Higher Burden Reduces Proper Time for Equal Depth

Theorem 3.5 (Higher Burden Reduces Proper Time for Equal Depth).
Let $\gamma_1 = (e_0, \ldots, e_n)$ and $\gamma_2 = (e'_0, \ldots, e'_n)$ be two causal chains of equal depth n. If $B_R(e_k) > B_R(e'_k)$ for every k, then:
$$ \boxed{ \tau_R[\gamma_1] < \tau_R[\gamma_2]. } $$

Proof.
By Theorem 3.3, $B_R(e_k) > B_R(e'_k)$ implies $\alpha(B_R(e_k)) < \alpha(B_R(e'_k))$ for every k. Therefore:
$$ \epsilon \, \alpha(B_R(e_k)) < \epsilon \, \alpha(B_R(e'_k)) \quad \forall k. $$

Summing:
$$ \sum_{k=0}^{n} \epsilon \, \alpha(B_R(e_k)) < \sum_{k=0}^{n} \epsilon \, \alpha(B_R(e'_k)). $$

Therefore:
$$ \tau_R[\gamma_1] < \tau_R[\gamma_2]. \qquad \blacksquare $$

Interpretation.
For equal causal depth, the chain passing through higher burden accumulates less proper time. This is the first time-dilation-like result:

> **For equal causal depth, uniformly higher burden implies lower accumulated proper time.**

3.3.7 Relation to Unburdened Causal Time

The unburdened causal time is:
$$ \tau_0[\gamma] = (n+1)\epsilon. $$

The burden-weighted proper time satisfies:
$$ \tau_R[\gamma] = \sum_{k=0}^{n} \epsilon \, \alpha(B_R(e_k)) \leq \sum_{k=0}^{n} \epsilon = \tau_0[\gamma]. $$

Thus:
$$ \tau_R[\gamma] \leq \tau_0[\gamma]. $$

The ratio:
$$ \frac{\tau_R[\gamma]}{\tau_0[\gamma]} = \frac{1}{n+1} \sum_{k=0}^{n} \alpha(B_R(e_k)) = \bar{\alpha}_\gamma $$
gives the average clock suppression along the chain.

3.3.8 Continuum Approximation

If causal chains become dense in an emergent continuum limit, the sum may be approximated by an integral.

Let s be a parameter along a causal path γ, and let $B(s)$ be the burden profile along the path. Then:
$$ \tau[\gamma] \approx \int_\gamma \alpha(B(s)) \, d\sigma = \int_\gamma \frac{d\sigma}{1 + \lambda B(s)}, $$
where $d\sigma$ is an elementary causal-depth measure.

Using the rational clock factor:
$$ \tau[\gamma] \approx \int_\gamma \frac{d\sigma}{1 + \lambda B(s)}. $$

This continuum expression is an approximation. The discrete causal-chain definition is more primitive in the present framework.

> **The integral form is an effective continuum limit of a causal-chain sum.**

3.3.9 Guardrails for This Subsection

1. **Proper time is chain-relative.** $\tau[\gamma]$ is assigned to a causal chain γ. It is not a universal external time coordinate.
2. **The step scale ϵ is not automatically Planck time.** ϵ is a model parameter. It should not be identified with a known fundamental constant unless a later calibration is provided.
3. **Burden-weighted proper time is not yet coordinate time.** $\tau[\gamma]$ is local/path proper time, not a global t.

4. **Clock suppression is not yet full gravity.** The result $B_1 > B_2 \Rightarrow \alpha_1 < \alpha_2$ shows burden suppresses clocks. This resembles gravitational time dilation, but a full gravitational theory requires spatial metric response, source structure, conservation laws, and field equations.

3.4 Non-Uniform Burden as Temporal Geometry

3.4.1 Purpose of This Subsection

The previous subsection defined burden-weighted proper time along a causal chain and proved that higher burden reduces clock accumulation. The present subsection develops the next consequence: when burden varies from region to region, clock rates become non-uniform. This non-uniformity is the first form of **temporal geometry** in the framework.

> **Non-uniform burden induces non-uniform time flow.**

This does not yet require a full spacetime metric. It requires only causal chains, emergent locality, and a burden-dependent clock factor. The result is a relational version of time-rate variation: different regions accumulate different amounts of proper time over comparable causal depth.

3.4.2 From Local Burden to Local Clock Rate

Let burden be assigned to emergent locations. If $x \in \mathcal{X}_\omega$ is an emergent point in the correlation geometry (Definition 2.5), let $B(x) \geq 0$ denote the local burden at x.

Define the local clock factor:
$$ \boxed{ \alpha(x) = \frac{1}{1 + \lambda B(x)}, \qquad \lambda > 0. } $$

This is the spatially local version of the event-based formula $\alpha(e) = 1/(1 + \lambda B(e))$.

If burden is uniform, $B(x) = B_0$ for all x, and the clock factor is also uniform:
$$ \alpha(x) = \frac{1}{1 + \lambda B_0} \quad \forall x. $$

If burden is non-uniform, $B(x_1) \neq B(x_2)$ generally implies $\alpha(x_1) \neq \alpha(x_2)$.

Thus, local burden variation produces local clock-rate variation.

The core relation is:
$$ \boxed{ B(x_1) > B(x_2) \quad \Longrightarrow \quad \alpha(x_1) < \alpha(x_2). } $$

3.4.3 Definition — Temporal Geometry

Definition 3.5 (Temporal Geometry).

The **temporal geometry** of an emergent region is the assignment:
$$ x \mapsto \alpha(x), $$
where $\alpha(x) = 1/(1 + \lambda B(x))$ is the local clock factor induced by the burden field $B(x)$.

Interpretation.
Temporal geometry is the spatial pattern of clock factors. It is not yet full spacetime curvature. It is the first layer of clock-rate structure.

> **Temporal geometry is the spatial pattern of clock-rate suppression induced by burden.**

3.4.4 Theorem — Non-Uniform Burden Induces Non-Uniform Proper Time

Theorem 3.6 (Non-Uniform Burden Induces Non-Uniform Proper Time).
Let R_1 and R_2 be two regions in $\mathcal{X}_\omega$ with constant burdens B_1 and B_2 respectively. Let $\alpha_i = 1/(1 + \lambda B_i)$ for $i = 1, 2$. Consider two causal chains γ_1 and γ_2, each with n links, where γ_i lies entirely in R_i. If $B_1 > B_2$, then:
$$ \boxed{ \tau[\gamma_1] < \tau[\gamma_2]. } $$

Proof.
Since $B_1 > B_2$ and $\lambda > 0$, Theorem 3.3 gives $\alpha_1 < \alpha_2$. For equal link-count n, the proper time along each chain is:
$$ \tau[\gamma_i] = n\epsilon \, \alpha_i. $$

Therefore:
$$ \tau[\gamma_1] = n\epsilon \, \alpha_1 < n\epsilon \, \alpha_2 = \tau[\gamma_2], $$
because $n\epsilon > 0$. $\blacksquare$

Interpretation.
Over equal causal depth, higher-burden regions accumulate less proper time. This is the relational-clock analogue of gravitational time dilation.

3.4.5 Burden Gradients

If burden varies continuously across emergent space, one may write:
$$ B = B(x), \qquad \alpha = \alpha(x) = \frac{1}{1 + \lambda B(x)}. $$

A burden gradient $\nabla B(x)$ then induces a clock-factor gradient $\nabla \alpha(x)$.

Using the rational clock factor:
$$ \alpha(x) = (1 + \lambda B(x))^{-1}. $$

Differentiating formally:
$$ \nabla \alpha(x) = -\frac{\lambda \nabla B(x)}{(1 + \lambda B(x))^2}. $$

Thus, where burden increases, the clock factor decreases.

> **Where burden increases, clock rate decreases.**

The continuum gradient expression is useful later, but it should not be treated as primitive. The foundational statement is the discrete comparison:
$$ B(x_1) > B(x_2) \quad \Longrightarrow \quad \alpha(x_1) < \alpha(x_2). $$

3.4.6 Effective Potential from Clock Rate

It is useful to express the clock factor in terms of an effective potential.

Definition 3.6 (Burden-Clock Potential).
Define the **burden-clock potential**:
$$ \boxed{ \Phi_C(x) = c_{\mathrm{eff}}^2 \log \alpha(x). } $$

Since $\alpha(x) = 1/(1 + \lambda B(x))$:
$$ \Phi_C(x) = c_{\mathrm{eff}}^2 \log \left(\frac{1}{1 + \lambda B(x)} \right) = -c_{\mathrm{eff}}^2 \log(1 + \lambda B(x)). $$

Since $1 + \lambda B(x) \geq 1$:
$$ \log(1 + \lambda B(x)) \geq 0, $$
and therefore:
$$ \Phi_C(x) \leq 0. $$

Interpretation.
Higher burden makes $\Phi_C(x)$ more negative. Thus:
> **Higher burden corresponds to a deeper effective temporal potential.**

3.4.7 Theorem — Weak-Burden Expansion

Theorem 3.7 (Weak-Burden Expansion).
If burden is small in the sense that $\lambda B(x) \ll 1$, then the burden-clock potential is approximately proportional to negative burden:
$$ \boxed{ \Phi_C(x) \approx -c_{\mathrm{eff}}^2 \lambda B(x). } $$
Similarly, the clock factor is approximately:
$$ \alpha(x) \approx 1 - \lambda B(x). $$

Proof.
Using the Taylor expansion of $\log(1 + y)$ for $y \ll 1$:
$$ \log(1 + \lambda B(x)) \approx \lambda B(x) - \frac{\lambda^2 B(x)^2}{2} + \cdots $$

To first order:
$$ \log(1 + \lambda B(x)) \approx \lambda B(x). $$

Therefore:
$$ \Phi_C(x) = -c_{\mathrm{eff}}^2 \log(1 + \lambda B(x)) \approx -c_{\mathrm{eff}}^2 \lambda B(x). $$

Similarly, using the geometric series expansion for $(1 + \lambda B(x))^{-1}$:

$$ \alpha(x) = \frac{1}{1 + \lambda B(x)} \approx 1 - \lambda B(x). \qquad \blacksquare $$

> **In the weak-burden regime, the temporal potential is approximately proportional to negative burden: $\Phi_C(x) \approx -c_{\mathrm{eff}}^2 \lambda B(x)$.**

This gives a simple linear bridge between burden and clock potential in the weak regime.

3.4.8 Redshift-Like Ratio

Consider two regions R_1 and R_2 with clock factors α_1 and α_2. The ratio of clock rates is:
$$ \frac{\alpha_1}{\alpha_2} = \frac{1 + \lambda B_2}{1 + \lambda B_1}. $$

If $B_1 > B_2$, then:
$$ \frac{\alpha_1}{\alpha_2} < 1. $$

Thus, the high-burden region has a lower clock rate relative to the low-burden region.

One may define a redshift-like quantity:
$$ 1 + z_{12} = \frac{\alpha_2}{\alpha_1} = \frac{1 + \lambda B_1}{1 + \lambda B_2}. $$

Then $z_{12} > 0$ when $B_1 > B_2$.

> **Signals from higher-burden regions appear slower relative to lower-burden references.**

This remains an analogy until signal propagation and observational comparison are defined. It is safer to say "redshift-like" rather than "redshift" at this stage.

3.4.9 Guardrails for This Subsection

1. **Non-uniform burden gives lapse-like temporal geometry, not full curvature by itself.** Full curvature requires a full metric structure.
2. **The burden potential is not yet the Newtonian potential.** Φ_C is an effective burden-clock potential. Identification with Newtonian gravity requires weak-field matching, probe dynamics, and empirical calibration.
3. **Redshift-like ratios require signal interpretation.** The ratio α_1/α_2 gives a clock-rate comparison. Calling this a redshift requires signal propagation, frequencies, transmission, and observation.
4. **Continuum gradients are effective.** Expressions such as $\nabla B(x)$ and $\nabla \alpha(x)$ assume continuum differentiability. They are not available at the most primitive relational level. The foundational statement is finite and order-based: $B(x_1) > B(x_2) \Rightarrow \alpha(x_1) < \alpha(x_2)$.

3.5 Scalar Burden-to-Lapse Bridge

3.5.1 Purpose of This Subsection

The previous subsections established that non-uniform burden produces non-uniform clock rates and an effective potential. The present subsection formalises the connection between the burden field and an effective **lapse function**, which is the scalar temporal component of an effective metric.

> **Burden becomes temporal geometry through a lapse-like scalar field.**

This is still not the full burden-to-metric dictionary. It concerns only the scalar time-rate part of the effective geometry. Spatial metric response, shift terms, burden flux, burden stress, and tensorial source structure belong to later sections.

3.5.2 From Clock Factor to Lapse

In ordinary geometric language, a lapse function controls the relation between a coordinate-like ordering parameter and local proper time. The present framework does not assume such a coordinate time at the foundation. Nevertheless, once causal order and extension order have been introduced, one may define a relational ordering parameter σ that labels causal depth or admissible extension increments.

This parameter is not physical time by itself. It is an order label.

The local proper-time increment is then defined by:
$$ \boxed{ d\tau = \alpha(x) \, d\sigma. } $$

Here, $\alpha(x)$ acts as a **lapse-like conversion factor**. It tells us how much local proper time accumulates per unit of relational order.

Using the burden-clock relation:
$$ \boxed{ \alpha(x) = \frac{1}{1 + \lambda B(x)}. } $$

Thus:
$$ \boxed{ d\tau = \frac{d\sigma}{1 + \lambda B(x)}. } $$

This equation is the **scalar burden-to-lapse bridge** in its simplest form.

> **Local burden suppresses the conversion of causal or extension order into proper time.**

3.5.3 Definition — Burden Lapse

Definition 3.7 (Burden Lapse).
Let $B(x) \geq 0$ be a scalar burden field over an emergent correlation-geometric region, where $x \in \mathcal{X}_\omega$. Let $\lambda > 0$. The **burden lapse** is the scalar field:
$$ \boxed{ \alpha_B(x) = \frac{1}{1 + \lambda B(x)}. } $$

When no ambiguity is likely, we write $\alpha(x)$ instead of $\alpha_B(x)$.

The burden lapse satisfies:
$$ 0 < \alpha_B(x) \leq 1. $$

It equals one precisely where burden vanishes:
$$ \alpha_B(x) = 1 \quad \Longleftrightarrow \quad B(x) = 0. $$

It decreases monotonically as burden increases.

3.5.4 Effective Temporal Line Element

In a continuum approximation, one may encode the lapse relation in a temporal line element.

Let $d\sigma$ represent an infinitesimal ordering increment, and let c_{eff} be an effective conversion speed or signal scale. Then the time-like part of an effective line element may be written as:
$$ \boxed{ ds^2_{\mathrm{time}} = -\alpha_B(x)^2 \, c_{\mathrm{eff}}^2 \, d\sigma^2. } $$

Substituting the burden lapse:
$$ \boxed{ ds^2_{\mathrm{time}} = -\frac{c_{\mathrm{eff}}^2}{(1 + \lambda B(x))^2} \, d\sigma^2. } $$

This expression should be interpreted carefully. It is not yet a complete spacetime metric. It contains only the lapse-like scalar temporal factor. A full metric would require spatial components, possible shift terms, compatibility with causal structure, and continuum regularity.

> **The scalar burden lapse supplies the time-time part of an effective metric ansatz.**

3.5.5 Relation to Proper Time Along Chains

The scalar lapse field connects directly to burden-weighted proper time.

For a causal chain passing through locations $x_0, x_1, \ldots, x_n$, the proper time is:
$$ \tau[\gamma] = \sum_{k=0}^{n} \epsilon \, \alpha_B(x_k). $$

Using the lapse formula:
$$ \tau[\gamma] = \sum_{k=0}^{n} \frac{\epsilon}{1 + \lambda B(x_k)}. $$

Thus, the scalar lapse is the local factor appearing in the chain-sum definition of proper time.

In the continuum approximation:
$$ \tau[\gamma] \approx \int_\gamma \alpha_B(x) \, d\sigma = \int_\gamma \frac{d\sigma}{1 + \lambda B(x)}. $$

So the burden lapse is not an additional object unrelated to the previous subsection. It is the local field version of the same clock-suppression rule.

3.5.6 The Full Effective Metric Ansatz

Combining the temporal lapse with the spatial correlation metric from Section 2, one may write a first weak-field effective metric ansatz:

Definition 3.8 (Effective Burden Metric Ansatz).
$$ \boxed{ ds^2_{\mathrm{eff}} = -\alpha_B(x)^2 \, c_{\mathrm{eff}}^2 \, d\sigma^2 + h_{ij}(x) \, dx^i dx^j, } $$

where:
- $\alpha_B(x) = 1/(1 + \lambda B(x))$ is the burden lapse,
- $h_{ij}(x)$ is the emergent spatial metric from correlation geometry,
- $d\sigma$ is the causal-depth or extension-order parameter.

This is the first effective metric ansatz in the framework. It is not yet derived from a full field equation. It is a constitutive relation: burden density controls the temporal component, while correlation geometry controls the spatial component.

3.5.7 Guardrails for This Subsection

1. **The lapse is not primitive time.** $\alpha_B(x)$ is a conversion factor from relational order to proper time, not a background clock.
2. **Scalar lapse is not full metric.** The expression $ds^2_{\mathrm{time}} = -\alpha_B^2 c_{\mathrm{eff}}^2 d\sigma^2$ contains only the time-like scalar part. A full metric requires spatial components, shift terms, and curvature.
3. **The potential is not automatically Newtonian gravity.** $\Phi_C = c_{\mathrm{eff}}^2 \log \alpha_B$ is an effective temporal potential. Identification with Newtonian gravity requires weak-field matching, probe dynamics, and empirical calibration.
4. **Gradient equations are effective.** Expressions such as $\nabla \Phi_C$ and ∇B require continuum differentiability. The foundational statement is discrete and comparative.

3.6 Guardrails and Summary of Section 3

3.6.1 Guardrails for Section 3

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **Time is still emergent.** Even after defining burden-weighted proper time, time is not a primitive background parameter. It is reconstructed along causal chains and is path-dependent.
2. **Burden is not violation.** $B > 0 \not\Rightarrow \mathfrak{M} > 0$. A region may be exactly physical ($\mathfrak{M} = 0$) and still carry high structural burden. Burden measures the cost of maintaining constraint compatibility, not the failure of it.
3. **Clock suppression is not yet a full metric equation.** $\alpha(B) = 1/(1+\lambda B)$ defines a scalar clock factor. It provides the lapse component of a metric, but does not by itself define the full metric tensor $g_{\mu\nu}$ or its curvature.

4. **The burden-clock relation is a model choice.** The specific rational form $\alpha(B) = 1/(1+\lambda B)$ is a mathematically simple, monotonic ansatz. The theorem-safe content is that *any* positive, strictly decreasing clock factor yields burden-induced clock suppression.
5. **Non-uniform burden gives lapse-like temporal geometry, not full curvature.** Spatial metric response, shift terms, and tensorial stress belong to the gravitational layer (Section 5).
6. **The burden potential is not yet the Newtonian potential.** $\Phi_C = c_{\mathrm{eff}}^2 \log \alpha_B$ is an effective temporal potential. Identification with Newtonian gravity requires weak-field matching, probe dynamics, and empirical calibration.
7. **Redshift-like ratios require signal interpretation.** The ratio α_1/α_2 gives a clock-rate comparison. Calling this a "redshift" requires a rigorous derivation of signal propagation, frequencies, and observation.
8. **Continuum gradients are effective.** Expressions such as $\nabla B(x)$ and $\nabla \alpha(x)$ assume continuum differentiability. They are not available at the most primitive relational level, where the foundational statement is discrete and order-based.

3.6.2 Summary of Section 3

Section 3 reconstructed emergent temporal duration from the interplay of causal order and constraint burden:

1. **Constraint burden** $B(R)$ was rigorously defined from the primitive algebraic data as a scalar functional measuring the commutator cost of localization: $B(R) = \sum_\alpha w_\alpha \omega([\hat{C}_\alpha, \Pi_R]^\dagger [\hat{C}_\alpha, \Pi_R])$. It was distinguished absolutely from constraint violation ($\mathfrak{M}=0$). It was decomposed into three structurally distinct source channels: mode (ρ_{mode}), interaction (ρ_{int}), and relational (ρ_{rel}) burden.
2. The **local clock factor** $\alpha(B) = 1/(1 + \lambda B)$ was introduced. It was proven to satisfy $0 < \alpha \leq 1$ and to decrease monotonically with burden (Theorem 3.3).
3. **Burden-weighted proper time** was formally defined along causal chains: $\tau[\gamma] = \sum_{e \in \gamma} \epsilon \, \alpha(B(e))$.
4. The theorem **"higher burden reduces proper time for equal causal depth"** was proven (Theorem 3.5), establishing the first time-dilation-like result.
5. **Non-uniform burden** was shown to induce non-uniform clock accumulation, producing the first layer of **temporal geometry** (Theorem 3.6).
6. An **effective burden-clock potential** $\Phi_C = c_{\mathrm{eff}}^2 \log \alpha$ was derived, establishing that higher burden corresponds to a deeper temporal potential.
7. The **weak-burden limit** $\Phi_C \approx -c_{\mathrm{eff}}^2 \lambda B$ was established (Theorem 3.7), providing a linear bridge between burden and temporal potential.
8. The **scalar burden-to-lapse bridge** $d\tau = \alpha(x) d\sigma$ was formalised, yielding the first effective metric ansatz: $ds^2_{\mathrm{eff}} = -\alpha_B^2 c_{\mathrm{eff}}^2 d\sigma^2 + h_{ij} dx^i dx^j$.
9. The **Causal Reconciliation Principle** was stated: both the effective speed of light and the rate of time are bounded by the degree to which relational structures can reconcile with causal dependency.

The conceptual chain of this section is:

$$ \boxed{ \text{causal order} + \text{constraint burden} \quad \Longrightarrow \quad \text{burden-weighted proper time} \quad \Longrightarrow \quad \text{temporal geometry} \quad \Longrightarrow \quad \text{effective lapse}. } $$

This completes the reconstruction of emergent temporal structure. The framework now possesses an emergent 3D spatial metric (Section 2) and an emergent scalar temporal lapse (Section 3), bound together by a causal partial order (Section 1). This sets the stage for the emergence of matter, which will populate this geometric substrate.

Section 4 — Fields, Particles, and Interactions

4.0 Purpose of Fields, Particles, and Interactions

4.0.1 Transition from Geometry to Matter

Section 2 reconstructed emergent spatial structure from the correlation kernel K_ω, proving that correlation distance becomes a pseudometric under multiplicative consistency, and that emergent points arise as equivalence classes of perfectly correlated observables. Section 3 reconstructed emergent time from causal depth weighted by constraint burden, deriving the burden-clock suppression theorem and the effective temporal geometry.

The framework now possesses:
1. A primitive relational algebra $\mathcal{A}$ with constraint ideal $\mathcal{I}_C$.
2. A physical sector defined by $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$.
3. An emergent spatial metric space $(\mathcal{X}_\omega, \widetilde{d}_\omega)$.
4. An emergent temporal structure $\tau[\gamma] = \sum_{e \in \gamma} \epsilon / (1 + \lambda B(e))$.
5. An effective metric ansatz $ds^2 = -\alpha_B^2 c_{\mathrm{eff}}^2 d\sigma^2 + h_{ij} dx^i dx^j$.

However, the framework does not yet contain fields, particles, or interactions. The emergent geometry is a relational substrate, but it is not yet populated by matter-like excitations.

The purpose of this section is to reconstruct these structures from the relational constraint framework.

> **Fields and particles are not primitive occupants of spacetime; they are stable zero-preserving modes of relational structure.**

In ordinary field theory, one begins with a spacetime manifold and defines fields on it. In this framework, that order is reversed. Emergent space is first reconstructed from correlation geometry. Emergent time is reconstructed from causal order and burden. Only after these structures are available can one define field-like and particle-like behaviour.

4.0.2 What This Section Introduces

Section 4 introduces the following:

1. **Fields** — stable, zero-preserving patterns of values over the emergent correlation geometry.
2. **Field modes** — eigenmodes of the correlation Laplacian that satisfy the field-sector zero condition.
3. **Particles** — localised, stable, identifiable, zero-preserving field modes.
4. **Particle identity** — persistence of a localised mode through admissible extension.
5. **Interactions** — non-additive constraint burden or constraint-response effects among modes.
6. **Interaction vertices** — minimal events where multiple stable modes jointly preserve zero.

These structures are still pre-geometric in the sense that they do not assume a background spacetime manifold. They describe how matter-like structure can emerge from the relational constraint framework.

4.0.3 Why Fields Come After Space and Direction

A field normally means something like $\phi(x)$, where x is a spacetime point. But in this theory, x is not primitive. We only obtained emergent points at Layer 5 as correlation-equivalence classes:
$$ \mathcal{X}_\omega = \mathcal{A}_{\mathrm{loc}} / \sim_\omega. $$

So we cannot define fields before defining emergent points. Likewise, field variation requires directions. But direction is not primitive either. We only obtained emergent directions at the end of Section 2 as profile-difference equivalence classes.

Only now can we define:
- field value,
- field variation,
- field equations.

This ordering is important.

> **Not: space first, then fields; but: relational correlations first, emergent space second, field modes third.**

4.0.4 Fields as Relational Mode Assignments

Once an emergent point set $\mathcal{X}_\omega$ exists, one may define a field-like object as an assignment of relational data to emergent locations or regions.

If $\mathcal{X}_\omega$ is the emergent point set, a scalar-like field may be represented schematically as:
$$ \phi: \mathcal{X}_\omega \to V, $$
where V is a value space.

But this notation is effective. It should not be read as a primitive definition. The deeper interpretation is that ϕ summarises a family of relational modes supported across correlation-equivalence classes.

> **A field is a stable assignment of zero-preserving mode data over emergent correlation geometry.**

This allows field-like behaviour to arise without assuming fields as fundamental background objects.

4.0.5 Particles as Stable Localised Modes

Particles are not introduced as primitive point objects. They arise as special stable field configurations with localisation, persistence, and identity.

A particle is a field excitation that is:
1. localised in correlation geometry,
2. stable under admissible extension,
3. zero-preserving,
4. identifiable across causal progression.

> **A particle is a stable, localised, zero-preserving mode.**

This definition preserves the framework's hierarchy. The theory first defines physical admissibility, then zero-preserving observables, then causal dependency, then emergent space, then burden-weighted time, then field modes. Particle identity appears only after all of these structures are available.

4.0.6 Interactions as Non-Additive Burden

With particles defined, the next question is how they interact.

In this framework, interactions are not introduced as primitive forces. They arise when the joint constraint cost of multiple modes is not equal to the sum of their individual costs.

> **Interaction is non-additivity of constraint burden.**

If two modes ϕ and χ combine, and the burden of the combined configuration differs from the sum of separate burdens, then the modes interact.

This gives a general relational interpretation of interaction without importing a specific force law.

4.0.7 Structural Sequence of Section 4

The section proceeds in the following order:

1. **Fields** are defined as stable zero-preserving mode assignments over $\mathcal{X}_\omega$.
2. **Field modes** are constructed as eigenmodes of the correlation Laplacian.
3. **Particles** are defined as localised, stable, identifiable, zero-preserving field modes.
4. **Particle identity** is formalised through coherent tracking across causal antichains.
5. **Interactions** are defined as non-additive constraint burden.
6. **Interaction vertices** are introduced as multi-mode zero-closure events.

The conceptual sequence is:
$$ \boxed{ \text{correlation geometry} \quad \longrightarrow \quad \text{field modes} \quad \longrightarrow \quad \text{localised stable modes} \quad \longrightarrow \quad \text{particles} \quad \longrightarrow \quad \text{interactions}. } $$

4.1 Fields as Stable Zero-Preserving Modes

4.1.1 Purpose of This Subsection

The previous subsection established that matter must emerge as stable configurations over the emergent spatial geometry $\mathcal{X}_\omega$. To formalize this, we require an operator that measures how relational data varies across the correlation geometry.

In standard physics, fields are defined on a manifold and their dynamics are governed by differential operators like the Laplacian. Here, no manifold or differential calculus is assumed. Instead, we construct a **correlation Laplacian** directly from the emergent metric structure, which measures the deviation of a relational mode from its correlation-weighted neighborhood average.

4.1.2 The Emergent Spatial Arena

From Section 2, localisable zero-preserving observables $A, B \in \mathcal{A}_{\mathrm{loc}}$ were equipped with a state-dependent correlation kernel $K_\omega(A, B)$. The corresponding logarithmic distance was:
$$ d_\omega(A, B) = -\ell_0 \log K_\omega(A, B). $$

Under multiplicative consistency (Assumption 2.5), d_ω is a pseudometric. Quotienting by zero-distance equivalence gives an emergent point set:
$$ \mathcal{X}_\omega = \mathcal{A}_{\mathrm{loc}} / \sim_\omega. $$
The induced metric on equivalence classes is:
$$ D_\omega([A]_\omega, [B]_\omega) = d_\omega(A, B). $$

This metric space (or approximate metric space in the coarse-grained case) is the proto-spatial arena on which field modes may be described.

> **Field modes live on emergent correlation geometry, not on primitive background space.**

4.1.3 Local Mode Data

Let $x \in \mathcal{X}_\omega$ be an emergent point or coarse-grained region. A field mode assigns to x some local relational data.

In an effective notation, a scalar-like mode may be written as:
$$ \phi(x) \in V, $$
where V is a value space (e.g., $\mathbb{R}$ or $\mathbb{C}$).

But this notation is shorthand. The deeper meaning is that $\phi(x)$ represents mode data associated with the equivalence class of localisable observables corresponding to x. If $x = [A]_\omega$, then $\phi(x)$ should be understood as relational data supported by, or associated with, the class $[A]_\omega$.

> **$\phi([A]_\omega)$ is effective notation for mode data attached to a correlation-equivalence class.**

4.1.4 Definition — Field on Emergent Points

Definition 4.1 (Field).
Let $(\mathcal{X}_\omega, D_\omega)$ be an emergent correlation-geometric space, exact or approximate. Let V be a value space representing possible local mode data. A **field** is an assignment:
$$ \boxed{ \phi: \mathcal{X}_\omega \to V } $$
such that the associated relational operation or configuration is zero-preserving.

4.1.5 Relational Weights

To define field dynamics, we need an operator that measures spatial variation. We construct this using the relational metric D_ω.

Definition 4.2 (Relational Weights).
For emergent points $x_i, x_j \in \mathcal{X}_\omega$, define relational weights:
$$ \boxed{ W_{ij} = \exp\left(-\frac{D_\omega(x_i, x_j)}{\ell_0} \right). } $$

Because $D_\omega(x_i, x_j) = -\ell_0 \log K_\omega(i, j)$, this is equivalently:
$$ W_{ij} = K_\omega(i, j). $$
Thus, the relational weights are precisely the correlation kernel evaluated on the emergent points. Strongly correlated points have high weight; distant points have weight approaching zero.

4.1.6 Definition — Correlation Laplacian

Definition 4.3 (Correlation Laplacian).
The **correlation Laplacian** is the operator:
$$ \boxed{ (\Delta_\omega \phi)(x_i) = \sum_j W_{ij} \left[\phi(x_j) - \phi(x_i) \right]. } $$

This operator measures the deviation of $\phi(x_i)$ from its correlation-neighbourhood average. It generalises the usual graph Laplacian without presupposing a metric manifold or calculus.

For later dynamical equations, it is useful to define the positive correlation Laplacian:
$$ L_\omega = -\Delta_\omega. $$
If the geometry is discrete, L_ω is a positive semi-definite matrix. If the continuum limit is approached, it approximates the standard Laplace-Beltrami operator.

4.1.7 Field Modes as Eigenmodes

A natural class of modes satisfies the eigenvalue relation:
$$ \boxed{ \Delta_\omega \phi_n = \lambda_n \phi_n. } $$
Here λ_n is the relational spectral value of the mode. Equivalently, $L_\omega \phi_n = -\lambda_n \phi_n$.

When combined with emergent time τ (Section 3), field modes are expected to satisfy a relational wave equation of schematic form:
$$ \partial_\tau^2 \phi + c_{\mathrm{eff}}^2 L_\omega \phi + m_{\mathrm{eff}}^2 \phi = 0. $$
In this effective equation, the spectral gap (the lowest non-zero eigenvalue of L_ω) corresponds to an effective mass m_{eff} for the excitation.

Remark 4.1 (Status of the Field Equation). The wave equation above is presented as a *candidate equation* or structural target, not a derived theorem. A rigorous derivation linking the primitive constraint algebra $\mathcal{I}_C$ to this specific differential operator remains an open target (Target 2, Section 9.2).

4.1.8 The Zero-Preservation Condition

A field is physical only if it preserves the master zero constraint. In represented Hilbert-space language, if the mode is represented by an operator or family of operators $\hat{\phi}(f)$ (smeared over test data f), then physical admissibility requires:
$$ \boxed{ \hat{\phi}(f) \, \mathscr{H}_{\mathrm{phys}}^\omega \subseteq \mathscr{H}_{\mathrm{phys}}^\omega. } $$
Equivalently:
$$ \hat{\phi}(f) \, \ker \hat{\mathfrak{M}}_\omega \subseteq \ker \hat{\mathfrak{M}}_\omega. $$

In effective classical language, if the field mode deforms a physical configuration x into x_ϕ, then:
$$ \boxed{ \mathfrak{M}(x) = 0 \quad \Longrightarrow \quad \mathfrak{M}(x_\phi) = 0. } $$

Thus, field modes are physical only if they preserve constraint-zero admissibility.

> **Field admissibility first; field dynamics second.**

This preserves the logic of the framework: constraint compatibility is prior to dynamics.

4.1.9 Stable Field Modes

A field mode is **stable** if small perturbations remain bounded under emergent-time flow.

Definition 4.4 (Stable Field Mode).
A field mode ϕ_n is stable if there exists a finite constant $C > 0$ such that:
$$ \boxed{ |\phi_n(\tau)| \leq C |\phi_n(0)| } $$
over the relevant emergent-time interval.

Stability means the mode does not blow up under evolution. Unstable modes would violate the continuity of admissible extension (Section 1.5) and are therefore excluded from the persistent physical sector.

4.1.10 Guardrails for This Subsection

1. **Fields are not fundamental.** They are derived from the emergent correlation geometry $\mathcal{X}_\omega$.
2. **The correlation Laplacian is a toy operator.** It is the natural field operator on the emergent geometry, but it is not yet a derived continuum differential operator.
3. **The wave equation is a candidate.** $\Box_{\mathrm{RCF}} \phi = 0$ needs a rigorous continuum proof to match standard field theory.
4. **Field support is emergent.** Support statements refer to emergent regions $R \subseteq \mathcal{X}_\omega$, not subsets of a primitive spacetime manifold.

—
4.2 Particles as Localised Stable Field Modes

4.2.1 Purpose of This Subsection

The previous subsection defined field modes as stable zero-preserving patterns over the emergent correlation geometry $\mathcal{X}_\omega$. A field mode alone, however, is not yet a particle. To behave like a particle, a mode must satisfy additional conditions: it must be spatially localised, persist stably over time, and maintain its identity as it propagates across causal antichains.

The purpose of this subsection is to define rigorously when a field excitation becomes particle-like.

> **A particle is a stable, localised, identifiable, zero-preserving field mode.**

4.2.2 Why Particles Come After Fields

In standard physics, particles are often treated as primitive point objects moving through a background space. But in this framework:
- Space is emergent (Section 2).
- Time is emergent (Section 3).
- Fields are emergent (Section 4.1).
Therefore, particles cannot be fundamental.

The correct hierarchical order is:
$$ \text{Space} \quad \longrightarrow \quad \text{Fields} \quad \longrightarrow \quad \text{Stable Modes} \quad \longrightarrow \quad \text{Localised Stable Modes} \quad \longrightarrow \quad \text{Particles}. $$

A particle is not "a little object inside space." It is a persistent relational pattern that becomes object-like only after emergent space, time, and field configurations have been reconstructed.

4.2.3 Definition — Particle-Like Excitation

Definition 4.5 (Particle-Like Excitation).
Let $\delta\phi$ be a field excitation over a background physical configuration ϕ_0 on $\mathcal{X}_\omega$. A **particle-like excitation** is an excitation satisfying four structural conditions:

1. **Zero-Preservation:** The excitation preserves the physical sector. In Hilbert space language: $\delta_\phi |\Psi\rangle \in T_\Psi \ker \hat{\mathfrak{M}}_\omega$ (or $\mathfrak{M}[\phi_0 + \delta\phi] \approx 0$ in the effective classical limit).
2. **Localisation:** The excitation has finite spread in the correlation geometry.
3. **Stability:** The amplitude remains bounded over proper time (Definition 4.4).
4. **Identifiability:** The excitation can be coherently tracked across successive causal antichains.

4.2.4 The Localisation Functional

To formalise the localisation condition, we define a functional that measures the spatial spread of an excitation relative to a candidate centre.

Definition 4.6 (Localisation Functional).
Let $\Sigma \subset \mathcal{E}$ be a causal antichain (Definition 1.11), which serves as an emergent spatial cross-section. Let $\delta\phi_\Sigma$ be a field excitation defined on Σ. Let $x_0 \in \mathcal{X}_\omega$ be a candidate localisation centre. Define:
$$ \boxed{ L[\delta\phi_\Sigma; x_0] = \frac{\sum_{x \in \Sigma} D_\omega(x, x_0)^2 |\delta\phi_\Sigma(x)|^2}{\sum_{x \in \Sigma} |\delta\phi_\Sigma(x)|^2}. } $$
Here, $D_\omega(x, x_0)$ is the emergent correlation distance (Definition 2.5).

Definition 4.7 (Localised Excitation).
The excitation $\delta\phi_\Sigma$ is **localised** if there exists some $x_0 \in \mathcal{X}_\omega$ such that:
$$ \boxed{ L[\delta\phi_\Sigma; x_0] < \infty. } $$
Furthermore, it is *sharply* localised if this value is sufficiently small compared to the scale ℓ_0.

4.2.5 Particle Identity Across Antichains

Identifiability requires that the excitation can be tracked as a coherent object through emergent time. Since time is reconstructed via causal depth (Section 1.3), tracking an object through time means tracking its localisation centre across a sequence of causally compatible antichains.

Let $\Sigma_s, \Sigma_{s+1}, \ldots, \Sigma_{s+k}$ be a sequence of antichains ordered by the open extension principle (Section 1.5). A localised excitation has **persistent identity** if there exists a sequence of excitations $\delta\phi_{\Sigma_s}, \delta\phi_{\Sigma_{s+1}}, \ldots$ such that:
1. Each $\delta\phi_{\Sigma_t}$ is localised with centre $X_t \in \mathcal{X}_\omega$.
2. Each $\delta\phi_{\Sigma_t}$ is stable.
3. The centres X_t form a coherent track respecting the emergent causal speed limit c_{RCF} (Theorem 1.7). That is, the relational distance traversed per causal step does not exceed ℓ_0.
4. The internal mode class is preserved up to gauge equivalence: $\delta\phi_{\Sigma_{t+1}} \sim_{\mathrm{mode}} \delta\phi_{\Sigma_t}$.

4.2.6 Theorem — Particle Identity

Theorem 4.1 (Particle Identity).
Let $\delta\phi$ be a physical field mode on $(\mathcal{X}_\omega, D_\omega)$. Suppose:
1. $\delta\phi$ is zero-preserving (preserves $\ker \hat{\mathfrak{M}}_\omega$).
2. $\delta\phi$ is localised on each antichain Σ_t in the sequence: $L[\delta\phi_{\Sigma_t}; X_t] < \infty$.
3. The localisation centres X_t form a coherent finite-speed track.
4. The internal mode class is preserved: $\delta\phi_{\Sigma_{t+1}} \sim_{\mathrm{mode}} \delta\phi_{\Sigma_t}$.

Then $\delta\phi$ defines a particle-like excitation.

Proof.
Condition 1 ensures the excitation remains within the physical sector. Condition 2 implies persistent localisation in the emergent metric space. Condition 3 ensures that the trajectory of the centre respects the pre-geometric causal structure and does not violate the bounded relational step length (Assumption 1.2). Condition 4 ensures identifiability across emergent time, meaning the internal relational data is not destroyed or radically transmuted during extension. Together, these properties satisfy the structural requirements for a persistent, trackable, localised object. $\blacksquare$

4.2.7 Interpretation

This theorem states that particles are not primitive points. They are:
$$ \boxed{ \text{particle} = \text{trackable localised stable field mode}. } $$

A particle trajectory is not a continuous line drawn in a background space. It is a discrete sequence of localised mode continuations through emergent correlation geometry.

> **Particle identity is relational continuity, not primitive objecthood.**

4.2.8 Guardrails for This Subsection

1. **Particles are not fundamental.** They are derived from stable, localised modes of the emergent field.
2. **Localisation is emergent.** $L[\delta\phi_\Sigma; x_0] < \infty$ is defined using the emergent metric D_ω, not a background coordinate chart.
3. **Tracking is causal, not temporal.** Identity is tracked across causal antichains (Σ_t), not by parameterising a curve with an external time parameter t.
4. **Mode preservation is structural.** The equivalence $\sim_{\mathrm{mode}}$ means the spectral data of the excitation is preserved; it does not mean the excitation is mathematically static.

4.3 Interactions as Non-Additive Constraint Burden

4.3.1 Purpose of This Subsection

With particles defined as stable, localised, zero-preserving field modes, the next structural question is: how do they interact?

In standard physics, interactions are typically introduced as primitive forces mediated by exchange particles or imposed via coupling terms in a Lagrangian. The Relational Constraint Framework does not begin with a Lagrangian or fundamental force laws. Instead, it defines physicality as constraint compatibility.

Therefore, interactions must emerge from the constraint structure itself. When multiple modes coexist, they jointly stress the relational algebra. The framework defines interaction precisely as the failure of the joint constraint-maintenance burden to be additive.

> **Interaction is non-additivity of constraint burden.**

4.3.2 Definition — Interaction Functional

Let $B(R)$ be the constraint burden functional (Definition 3.1). For two field modes ϕ and χ localized in regions R_ϕ and R_χ, the **interaction functional** is defined as the non-additive part of the burden:
$$ \boxed{ \mathcal{I}[\phi, \chi] := B(R_{\phi+\chi}) - B(R_\phi) - B(R_\chi). } $$

4.3.3 Theorem — Interaction Non-Additivity

Theorem 4.2 (Interaction Non-Additivity).
$\mathcal{I}[\phi, \chi] = 0$ if and only if the joint constraint-maintenance burden is additive.
$\mathcal{I}[\phi, \chi] \neq 0$ if and only if the joint configuration carries additional or reduced mutual constraint cost.

Proof.

Substituting the definition of burden (Definition 3.1) into the interaction functional, we expand the commutator $[\hat{C}_\alpha, \Pi_{R_{\phi+\chi}}]$. Because the joint localization operator $\Pi_{R_{\phi+\chi}}$ contains cross-terms linking ϕ and χ, the expansion yields:
$$ \mathcal{I}[\phi, \chi] = \sum_\alpha w_\alpha \omega\left([\hat{C}_\alpha, \Pi_\phi]^\dagger [\hat{C}_\alpha, \Pi_\chi] \right) + \text{c.c.} $$
If $\mathcal{I} = 0$, the cross-commutators vanish in the physical state, meaning the modes do not structurally interfere in the constraint algebra. If $\mathcal{I} \neq 0$, the presence of ϕ alters the constraint commutator cost of χ, and vice versa. The modes interact. $\blacksquare$

4.3.4 Sign of Interaction

The sign of the interaction functional determines the qualitative nature of the relational pressure between the modes:

- **Repulsion / Incompatibility ($\mathcal{I} > 0$):** If $\mathcal{I}[\phi, \chi] > 0$, the joint configuration carries *additional* reconciliation burden. The relational algebra finds it harder to maintain zero-closure with both modes present. This creates a structural pressure to separate or decorrelate, analogous to repulsive forces.
- **Binding / Compatibility ($\mathcal{I} < 0$):** If $\mathcal{I}[\phi, \chi] < 0$, the joint configuration carries *reduced* burden. The modes mutually assist in satisfying the constraints, creating a structural pressure to cohere. This is analogous to binding energy or attractive forces.

> **Binding is burden reduction; repulsion is burden increase.**

4.3.5 Many-Body Generalisation

The two-body interaction functional naturally generalises to n modes. For a collection of modes $\phi_1, \ldots, \phi_n$, the total non-additive burden is:

Definition 4.8 (Many-Body Interaction).
$$ \boxed{ \mathcal{I}_n[\phi_1, \ldots, \phi_n] := B\left(R_{\sum \phi_a}\right) - \sum_{a=1}^n B(R_{\phi_a}). } $$

If $\mathcal{I}_n \neq 0$, the collection exhibits many-body interaction. Note that true n-body interactions (where $\mathcal{I}_n$ cannot be decomposed into a sum of lower-order $\mathcal{I}_2$ terms) emerge naturally if the underlying constraint algebra $\mathcal{I}_C$ is sufficiently non-linear.

4.3.6 Locality of Interactions

Interactions should be local relative to the emergent correlation geometry. The relational weights $W_{ij} = \exp(-D_\omega(x_i, x_j)/\ell_0)$ (Definition 4.2) provide the natural measure of proximity.

A local interaction cost for two modes localised around regions i and j may be written schematically as:

Definition 4.9 (Local Interaction Cost).
$$ \boxed{ \mathcal{M}_{\mathrm{int}}[\phi, \chi] = g^2 \sum_{i,j} W_{ij} |\phi_i|^2 |\chi_j|^2, } $$
where g is an effective coupling constant derived from the algebraic structure.

Because W_{ij} decays exponentially with correlation distance D_ω, this formalises the statement: interaction strength is controlled by proximity in the emergent geometry. Modes that are relationally distant interact weakly.

> **Interaction strength decays with relational distance.**

4.3.7 Interaction Vertices

An **interaction vertex** is a minimal event where multiple stable modes jointly preserve zero. It is the pre-geometric analogue of a scattering event or a Feynman vertex.

Abstractly, it is represented by a multi-mode compatibility condition. Let $\phi_1, \ldots, \phi_n$ be modes converging at an event $e \in \mathcal{E}$. The vertex is admissible if the combined constraint generated by their intersection vanishes:

Definition 4.10 (Interaction Vertex).
An interaction vertex is a primitive event $e \in \mathcal{E}$ satisfying a multi-mode zero-closure condition:
$$ \boxed{ C_{\mathrm{vertex}}(\phi_1, \phi_2, \ldots, \phi_n) = 0. } $$

An interaction event is physical precisely when the multi-mode compatibility constraint vanishes, ensuring that the master constraint $\hat{\mathfrak{M}}_\omega$ remains zero at that node.

> **An interaction vertex is a multi-mode zero-closure event.**

4.3.8 Guardrails for This Subsection

1. **Interaction is not automatically a fundamental force.** Non-additive burden is an interaction *source*, not yet a dynamical force equation or gauge boson exchange.
2. **Non-additivity must remain admissible.** Physical interaction means non-additivity *inside* the zero sector. The joint configuration must still satisfy $\mathfrak{M} = 0$.
3. **Negative burden interaction is not automatically attraction.** Binding-like behaviour (reduced burden) provides the *potential* for attraction, but actual motion requires a dynamical selection rule or least-burden extension principle.
4. **Positive burden interaction is not automatically repulsion.** Similarly, increased burden implies structural incompatibility, which manifests as repulsion only once dynamical evolution rules are specified.

5. **Interaction locality depends on emergent geometry.** If the correlation geometry is approximate or state-dependent, the locality of interactions is also approximate or state-dependent.

4.4 Guardrails and Summary of Section 4

4.4.1 Guardrails for Section 4

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **Fields are not fundamental.** They are derived mathematically from the emergent correlation geometry $(\mathcal{X}_\omega, D_\omega)$. They do not exist independently of a physical state ω.
2. **Particles are not primitive point objects.** They are stable, localised, trackable excitations of field modes. Their identity is a consequence of relational continuity, not inherent substance.
3. **The correlation Laplacian is a structural operator.** While it generalises the standard Laplacian, its continuum limit and exact correspondence to standard field theory operators remain a formal target rather than a proven theorem.
4. **Interactions are not introduced as external forces.** They arise purely from the non-additivity of constraint burden ($\mathcal{I}[\phi, \chi] \neq 0$). Standard force laws and gauge couplings must be derived from this underlying non-additivity, not postulated.
5. **Interaction vertices are zero-closure events.** They are not collision points in a background spacetime, but primitive events where multiple modes jointly satisfy the master constraint.
6. **No Standard Model yet.** The framework derives the *structure* of fields, particles, and interactions. Deriving the specific gauge groups, particle spectra, and coupling constants of the Standard Model remains an open target (Target 4, Section 9.2).

4.4.2 Summary of Section 4

Section 4 reconstructed the matter layer of the Relational Constraint Framework, populating the emergent geometry with fields, particles, and interactions:

1. **Fields** were defined as stable zero-preserving mode assignments $\phi: \mathcal{X}_\omega \to V$ over the emergent point set.
2. The **correlation Laplacian** Δ_ω was introduced using relational weights $W_{ij} = \exp(-D_\omega/\ell_0)$, providing the natural operator to measure field variation on the emergent geometry.
3. **Stable field modes** were defined as bounded eigenmodes of the correlation Laplacian, satisfying $\Delta_\omega \phi_n = \lambda_n \phi_n$.
4. **Particles** were rigorously defined as localised, stable, identifiable, zero-preserving field modes. A localisation functional $L[\delta\phi_\Sigma; x_0]$ was introduced to measure their spatial spread.

5. **Particle identity** was formalised (Theorem 4.1) by proving that persistent tracking across causal antichains Σ_t constitutes a relational trajectory, without assuming a background spacetime manifold.
6. **Interactions** were defined as the non-additivity of constraint burden: $\mathcal{I}[\phi, \chi] = B(R_{\phi+\chi}) - B(R_\phi) - B(R_\chi)$. Repulsion corresponds to positive burden (incompatibility), while binding corresponds to negative burden (mutual compatibility).
7. **Interaction vertices** were introduced as multi-mode zero-closure events, representing the pre-geometric analogue of scattering.

The conceptual chain of this section is:
$$ \boxed{ \text{correlation geometry} \to \text{field modes} \to \text{localised stable modes} \to \text{particles} \to \text{interactions}. } $$

This completes the reconstruction of matter. The framework now possesses an emergent $3+1$ geometry populated by interacting, stable modes. The next step is to examine how the macroscopic distribution of this matter—that is, the distribution of constraint burden—shapes the global geometry itself, leading to the emergence of gravity.

Section 5 — Gravity as Geometry of Constraint Burden

5.0 Purpose of the Burden Tensor and Gravity Source Structure

5.0.1 Transition from Matter to Gravitational Source

Section 4 reconstructed fields and particles as stable zero-preserving modes on the emergent correlation geometry. Interactions were defined as the non-additive burden generated when these modes compose.

However, the burden-clock bridge established in Section 3 only utilized a *scalar* burden density $\rho_B(x)$ to suppress the local clock rate (the lapse function α_B). While this provided the first temporal geometry and a mechanism for time dilation, a scalar field alone is insufficient to generate a full, dynamically consistent gravitational theory. General relativity requires a tensorial source—the stress-energy tensor $T_{\mu\nu}$—to source spatial curvature, frame-dragging, and anisotropic stresses.

The purpose of this section is to promote scalar constraint burden into a tensorial effective source, map it to an effective spacetime metric, and establish the conditions under which the framework recovers an Einstein-like field equation.

> **Gravity is not introduced as a fundamental force. It emerges when variations in constraint burden reshape the emergent time and space.**

5.0.2 The Causal Reconciliation Principle and Gravity

The underlying physical mechanism for this geometric response is the **Causal Reconciliation Principle** (Section 1.6). The rate of time and the speed of causal propagation are bounded by how quickly relational structures can reconcile with causal dependency.

When a region contains a high concentration of stable modes (matter) or complex interactions, the local reconciliation cost (burden) increases. This slows the local clock rate (temporal curvature) and alters the relational correlation distances (spatial curvature). Gravity, in this framework, is the macroscopic geometric response of the emergent spacetime to the microscopic distribution of constraint-reconciliation load.

5.0.3 What This Section Establishes

This section establishes the following:
1. The **component-selective burden tensor** $\Theta^{(B)}_{\mu\nu}$ is constructed from mode, interaction, and relational sources.
2. The **active-source conservation theorem** proves that the total active burden must be covariantly divergence-free.
3. The **burden-to-metric dictionary** maps the burden tensor components to an effective ADM-like metric.
4. The **Einstein-like closure theorem** proves that, under smoothness and Lorentz-compatibility assumptions, the minimal geometric response to a conserved burden source takes the form of Einstein's field equations.

5.1 Component-Selective Burden Sources

5.1.1 From Scalar Density to Tensorial Source

In Section 3, the burden density was treated as a scalar $\rho_B(x)$ that dictated the lapse function $\alpha_B(x)$. To source a full spacetime geometry, we must distinguish not only how much burden is present, but how it flows and how it exerts directional stress on the relational network.

In standard general relativity, the stress-energy tensor $T_{\mu\nu}$ encodes energy density, momentum density (flux), and anisotropic pressure (stress). The Relational Constraint Framework requires an analogous object derived entirely from constraint burden.

Definition 5.1 (Burden Tensor).
Let the coarse-grained emergent domain admit local effective coordinates $x^\mu = (\tau, x^1, \dots, x^d)$. The **burden tensor** is the effective source object:
$$ \boxed{ \Theta^{(B)}_{\mu\nu} = \begin{pmatrix} \rho_B & J^{(B)}_j \\ J^{(B)}_i & \Pi^{(B)}_{ij} \end{pmatrix}, } $$
where:
- $\Theta^{(B)}_{00} = \rho_B$ is the **burden density** (the cost of maintaining local zero-closure, as defined in Section 3.1),
- $\Theta^{(B)}_{0i} = \Theta^{(B)}_{i0} = J^{(B)}_i$ is the **burden flux** (the flow of reconciliation cost through emergent space),

- $\Theta^{(B)}_{ij} = \Pi^{(B)}_{ij}$ is the **relational burden stress** (the anisotropic directional pressure exerted by the constraint structure).

5.1.2 The Three Source Channels

The framework refines the burden tensor from a single homogeneous block into a source-origin decomposition. In Section 3.1 (Definition 3.2), the total burden density was decomposed as:
$$ \rho_B(x) = \rho_{\mathrm{mode}}(x) + \rho_{\mathrm{int}}(x) + \rho_{\mathrm{rel}}(x). $$

Consequently, the full burden tensor decomposes into three distinct structural channels:

Definition 5.2 (Component-Selective Burden Tensor).
$$ \boxed{ \Theta^{(B)}_{\mu\nu} = a_{\mathrm{mode}}\Theta^{(\mathrm{mode})}_{\mu\nu} + a_{\mathrm{int}}\Theta^{(\mathrm{int})}_{\mu\nu} + a_{\mathrm{rel}}\Theta^{(\mathrm{rel})}_{\mu\nu}, } $$
where $a_{\mathrm{mode}}, a_{\mathrm{int}}, a_{\mathrm{rel}} \in [0,1]$ are activation weights representing the coupling of each channel to the effective geometry.

Interpretation of the Channels:
1. **Mode burden ($\Theta^{(\mathrm{mode})}_{\mu\nu}$):** Generated by the existence and occupation of stable field or particle-like modes (Section 4.2). This is the structural analogue of ordinary visible matter and radiation.
2. **Interaction burden ($\Theta^{(\mathrm{int})}_{\mu\nu}$):** Arises from non-additivity in composition (Definition 4.8). A bound system, such as a star or a neutron star, carries interaction burden given by $B_{\mathrm{int}}(R_1, R_2) = B(R_1 \cup R_2) - B(R_1) - B(R_2)$. This corresponds to binding energy.
3. **Relational burden ($\Theta^{(\mathrm{rel})}_{\mu\nu}$):** Generated by the correlation network itself (Section 2). It measures the burden required to maintain relational consistency across configurations. Unlike the first two, it does not belong to a single localised object; it belongs to the network topology.

> **RCF burden sourcing is component-selective: mode, interaction, and relational burden may contribute separately or together, while the total active source must remain conserved.**

5.1.3 Symmetry of the Burden Tensor

For the burden tensor to act as a valid source for an effective geometric field equation (which will be derived from a symmetric metric tensor $g_{\mu\nu}$), it must be symmetric.

Theorem 5.1 (Symmetry of the Burden Tensor).
If the relational burden stress is symmetric, $\Pi^{(B)}_{ij} = \Pi^{(B)}_{ji}$, and the burden flux is represented symmetrically, $\Theta^{(B)}_{0i} = \Theta^{(B)}_{i0} = J^{(B)}_i$, then the burden tensor is symmetric:
$$ \boxed{ \Theta^{(B)}_{\mu\nu} = \Theta^{(B)}_{\nu\mu}. } $$

Proof.

Trivial from the definition: $\Theta^{(B)}_{00} = \rho_B$ is a scalar and trivially symmetric; $\Theta^{(B)}_{0i} = \Theta^{(B)}_{i0}$ by definition of the flux components; and $\Theta^{(B)}_{ij} = \Pi^{(B)}_{ij} = \Pi^{(B)}_{ji} = \Theta^{(B)}_{ji}$ by the assumption of symmetric relational stress. $\blacksquare$

Remark 5.1. The assumption of symmetric stress ($\Pi^{(B)}_{ij} = \Pi^{(B)}_{ji}$) is the structural equivalent of angular momentum conservation in standard field theory. It implies that the constraint algebra does not exert intrinsic microscopic torques on the emergent geometry.

5.1.4 Guardrails for This Subsection

1. **The burden tensor is not the stress-energy tensor.** $\Theta^{(B)}_{\mu\nu}$ is a relational cost measure. While it structurally mimics $T_{\mu\nu}$, its physical origin is the maintenance of admissibility, not the conservation of Noether charges.
2. **Components are algebraically derived.** The flux J_i and stress Π_{ij} are not postulated; they are defined by the spatial gradients and directional variations of the reconciliation cost in the emergent geometry.
3. **Activation weights are model-dependent.** The weights $a_{\mathrm{mode}}, a_{\mathrm{int}}, a_{\mathrm{rel}}$ must be specified in a concrete model or derived from the spectral properties of the correlation Laplacian.

5.2 Active-Source Conservation

5.2.1 Purpose of Conservation

For the burden tensor $\Theta^{(B)}_{\mu\nu}$ to act as a valid source for an effective geometric field equation, it must satisfy a conservation law. In standard General Relativity, the Bianchi identities force the Einstein tensor to be divergence-free, which in turn forces the stress-energy tensor to be divergence-free ($\nabla^\mu T_{\mu\nu} = 0$). This expresses local energy-momentum conservation.

The Relational Constraint Framework must satisfy an analogous condition. If geometry is the macroscopic response to constraint burden, then the flow and distribution of that burden cannot be arbitrary. The total active burden source must be covariantly conserved.

> **Zero-preserving geometry structurally demands zero-preserving burden flow.**

5.2.2 Theorem — Active-Source Conservation

Theorem 5.2 (Active-Source Conservation).
Let the total active burden source be:
$$ \Theta^{(B)}_{\mu\nu} = a_{\mathrm{mode}}\Theta^{(\mathrm{mode})}_{\mu\nu} + a_{\mathrm{int}}\Theta^{(\mathrm{int})}_{\mu\nu} + a_{\mathrm{rel}}\Theta^{(\mathrm{rel})}_{\mu\nu}. $$

If the emergent geometry is smooth and Lorentz-compatible, and if the geometric side of the field equation is divergence-free by the Bianchi identity, then the total active source must satisfy:
$$ \boxed{ \nabla_B^\mu \Theta^{(B)}_{\mu\nu} = 0. } $$

Proof.
Assume the effective metric g_B (to be formally defined in Section 5.3) satisfies a candidate geometric equation of the form:
$$ G_{\mu\nu}[g_B] + \Lambda_B g^{(B)}_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}, $$
where $G_{\mu\nu}$ is the Einstein tensor, Λ_B is an effective cosmological constant, and κ_B is an effective coupling constant.

By the contracted Bianchi identities of standard differential geometry, the Einstein tensor satisfies:
$$ \nabla_B^\mu G_{\mu\nu} = 0. $$
By metric compatibility ($\nabla_B g = 0$), we also have:
$$ \nabla_B^\mu g^{(B)}_{\mu\nu} = 0. $$

Taking the covariant divergence of the candidate field equation yields:
$$ \nabla_B^\mu \left(G_{\mu\nu} + \Lambda_B g^{(B)}_{\mu\nu} \right) = \nabla_B^\mu \left(\kappa_B \Theta^{(B)}_{\mu\nu} \right). $$

Since the left-hand side vanishes identically by the Bianchi identity, and κ_B is a constant, we obtain:
$$ 0 = \kappa_B \nabla_B^\mu \Theta^{(B)}_{\mu\nu}. $$

Assuming $\kappa_B \neq 0$, it follows strictly that:
$$ \nabla_B^\mu \Theta^{(B)}_{\mu\nu} = 0. \qquad \blacksquare $$

5.2.3 Component Exchange

The conservation law applies to the *total* active source. It does not require each individual component channel (mode, interaction, relational) to be independently conserved.

The framework explicitly allows for exchange between channels:
$$ \nabla_B^\mu \Theta^{(\mathrm{int})}_{\mu\nu} = - \nabla_B^\mu \Theta^{(\mathrm{mode})}_{\mu\nu} - \nabla_B^\mu \Theta^{(\mathrm{rel})}_{\mu\nu}. $$

This means interaction burden can be converted into mode burden (e.g., binding energy being released as particles), or absorbed by relational network adjustments, provided the total sum closes.

> **The total active burden is conserved, but its components may exchange.**

This mirrors the physics of standard thermodynamics and field theory, where energy can change form (e.g., rest mass to kinetic energy to binding energy) while the total stress-energy remains conserved.

5.2.4 Guardrails for This Subsection

1. **Conservation is conditional on the continuum limit.** The covariant divergence ∇_B^μ requires a smooth differentiable manifold. At the microscopic relational level, conservation is expressed as discrete zero-preservation across causal links, not as a differential equation.
2. **The theorem works backwards.** This theorem shows that *if* a geometric field equation holds, the source *must* be conserved. It does not yet derive the field equation itself; that is done in Section 5.4.
3. **Noether's theorem is not invoked.** The conservation here is not derived from spacetime translation symmetry. It is derived from the structural requirement that the geometric response (Einstein tensor) must have zero divergence, which traces back to the admissibility of the relational algebra.

5.3 Burden Tensor to Effective ADM Metric Mapping

5.3.1 The ADM Dictionary

To explicitly connect the burden tensor $\Theta^{(B)}_{\mu\nu}$ to spacetime geometry, we utilize the Arnowitt-Deser-Misner (ADM) decomposition. The ADM formalism foliates spacetime into spatial slices and uses the spatial metric, lapse, and shift to encode the full 4D geometry.

In the Relational Constraint Framework, ADM is not a primitive starting point. It is a continuum bookkeeping layer applied *after* the emergent geometry has stabilized. It provides the natural language to connect our independently derived spatial metric (Section 2) and temporal lapse (Section 3) into a single effective line element.

Definition 5.3 (Burden-Metric Dictionary).
Let $\rho_B, J_B^i,$ and Π_B^{ij} be the projections of the burden tensor $\Theta^{(B)}_{\mu\nu}$. Define the burden lapse, shift, and spatial metric as:
1. **Density to Lapse:** $\rho_B \to N_B$, where $N_B = \frac{1}{1 + \lambda \rho_B}$.
2. **Flux to Shift:** $J_B^i \to N_B^i$, where $N_B^i = \sigma J_B^i$.
3. **Stress to Spatial Metric:** $\Pi_B^{ij} \to h^{(B)}_{ij}$, where $h^{(B)}_{ij} = h^{(0)}_{ij} + \eta \Pi^{(B)}_{ij} + \zeta \rho_B h^{(0)}_{ij}$.

Here, σ, η, ζ are model-dependent effective coupling coefficients, and $h^{(0)}_{ij}$ is the baseline correlation metric from Section 2 (Definition 2.5).

5.3.2 Theorem — Effective Metric Ansatz

Theorem 5.3 (Effective ADM Metric).
Given the burden-metric dictionary (Definition 5.3), the effective ADM-like line element is:
$$ \boxed{ ds_B^2 = -N_B^2 d\tau^2 + h^{(B)}_{ij} \left(dx^i + N_B^i d\tau \right) \left(dx^j + N_B^j d\tau \right). } $$

Proof.
The standard ADM line element is given by $ds^2 = -N^2 d\tau^2 + h_{ij}(dx^i + N^i d\tau)(dx^j + N^j d\tau)$. By substituting the burden definitions N_B, $N_B^i,$ and $h^{(B)}_{ij}$ for the generic ADM variables N, $N^i,$ and h_{ij}, we obtain the effective burden metric. $\blacksquare$

5.3.3 Theorem — Metric Boundedness at Saturation

A crucial feature of this mapping is that the emergent geometry possesses a strict lower bound, preventing the literal infinite collapse of spatial volumes.

Theorem 5.4 (Metric Boundedness at Saturation).
Let $h^{(B)}_{ij}$ be the effective spatial metric derived from the burden stress $\Pi^{(B)}_{ij}$. Under pressure saturation ($\rho_B \to \rho_B^{\max}$, to be formally derived in Section 6.1), the emergent geometry imposes a strict lower bound on the metric determinant:
$$ \boxed{ \det(h^{(B)}_{ij}) \geq \ell_0^{2d} > 0, } $$
where d is the spatial dimension and ℓ_0 is the fundamental correlation length.

Proof Sketch.
By Assumption 6.1 (Minimum Relational Size), the relational size L_R of any physically realizable region is bounded below by ℓ_0. Since $h^{(B)}_{ij}$ is derived from correlation distances D_ω, and $D_\omega(x,y) \geq \ell_0$ for distinct emergent points, the eigenvalues of the spatial metric cannot vanish. Therefore, the volumetric measure $\sqrt{\det(h)}$ cannot collapse to zero, preventing a literal point-singularity. The excess burden must instead manifest as a divergence in the radial metric component h_{rr} and a vanishing lapse α_B. $\blacksquare$

5.3.4 Component Mapping Summary

The ADM dictionary establishes a rigorous structural parallel between the relational constraint framework and standard spacetime geometry:

RCF Burden Object	Effective Metric Role	Interpretation
ρ_B	$N_B \to g_{00}$	Clock suppression / temporal curvature
J_B^i	$N_B^i \to g_{0i}$	Burden flow / frame-dragging shift
Π_B^{ij}	$h^{(B)}_{ij} \to g_{ij}$	Anisotropic relational stress / spatial curvature

This dictionary establishes that scalar burden density controls the rate of time (lapse), burden flux controls spatial displacement between slices (shift), and burden stress controls the spatial metric deformation (curvature).

5.3.5 Guardrails for This Subsection

1. **The ADM mapping is an ansatz.** The constitutive relation mapping burden to lapse/shift/metric is a model-dependent dictionary. The specific linear and rational forms chosen here are the simplest ansätze that recover the correct limits.

2. **The baseline metric $h^{(0)}_{ij}$ is emergent.** The spatial metric is not fundamentally flat Minkowski space; it is the emergent correlation metric defined by the state ω.
3. **Coupling coefficients require derivation.** The constants σ, η, ζ must ultimately be derived from the microscopic spectral properties of the constraint algebra, rather than assumed.
4. **Singularities are projection artifacts.** Theorem 5.4 guarantees that volume does not vanish to infinity. The "singularity" of standard GR is replaced by bounded density plus divergent geometry.

5.4 Einstein-Like Burden Geometry Consistency

5.4.1 The Closure Target

With a symmetric, conserved burden tensor $\Theta^{(B)}_{\mu\nu}$ (Theorems 5.1 and 5.2) and a smooth, Lorentz-compatible effective metric g_B (Theorem 5.3), the framework is positioned to state its gravitational field equation.

In standard general relativity, Einstein's field equations are postulated based on geometric intuition and conservation requirements. In the Relational Constraint Framework, we do not postulate the equation. Instead, we ask: what is the minimal, most general geometric response to a conserved burden source in a smooth 4D Lorentzian geometry?

5.4.2 Theorem — Einstein-Like Closure

Theorem 5.5 (Einstein-Like Closure, Conditional Uniqueness Form).
Assume the emergent geometry is smooth, 4-dimensional, Lorentz-compatible, and governed by a symmetric rank-2 tensor equation sourced by the conserved burden tensor $\Theta^{(B)}_{\mu\nu}$. Then the minimal geometric field equation must take the form:
$$ \boxed{ G_{\mu\nu}[g_B] + \Lambda_B g^{(B)}_{\mu\nu} = \kappa_B \, \Theta^{(B)}_{\mu\nu}, } $$
where $G_{\mu\nu}$ is the Einstein tensor, Λ_B is an effective cosmological constant, and κ_B is an effective coupling constant.

Proof Sketch.
The proof relies on Lovelock's theorem from differential geometry. Lovelock's theorem states that in a four-dimensional smooth manifold, the only natural, symmetric, divergence-free rank-2 tensors that depend solely on the metric and its first two derivatives are linear combinations of the Einstein tensor $G_{\mu\nu}$ and the metric tensor $g_{\mu\nu}$ itself.

Given our assumptions:
1. The geometric response must be a symmetric rank-2 tensor (to match $\Theta^{(B)}_{\mu\nu}$).
2. It must be divergence-free (to match $\nabla_B^\mu \Theta^{(B)}_{\mu\nu} = 0$).
3. It must depend only on the metric and its derivatives (natural tensor).
4. We assume a second-order theory (no higher-than-second derivatives of the metric, ensuring standard causality and well-posedness).

By Lovelock's theorem, the left-hand side of the field equation must therefore be of the form $G_{\mu\nu} + \Lambda_B g_{\mu\nu}$. The conservation of the burden tensor ($\nabla_B^\mu \Theta^{(B)}_{\mu\nu} = 0$) is exactly the condition required for consistency with the geometric identity $\nabla_B^\mu (G_{\mu\nu} + \Lambda_B g_{\mu\nu}) = 0$. $\blacksquare$

5.4.3 Status of the Equation

It is critical to maintain theorem-safe status language regarding this result.

1. **The equation is a conditional reduction theorem.** It states that *if* the framework recovers a smooth, second-order metric theory in the continuum limit, that theory must be Einstein-like. Lovelock's theorem mathematically forces the form of the equation.
2. **It is not yet a dynamical derivation from the master constraint.** The full derivation of κ_B (the effective gravitational constant) and the exact mapping of the algebraic master constraint $\hat{\mathfrak{M}}_\omega = 0$ to the differential Einstein tensor remains an open target.
3. **Lovelock compatibility protects the left-hand side.** The framework does not need to arbitrarily postulate the Einstein tensor; it is forced by the uniqueness theorem under the stated assumptions.

> **Gravity is the equation of state of zero-reconciliation.**

The emergence of the Einstein tensor is not an input, but a mathematical necessity of coupling a conserved source to a smooth 4D geometry. The distinct contribution of RCF is identifying that the source $\Theta^{(B)}_{\mu\nu}$ is fundamentally constraint burden, not primitive stress-energy.

5.4.4 Guardrails for This Subsection

1. **Gravity is not a fundamental force.** It is the macroscopic geometric response to constraint burden.
2. **The burden tensor is not automatically stress-energy.** $\Theta^{(B)}_{\mu\nu}$ is stress-energy-like, but identification with physical $T_{\mu\nu}$ requires proof that matter excitations generate proportional constraint burden (Target 4 in Section 9).
3. **The ADM mapping is an ansatz.** The constitutive relation mapping burden to lapse/shift/metric is a model-dependent dictionary, not a primitive law.
4. **Einstein-like closure is an open target.** The equation $G_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$ is a conditional reduction theorem, not a completed derivation from the algebra.
5. **Component activation is model-dependent.** The weights $a_{\mathrm{mode}}, a_{\mathrm{int}}, a_{\mathrm{rel}}$ must be derived or specified in a concrete model.

5.5 Guardrails and Summary of Section 5

5.5.1 Guardrails for Section 5

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **Gravity is not a fundamental force.** It is the macroscopic geometric response of emergent spacetime to the distribution of constraint burden.
2. **The burden tensor is not automatically the stress-energy tensor.** $\Theta^{(B)}_{\mu\nu}$ is a relational cost measure. While it structurally mimics $T_{\mu\nu}$ and sources geometry in the same way, a rigorous theorem proving that standard matter excitations generate proportional constraint burden remains an open target.
3. **The ADM mapping is an ansatz.** The constitutive relation mapping burden density to lapse, flux to shift, and stress to spatial metric is a model-dependent dictionary. It is the simplest mapping that recovers the correct limits, but it is not a primitive law of the algebra.
4. **Einstein-like closure is a conditional reduction theorem.** The equation $G_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$ is forced by Lovelock's theorem *if* a smooth 4D metric theory is recovered. The dynamical derivation of this equation directly from the algebraic master constraint $\hat{\mathfrak{M}}_\omega = 0$ remains an open target.
5. **Component activation is model-dependent.** The weights $a_{\mathrm{mode}}, a_{\mathrm{int}}, a_{\mathrm{rel}}$ dictating how much each burden channel sources geometry must be specified in a concrete model or derived from the spectral properties of the correlation Laplacian.
6. **Singularities are structurally avoided.** By Theorem 5.4, the fundamental scale ℓ_0 prevents the metric determinant from vanishing, meaning geometric infinities (like $r=0$ in Schwarzschild) are projection artifacts of the continuum limit, not physical points of infinite density.

5.5.2 Summary of Section 5

Section 5 established the gravitational layer of the Relational Constraint Framework, bridging the macroscopic geometry to microscopic constraint reconciliation costs:

1. The **burden tensor** $\Theta^{(B)}_{\mu\nu}$ was defined, promoting the scalar burden density into a tensorial object with density, flux, and stress components. It was decomposed into three distinct structural channels: mode, interaction, and relational burden.
2. The **active-source conservation theorem** was proven, showing that the total active burden must be covariantly divergence-free ($\nabla_B^\mu \Theta^{(B)}_{\mu\nu} = 0$) as a structural consequence of the geometric Bianchi identities.
3. The **burden-to-metric dictionary** was constructed using the ADM formalism, mapping burden density to the lapse function, burden flux to the shift vector, and burden stress to the spatial metric.
4. The **metric boundedness theorem** (Theorem 5.4) was proven, showing that the ℓ_0-floor prevents the volumetric collapse of space, structurally avoiding physical singularities.
5. The **Einstein-like closure theorem** was proven: under the assumptions of a smooth, 4-dimensional, Lorentz-compatible continuum limit, Lovelock's theorem forces the minimal geometric response to the conserved burden tensor to take the form of Einstein's field equations, $G_{\mu\nu} + \Lambda_B g_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$.

The conceptual chain of this section is:
$$ \boxed{ \text{constraint burden} \quad \longrightarrow \quad \text{burden tensor} \quad \longrightarrow \quad \text{effective ADM metric} \quad \longrightarrow \quad \text{Einstein-like closure}. } $$

This completes the reconstruction of gravitational geometry. The framework now possesses an emergent $3+1$ spacetime governed by an Einstein-like equation, where the source of curvature is the relational cost of maintaining zero-closure. The next step is to examine the extreme limit of this cost: the black-hole-like regimes where reconciliation saturates and geometry projects infinity onto a boundary.

Section 6 — Black Holes as Unreconciled Relational Sectors

6.0 Purpose of Black-Hole-Like Domains in RCF

6.0.1 Reframing the Black Hole

In standard general relativity, a black hole is defined as a region of spacetime from which nothing, not even light, can escape to infinity. It is bounded by an event horizon, and its interior typically contains a curvature singularity where the continuum description of spacetime breaks down.

The Relational Constraint Framework does not begin with a pre-existing spacetime manifold, nor does it assume a background metric. Therefore, it cannot define a black hole primitive as a causal boundary in a pre-given geometry, nor can it accept a literal infinite-density singularity as a physical object, since points and metrics are emergent.

Instead, the framework must reconstruct black-hole-like behaviour from the fundamental dynamics of the constraint algebra, burden, and causal dependency.

The purpose of this section is to demonstrate that extreme concentrations of constraint burden naturally produce domains where relational structures cannot reconcile with causal dependency. In these domains, causal propagation is suppressed, local time flattens, and exterior accessibility collapses.

> **A black hole is not a hole in spacetime, nor a literal singularity. It is a strongly burden-loaded, unreconciled relational sector where exterior reconstruction fails.**

6.0.2 The Causal Reconciliation Principle in the Extreme Limit

In Section 1.6 and Section 3.0, we established the **Causal Reconciliation Principle**: the effective speed of causal propagation and the rate of proper time are bounded by the degree to which relational structures can reconcile with causal dependency.

When a region contains a high concentration of stable modes (matter) or complex interactions, the local reconciliation cost (burden) increases. This slows the local clock rate (temporal curvature) and alters the relational correlation distances (spatial curvature).

A black-hole-like domain represents the extreme limit of this process. It is a sector where the local burden $B_{\mathrm{active}}(R)$ becomes so large that the reconciliation rate approaches zero. The relational structure simply cannot process constraint updates fast enough to keep pace with the external universe.

6.0.3 What This Section Establishes

This section establishes the following:

1. **The ℓ_0-Floor and Pressure Saturation:** The fundamental scale ℓ_0 imposes a natural ceiling on burden density, preventing literal infinite collapse.
2. **Gravity Basins vs. Black Holes:** Excess burden alone creates a gravity basin; compactness-driven pressure saturation creates a black hole.
3. **The Unreconciled Region Principle:** A black-hole-like region is defined by the failure of relational structures to reconcile with causal dependency, leading to extreme flux suppression.
4. **Local Dimensional Suppression:** A local version of Theorem 2.10 proves that the radial inference channel collapses at the horizon, reducing exterior-accessible dimension from 3 to 2.
5. **Boundary Recovery:** If the boundary record map is injective, the lowest-burden admissible sector of the interior is recoverable.
6. **Entropy-Area Scaling:** The entropy of the region is a coarse-grained count of admissible boundary records, scaling with effective boundary area.
7. **Lowest-Burden Emission:** Relaxation occurs by emitting the simplest admissible carrier sector, not the original infallen matter class.
8. **Hawking-Like Appearance:** Under coarse-grained observation, lowest-burden emission appears approximately thermal.

6.0.4 Guardrails for This Section

1. **A black hole is not a literal hole.** It is a maximum-constrained, unreconciled relational sector.
2. **A horizon is not a material surface.** It is a boundary of exterior accessibility failure.
3. **A singularity is not a primitive infinite-density point.** It is a projection divergence at maximum internal complexity. The framework does not admit literal infinities in physical configurations.
4. **$\alpha_R \to 0$ is asymptotic suppression, not exact zero.** For any finite burden $B < \infty$, the clock factor $\alpha_B > 0$. The clock slows asymptotically, but does not literally stop.

6.1 The ℓ_0-Floor and Pressure Saturation

6.1.1 Purpose of This Subsection

Before defining black-hole-like domains, we must establish the physical mechanism that prevents literal infinite collapse. In standard general relativity, a black hole interior collapses to a point of infinite density because there is no mechanical force capable of resisting gravity after the horizon forms. The Relational Constraint Framework rejects this: the primitive algebra $\mathcal{A}$ possesses a fundamental non-zero scale, and this scale imposes a structural ceiling on compression.

This subsection defines the burden density ρ_B as a quantity distinct from the compactness ratio $\mathcal{S}_R$, proves that the ℓ_0-floor imposes a maximum density, and formalizes the pressure saturation mechanism.

6.1.2 Definition — Burden Density

In Section 3.1, constraint burden $B(R)$ was defined as the total relational reconciliation cost for a region R. To discuss collapse, we need the burden per unit volume.

Definition 6.1 (Burden Density).
Let R be a coarse-grained region with total active burden $B_{\mathrm{active}}(R)$ and emergent relational size L_R (defined via the correlation metric D_ω). The **burden density** is:
$$ \boxed{ \rho_B(R) = \frac{B_{\mathrm{active}}(R)}{L_R^3}. } $$

Remark 6.1 (Distinction from compactness). The burden density ρ_B must not be confused with the burden compactness ratio $\mathcal{S}_R = \gamma B(R) / L_R$ (to be formally defined in Section 6.2). The compactness ratio $\mathcal{S}_R$ has dimensions of $[\text{length}]^{-1}$ (analogous to M/R in GR), while ρ_B has dimensions of $[\text{length}]^{-3}$ (analogous to energy density). They capture different physics: ρ_B measures the local compression load, while $\mathcal{S}_R$ measures the global compactness relative to the horizon threshold.

6.1.3 The ℓ_0-Floor

The correlation distance was defined (Definition 2.4) as:
$$ d_\omega(A, B) = -\ell_0 \log K_\omega(A, B), $$
where $\ell_0 > 0$ is a fundamental length scale. This scale is not a postulate added for black-hole physics; it is already committed to by the framework's correlation geometry.

Assumption 6.1 (Minimum Relational Size).
The emergent relational size L_R of any physically realizable region R is bounded below by the fundamental scale:
$$ \boxed{ L_R \geq \ell_0 > 0. } $$

Interpretation. The correlation geometry cannot resolve distances below ℓ_0. A region cannot be compressed to a size smaller than the fundamental pixel of the emergent metric. This is the pre-geometric analogue of a minimum volume or exclusion principle.

6.1.4 Lemma — Maximum Burden Density

Lemma 6.1 (Maximum Burden Density).
For any region R with finite total burden $B_{\mathrm{active}}(R) < \infty$, the burden density is bounded above:
$$ \boxed{ \rho_B(R) \leq \rho_B^{\max} := \frac{B_{\mathrm{active}}(R)}{\ell_0^3}. } $$

Proof.
By Assumption 6.1, $L_R \geq \ell_0$. Therefore:
$$ \rho_B(R) = \frac{B_{\mathrm{active}}(R)}{L_R^3} \leq
\frac{B_{\mathrm{active}}(R)}{\ell_0^3} = \rho_B^{\max}. $$
Equality holds when $L_R = \ell_0$, i.e., when the region has been compressed to the minimum relational size. $\blacksquare$

Interpretation. The framework does not need to import a "maximum pressure" axiom from outside. The ceiling on ρ_B falls out automatically from a scale (ℓ_0) the framework has already committed to. Once L_R saturates at ℓ_0 and ρ_B hits its ceiling, further compaction cannot produce a bigger density number — the algebra will not generate one.

6.1.5 Pressure Saturation

In standard thermodynamics, the ideal gas law $PV = nRT$ dictates that as volume decreases, pressure rises. In general relativity, the Tolman-Oppenheimer-Volkoff limit reveals a deeper fact: pressure itself sources gravity (it appears in $T_{\mu\nu}$), so past a certain compactness, increasing pressure to resist collapse actually accelerates the collapse. No equation of state, however stiff, can hold up a sufficiently compact star.

The Relational Constraint Framework provides a structural analogue. In RCF:

- **Volume** corresponds to L_R^3 (the emergent spatial measure).
- **Pressure** corresponds to the burden density $\rho_B(R) = B(R)/L_R^3$ (the relational stress resisting compression).
- **Temperature** corresponds to the burden lapse gradient $\nabla \alpha_B$ near the boundary (how rapidly the clock rate changes).

As burden accumulates in a region, the volume contracts ($L_R \downarrow$) and the burden density rises ($\rho_B \uparrow$). This is ordinary gravitational compression. However, when L_R reaches ℓ_0, the volume cannot contract further. The burden density hits $\rho_B^{\max}$.

Definition 6.2 (Pressure Saturation).
A region R has reached **pressure saturation** when:
$$ \boxed{ L_R = \ell_0, \qquad \rho_B(R) = \rho_B^{\max}. } $$

At pressure saturation, the system has exhausted its capacity to store further burden as ordinary 3D volumetric compression. The pressure has not been "suppressed" (reduced to zero); it has saturated (reached its absolute maximum). What is suppressed is the **compressibility**: the ability of the volume to shrink further.

> **Pressure does not fail at the horizon. It saturates. The volume channel hits a ceiling, and the unresolved excess must project into geometry.**

6.1.6 Proposition — Geometric Projection of Saturated Burden

Proposition 6.1 (Finite Algebra, Unbounded Geometry).
Let R be a region at pressure saturation ($L_R = \ell_0$, $\rho_B = \rho_B^{\max}$). If additional burden is injected into R (so $B_{\mathrm{active}}(R)$ increases), the burden density ρ_B cannot increase further. Instead, the excess burden manifests as geometric divergence: the burden lapse α_R continues to fall, and the effective ADM curvature (Section 5.3) continues to climb, while ρ_B remains bounded.

Proof Sketch.
By Lemma 6.1, ρ_B is capped at $\rho_B^{\max}$. When additional burden B_{in} is injected, the algebra cannot represent it as increased density (since L_R is already at ℓ_0). The only remaining pipeline is the burden-to-metric dictionary (Definition 5.3):

1. **Lapse continues to fall:** $\alpha_B = 1/(1 + \lambda B(R)) \to 0$ as $B(R) \to \infty$, even though ρ_B is bounded.
2. **Radial metric diverges:** The ADM spatial metric $h^{(B)}_{ij}$ responds to the burden stress $\Pi^{(B)}_{ij}$. As the system saturates, the angular components $h_{\theta\theta}, h_{\phi\phi}$ freeze at their minimum finite values (proportional to ℓ_0^2), but the radial component h_{rr} diverges to accommodate the excess.

The algebra generates finite density but unbounded geometry. This is the projection mechanism: the "infinity" of the classical singularity is not a physical point of infinite density; it is a geometric divergence projected outward onto the boundary. $\blacksquare$

6.1.7 Guardrails for This Subsection

1. **Pressure saturates; it does not vanish.** The correct picture is a ceiling, not a floor. The term "pressure saturation" replaces the misleading "pressure suppression."
2. **The ceiling is not an axiom.** It derives from ℓ_0, a scale already present in the correlation geometry.
3. **The singularity is a projection artifact.** The infinite density of standard GR is replaced by bounded density plus divergent geometry. The infinity is in the representation, not the algebra.

6.2 Gravity Basins vs. Unreconciled Sectors

6.2.1 The Distinction Between Accumulation and Horizon Formation

A critical clarification must be made regarding the accumulation of constraint burden. Excess burden accumulation does *not* automatically imply black hole formation.

When a region accumulates high mode and interaction burden (e.g., a massive cluster of galaxies), the local clock rate suppresses ($\alpha_B < 1$) and the spatial metric curves. This generates a **Relational Gravity Basin** — a deep gravitational well that draws surrounding relational structures inward. A cosmological example is the Great Attractor: an enormous mass concentration whose gravitational attraction dominates our region of the universe, but which is not a black hole.

In a gravity basin, while the reconciliation cost is high, the relational geometry is still capable of supporting 3D volumetric compression. Outward relational flux and causal propagation are still possible. The region is highly burdened, but it is not an unreconciled sector.

For a black-hole-like domain to form, high burden accumulation must drive the system to pressure saturation (Section 6.1). The distinction is:

> **Excess burden accumulation creates a gravity basin; compactness-driven pressure saturation creates a black hole.**

6.2.2 The Burden Compression Ratio

To formalize this distinction, we must define the compactness parameter and give it full derivational treatment.

Definition 6.3 (Burden Compression Ratio).
Let R be a coarse-grained region with total active burden $B_{\mathrm{active}}(R)$ and relational size L_R. The **burden compression ratio** is:
$$ \boxed{ \mathcal{S}_R = \frac{\gamma \, B_{\mathrm{active}}(R)}{L_R}, } $$
where γ is a burden-to-geometry coupling constant.

Interpretation. The ratio $\mathcal{S}_R$ is dimensionally analogous to the Schwarzschild compactness parameter $2GM/(c^2 R)$ in general relativity. It measures the total burden load relative to the size of the region, not the density. A diffuse high-burden region (large $B(R)$, large L_R) can have a small $\mathcal{S}_R$ despite having enormous total burden.

Why B/L (linear in each)? The form $\mathcal{S}_R \propto B(R)/L_R$ is chosen because:
1. In the weak-field limit (Section 3.4), the effective temporal potential $\Phi_C \approx -c_{\mathrm{eff}}^2 \lambda B$. This is a total quantity, not a density.
2. The gravitational "well depth" of a region scales with total burden, while the "escape capacity" scales with the size of the boundary (proportional to L_R in the spherically symmetric case).
3. The ratio B/L therefore measures whether the total burden is large enough relative to the boundary size to trap relational flux.

The constant γ is not yet derived from the microscopic algebra; its derivation from the spectral properties of the correlation Laplacian remains an open target (Target 3, Section 9.2).

6.2.3 Definition — Relational Gravity Basin

Definition 6.4 (Relational Gravity Basin).
Let R be a coarse-grained region with high active burden $B_{\mathrm{active}}(R) \gg 0$ and burden compression ratio $\mathcal{S}_R < 1$. R is a **relational gravity basin** if:
$$ \boxed{ \mathcal{S}_R < 1, \qquad \rho_B(R) < \rho_B^{\max}, \qquad \alpha_R < 1, \qquad \mathcal{P}_{\partial R}(\Phi_{\mathrm{out}}) > 0. } $$

Here, $\mathcal{P}_{\partial R}(\Phi_{\mathrm{out}})$ is the outward admissible relational flux through the boundary. A gravity basin is a deep geometric well, but causal propagation outward is still possible. The burden density has not reached saturation.

Canonical Example: The Great Attractor. The Great Attractor region is an enormous mass concentration whose gravitational attraction dominates our local region of the universe. It has extremely high total burden $B(R)$, but it is distributed over a vast relational volume (L_R is large). Consequently, $\mathcal{S}_R \ll 1$ and $\rho_B \ll \rho_B^{\max}$. The region bends time and space — causing infall of surrounding structures — but pressure has not saturated, outward flux is non-zero, and the full 3D correlation geometry is intact. It is a gravity basin, not a black hole.

6.2.4 Definition — The Black-Hole-Like Condition

A black hole forms when the burden compression ratio exceeds the structural capacity of the relational geometry, driving the system to pressure saturation and completely suppressing outward flux.

Definition 6.5 (Black-Hole-Like Domain).
A region R is **black-hole-like** if its burden stress reaches the absolute structural maximum, causing a geometric projection (pinch-off):
$$ \boxed{ \mathcal{S}_R \geq 1, \qquad \rho_B(R) \to \rho_B^{\max}, \qquad \alpha_R \ll 1, \qquad \mathcal{P}_{\partial R}(\Phi_{\mathrm{out}}) \to 0. } $$

When $\mathcal{S}_R \geq 1$, the burden is compact enough that the relational geometry cannot structurally support it without saturating. The burden density hits $\rho_B^{\max}$, the outward flux $\mathcal{P}_{\partial R}(\Phi_{\mathrm{out}})$ vanishes, and the region becomes an unreconciled sector.

> **Excess burden accumulation produces gravitational takeover; black-hole formation occurs only when compact burden drives the pressure/compression channel to saturation. At saturation, pressure is not suppressed downward; rather, compressibility and normal-volume accessibility are suppressed. The apparent infinite collapse is therefore projected onto the emergent geometry as a two-dimensional boundary structure.**

6.2.5 The Dual Horizon Ratios

The strength of the boundary projection is governed by the relational isolation of the region. Let $W_{\mathrm{in}}(R)$ be the total internal relational cohesion (sum of correlation weights strictly inside R), and $W_{\mathrm{out}}(R)$ be the total exterior relational coupling (sum of correlation weights crossing the boundary ∂R).

Definition 6.6 (Dual Horizon Ratios).
1. **Internal Isolation Ratio:** $\chi_R = W_{\mathrm{in}}(R) / W_{\mathrm{out}}(R)$
2. **Exterior Accessibility Ratio:** $\varepsilon_R = W_{\mathrm{out}}(R) / W_{\mathrm{in}}(R) = 1/\chi_R$

Theorem 6.2 (Dual Horizon Limits).
The relational horizon limit is characterized by:
$$ \varepsilon_R \to 0 \quad \Longleftrightarrow \quad \chi_R \to \infty. $$
In this regime, exterior accessibility fails. However, this failure of reconstruction does not imply internal causal annihilation:
$$ \boxed{ \varepsilon_R \to 0 \quad \not\Rightarrow \quad \prec_R = \varnothing. } $$

Proof.
The causal dependency order $\prec_R$ is an internal structural property of the event set $\mathcal{E}_R$, generated by constraint closure inclusion. The exterior accessibility ratio ε_R depends on the state-dependent correlation kernel K_ω evaluated across the boundary. By the Two-Link Principle (Principle 1.1), causal links and relational correlation links are structurally distinct and logically independent. Suppressing the correlation link does not entail destruction of the internal causal links. $\blacksquare$

6.2.6 Guardrails for This Subsection

1. **A gravity basin is not a black hole.** High burden without compactness produces infall and time dilation, but not a horizon.
2. **$\mathcal{S}_R$ requires calibration.** The coupling constant γ is not yet derived from the algebra. This is Target 3 in Section 9.2.
3. **Pressure saturation is the mechanism.** The correct thermodynamic picture is a gas compressed to its hard-core limit: pressure hits a ceiling, compressibility vanishes, and excess must go elsewhere (into geometry).

6.3 Local Dimensional Suppression at the Horizon

6.3.1 Purpose of This Subsection

If the interior of a black-hole-like domain has reached pressure saturation (Definition 6.5), we must ask: what happens to the geometry at this limit?

In standard physics, the holographic principle states that the information of a 3D volume is encoded on a 2D surface. The Relational Constraint Framework can now *derive* this from the **Dimensional Closure Theorem (Theorem 2.10)**. When pressure saturates, the 3D volume cannot shrink to zero (which would imply a literal infinite singularity). Instead, the radial inference channel collapses, pinching the exterior-accessible geometry from 3D to 2D.

However, Theorem 2.10 as stated computes a single global D for the whole emergent geometry. To apply it at a horizon, we need a **local version** evaluated at the boundary ∂R.

6.3.2 The Radial-Tangential Decomposition

To apply dimensional closure locally, we must decompose the direction inference channel into components relative to the boundary normal.

Definition 6.7 (Radial-Tangential Decomposition of Direction).
Let $x \in \partial R$ be an emergent point on the boundary of a black-hole-like domain. Let $\hat{n}_r$ denote the emergent direction (Definition 2.11) pointing from x into the interior of R (the **radial** direction). Let $\hat{u}, \hat{v}$ denote emergent directions tangent to ∂R at x (the **tangential** directions).

The **radial component** of the direction channel at x is the profile difference:
$$ \Delta^{\mathrm{rad}}_{x \to y}(z) := d_\omega(y, z) - d_\omega(x, z), $$
where y is a probe point displaced from x in the $\hat{n}_r$ direction (i.e., y is interior to R).

The **tangential component** is defined analogously for displacements along $\hat{u}$ or $\hat{v}$.

6.3.3 Proposition — Local Rank Collapse at the Horizon

Proposition 6.3 (Local Dimensional Suppression).
Let R be a black-hole-like domain satisfying pressure saturation ($\mathcal{S}_R \geq 1$, $\rho_B \to \rho_B^{\max}$, $\alpha_R \ll 1$). For an exterior observer attempting to infer the interior of R across the boundary ∂R, the exterior-accessible spatial inference rank r_{acc} at ∂R drops from 3 to 2.

Specifically, the radial component of the direction channel collapses, while the tangential components and the existence/position channels (restricted to the boundary) survive.

Proof.

By Theorem 6.2, pressure saturation drives the exterior accessibility ratio to zero: $\varepsilon_R \to 0$. This means the correlation kernel across the boundary vanishes:
$$ K_\omega(x_{\mathrm{ext}}, x_{\mathrm{int}}) \to 0 \qquad \text{for } x_{\mathrm{int}} \in R, \; x_{\mathrm{ext}} \notin R. $$

By Definition 2.4, this implies the correlation distance to the interior diverges:
$$ d_\omega(x_{\mathrm{ext}}, x_{\mathrm{int}}) = -\ell_0 \log K_\omega(x_{\mathrm{ext}}, x_{\mathrm{int}}) \to +\infty. $$

Step 1: Position channel attenuates. The scalar position channel (measuring d_ω) ceases to provide meaningful gradient information about the interior. It only registers the boundary itself. The exterior observer cannot distinguish depth into the region.

Step 2: Radial direction collapses. The radial component of the direction channel (Definition 6.7) is computed via profile differences: $\Delta^{\mathrm{rad}}_{x \to y}(z) = d_\omega(y, z) - d_\omega(x, z)$, where y is an interior probe. Because the correlation kernel K_ω vanishes for interior probes, all interior points y project to the same null state. Thus, for any exterior probe z:
$$ d_\omega(y, z) \approx d_\omega(x, z) \qquad \text{for all interior } y, $$
meaning $\Delta^{\mathrm{rad}}_{x \to y}(z) \approx 0$. The radial direction channel mathematically collapses.

Step 3: Tangential components survive. The tangential directions lie entirely on ∂R, where the correlation kernel remains non-zero (the boundary is still relationally accessible). Thus, Δ^{tan} retains non-trivial profile differences.

Step 4: Rank reduction. With the radial direction channel collapsed, the maximum accessible spatial inference rank at the boundary drops:
$$ r_{\mathrm{acc}}(\partial R) \leq 2 < 3 = r_{\min}. $$

By the under-closure condition of Theorem 2.10 ($D < 3 \implies \Xi_C > 0$), a 3D volumetric inference is impossible. The exterior observer can only infer Existence and the scalar radial Position of the boundary, plus tangential direction. The effective accessible dimension is 2. $\blacksquare$

6.3.4 Theorem — Dimensional Suppression to 2D

Theorem 6.3 (Dimensional Suppression at Maximum Pressure).
Let R be a black-hole-like domain where the burden stress has reached pressure saturation ($\rho_B \to \rho_B^{\max}$, $\mathcal{S}_R \geq 1$). Then the boundary ∂R is mathematically forced to be a 2D surface.

Proof.
By Proposition 6.3, the exterior-accessible inference rank at ∂R is $r_{\mathrm{acc}} = 2$. By the local application of the Dimensional Closure Theorem (Theorem 2.10), a spatial geometry with inference rank 2 cannot support 3D volumetric structure. Because the geometry cannot support 3D volume at maximum pressure, and it cannot collapse to a 0D point (which would violate the existence of the underlying algebraic events), the inference structure is forced to project onto a 2D surface. $\blacksquare$

6.3.5 Interpretation: The Mechanism of Holography and Singularity Avoidance

This theorem provides the rigorous mathematical mechanism for why a black hole's boundary is a 2D surface, deriving it directly from the framework's dimensional selection logic and thermodynamic pressure limits.

The 3D volume does not disappear into an infinite singularity. When the relational pressure (burden density) hits the absolute structural maximum $\rho_B^{\max}$ (imposed by the ℓ_0-floor), the 3rd inference channel (radial direction) is pinched off because the extreme stress prevents the correlation kernel from resolving interior depth. The "infinite" density is therefore projected onto a 2D surface because the observer's inference rank has been reduced to 2.

The metric signature of saturation. At pressure saturation:
1. The angular metric components freeze at their minimum finite values: $h_{\theta\theta} \to \ell_0^2$, $h_{\phi\phi} \to \ell_0^2 \sin^2\theta$.
2. The determinant of the spatial metric remains positive and finite: $\det(h) > 0$.
3. The radial metric component diverges: $h_{rr} \to \infty$.
4. The burden lapse vanishes: $\alpha_B \to 0$.

The proper radial distance to the center becomes:
$$ \Delta R = \int_0^{r_s} \sqrt{h_{rr}} \, dr \to \infty. $$

The black hole interior is a finite 3D volume (bounded by the saturated angular area) but an infinite radial depth. The "singularity" is not a point; it is the asymptotic infinite-depth endpoint of the radial coordinate. The infinity of the classical GR singularity is projected onto the boundary horizon because the proper time to reach the radial endpoint is infinite, effectively freezing the interior structure on the horizon.

> **A black hole boundary is 2D because maximum structural pressure collapses the radial inference channel, projecting the internal compression onto the geometry as a surface. The singularity is not a point of infinite density; it is a projection artifact of attempting to describe a saturated geometry with ordinary 3D coordinates.**

6.3.6 Guardrails for This Subsection

1. **The rank collapse is local, not global.** Proposition 6.3 evaluates inference rank at ∂R, not at "the universe." The 3D bulk may still exist as an internal/unresolved relational structure; it is the exterior-accessible correlation geometry that becomes 2D.
2. **The radial-tangential decomposition is new.** Definition 6.7 introduces this split for the first time. It is needed before "the radial part degenerates" is a provable statement.
3. **The singularity is a representation artifact.** The framework does not admit literal infinities in physical configurations. The infinity is in the geometric representation (divergent h_{rr}), not in the algebra (bounded ρ_B).

6.4 Boundary Recovery in the Strong-Sector Regime

6.4.1 The Boundary Record Map

If the interior of a black-hole-like domain is hidden by pressure saturation and dimensional suppression (Section 6.3), the next physical question is whether the exterior boundary data can reconstruct the interior state. This is the RCF analogue of the holographic principle.

Because the radial inference channel has collapsed, the exterior observer cannot probe the 3D interior directly. However, the interior zero-preserving configurations may still leave a "shadow" or "record" on the exterior-accessible 2D boundary.

Definition 6.8 (Boundary Record Map).
Let $\mathcal{Z}(R)$ be the set of admissible interior zero-preserving configurations of R (configurations satisfying $\mathfrak{M} = 0$). Let $\mathcal{B}_{\partial R}$ be the space of exterior-accessible boundary data. The **boundary record map** is:
$$ \boxed{ \mathfrak{B}: \mathcal{Z}(R) \to \mathcal{B}_{\partial R}, } $$
which assigns to each interior configuration its boundary-preserved observable content.

6.4.2 Definition — Faithful Boundary Recovery

For the boundary data to be a faithful representation of the interior, the map $\mathfrak{B}$ must not collapse distinct physical configurations into the same boundary record.

Definition 6.9 (Faithful Modulo Equivalence).
The boundary map $\mathfrak{B}$ is **faithful modulo equivalence** if:
$$ \mathfrak{B}(\mathcal{H}_1) = \mathfrak{B}(\mathcal{H}_2) \quad \Longrightarrow \quad \mathcal{H}_1 \sim \mathcal{H}_2, $$
where $\sim$ is the framework’s physical equivalence relation on admissible histories (i.e., they differ only by null directions or gauge redundancy, meaning they represent the same physical state).

6.4.3 Theorem — Boundary Recovery

Theorem 6.4 (Boundary Recovery).
Assume the strong black-hole-like regime ($\chi_R \gg 1, \varepsilon_R \ll 1, \alpha_R \ll 1$). If the boundary record map $\mathfrak{B}$ is complete and injective on the surviving strong-sector equivalence classes, then the boundary data determine the accessible interior equivalence class up to the constraint quotient.

Proof.
If $\mathfrak{B}$ is injective on the surviving equivalence classes, then distinct interior classes cannot collapse to the same boundary record. Therefore, given a boundary record $b \in \mathcal{B}_{\partial R}$, there exists at most one interior equivalence class $[\mathcal{H}]_{\mathrm{ext}}$ mapping to b. The interior is recoverable, not by direct access, but by relational reconstruction from the boundary code. $\blacksquare$

6.4.4 Interpretation

This theorem preserves the framework's crucial guardrail regarding black hole interiors: **recovery is into the lowest-burden admissible sector, not a return of the original matter class.**

The full 3D relational interior is still there, but the accessible channel is so suppressed that the boundary presentation dominates. The 3D content is preserved in the underlying relational structure, but operationally hidden by projection. An exterior observer can reconstruct the *state* of the interior, but cannot extract the specific relational modes (e.g., the specific particles) that fell in, because those are high-burden configurations trapped behind the projection.

> **The interior is recoverable structurally, but not accessible dynamically.**

6.4.5 Guardrails for This Subsection

1. **Recovery is not resurrection.** Theorem 6.4 states that the interior state is mathematically encoded on the boundary. It does not imply that an exterior observer can physically extract the infallen matter.
2. **Injectivity is a conditional assumption.** The theorem holds *if* $\mathfrak{B}$ is injective. Proving that the constraint algebra guarantees this injectivity for generic black-hole-like domains remains an open target.
3. **No background manifold is assumed.** The "boundary" ∂R is not a 2D surface embedded in a pre-existing 4D spacetime; it is the operational threshold where the exterior accessibility ratio ε_R drops below measurement resolution.

6.5 Entropy-Area Scaling and Lowest-Burden Emission

6.5.1 Boundary Entropy as Record Count

If the interior of a black-hole-like domain is hidden behind a projection boundary (Section 6.2) and the accessible geometry is reduced to 2D (Section 6.3), the entropy of the region must reflect the number of admissible boundary records available to an exterior observer. It cannot scale with the interior volume, because the interior degrees of freedom are flux-suppressed and operationally inaccessible.

Theorem 6.5 (Boundary Entropy as Coarse-Grained Record Count).
Let ∂R be the 2D boundary of a strongly burden-loaded region. If the admissible boundary microstates grow exponentially with the effective boundary area, then the boundary entropy satisfies:
$$ \boxed{ S_{\partial R} \propto \operatorname{Area}(\partial R). } $$

Proof Sketch.
Let $N_{\partial R}$ be the number of admissible boundary record states. By the definition of entropy, $S_{\partial R} = \log N_{\partial R}$.
By Theorem 6.3, the exterior-accessible inference rank at ∂R is 2. The boundary records correspond to localised zero-preserving modes on this 2D emergent metric space (induced by the spatial metric h_{ij} on ∂R). If these modes scale exponentially with the 2D geometric area—$N_{\partial R} \sim e^{\alpha \operatorname{Area}(\partial R)}$ for some constant $\alpha > 0$—then:
$$ S_{\partial R} = \log N_{\partial R} \sim \alpha \operatorname{Area}(\partial R). \qquad \blacksquare $$

Why area, not volume. The entropy is governed by boundary area rather than interior volume because the radial inference channel has collapsed (Proposition 6.3). The exterior observer can only count records along the remaining 2 tangential dimensions. The dimensional suppression theorem provides the *derived reason* entropy scales with area: the exterior-accessible inference rank at ∂R is literally 2, not 3.

6.5.2 Lowest-Burden Emission

As the unreconciled region relaxes, it must emit structure to re-equilibrate with the exterior. The framework dictates that relaxation is driven by burden minimization. The region sheds its structural load through the limited flux channels, and the most admissible output is the one carrying the least structural burden.

Theorem 6.6 (Lowest-Burden Emission).
Let R be a strongly burden-loaded region undergoing relaxation. The emitted sector is not the original infallen matter class, but the **lowest-burden admissible output** compatible with the projection-flux boundary.

This output may take either of two forms:
1. **Dimension-independent carriers:** Temperature-linked or energy-like emission that does not carry 3D relational structure (e.g., pure burden-flux radiation).
2. **Boundary-projected 3D relational content:** Three-dimensional relational excitations that are temporarily reduced by projection and then reconstitute after emission.

Proof Sketch.
The projection principle acts by suppressing accessible flux channels. The strongest projection-compatible output is the one carrying the least structural burden, as high-burden modes are trapped by their own reconciliation cost. The region relaxes by shedding this simplest admissible carrier sector. $\blacksquare$

> **A black-hole-like region relaxes by emitting the lowest-burden admissible sector, whether dimension-independent or boundary-projected from 3D relational content.**

6.5.3 Hawking-Like Appearance

Under coarse-grained observation, the lowest-burden emission appears thermal.

Theorem 6.7 (Hawking-Like Emission Under Coarse Graining).
Let a black-hole-like region relax by emitting the lowest-burden admissible sector. If the boundary flux channels are strongly suppressed and the emission is observed only at coarse-grained resolution, the outgoing spectrum is approximately thermal.

Proof Sketch.
Thermal spectra appear when transition rates obey approximate detailed balance. In the relational framework, the boundary relaxation process is driven by burden-gradient smoothing. If the outward channel is exponentially suppressed relative to the inward channel (due to $\alpha_R \ll 1$), the stationary distribution over the boundary record states becomes Gibbs-like. Consequently, the emission law is Hawking-like in form. $\blacksquare$

6.5.4 Guardrails for This Subsection

1. **Area scaling requires a mode assumption.** The proof assumes boundary records scale exponentially with area. A rigorous derivation from the spectral gap of the boundary correlation Laplacian remains an open target.
2. **Recovery is not resurrection.** Theorem 6.6 explicitly states that the original matter class is not what is emitted. The information paradox is structurally avoided because the specific relational modes remain trapped in the interior, while only the burden is radiated away.
3. **Hawking-like emission is a target.** The exact calculation of the Hawking temperature T_H from the microscopic constants ($\epsilon, \ell_0, \lambda$) requires a completed continuum limit.

6.6 Guardrails and Summary of Section 6

6.6.1 Guardrails for Section 6

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **A black hole is not a literal hole.** It is a maximum-constrained, unreconciled relational sector where pressure saturation prevents further 3D compression.
2. **A gravity basin is not a black hole.** Excess burden without compactness produces infall and time dilation, but not a horizon. The Great Attractor is the canonical example.
3. **Pressure saturates; it does not vanish.** The correct thermodynamic picture is a ceiling ($\rho_B^{\max}$ imposed by ℓ_0), not a floor. What vanishes is compressibility, not pressure.
4. **The singularity is a projection artifact.** The framework does not admit literal infinities in physical configurations. The infinity is in the geometric representation (divergent h_{rr}), not in the algebra (bounded ρ_B).
5. **A horizon is not a material surface.** It is an operational boundary where $\varepsilon_R \ll 1$.
6. **$\alpha_R \to 0$ is asymptotic suppression.** For any finite burden $B < \infty$, the clock factor $\alpha_B > 0$.
7. **Interior causal structure may persist.** Exterior reconstruction failure does not imply internal causal annihilation.
8. **Recovery is not resurrection.** Boundary recovery reconstructs the lowest-burden admissible sector, not the original infallen matter class.
9. **Hawking-like emission is a target.** The thermal appearance follows from coarse-grained boundary relaxation, not from a primitive postulate.

6.6.2 Summary of Section 6

Section 6 reframed black holes within the Relational Constraint Framework, deriving their structural and thermodynamic properties from the extreme limit of constraint burden:

1. **The ℓ_0-Floor and Pressure Saturation** established that the fundamental scale ℓ_0 imposes a natural ceiling on burden density ($\rho_B^{\max}$), preventing literal infinite collapse. Pressure saturates at this ceiling; it does not vanish.

2. **Gravity Basins vs. Black Holes** formally distinguished excess burden accumulation (gravity basin, e.g., the Great Attractor) from compactness-driven pressure saturation (black hole). The burden compression ratio $\mathcal{S}_R$ was given full derivational treatment.
3. **Local Dimensional Suppression** proved (Proposition 6.3, Theorem 6.3) that pressure saturation collapses the radial inference channel at the horizon, reducing exterior-accessible dimension from 3 to 2. The radial-tangential decomposition was formally introduced.
4. **Singularity Avoidance** was established: the infinity of the classical GR singularity is replaced by bounded density ($\rho_B \leq \rho_B^{\max}$) plus divergent radial geometry ($h_{rr} \to \infty$). The singularity is a projection artifact.
5. **Boundary Recovery** proved that if the boundary record map is injective, the interior equivalence class is recoverable structurally.
6. **Entropy-Area Scaling** was derived as a coarse-grained count of admissible boundary records on the 2D suppressed surface.
7. **Lowest-Burden Emission** established that relaxation emits the simplest admissible carrier sector, structurally resolving the information paradox.
8. **Hawking-Like Appearance** showed that lowest-burden emission becomes approximately thermal under coarse-grained observation.

The conceptual chain of this section is:
$$ \boxed{ \text{burden accumulation} \to \text{compactness } (\mathcal{S}_R \geq 1) \to \text{pressure saturation } (\rho_B^{\max}) \to \text{radial channel collapse} \to \text{2D boundary} \to \text{area-entropy} \to \text{lowest-burden emission} \to \text{Hawking-like appearance}. } $$

This completes the black-hole layer. The framework now possesses an emergent spacetime populated by matter, governed by an Einstein-like geometric response, and capable of extreme unreconciled sectors where dimensional suppression projects infinity onto geometry.

Section 7 — Records, Classicality, and the Born Rule

7.0 Purpose of Records, Classicality, and Probability

7.0.1 Transition from Quantum Structure to Definite Outcomes

Section 6 reframed black-hole-like domains as unreconciled relational sectors where extreme burden suppresses exterior accessibility. The framework now possesses the full architectural spine: a primitive algebraic foundation, causal dependency, emergent spatial geometry, burden-weighted time, matter modes, gravitational sourcing, and horizon structure.

However, the framework does not yet explain how the definite, classical world of observable facts emerges from the underlying zero-preserving relational algebra. If physical reality is fundamentally a superposition of constraint-compatible relational modes, why do observers experience definite outcomes, stable histories, and predictable frequencies?

The purpose of this section is to reconstruct classicality and probability from the relational constraint structure, without importing external measurement postulates.

> **Classicality emerges when certain zero-preserving records become stable, redundant, and effectively irreversible under constraint-preserving interactions.**

7.0.2 Why Probability Must Emerge from Admissibility

In standard quantum mechanics, the Born rule is postulated as an external probability measure applied to a Hilbert space. In the Relational Constraint Framework, Hilbert space is a derived GNS representation, and the primitive physical condition is the vanishing of the master constraint: $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$.

Therefore, probability cannot be bolted on as a separate axiom. It must emerge as the natural measure of how the zero-constraint is distributed across stable structural alternatives (record sectors).

The central claim of this section is that the Born weights are not an extra probabilistic layer imposed on the theory; they are the record-sector decomposition of the physical state's zero-constraint support.

> **The master constraint remains zero, but the zero is read through stable record weights, so the probability rule is anchored back to admissibility rather than floating free as interpretation.**

7.0.3 What This Section Establishes

This section establishes the following:

1. **Stable Record Separation:** Decoherence-like burden separation causes distinct record sectors to become approximately orthogonal, based on a rigorous quantitative threshold.
2. **Redundant Record Robustness:** Classicality arises when records are encoded across many environmental fragments, making them robust against local disturbance.
3. **Normalized Record Weights:** Positive linear functionals on stable record effects yield mathematically normalized weights ($p_i \ge 0, \sum p_i = 1$).
4. **The Born Rule as Sectorwise Zero-Decomposition:** The master constraint zero condition is shown to distribute over record sectors weighted by squared amplitudes ($|c_i|^2$).
5. **Zero-Envariance (Z-Envariance):** Symmetries of the zero-preserving state force the branch measure to be proportional to squared amplitude, uniquely selecting the Born form.

7.0.4 Guardrails for This Section

1. **No external measurement postulate.** The framework does not assume wavefunction collapse or an external observer triggering probability. Probability is the sectorwise reading of admissibility.
2. **Classicality is emergent.** Records become classical only when they are stable, redundant, and zero-preserving. There are no primitive classical objects.

3. **Decoherence is internal.** Suppression of interference arises from the high burden of cross-sector reconciliation, not from an external thermodynamic arrow or environmental postulate.
4. **The Born rule is a theorem target.** It is derived as the unique measure consistent with Z-envariance and sectorwise zero-decomposition, not postulated at the foundation.

7.1 Stable Record Separation

7.1.1 The Problem of Definite Outcomes

The physical state ω evaluates relational expressions. In the GNS representation (Section 0.4), this appears as a cyclic vector Ω_ω in a Hilbert space $\mathscr{H}_\omega$. If the state is a superposition of different macroscopic configurations, standard quantum mechanics asks how one definite outcome is selected.

RCF does not begin with a "measurement postulate" that collapses the state. Instead, it asks: under what conditions do distinct relational configurations become effectively independent, preventing interference?

The answer lies in the **constraint burden** (Section 3.1). When different macroscopic configurations interact with their environment, they generate distinct correlation profiles. Maintaining coherence between these distinct profiles requires reconciling vastly different causal dependencies, which imposes a high structural burden. The framework naturally suppresses this high-burden cross-talk.

7.1.2 Definition — Record Sector

To formalize this, we define record sectors within the represented physical Hilbert space.

Definition 7.1 (Record Sector).
Let the physical sector $\mathscr{H}_{\mathrm{phys}}^\omega = \ker \hat{\mathfrak{M}}_\omega$ (Section 1.1) decompose into a direct sum of closed subspaces corresponding to stable, distinguishable macroscopic relational configurations (records). A **record sector** $\mathscr{H}_i^\omega \subset \mathscr{H}_{\mathrm{phys}}^\omega$ is such a subspace.

Let P_i be the orthogonal projector onto $\mathscr{H}_i^\omega$. A family of record sectors $\{\mathscr{H}_i^\omega\}$ is **exhaustive** if their projectors resolve the identity on the physical sector:
$$ \sum_i P_i = I_{\mathrm{phys}}. $$

7.1.3 Quantifying Cross-Sector Burden

To make the decoherence mechanism rigorous, we must quantitatively define the "high burden" of cross-correlation. Let Π_i and Π_j be the algebraic projection operators corresponding to macroscopic record sectors i and j.

Definition 7.2 (Intra- and Cross-Sector Burden).
The **intra-sector burden** is the cost of maintaining admissibility within a single record sector:
$$ B_{\mathrm{intra}} = \sum_\alpha w_\alpha \, \omega\left([\hat{C}_\alpha, \Pi_i]^\dagger [\hat{C}_\alpha, \Pi_i] \right). $$
The **cross-sector burden** is the cost of maintaining coherence between distinct sectors:
$$ B_{\mathrm{cross}}(i,j) = \sum_\alpha w_\alpha \, \omega\left([\hat{C}_\alpha, \Pi_{ij}]^\dagger [\hat{C}_\alpha, \Pi_{ij}] \right), $$
where Π_{ij} is the off-diagonal transition operator mapping sector i to sector j.

7.1.4 Theorem — Stable Record Separation (Quantitative Decoherence)

Theorem 7.1 (Stable Record Separation).
Suppose the physical state decomposes into record-bearing classes such that the cross-sector burden is strictly dominant: $B_{\mathrm{cross}}(i,j) \gg B_{\mathrm{intra}}$ for $i \neq j$. Then, under coarse-graining, the off-diagonal correlation terms between distinct record sectors become vanishingly small.

Specifically, if P_i and P_j project onto distinct record sectors ($i \neq j$), then for any localisable observable $A \in \mathcal{A}_{\mathrm{loc}}$:
$$ \boxed{ \lim_{B_{\mathrm{cross}}/B_{\mathrm{intra}} \to \infty} \langle \Omega_\omega, P_i \, \pi_\omega(A) \, P_j \, \Omega_\omega \rangle_\omega = 0. } $$

Proof Sketch.
By the Causal Reconciliation Principle (Section 3.0), maintaining relational admissibility (zero-constraint closure) across two macroscopically distinct branches requires reconciling their vastly different causal dependency networks. This generates a massive B_{cross}.

Because the master constraint $\hat{\mathfrak{M}}_\omega$ strictly enforces zero-violation, the algebraic paths that would sustain these high-burden cross-correlations are dynamically suppressed. The relational network sheds this prohibitive load by decoupling the sectors. Once null directions are removed by the GNS quotient, the remaining physical inner product between distinct branches becomes vanishingly small. The branches therefore behave like quasi-classical alternatives rather than coherently interfering amplitudes. $\blacksquare$

7.1.5 Interpretation

This is the RCF analogue of decoherence. Interference is not lost by an external "wavefunction collapse" or by an ad hoc environmental postulate; it is suppressed *internally* because maintaining cross-sector consistency violates the least-burden reconciliation principle.

What remains is a set of independent, stable record sectors. The physical state Ω_ω can be effectively decomposed as:
$$ \Omega_\omega \approx \sum_i c_i \, \Omega_{\omega, i}, $$

where $\Omega_{\omega, i} = P_i \Omega_\omega$ are the sector-specific state vectors, and $\langle \Omega_{\omega, i}, \Omega_{\omega, j} \rangle_\omega \approx 0$ for $i \neq j$.

> **Decoherence is the internal suppression of high-burden cross-sector reconciliation.**

7.1.6 Guardrails for This Subsection

1. **Decoherence does not select an outcome.** Theorem 7.1 explains why interference disappears, yielding a set of distinct alternatives. It does not yet explain why one alternative is experienced with a specific probability; that requires the Born rule derivation (Section 7.4).
2. **Records are emergent.** The projectors P_i are not fundamental operators; they are coarse-grained macroscopic configurations that are stable under zero-preserving interactions.
3. **Approximate orthogonality.** The suppression is asymptotic (≈ 0). Exact orthogonality is an idealization of the continuum/large-scale limit.

7.2 Redundant Record Robustness (Classicality)

7.2.1 From Decoherence to Objectivity

Theorem 7.1 showed that high cross-sector burden suppresses interference, yielding distinct, stable record sectors. However, decoherence alone does not fully explain the emergence of classicality. A world is classical when its facts are objective—meaning they are stable against local disturbances and can be accessed by multiple observers without being destroyed.

In the Relational Constraint Framework, we do not postulate an external "observer" or a classical environment. Instead, we formalize this using internal relational fragments. A record becomes classical when it is redundantly encoded across many independent fragments of the relational network.

7.2.2 Definition — Environmental Fragments and Redundancy

Let R be a zero-preserving record associated with a sector $\mathscr{H}_i^\omega$. Let the rest of the relational network be partitioned into a set of independent subsystems (fragments) $\{E_k\}$ that are relationally coupled to R.

We measure the correlation between the record R and a fragment E_k using the state-dependent correlation kernel K_ω (Definition 2.2), or an equivalent operational distinguishability measure.

Definition 7.3 (Redundant Encoding).
Let $\eta \in (0, 1]$ be a recognition threshold. The record R is **redundantly encoded** if many fragments independently carry recoverable information about R above this threshold.

The **redundancy degree** of R is the number of such fragments:
$$ \boxed{ \mathcal{R}_\eta(R) = \#\{E_k : \mathrm{Corr}_\omega(E_k, R) \geq \eta\}. } $$

7.2.3 Definition — m-Robust Record

A record is objectively classical if it cannot be easily erased by a local perturbation. We formalize this as m-robustness.

Definition 7.4 (m-Robust Record).
A record R is **m-robust** if deleting, disturbing, or decohering any set of at most m encoding fragments still leaves at least one fragment from which the record value can be completely recovered.

Mathematically, if $D \subset \{E_k\}$ is a set of disturbed fragments with $|D| \leq m$, then there exists some surviving fragment $E_j \notin D$ such that $\mathrm{Corr}_\omega(E_j, R) \geq \eta$.

7.2.4 Theorem — Redundant Record Robustness

Theorem 7.2 (Redundant Record Robustness).
Let R be a zero-preserving record with redundancy degree $\mathcal{R}_\eta(R) = N_R$. Assume each of the N_R fragments independently encodes the same record value above the threshold η. If $N_R > m$, then R is m-robust.

Proof.
Let D be any set of disturbed fragments with $|D| \leq m$.
At worst, all m disturbed fragments belong to the set of N_R record-encoding fragments.
The number of surviving record-encoding fragments is therefore at least:
$$ N_R - |D| \geq N_R - m. $$
Since $N_R > m$, we have $N_R - m > 0$.
Therefore, $N_R - |D| \geq 1$. There exists at least one fragment $E_j \notin D$ such that $\mathrm{Corr}_\omega(E_j, R) \geq \eta$. The record value is recoverable. By Definition 7.4, R is m-robust. $\blacksquare$

7.2.5 Interpretation

Classicality is not a primitive property of macroscopic objects; it is an emergent property of redundant encoding.

A classical fact is a stable, redundant, zero-preserving record. Objectivity arises because multiple observers can access the same fact through different fragments (different local observables E_k) without directly disturbing the original record sector R.

> **Objectivity is structural redundancy.**

7.2.6 Guardrails for This Subsection

1. **Fragments are internal.** The "environment" E_k is not an external classical background; it is a subsystem of the emergent relational network itself.
2. **Threshold η is operational.** The value η represents the minimum correlation required for an operational observer to confidently distinguish the record. It is a coarse-graining parameter, not a fundamental constant.
3. **Robustness is finite.** A record is m-robust, not infinitely robust. Sufficiently large disturbances ($m > N_R$) can still destroy the record.

7.3 Normalized Record Weights

7.3.1 The Formal Probability Layer

Now that stable, independent record sectors exist (Section 7.1) and have become objectively classical via redundant encoding (Section 7.2), we must assign weights to them. An observer tracking the relational network will experience one of these sectors. The framework requires a non-negative measure on these sectors compatible with the algebraic state.

Because the primitive physical object is the positive linear functional ω, the natural source of these weights is the state evaluation of positive operators corresponding to the record sectors.

7.3.2 Definition — Positive Effect Family

Let $\{\mathscr{H}_i^\omega\}$ be the exhaustive family of stable record sectors in the physical Hilbert space $\mathscr{H}_{\mathrm{phys}}^\omega$.

Definition 7.5 (Positive Effect Family).
Let $\{E_i\}$ be a family of positive operators (effects) in the physical observable algebra $\mathcal{A}_{\mathrm{obs}}$ that resolve the identity on the physical sector:
$$ \boxed{ \sum_i E_i = \mathbf{1}_{\mathrm{phys}}. } $$
Each E_i corresponds to the operational act of interrogating the physical sector to determine if record i is present. Because $E_i \in \mathcal{A}_{\mathrm{obs}}$, it is zero-preserving (Definition 1.3).

7.3.3 Definition — Normalized Record Weights

The algebraic state ω evaluates the likelihood of these records.

Definition 7.6 (Record Weights).
Let ω be a normalised physical state ($\omega(\mathbf{1}) = 1$). The **record weight** (or branch weight) assigned to sector i is:
$$ \boxed{ p_i := \omega(E_i). } $$

7.3.4 Theorem — Normalization and Positivity

Theorem 7.3 (Normalization and Positivity).

The record weights p_i satisfy the mathematical axioms of a probability measure:
$$ \boxed{ p_i \geq 0 \qquad \text{and} \qquad \sum_i p_i = 1. } $$

Proof.
1. **Positivity:** Since E_i is a positive operator ($E_i \geq 0$) and ω is a positive linear functional ($\omega(A^\dagger A) \geq 0$ for all A), it follows immediately that $p_i = \omega(E_i) \geq 0$.
2. **Normalization:** By linearity of the state ω and the exhaustivity of the effect family ($\sum_i E_i = \mathbf{1}_{\mathrm{phys}}$):
$$ \sum_i p_i = \sum_i \omega(E_i) = \omega\left(\sum_i E_i\right) = \omega(\mathbf{1}_{\mathrm{phys}}). $$
Since the state is normalised, $\omega(\mathbf{1}) = 1$. Therefore:
$$ \sum_i p_i = 1. \qquad \blacksquare $$

7.3.5 Interpretation

This theorem closes the formal probability piece. Positivity makes the weights non-negative, and normalization makes them sum to one.

However, this theorem alone does not prove the Born rule ($p_i = |c_i|^2$); it only proves that a consistent probability-like measure exists. The weights p_i could, in principle, be any function of the state amplitudes that satisfies positivity and normalization.

The remaining issue is to anchor these weights back to the foundational zero-constraint and uniquely specify their functional form.

> **A normalized probability measure exists structurally, but its specific functional form remains to be derived.**

7.3.6 Guardrails for This Subsection

1. **Normalized weights $\neq$ full Born interpretation.** The mathematical structure of probability ($p_i \ge 0, \sum p_i = 1$) is established, but the leap to empirical frequencies requires the stability and accessibility of records.
2. **Effects are observables.** The operators E_i are not external probes; they are zero-preserving observables within the algebra $\mathcal{A}_{\mathrm{obs}}$.
3. **No collapse postulate.** The state ω does not "collapse" into one sector. The weights p_i simply reflect how the zero-constraint is distributed across the stable alternatives.

7.4 The Born Rule as Sectorwise Zero-Decomposition

7.4.1 The Core Synthesis

The most critical theoretical move in RCF probability theory is connecting the record weights p_i (assigned to the decohered sectors) back to the master constraint $\hat{\mathfrak{M}}_\omega$.

In standard quantum mechanics, probabilities are imposed after the fact as squared amplitudes. In RCF, physical admissibility *is* the vanishing of the master constraint: $\omega(\hat{\mathfrak{M}}_\omega) = 0$. If the universe branches into decohered record sectors (Section 7.1), the zero-constraint must be distributed across those sectors.

The weights p_i are not arbitrary; they represent how the zero-constraint is partitioned among the stable alternatives.

Theorem 7.4 (Born Weights as Sectorwise Zero-Decomposition).
Let the physical state decompose into decohered record sectors $\mathscr{H}_i^\omega$ (satisfying the conditions of Theorem 7.1). Let $\hat{C}_i$ be the component of the primitive constraint operator acting on sector i. The sector-weighted master constraint is:
$$ \boxed{ \mathcal{M}_B = \sum_i p_i \hat{C}_i^\dagger \hat{C}_i, \qquad p_i \geq 0, \quad \sum_i p_i = 1. } $$
The physical state condition $\omega(\mathcal{M}_B) = 0$ implies that the zero-constraint is distributed across the stable record sectors according to the weights p_i.

Proof.
Because each $\hat{C}_i^\dagger \hat{C}_i$ is a positive operator and the weights p_i are non-negative, the weighted sum $\mathcal{M}_B$ is positive. The only way for the expectation value $\omega(\mathcal{M}_B)$ to vanish in a physical state is for the constrained sectors to vanish on the physical support.

The master constraint says what must vanish. The Born weights p_i say how the zero is distributed across stable record alternatives. The physical state says the universe occupies the zero-constraint support. Therefore, probability is not added on top of the theory; it is the sectorwise reading of constraint compatibility. $\blacksquare$

> **Born weights do not compete with the master constraint; they refine its zero condition across record sectors.**

7.4.2 Why This Uniquely Selects $|c_i|^2$ (Z-Envariance)

While Theorem 7.4 proves that weights must distribute the zero-constraint, it does not uniquely specify the functional form of p_i. To show that $p_i = |c_i|^2$, the framework employs **Zero-Envariance (Z-Envariance)**, adapting the envariance concept to the RCF constraint structure.

Definition 7.7 (Zero-Envariance).
A local transformation $U_{\mathcal{S}}$ on a system sector is **Z-envariant** if there exists a compensating transformation $U_{\mathcal{E}}$ on the environment (relational network) sector such that:
1. $(U_{\mathcal{S}} \otimes U_{\mathcal{E}}) |\Psi\rangle = |\Psi\rangle$ (the global physical state is unchanged).
2. The global state remains in the physical kernel: $(U_{\mathcal{S}} \otimes U_{\mathcal{E}}) |\Psi\rangle \in \ker \hat{\mathfrak{M}}_\omega$.

7.4.3 Lemma — Phase Invariance from Z-Envariance

Lemma 7.1 (Phase Invariance).
If branch phase rotations are Z-envariant, then the branch measure (probability) must be phase-invariant:
$$ \boxed{ p_i(c_i) = p_i(|c_i|). } $$

Proof.
A local phase rotation $U_{\mathcal{S}} |s_i\rangle = e^{i\theta} |s_i\rangle$ can be compensated by an inverse environmental phase $U_{\mathcal{E}} |E_i\rangle = e^{-i\theta} |E_i\rangle$. The joint action leaves the entangled state $|\Psi\rangle = \sum c_i |s_i\rangle |E_i\rangle$ and the master constraint structure unchanged. Since the branch measure is an objective invariant of the physical state (by Theorem 7.4), it cannot depend on the phase. Thus, p_i depends only on the magnitude $|c_i|$. $\blacksquare$

7.4.4 Lemma — Equal-Magnitude Branch Equality

Lemma 7.2 (Equal-Magnitude Branch Equality).
If two decohered branches have equal amplitude magnitude ($|c_1| = |c_2|$), and branch swaps are Z-envariant, then their measures are equal:
$$ \boxed{ p_1 = p_2. } $$

Proof.
By phase invariance (Lemma 7.1), choose phases so $c_1 = c_2$. A swap of the system states combined with a swap of the environment records leaves the global state invariant. Because the master constraint $\hat{\mathfrak{M}}_\omega$ acts symmetrically on identical sectors, the zero-constraint distribution cannot distinguish between them. Thus, their weights must be equal. $\blacksquare$

7.4.5 Theorem — The Born Rule

Theorem 7.5 (The Born Rule as Normalized Record Weights).
Let $|\Psi\rangle = \sum_i c_i |s_i\rangle |E_i\rangle$ be a decohered physical state in $\ker \hat{\mathfrak{M}}_\omega$. Assume:
1. Branch phase rotations are Z-envariant.
2. Equal-magnitude branch swaps are Z-envariant.
3. Branch measures are objective zero-closure invariants (Theorem 7.4).
4. Branch measures are additive over decohered orthogonal record-sector splits (Theorem 7.3).

Then the unique normalised branch measure is:
$$ \boxed{ p_i = \frac{|c_i|^2}{\sum_j |c_j|^2}. } $$
If the state is normalised ($\sum_j |c_j|^2 = 1$), then:
$$ \boxed{ p_i = |c_i|^2. } $$

Proof.
Let $p(x)$ be the measure function for a branch of amplitude magnitude $x = |c_i|$.

Split any branch i into N_i identical sub-branches of equal magnitude $\alpha = |c_i|/\sqrt{N_i}$, such that the total amplitude squared is preserved: $\sum_{k=1}^{N_i} \alpha^2 = N_i (|c_i|^2 / N_i) = |c_i|^2$.
By Lemma 7.2 (equal-magnitude equality), all sub-branches across the entire decomposition have equal measure $p(\alpha)$.
By additivity (Theorem 7.3), the total measure of the original branch i is the sum of the measures of its sub-branches:
$$ p_i = N_i \, p(\alpha) = N_i \, p\left(\frac{|c_i|}{\sqrt{N_i}}\right). $$
Let $y = x^2$. Then $p(x) = p(\sqrt{y})$. The equation becomes:
$$ p(\sqrt{y}) = N \, p\left(\sqrt{\frac{y}{N}}\right). $$
The unique regular solution to this functional equation is $p(\sqrt{y}) \propto y$. Therefore:
$$ p_i \propto |c_i|^2. $$
Applying normalization ($\sum p_i = 1$) yields:
$$ p_i = \frac{|c_i|^2}{\sum_j |c_j|^2}. $$
If the state vector is normalised such that $\sum_j |c_j|^2 = 1$, this reduces to $p_i = |c_i|^2$. $\blacksquare$

7.4.6 Guardrails for This Subsection

1. **The Born rule is not an external postulate.** It is derived as the unique sectorwise decomposition of the master constraint zero, forced by Z-envariance.
2. **Z-envariance requires decoherence.** The compensating environmental transformations assume that the branches are stably separated. The proof of the Born rule explicitly relies on the quantitative decoherence condition ($B_{\mathrm{cross}} \gg B_{\mathrm{intra}}$) established in Theorem 7.1. In the absence of decoherence, well-defined branches do not yet exist, and the probability calculus does not apply.
3. **Probability is still a measure of admissibility.** Even with the Born rule established, the physical meaning of p_i is the degree to which sector i participates in the zero-constraint support, not an ontological "collapse" into a single material reality.

7.5 Guardrails and Summary of Section 7

7.5.1 Guardrails for Section 7

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **No external measurement postulate.** The framework does not assume wavefunction collapse or an external observer triggering probability. Probability is derived internally as the sectorwise reading of admissibility.
2. **Classicality is emergent.** Records become classical only when they are stable, redundant, and zero-preserving (m-robust). There are no primitive classical objects or pointer states.
3. **Decoherence is internal.** Suppression of interference arises from the high burden of cross-sector reconciliation, not from an externally imposed thermodynamic arrow or environmental postulate.

4. **The Born rule is a theorem.** It is derived as the unique measure consistent with Z-envariance and sectorwise zero-decomposition, not postulated at the foundation as an axiom of quantum mechanics.
5. **Probability is still a measure of admissibility.** Even with the Born rule established, the physical meaning of p_i is the degree to which sector i participates in the zero-constraint support, not an ontological "collapse" into a single material reality.
6. **Z-envariance requires decoherence.** The compensating environmental transformations assume that the branches are stably separated. In the absence of decoherence, well-defined branches do not yet exist, and the probability calculus does not apply.

7.5.2 Summary of Section 7

Section 7 reconstructed classicality and probability entirely from the relational constraint framework, without importing standard quantum measurement axioms:

1. **Stable Record Separation** (Theorem 7.1) showed that high cross-sector burden suppresses interference, causing macroscopic record sectors to become effectively orthogonal. Decoherence is driven by the cost of maintaining constraint compatibility.
2. **Redundant Record Robustness** (Theorem 7.2) proved that classicality arises from m-robust encoding across environmental fragments. Objectivity is a structural property of redundant encoding, not a primitive feature of macroscopic objects.
3. **Normalized Record Weights** (Theorem 7.3) established the formal mathematical structure of probability ($p_i \ge 0, \sum p_i = 1$) from the evaluation of positive zero-preserving effects.
4. **The Born Rule as Sectorwise Zero-Decomposition** (Theorem 7.4) proved that the master constraint zero condition distributes over record sectors according to the weights p_i. Probability is not added on top of the theory; it is how the zero-constraint is partitioned.
5. **Zero-Envariance (Z-Envariance)** (Theorem 7.5) forced the branch measure to be proportional to squared amplitude, rigorously deriving the Born rule ($p_i = |c_i|^2$) as the unique normalized measure consistent with the symmetries of the physical state.

The conceptual chain of this section is:
$$ \boxed{ \text{master constraint } = 0 \quad \Rightarrow \quad \text{sector-weighted zero support} \quad \Rightarrow \quad \text{Born weights} \quad \Rightarrow \quad \text{operational probabilities}. } $$

This completes the probability and classicality layer. The framework now possesses an emergent spacetime, populated by interacting matter, governed by gravity, and capable of explaining definite observational outcomes. The final step is to apply this relational architecture to the universe as a whole.

Section 8 — Cosmology and the Open Universe

8.0 Purpose of Cosmology in RCF

8.0.1 Transition from Local Structure to Global Dynamics

Section 7 established how classicality and probability emerge from the zero-preserving relational substrate. The framework now possesses the full local architecture: physical states, observables, emergent space, emergent time, fields, particles, interactions, gravity, black holes, and records.

However, the universe is not merely a local laboratory. It is a global, historical structure. The framework must explain how the relational substrate unfolds on cosmological scales, why the universe expands, and how large-scale phenomena like cosmic acceleration and clustering emerge without being assumed as primitive background properties.

The purpose of this section is to reconstruct cosmology as the large-scale dynamics of the open relational extension.

> **Cosmic expansion is the large-scale unfolding of causal-relational structure under burden-weighted open extension, stabilized by records and expressed in emergent correlation geometry.**

8.0.2 The Effective Expansion Driver

In standard cosmology, expansion is driven by a scale factor $a(t)$ evolving according to a Friedmann equation sourced by a stress-energy tensor. In RCF, there is no primitive scale factor and no background time t.

Instead, the driver of global change is the **Open Extension Principle** (Postulate 1B). The universe extends because it admits zero-preserving continuations. The *rate* and *morphology* of this extension are conditioned by how quickly relational structures can reconcile with causal dependency (the Causal Reconciliation Principle) and how burden is distributed.

Definition 8.1 (Effective Expansion Driver).
The **effective expansion driver** is the coarse-grained rate at which an open causal chain extends through relationally conditioned structure, redistributes burden, and separates admissible correlation channels across the emergent large-scale geometry. It is a function of:
$$ \boxed{ \mathcal{E}_{\mathrm{eff}} = \mathcal{E}(\rho_{\mathrm{mode}}, \rho_{\mathrm{int}}, \rho_{\mathrm{rel}}, D_\omega, \mathrm{Ext}). } $$

Interpretation.
The causal chain governs open extension (what continuations are allowed). Relational structure conditions how that chain is realized (pace, accessibility, morphology). Expansion is driven by the causal chain, but its pace is shaped by burden redistribution and relational reconciliation.

> **The causal chain governs open extension; relational structure conditions its realization.**

8.0.3 What This Section Establishes

This section establishes the following:

1. **Cosmic Expansion as Relational Growth:** The scale factor of the universe is identified with the coarse-grained growth of correlation distance d_ω. Expansion is not motion through a void; it is the stretching of the correlation web.
2. **Dark-Energy-Like Expansion:** The expansive channel of open extension, dominated by mode burden release, mimics dark-energy-like behaviour.
3. **Dark-Matter-Like Binding:** The binding channel, dominated by relational burden (ρ_{rel}), mimics dark-matter-like halo response.
4. **Friedmann-like Dynamics:** Under assumptions of large-scale homogeneity and isotropy, the effective dynamics reduces to a Friedmann-like expansion law governed by the component-selective burden source.

8.0.4 Guardrails for This Section

1. **Expansion is not primitive motion.** It is the growth of correlation distance under open extension.
2. **The scale factor is derived.** $a(\tau)$ is reconstructed from the attenuation of relational weights, not assumed as a background function.
3. **Dark-sector claims are phenomenological targets.** ρ_{rel} as dark matter and open extension as dark energy are compelling structural analogues, but require model-specific matching to be elevated to theorems.
4. **Friedmann dynamics is conditional.** The reduction to a Friedmann-like law assumes large-scale homogeneity and isotropy, which are coarse-grained emergent symmetries, not primitive axioms.

8.1 The Effective Expansion Driver

8.1.1 Open Extension as the Cosmological Spine

In Section 1, the Open Extension Principle (Postulate 1B) stated that a physical universe is an extendible zero-preserving partial order. The extension label s was strictly distinguished from physical time.

In the cosmological regime, we must understand *why* the universe extends and *what* drives the pace of that extension. The universe does not extend because it is "programmed" to do so by a background differential equation. It extends because maintaining zero-closure under finite causal speed (Theorem 1.7) requires the relational network to continually open up new admissible pathways to distribute constraint corrections.

The **Causal Reconciliation Principle** (Section 1.6) states that constraint satisfaction cannot happen instantaneously. The universe uses admissible relational channels to distribute constraint correction across the causal chain. Open extension exists to preserve the zero constraints under finite propagation speed.

> **The causal chain governs open extension; relational structure conditions its realization.**

8.1.2 The Role of the Three Burden Sources in Cosmology

The three burden channels introduced in Section 3.1 (and formalized tensorially in Section 5.1) play distinct, mechanically identifiable roles in the cosmological engine. The total burden density is:
$$ \rho_B(x) = \rho_{\mathrm{mode}}(x) + \rho_{\mathrm{int}}(x) + \rho_{\mathrm{rel}}(x). $$

Their specific cosmological functions are:

1. **Mode Burden (ρ_{mode}):** Generated by the existence and occupation of stable field or particle-like modes. This is the structural analogue of ordinary visible matter and radiation. As the universe extends, mode burden is redistributed. The release or "unlocking" of mode burden from highly compressed early states drives the expansive, radiative side of open extension.
2. **Interaction Burden (ρ_{int}):** Arises from non-additivity in composition. It supports internal structural complexity and hierarchical binding (e.g., galaxies, clusters). It governs how structures form and maintain coherence against the backdrop of global expansion.
3. **Relational Burden (ρ_{rel}):** Generated by the correlation network itself. It measures the burden required to maintain relational consistency across vast configurations. Unlike the first two, it does not belong to a single localised object; it belongs to the topology of the correlation web. It acts as a network-level load that preserves coherent structure, generating binding responses that counteract pure expansion.

8.1.3 The Cosmological Duality

The interaction of these three channels creates a natural structural duality within the framework, partitioning the cosmological dynamics into two competing regimes.

Proposition 8.1 (Cosmological Duality).
The effective dynamics of the open extension is governed by the competition between an expansive sector and a binding sector:
$$ \boxed{ \text{Expansive Sector} \sim \mathcal{E}_{\mathrm{eff}}(\rho_{\mathrm{mode}}, \Delta \rho_{\mathrm{int}}) } $$
$$ \boxed{ \text{Binding Sector} \sim \rho_{\mathrm{rel}} } $$

Interpretation:
* **Expansive Sector:** Dominated by mode burden and the release of interaction burden. When modes unlock or interactions resolve, burden is released, accelerating the open extension and increasing the correlation distance between regions. This mimics dark-energy-like acceleration and standard radiative expansion.
* **Binding Sector:** Dominated by relational burden. As matter clusters, the correlation web becomes denser, increasing ρ_{rel}. This network-level load generates a halo-like geometric response (via the burden tensor $\Theta^{(B)}_{\mu\nu}$) that counteracts the expansive drive, binding structures together.

> **Dark energy is the expansive, unlocking side of open extension; dark matter is the binding, structure-preserving side that keeps relational systems from dissolving into pure spread.**

8.1.4 Guardrails for This Subsection

1. **Expansion is not a property of a background metric.** The expansive sector is driven by the release of microscopic reconciliation load, which accelerates the admissible extension rate.
2. **Binding is not a primitive force.** The binding sector is simply the geometric response to high relational network density, formalized via the burden tensor.
3. **The duality is structural.** The framework does not postulate a cosmological constant or cold dark matter particles. It postulates that the constraint algebra has three channels, and their macroscopic geometric effects naturally partition into expansion and binding.

8.2 Cosmic Expansion as Relational Growth

8.2.1 Redefining the Scale Factor

In standard cosmology, cosmic expansion is modelled by a time-dependent scale factor $a(t)$ stretching a pre-existing spatial metric. The spatial distance between two comoving observers is $d(t) = a(t) \, d_0$, where d_0 is a fixed coordinate distance.

In the Relational Constraint Framework, spatial distance is not primitive. It is reconstructed from the state-dependent correlation kernel K_ω (Section 2). Therefore, cosmic expansion cannot be modelled as the stretching of a background manifold. Instead, it must be understood as the **growth of correlation distance** among admissible event-regions.

If A and B represent two coarse-grained relational regions (e.g., two galaxy clusters), their emergent spatial separation is:
$$ D_\omega(A, B) = -\ell_0 \log K_\omega(A, B). $$
As the universe undergoes open extension (Section 1.5), the physical state ω evolves to ω_s at extension stage s. Cosmic expansion corresponds to the attenuation of the correlation weights $K_{\omega_s}(A, B)$ between distant regions over extension history.

8.2.2 Definition — Relational Scale Factor

To formalize this, we define the scale factor as the ratio of correlation distances across successive extension stages.

Definition 8.2 (Relational Scale Factor).
Let A and B be two coarse-grained relational regions. Let $D_{\omega_s}(A, B)$ be their correlation distance at extension stage s. The **relational scale factor** $a(\tau)$ is defined as the ratio of the large-scale average correlation distance at stage $s+1$ to that at stage s:

$$ \boxed{ a(\tau_{s+1}) \sim \frac{\langle D_{\omega_{s+1}}(A, B) \rangle}{\langle D_{\omega_s}(A, B) \rangle}. } $$

Here, τ_s is the burden-weighted proper time (Definition 3.4) accumulated along the universe's primary causal backbone up to stage s. The angle brackets $\langle \dots \rangle$ denote a coarse-grained spatial average over large, comoving relational regions.

8.2.3 Theorem — Relational Expansion Under Attenuation

We now prove that if open extension attenuates the relational weights between distant regions, the scale factor necessarily grows.

Theorem 8.1 (Relational Expansion Under Attenuation).
Let $(\mathcal{E}, \prec, K_\omega)$ be the relational structure of the universe. Suppose that under open extension from stage s to $s+1$, the persistent relational weights attenuate such that:
$$ W_{s+1}(i, j) = q_{ij} W_s(i, j) \qquad \text{with} \qquad 0 < q_{ij} \leq 1, $$
where $W_s(i,j) = K_{\omega_s}(i,j)$ is the correlation kernel between regions i and j.

Then the correlation distance grows as:
$$ \boxed{ D_{\omega_{s+1}}(i, j) = D_{\omega_s}(i, j) + \ell_0 \log\left(\frac{1}{q_{ij}}\right). } $$
Consequently, the relational scale factor $a(\tau)$ increases.

Proof.
By Definition 2.4, the correlation distance at stage s is:
$$ D_{\omega_s}(i, j) = -\ell_0 \log W_s(i, j). $$
Substituting the attenuated weight $W_{s+1}(i, j) = q_{ij} W_s(i, j)$ into the definition for stage $s+1$:
$$ D_{\omega_{s+1}}(i, j) = -\ell_0 \log \left(q_{ij} W_s(i, j) \right). $$
Using the logarithm property $\log(ab) = \log(a) + \log(b)$:
$$ D_{\omega_{s+1}}(i, j) = -\ell_0 \log q_{ij} - \ell_0 \log W_s(i, j). $$
Substituting $D_{\omega_s}(i, j)$ back in:
$$ D_{\omega_{s+1}}(i, j) = D_{\omega_s}(i, j) - \ell_0 \log q_{ij}. $$
Since $0 < q_{ij} \leq 1$, we have $\log q_{ij} \leq 0$, which implies $-\ell_0 \log q_{ij} \geq 0$. Rewriting this term:
$$ D_{\omega_{s+1}}(i, j) = D_{\omega_s}(i, j) + \ell_0 \log\left(\frac{1}{q_{ij}}\right). $$
Because $\ell_0 \log(1/q_{ij}) \geq 0$, the distance between any two regions i and j either remains the same (if $q_{ij}=1$) or strictly increases. Averaging over large-scale regions yields a strict increase in the average correlation distance, meaning $a(\tau_{s+1}) > a(\tau_s)$. $\blacksquare$

8.2.4 Interpretation

This theorem provides the structural mechanism for the Big Bang expansion. Expansion is not the motion of galaxies through a pre-existing void. It is the stretching of the relational correlation geometry itself.

As the universe extends, the reconciliation cost (burden) of maintaining perfect correlations across vast causal distances becomes prohibitive. The correlation weights K_ω naturally attenuate, causing the emergent spatial distance D_ω to grow logarithmically.

> **Expansion is the decoherence of the cosmic web.**

8.2.5 Guardrails for This Subsection

1. **Expansion is not primitive motion.** The scale factor $a(\tau)$ is not a function stretching a background manifold; it is a derived ratio of emergent correlation distances.
2. **Comoving coordinates are emergent.** The "regions" A and B are not points in a pre-existing coordinate chart; they are coarse-grained equivalence classes of zero-preserving events.
3. **The rate depends on burden.** The attenuation factor q_{ij} is not arbitrary; it is determined by the Causal Reconciliation Principle and the distribution of constraint burden across the cosmic web.

8.3 Dark-Energy-Like Expansion and Dark-Matter-Like Binding

8.3.1 The Cosmological Duality

The standard ΛCDM model of cosmology requires two distinct unknown components to explain large-scale dynamics: Dark Energy (to drive accelerated expansion) and Dark Matter (to provide anomalous binding and structure formation).

In the Relational Constraint Framework, these two phenomena are not introduced as new primitive particles or a cosmological constant. Instead, they emerge naturally from the component-selective burden tensor (Definition 5.2) and the Causal Reconciliation Principle. The cosmological duality proposed in Section 8.1 provides a structural origin for both:

> **Dark energy is the expansive, unlocking side of open extension; dark matter is the binding, structure-preserving side that keeps relational systems from dissolving into pure spread.**

8.3.2 Dark-Energy-Like Expansion

If the large-scale sector is dominated by the expansive channel—where open extension permits rapid growth of correlation distance and mode burden is released—the effective dynamics mimic dark-energy-like acceleration.

As the universe extends, maintaining tight correlations across vast causal distances becomes highly burdensome. The Causal Reconciliation Principle dictates that the network will shed this burden by attenuating the correlation weights K_ω (Theorem 8.1). If the mode burden ρ_{mode} is distributed such that it continuously drives this attenuation faster than the relational network can re-establish correlations, the scale factor $a(\tau)$ accelerates.

This expansive drive is mathematically encoded in the active burden tensor $\Theta^{(B)}_{\mu\nu}$ acting as a negative pressure source in the effective ADM metric (Definition 5.3). When the expansive sector dominates, the effective equation of state mimics a cosmological constant.

8.3.3 Dark-Matter-Like Binding via Relational Burden

Standard cosmology posits dark matter to explain why galaxies and clusters bind more strongly than visible matter allows. RCF provides a structural candidate for this: **relational burden** (ρ_{rel}).

Suppose matter (mode burden) clusters. As it clusters, the correlation web connecting these structures becomes denser. The framework predicts that $\rho_{\text{rel}} \propto \text{correlation density}$, not merely localized matter density.

Proposition 8.2 (Properties of Relational Burden Matching Dark Matter).
The relational burden channel ρ_{rel} possesses four structural properties that naturally mimic the phenomenology of dark matter:

1. **Correlation with Matter Clustering:** ρ_{rel} grows where ρ_{mode} grows, because matter clustering densifies the correlation network. Thus, it appears gravitationally linked to visible galaxies.
2. **Non-Luminous:** ρ_{rel} does not belong to individual localised objects (particles); it belongs to the correlation web topology. Therefore, it does not couple to the electromagnetic sector directly and is structurally non-luminous.
3. **Halo Extension:** The correlation kernel K_ω decays smoothly with distance. Therefore, the support of ρ_{rel} extends strictly beyond the localized support of ρ_{mode}. Mathematically, the relational radius $R_{\rho_{\text{rel}}} > R_{\rho_{\text{mode}}}$, naturally producing halo-like structures without requiring exotic extended mass profiles.
4. **Gravitational Response:** ρ_{rel} contributes directly to the burden tensor $\Theta^{(B)}_{\mu\nu}$ (Definition 5.2), which sources the effective spatial metric $h^{(B)}_{ij}$. Thus, it generates an extra geometric binding response.

> **Relational burden is a candidate effective source for dark-matter-like phenomenology, without introducing exotic particles.**

8.3.4 Theorem — Halo Geometric Response

Theorem 8.2 (Halo Geometric Response).
Let a localized matter cluster be defined by a high density of mode burden ρ_{mode} within a relational radius R_M. The surrounding correlation network develops a relational burden density $\rho_{\text{rel}}(x)$ that decays smoothly with distance $D_\omega(x, R_M)$ but remains non-zero for $D_\omega > R_M$.

This ρ_{rel} sources an effective spatial curvature via the burden-to-metric dictionary (Definition 5.3) that mimics an extended dark matter halo.

Proof Sketch.
Mode burden generates highly localized correlations. The relational burden is the cost of maintaining the broader network consistency. By Definition 2.2, the kernel K_ω is strictly positive (though potentially small) even outside the immediate matter cluster. Therefore, the relational stress $\Pi^{(B)}_{ij}$, and its corresponding burden density ρ_{rel}, has a support that strictly contains the support of ρ_{mode}.

When mapped to the effective spatial metric $h^{(B)}_{ij} = h^{(0)}_{ij} + \eta \Pi^{(B)}_{ij} + \zeta \rho_B h^{(0)}_{ij}$, this extended ρ_{rel} sources an extended metric well (a halo) that alters the geodesics of test particles outside the visible matter radius R_M. $\blacksquare$

8.3.5 Guardrails for This Subsection

1. **$\rho_{\text{rel}} \neq \rho_{\text{DM}}$ as a theorem.** The framework cannot yet claim exact quantitative matching to galaxy rotation curves or cluster lensing profiles. It claims that ρ_{rel} is a structural candidate whose properties naturally align with dark matter phenomenology.
2. **Dark energy is not a cosmological constant.** The expansive sector is driven by the dynamics of open extension and burden release. While it mimics a cosmological constant in the continuum limit, its microscopic origin is the reconciliation cost of the causal web.
3. **Model dependence is required.** To elevate these structural analogues to rigorous predictions, a concrete model of the correlation kernel K_ω for large-scale structure must be specified.

8.4 Friedmann-like Cosmological Emergence

8.4.1 The Conditional Friedmann Theorem

In standard cosmology, the dynamics of the universe's scale factor $a(t)$ are governed by the Friedmann equations, derived from Einstein's field equations under the assumption of large-scale homogeneity and isotropy (the Cosmological Principle).

The Relational Constraint Framework does not assume the Friedmann equations or a background Robertson-Walker metric. However, if the emergent effective geometry (Section 5) satisfies the conditions of smoothness, Lorentz-compatibility, and large-scale coarse-grained symmetries, the framework must recover an analogous dynamical law.

Theorem 8.3 (Friedmann-like Cosmology, Conditional Form).
If the emergent large-scale sector is homogeneous, isotropic, and governed by the covariantly conserved active burden tensor $\Theta^{(B)}_{\mu\nu}$ (Theorem 5.2), then the coarse-grained dynamics reduces to a Friedmann-like expansion law:
$$ \boxed{ H_{\mathrm{eff}}^2 = F(\rho_B, \Pi_B, D_\omega, \ldots), } $$

where $H_{\mathrm{eff}} = \frac{\dot{a}}{a}$ is the effective Hubble parameter defined with respect to the relational scale factor $a(\tau)$ (Definition 8.2).

Proof Sketch.
1. **Metric Form:** Homogeneity and isotropy eliminate preferred spatial directions and locations in the large-scale limit. This forces the effective ADM metric (Theorem 5.3) to reduce to the Robertson-Walker form, characterized solely by the scale factor $a(\tau)$ and a constant spatial curvature k.
2. **Einstein-like Closure:** By Theorem 5.4, the geometric response is governed by $G_{\mu\nu} + \Lambda_B g_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$.
3. **Dynamical Equations:** Substituting the Robertson-Walker metric into the Einstein tensor yields the standard Hamiltonian and evolution constraints (the Friedmann equations).
4. **Source Mapping:** The active burden tensor $\Theta^{(B)}_{\mu\nu}$ provides the effective energy density ρ_B and effective pressure p_B (derived from the trace of the spatial burden stress $\Pi^{(B)}_{ij}$). The covariant conservation $\nabla_B^\mu \Theta^{(B)}_{\mu\nu} = 0$ (Theorem 5.2) provides the continuity equation linking the change in ρ_B to the expansion work done by p_B.

Therefore, the dynamics of $a(\tau)$ are completely determined by the total active burden density and its effective equation of state. $\blacksquare$

8.4.2 The Effective Source Law

While a full rigorous derivation of the exact Friedmann equations directly from the microscopic algebraic master constraint remains an open target, the framework posits an effective, conditional source law for cosmological modelling.

In the weak-field, low-velocity limit, the temporal component of the Einstein-like closure yields a Poisson-like equation for the effective gravitational potential Φ. Within RCF, this potential is sourced by the three component channels of constraint burden:

Proposition 8.3 (Effective Cosmological Source Law).
In the weak-field continuum limit, the effective potential Φ obeys:
$$ \boxed{ \nabla^2 \Phi = 4\pi G \left(\rho_{\mathrm{mode}} + \rho_{\mathrm{int}} + \beta \rho_{\mathrm{rel}} \right), } $$
where:
* ρ_{mode} acts as standard visible matter density.
* ρ_{int} acts as binding energy (e.g., massive astrophysical objects).
* β is a scaling coefficient relating relational network density to effective gravitational mass.
* G is the effective gravitational constant (related to κ_B).

8.4.3 Interpretation

This theorem and source law complete the cosmological bridge. The universe expands because the correlation web attenuates under open extension (Theorem 8.1). The *rate* of this expansion is governed by a Friedmann-like equation, where the source driving the expansion is not primitive stress-energy, but the total constraint burden.

Furthermore, the component decomposition of the burden provides a natural structural mapping to the dark sector. The expansive release of mode burden drives acceleration (dark energy analogue), while the extended halo of relational burden provides the extra geometric binding (dark matter analogue).

> **Cosmic dynamics is the large-scale equation of state of zero-reconciliation.**

8.4.4 Guardrails for This Subsection

1. **Homogeneity and isotropy are emergent.** The Friedmann reduction assumes that, at sufficient coarse-graining, the correlation geometry becomes statistically homogeneous. This is an observational and structural assumption, not a primitive axiom.
2. **The exact Hubble parameter is open.** The precise functional form of $F(\rho_B, \Pi_B, \dots)$ in Theorem 8.3 requires a completed derivation of the effective coupling κ_B from the microscopic scales ($\epsilon, \ell_0, \lambda$).
3. **The continuity equation is burden conservation.** The standard cosmological continuity equation ($\dot{\rho} + 3H(\rho + p) = 0$) is mathematically enforced by the active-source conservation theorem (Theorem 5.2).

8.5 Guardrails and Summary of Section 8

8.5.1 Guardrails for Section 8

To prevent overinterpretation of the structures established in this section, the following guardrails must be strictly observed:

1. **Expansion is not primitive motion.** Cosmic expansion is not the movement of galaxies through a pre-existing void. It is the growth of correlation distance D_ω under the open extension of the relational network.
2. **The scale factor is derived.** The scale factor $a(\tau)$ is reconstructed from the attenuation of relational weights K_ω. It is not an assumed background function stretching a primitive manifold.
3. **Dark-sector claims are phenomenological targets.** Identifying the expansive sector with dark energy and relational burden (ρ_{rel}) with dark matter are compelling structural analogues. However, they remain quarantined speculations until rigorous, model-specific quantitative matching to rotation curves, lensing, and CMB spectra is achieved.
4. **Friedmann dynamics is conditional.** The reduction to a Friedmann-like expansion law assumes large-scale homogeneity and isotropy. These are not primitive axioms of the framework, but coarse-grained emergent symmetries of the correlation geometry.

8.5.2 Summary of Section 8

Section 8 reconstructed cosmology as the large-scale dynamics of the open relational extension:

1. **The Effective Expansion Driver** was defined as the coupled action of open extension, causal reconciliation, and burden redistribution. The causal chain governs the possibility of extension, while burden distribution conditions its pace.
2. **Cosmic Expansion** was formalized as the growth of correlation distance under weight attenuation. The relational scale factor $a(\tau)$ was rigorously defined, and it was proven (Theorem 8.1) that attenuation of the kernel K_ω necessarily forces the scale factor to increase.
3. **Dark-Energy-Like Expansion** was identified with the expansive channel of open extension. The release of mode burden and the prohibitive cost of maintaining long-range correlations drive an accelerated attenuation of the web, mimicking a cosmological constant.
4. **Dark-Matter-Like Binding** was identified with relational burden ρ_{rel}. It was shown (Proposition 8.2, Theorem 8.2) that the structural properties of relational burden naturally mimic dark matter phenomenology, including its correlation with visible matter, non-luminosity, and extended halo geometry.
5. **Friedmann-like Dynamics** was conditionally derived (Theorem 8.3). Under assumptions of emergent homogeneity and isotropy, the effective burden tensor maps to a Robertson-Walker metric, and the dynamics of the scale factor reduce to a Friedmann-like expansion law sourced by the total active constraint burden.

The conceptual chain of this section is:
$$ \boxed{ \text{causal admissibility} \to \text{relational correlation geometry} \to \text{burden-weighted clocking} \to \text{open extension} \to \text{cosmic expansion}. } $$

This completes the cosmological layer. The framework has now reconstructed the major pillars of physical reality—space, time, matter, gravity, black holes, probability, and cosmic expansion—entirely from a primitive algebraic foundation of relational constraints. The final step is to perform a rigorous audit of these claims, clearly separating proven theorems from open targets and quarantined speculations.

Section 9 — Open Targets and Quarantined Phenomenology

9.0 Purpose of the Audit and Consolidation Pass

9.0.1 The Final Synthesis Block

The preceding sections have constructed the complete architectural spine of the Relational Constraint Framework (RCF). We began with a primitive unital $(*)$-algebra $\mathcal{A}$ and a constraint ideal $\mathcal{I}_C$, derived the physical state space and emergent Hilbert space, reconstructed causal order and spatial geometry, derived burden-weighted time, recovered fields and particles, established the Einstein-like gravitational closure, reframed black holes as unreconciled flux-suppressed sectors, derived the Born rule from zero-envariance, and reconstructed cosmology as open relational extension.

The framework is now a coherent, mathematically structured research programme. However, the discipline of the framework demands absolute clarity regarding the epistemic status of each claim. A theoretical structure is only as strong as its honesty about its own boundaries.

The purpose of this final section is to perform an audit and consolidation pass. It separates definitions, conditional theorems, open targets, and quarantined speculation. It ensures that the framework says exactly what it proves, and no more.

> **The foundation must say exactly what it proves, and no more.**

9.0.2 The Status Tag System

Throughout the manuscript, claims have been labelled. We now formally consolidate them into four strict categories:

1. **Definition:** A formal object or rule introduced without need for proof.
2. **Conditional Theorem:** A result that follows logically from definitions under explicitly stated assumptions.
3. **Open Theorem Target:** A precise mathematical statement that the framework aims to prove, but which currently relies on unproven assumptions or missing microscopic derivations.
4. **Quarantined Speculation:** A broad physical interpretation or phenomenological claim that is not yet theorem-ready and must be kept out of the core deductive stack.

This section maps the entire framework into these categories and identifies the precise mathematical locks that remain to be picked.

9.1 The Consolidated Claim-Status Map

The structural hierarchy of the Relational Constraint Framework can now be audited layer by layer. This map identifies the exact epistemic status of the core objects, theorems, and assumptions at each stage of the derivation.

Layer 0: The Algebraic Foundation
* **Status:** ✅ Established (Proven / Definition)
* **Objects:** $\mathcal{A}, \mathcal{I}_C, \omega, (\mathscr{H}_\omega, \pi_\omega, \Omega_\omega), \hat{\mathfrak{M}}_\omega, \mathfrak{M}(x)$.
* **Theorems:** Master-zero equivalence (Theorem 0.9), GNS state reconstruction (Theorem 0.5), ideal-null constraint annihilation (Theorem 0.7).
* **Assumptions:** Properness of $\mathcal{I}_C$ (Assumption 0.3), existence of positive physical states (Proposition 0.1), ideal-null physicality for full ideal inclusion (Definition 0.9).
* **Audit Note:** This layer is mathematically rigorous and standard within algebraic quantum field theory, adapted here for purely relational constraints. The distinction between primitive quadratic vanishing and ideal-null physicality is explicitly maintained. Boundedness of GNS operators is flagged as requiring a C^*-upgrade or domain assumptions.

Layer 1: The Zero-Preserving Causal Foundation
* **Status:** ✅ Proven Conditionally

* **Objects:** Zero-preserving observables ($\mathcal{A}_{\mathrm{obs}}$, Definition 1.6), Causal dependency ($\prec$, Definition 1.9), Extension set (Ext, Definition 1.17), Causal speed limit (c_{RCF}, Theorem 1.7).
* **Theorems:** Strong observables preserve kernel (Theorem 1.2), causal dependency is a strict partial order (Theorem 1.6), emergent causal speed limit (Theorem 1.7).
* **Assumptions:** Bounded relational step length ℓ_C (Assumption 1.2, unified with ℓ_0), necessity composition for transitivity (Assumption 1.1).
* **Audit Note:** The observable algebra was redefined using state-preservation (Definition 1.3) because the standard commutator condition $[A, \hat{C}_\alpha] \in \mathcal{I}_C$ is vacuous for two-sided ideals. The causal speed limit relies explicitly on the unified locality scale ℓ_0.

Layer 2: Correlation Geometry and Emergent Space
* **Status:** ✅ Proven Conditionally
* **Objects:** Correlation kernel (K_ω, Definition 2.2), Correlation distance (d_ω, Definition 2.4), Emergent points ($\mathcal{X}_\omega$, Definition 2.5), Emergent direction ($T_A^{\mathrm{em}} \mathcal{X}_\omega$, Definition 2.11).
* **Theorems:** Pseudometricity under multiplicative consistency (Theorem 2.6), metric quotient (Theorem 2.7), approximate triangle inequality (Theorem 2.8), $D=3$ dimensional closure (Theorem 2.10).
* **Assumptions:** Multiplicative correlation consistency (Assumption 2.5), exact/approximate metricity (Assumption 2.6), independence of existence/position/direction inference channels (Assumption 2.7).
* **Audit Note:** The translation from algebraic correlation to metric geometry is mathematically rigorous *given* the multiplicative consistency of the kernel. Assumption 2.5 is explicitly postulated, as it does not follow from Cauchy-Schwarz. The $D=3$ closure theorem is structurally proven based on the algebraic types of the inference channels. The $D=4$ projection is labeled as Conjecture 2.2.

Layer 3: Emergent Time and Burden-Clock Geometry
* **Status:** ✅ Proven Conditionally
* **Objects:** Constraint burden (B_R, Definition 3.1), Local clock factor (α, Definition 3.3), Burden-weighted proper time ($\tau[\gamma]$, Definition 3.4), Burden lapse (α_B, Definition 3.7).
* **Theorems:** Burden suppresses clock rate (Theorem 3.3), non-uniform burden induces non-uniform proper time (Theorem 3.6), scalar burden-to-lapse bridge (Definition 3.8).
* **Assumptions:** Rational form of $\alpha(B) = 1/(1+\lambda B)$ (model choice).
* **Audit Note:** Burden is rigorously defined as a commutator expectation value (Definition 3.1), anchoring it entirely in primitive algebraic data. The time-dilation mechanism is mathematically sound. The specific rational form of the clock factor is an ansatz, but the monotonic suppression theorem holds for any positive decreasing function.

Layer 4: Fields, Particles, and Interactions
* **Status:** ⚠ Toy-level / Conditional
* **Objects:** Fields (ϕ, Definition 4.1), Correlation Laplacian (Δ_ω, Definition 4.3), Particle-like excitations (Definition 4.5), Interaction functional ($\mathcal{I}$, Definition 4.8).
* **Theorems:** Particle identity (Theorem 4.1), interaction non-additivity (Theorem 4.2).

* **Assumptions:** Continuum limit of Δ_ω, identification of stable eigenmodes with physical matter.
* **Audit Note:** The definitions are structurally consistent, but the correlation Laplacian is currently a toy operator. A rigorous derivation linking algebraic constraints to specific field equations (e.g., Klein-Gordon, Dirac) is missing.

Layer 5: Gravity as Geometry of Constraint Burden
* **Status:** ⚠ Conditional Reduction Theorem
* **Objects:** Burden tensor ($\Theta^{(B)}_{\mu\nu}$, Definition 5.2), Effective ADM metric (ds_B^2, Theorem 5.3).
* **Theorems:** Active-source conservation (Theorem 5.2), metric boundedness at saturation (Theorem 5.4), Einstein-like closure (Theorem 5.5).
* **Assumptions:** Smooth 4D Lorentzian continuum limit, Lovelock theorem applicability, constitutive burden-metric dictionary (Definition 5.3).
* **Audit Note:** The recovery of Einstein's field equations is a standard uniqueness theorem (Lovelock) *if* a smooth 4D metric is assumed. The gap is deriving the smooth metric and the exact coupling κ_B from the microscopic algebra. Theorem 5.4 structurally prevents physical singularities by bounding the metric determinant via ℓ_0.

Layer 6: Black Holes as Unreconciled Relational Sectors
* **Status:** ⚠ Open Theorem Targets
* **Objects:** Pressure saturation ($\rho_B^{\max}$, Definition 6.2), Burden compression ratio ($\mathcal{S}_R$, Definition 6.3), Dual horizon ratios (χ_R, ε_R, Definition 6.6), Local radial-tangential decomposition (Definition 6.7).
* **Theorems:** Dual horizon limits (Theorem 6.2), local dimensional suppression (Proposition 6.3, Theorem 6.3), boundary recovery (Theorem 6.4), entropy-area scaling (Theorem 6.5), Hawking-like emission (Theorem 6.7).
* **Assumptions:** Injectivity of boundary record map, exponential scaling of boundary microstates.
* **Audit Note:** The structural reframing of black holes as unreconciled sectors is highly consistent. The pressure saturation mechanism (derived from the ℓ_0-floor) replaces the classical singularity with a geometric projection. The local rank-collapse proposition provides a derived mechanism for holography.

Layer 7: Records, Classicality, and the Born Rule
* **Status:** ⚠ Open Theorem Targets
* **Objects:** Record sectors ($\mathscr{H}_i^\omega$, Definition 7.1), Redundancy degree ($\mathcal{R}_\eta$, Definition 7.3), Z-Envariance (Definition 7.7).
* **Theorems:** Stable record separation (Theorem 7.1), normalized record weights (Theorem 7.3), Born rule as sectorwise zero-decomposition (Theorem 7.5).
* **Assumptions:** Z-envariance symmetry closure, quantitative decoherence threshold ($B_{\mathrm{cross}} \gg B_{\mathrm{intra}}$).
* **Audit Note:** The derivation of the Born rule via Z-envariance is mathematically rigorous *given* the existence of decohered branches. The gap is rigorously proving that high burden inevitably and universally causes this decoherence in all physical regimes.

Layer 8: Cosmology and the Open Universe
* **Status:** 🔴 Quarantined Phenomenology

* **Objects:** Effective expansion driver ($\mathcal{E}_{\mathrm{eff}}$, Definition 8.1), Relational scale factor ($a(\tau)$, Definition 8.2).
* **Theorems:** Relational expansion under attenuation (Theorem 8.1), Friedmann-like cosmology (Theorem 8.3, conditional).
* **Assumptions:** Large-scale homogeneity and isotropy.
* **Audit Note:** The mapping of RCF dynamics to Friedmann cosmology is highly conditional. The identification of relational burden with dark matter and open extension with dark energy are compelling structural analogues but lack quantitative derivation.

9.2 Priority Open Targets

The framework has crossed major theoretical thresholds, but it is not yet a completed theory of physics. It is a structured programme with a short, sharply named list of remaining mathematical locks.

To elevate RCF from a structural framework to a predictive physical theory, the following targets must be rigorously closed:

1. Microscopic Derivation of Multiplicative Correlation Consistency
Status: Open Target (Layer 2)
The Problem: The emergent spatial metric relies on Assumption 2.5: $K_\omega(A,C) \geq K_\omega(A,B)K_\omega(B,C)$. Currently, this is assumed as a macroscopic regularity to guarantee the triangle inequality (Theorem 2.6).
The Goal: This inequality must be proven directly from the dynamics of the constraint algebra $\mathcal{I}_C$ acting on the physical state ω. A likely route involves leveraging the Kadison-Schwarz inequality for positive linear functionals on $(*)$-algebras, demonstrating that state positivity inherently enforces multiplicative consistency for the normalized overlap kernel.

2. Smooth Continuum Emergence
Status: Open Target (Layers 2, 5, 8)
The Problem: The framework currently operates on discrete or coarse-grained correlation graphs. The use of differential geometry, the ADM decomposition (Theorem 5.3), and the Friedmann metric (Theorem 8.3) assume a smooth continuum limit.
The Goal: A rigorous theorem is needed to show that the emergent metric space $(\mathcal{X}_\omega, D_\omega)$ converges to a smooth 4D Lorentzian manifold in the large-scale limit. This requires proving that the correlation Laplacian Δ_ω converges to the standard Laplace-Beltrami operator, justifying the calculus used in the gravitational and cosmological layers.

3. Derivation of the Effective Gravitational Coupling κ_B
Status: Open Target (Layer 5)
The Problem: The Einstein-like closure (Theorem 5.5) forces the *form* of the geometric equation ($G_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$), but κ_B is not computed.

The Goal: The effective coupling κ_B (and the burden compactness constant γ) must be derived from the framework's microscopic constants ($\epsilon, \ell_0, \lambda$).

4. Exact Matter-Burden Proportionality
Status: Open Target (Layers 4, 5)
The Problem: The burden tensor $\Theta^{(B)}_{\mu\nu}$ is structurally stress-energy-like, and burden has been rigorously defined via commutators (Definition 3.1).
The Goal: Prove that the stable field modes of Section 4 generate a burden response strictly proportional to their physical stress-energy $T_{\mu\nu}$. If proven, the Einstein-like closure automatically becomes a derivation of standard General Relativity, as the burden tensor would be mathematically identified with the physical stress-energy tensor.

5. Formal Proof of the $D=3$ Inference Independence
Status: Open Target (Layer 2)
The Problem: The $D=3$ closure theorem (Theorem 2.10) relies on Assumption 2.7 (Spatial Inference Independence).
The Goal: Provide a formal algebraic proof that existence, position, and direction cannot collapse in $D<3$ and cannot redundantly overlap in $D>3$ without generating a positive closure defect ($\Xi_C > 0$) that violates the master constraint. This would elevate the $D=3$ selection from a conditional theorem to an absolute theorem.

6. Universal Validity of the Decoherence Threshold
Status: Open Target (Layer 7)
The Problem: The Born rule derivation (Theorem 7.5) relies on the condition $B_{\mathrm{cross}} \gg B_{\mathrm{intra}}$ to guarantee decohered branches.
The Goal: Prove that for any macroscopic physical state, this burden ratio inevitably scales to infinity in the coarse-grained limit, universally guaranteeing the existence of decohered branches required for Z-envariance.

9.3 Quarantined Phenomenology

The following physical claims are structurally supported by the framework as natural analogues or effective descriptions. However, they must remain quarantined from the core theorem stack until the priority open targets (Section 9.2) are closed and model-specific calculations are performed.

These are not wild speculations; they are the expected macroscopic shadows of the framework's microscopic structures. But rigor demands they be labeled as unproven phenomenological targets.

9.3.1 Dark Matter as Relational Burden (ρ_{rel})
The framework predicts that matter clustering increases correlation density, generating a non-local, non-luminous burden (ρ_{rel}) that sources extra geometry. As shown in Proposition 8.2 and Theorem 8.2, this naturally mimics dark matter halos, including their extension beyond visible matter and their gravitational response.

* **Why it is quarantined:** RCF cannot yet claim $\rho_{\mathrm{rel}} = \rho_{\mathrm{DM}}$ as a theorem. It is a candidate effective source. Elevating it requires deriving the exact scaling coefficient β (Proposition 8.3) and demonstrating quantitative matching to galaxy rotation curves and cluster lensing profiles.

9.3.2 Dark Energy as Open Extension
The expansive channel of open extension, driven by burden release and the prohibitive cost of maintaining long-range correlations, mimics dark-energy-like acceleration. In the continuum limit, it acts as an effective cosmological constant.
* **Why it is quarantined:** Identifying this RCF mechanism with the physical cosmological constant Λ requires deriving its exact magnitude and equation of state from the microscopic reconciliation rate (ϵ, ℓ_0) and the cosmic burden distribution. Until then, it remains a structural analogue.

9.3.3 Inflation Replacement (Primordial Common Closure)
RCF suggests that large-scale cosmic uniformity arises because spatially distant regions share common causal-constraint ancestry ($C_0 \prec A, B$) in the pre-geometric order (Section 1.4). This renders late-time causal contact (and thus traditional cosmic inflation) unnecessary for explaining the horizon problem.
* **Why it is quarantined:** While structurally compelling, this hypothesis remains quarantined until the framework can derive the exact scale-invariant perturbation spectrum of the Cosmic Microwave Background from primordial closure fluctuations. A structural alternative to the horizon problem is not yet a quantitative replacement for inflationary dynamics.

9.3.4 Standard Model Gauge Group Emergence
The framework defines interactions as non-additive constraint burden (Section 4.3) and gauge structure as relational comparison rules. This provides a general algebraic mechanism for forces.
* **Why it is quarantined:** Deriving the specific gauge groups ($SU(3) \times SU(2) \times U(1)$), fermionic statistics, chirality, and the specific particle spectrum of the Standard Model from the primitive constraint algebra is strictly quarantined as future work. The framework currently derives the *existence* of interactions, not the specific Standard Model.

9.4 Guardrails for the Final Synthesis

To ensure the Relational Constraint Framework remains mathematically robust and epistemically honest as it evolves, the following final guardrails must be strictly observed by both the author and future researchers. These rules protect the core deductive stack from contamination by unproven physical analogies.

1. **No Overclaiming Identity.**
Structural analogy is not mathematical identity. The fact that the burden tensor $\Theta^{(B)}_{\mu\nu}$ mimics the stress-energy tensor $T_{\mu\nu}$, or that relational burden mimics dark matter, does not mean they are proven to be exactly those physical quantities. Analogies guide research; theorems conclude it.

2. **No Smuggling Microscopic Assumptions.**
The open targets listed in Section 9.2 must be closed using *only* the primitive objects of the framework ($\mathcal{A}, \mathcal{I}_C, \omega$) and their rigorously derived consequences. One cannot invoke standard quantum field theory or general relativity to bridge a gap in the relational algebra, as that would defeat the purpose of the foundational hierarchy.

3. **Honesty About the Continuum Limit.**
The transition from discrete relational structure to a smooth manifold is one of the hardest problems in theoretical physics. The framework must acknowledge this gap rather than assuming it away. The use of differential geometry (Lovelock's theorem, ADM decomposition) is conditional on this limit being valid.

4. **Physicality is Constraint Compatibility.**
No matter how complex the emergent structures become—spacetime, black holes, particles, quantum probabilities—the foundational definition of physicality remains $\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0$. If a proposed structure violates this, it is unphysical, regardless of its geometric or dynamical elegance.

5. **Quarantined Speculation is Not Failure.**
The quarantined phenomenology of Section 9.3 (dark matter, dark energy, inflation replacement) represents the most exciting physical predictions of the framework. However, they must remain quarantined until the mathematical locks are picked. Presenting them as proven results would compromise the integrity of the entire manuscript.

9.5 Summary and Manuscript Closure

The Relational Constraint Framework has been formally established. It began with the claim that physical reality is not a collection of objects in a pre-existing background, but a constrained relational structure whose admissible configurations are those in which inconsistency vanishes.

From this algebraic foundation, the manuscript systematically reconstructed the pillars of physical reality:

* **Space** was reconstructed from correlation profiles. Emergent points arose as equivalence classes of perfectly correlated observables, and dimensionality was locked at $D=3$ by the closure requirements of relational inference.
* **Time** was reconstructed from causal depth weighted by constraint burden. The rate of time flow emerged as a local clock factor suppressed by the cost of maintaining admissibility.
* **Matter** was reconstructed as stable, localised zero-preserving modes of the correlation Laplacian. Interactions emerged as the non-additive burden generated when these modes compose.
* **Gravity** was reconstructed as the macroscopic geometric response to burden distribution. An Einstein-like field equation was forced by Lovelock's theorem under the assumption of a smooth continuum limit.

* **Black Holes** were reconstructed as unreconciled, flux-suppressed relational sectors. The classical singularity was replaced by a pressure saturation ceiling imposed by the fundamental length scale ℓ_0, projecting the interior infinity onto a 2D boundary.
* **Probability** was reconstructed as the sectorwise decomposition of the master constraint zero. The Born rule was derived as the unique normalized measure consistent with Z-envariance.
* **Cosmology** was reconstructed as the open extension of the relational network. Cosmic expansion emerged as the growth of correlation distance under the attenuation of relational weights.

The framework is no longer missing its spine or its legs. It is down to the joints where the last bolts of microscopic derivation still need tightening. The remaining work is no longer architectural rescue or conceptual redesign; it is rigorous theorem engineering.

The foundational commitment stands:

$$ \boxed{ \text{A physical structure is an admissible relational structure whose constraint violations vanish in the physical state.} } $$

This concludes the formal manuscript.